SURFING

WACE HUMAN BIOLOGY

Unit 2 Reproduction and Inheritance

• Kerri Humphreys •

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Contents

Introduction		iv
Words to Watch		iv
1	Assumed Knowledge	1
2	Discovering DNA	2
3	The Structure of DNA	5
4	Activity – Making a Model of DNA	7
5	DNA in the Nucleus and Organelles	9
6	Genome, DNA and Bioinformatics	11
7	DNA Replication	12
8	DNA Repairs Itself	14
9	The Beadle and Tatum Experiment	15
10	Introns and Exons	16
11	Gene Expression	17
12	Polypeptide Synthesis – Transcription	18
13	Polypeptide Synthesis – Translation	20
14	Activity – Making a Model of Protein Synthesis	22
15	Protein Structure	24
16	Polar and Non-Polar Amino Acids	26
17	The Role of Proteins in Human Biology	27
18	Phenotype and Epigenetics	29
19	Genetic Profiling and Screening	31
20	The Cell Cycle	32
21	Mitosis and Cytokinesis	34
22	Experiment – Investigating Mitosis	36
23	The Need for Mitosis	37
24	Stem Cells	38
25	Stem Cell Research	40
26	Tumours and Cancers	42
27	Meiosis	43
28	Processes in Meiosis	45
29	Non-Disjunction	47
30	Case Study – Down Syndrome	48
31	Experiment – Investigating Meiosis	50
32	Comparing Meiosis and Mitosis	51
33	Fertilisation	52
34	The Male Reproductive System	53
35	The Female Reproductive System	55
36	Male and Female Hormones	57
37	Spermatogenesis	58
38	Oogenesis	59
39	Menstrual and Ovarian Cycles	60
40	Conception and Pregnancy	61
41	Play – Cell Productions	63
42	Germ Layers and Embryonic Development	67
43	Birth	69
44	After the Birth	71
45	Contraception	72
46	Lifestyle Choices and Foetal Development	74
47	Mutagens	76
48	Sexually Transmitted Diseases	77
49	Reproductive Technologies	79
50	Genetic Screening	81
51	Screening Embryos	82
52	Pap Smears and Breast Screening	84
53	Prostate Cancer	86
54	Genome, Gene and Allele	87
55	Gregor Mendel and His Experiments	88
56	Writing Genotypes	90
57	Probability Models	91
58	Variation in Genotype	92
59	Karyotyping	93
60	Sex Chromosomes	95
61	Polygenic Inheritance	96
62	Pedigrees	98
63	Codominance	100
64	Multiple Alleles	101
65	Morgan and Sex-Linkage	102
66	Human Blood Groups and Rhesus Factor	104
67	Autosomal Genetic Diseases	106
68	Sex-Linked Genetic Diseases	108
69	DNA Profiling	109
70	Technology for DNA Profiling	111
71	Implications of Genetic Testing	113
72	The Human Genome Project	114
Topic Test		115
Answers		121
Index		155

Introduction

Each book in the *Surfing* series contains a summary, with occasional more detailed sections, of all the mandatory parts of the syllabus, along with questions and answers.

All types of questions – multiple choice, short response, structured response and free response – are provided. Questions are written in exam style so that you will become familiar with the concepts of the topic and answering questions in the required way.

Answers to all questions are included.

A topic test at the end of the book contains an extensive set of summary questions. These cover every aspect of the topic, and are useful for revision and exam practice.

Words To Watch

account, account for State reasons for, report on, give an account of, narrate a series of events or transactions.

analyse Interpret data to reach conclusions.

annotate Add brief notes to a diagram or graph.

apply Put to use in a particular situation.

assess Make a judgement about the value of something.

calculate Find a numerical answer.

clarify Make clear or plain.

classify Arrange into classes, groups or categories.

comment Give a judgement based on a given statement or result of a calculation.

compare Estimate, measure or note how things are similar or different.

construct Represent or develop in graphical form.

contrast Show how things are different or opposite.

create Originate or bring into existence.

deduce Reach a conclusion from given information.

define Give the precise meaning of a word, phrase or physical quantity.

demonstrate Show by example.

derive Manipulate a mathematical relationship(s) to give a new equation or relationship.

describe Give a detailed account.

design Produce a plan, simulation or model.

determine Find the only possible answer.

discuss Talk or write about a topic, taking into account different issues or ideas.

distinguish Give differences between two or more different items.

draw Represent by means of pencil lines.

estimate Find an approximate value for an unknown quantity.

evaluate Assess the implications and limitations.

examine Inquire into.

explain Make something clear or easy to understand.

extract Choose relevant and/or appropriate details.

extrapolate Infer from what is known.

hypothesise Suggest an explanation for a group of facts or phenomena.

identify Recognise and name.

interpret Draw meaning from.

investigate Plan, inquire into and draw conclusions about.

justify Support an argument or conclusion.

label Add labels to a diagram.

list Give a sequence of names or other brief answers.

measure Find a value for a quantity.

outline Give a brief account or summary.

plan Use strategies to develop a series of steps or processes.

predict Give an expected result.

propose Put forward a plan or suggestion for consideration or action.

recall Present remembered ideas, facts or experiences.

relate Tell or report about happenings, events or circumstances.

represent Use words, images or symbols to convey meaning.

select Choose in preference to another or others.

sequence Arrange in order.

show Give the steps in a calculation or derivation.

sketch Make a quick, rough drawing of something.

solve Work out the answer to a problem.

state Give a specific name, value or other brief answer.

suggest Put forward an idea for consideration.

summarise Give a brief statement of the main points.

synthesise Combine various elements to make a whole.

1 Assumed Knowledge

QUESTIONS

1. The diagram shows two types of cell division.

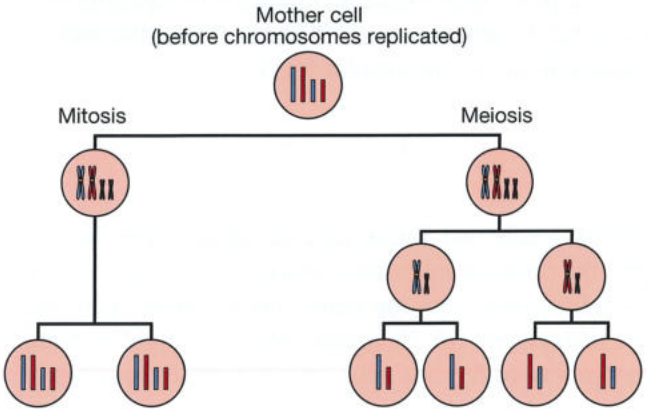


Figure 1.1 Mitosis and meiosis.

   (a) Define mitosis.
   (b) What is meiosis?
   (c) Identify a difference in the daughter cells produced by mitosis and meiosis.
2. How is information transferred when cells reproduce themselves?
3. What does DNA stand for?

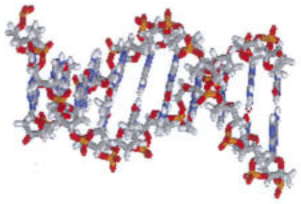


Figure 1.2 DNA.

4. Name the basic unit of DNA.
5. Where is DNA located in cells?
6. Outline the structure of the DNA molecule.
7. What is the relationship between genes and DNA?
8. Explain the advantages of DNA replicating exactly.
9. What is a mutation?
10. What is a pedigree?
11. The diagram shows a body system.

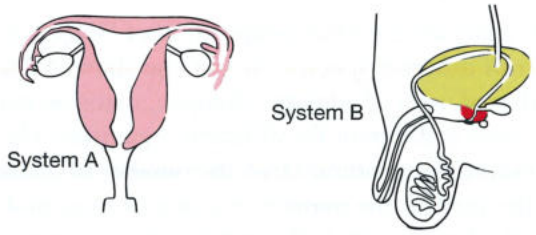


Figure 1.3 Two body systems.

Identify system A and system B.

12. What is biotechnology?
13. What is a somatic cell?
14. Define a stem cell.
15. What is a germ layer?
16. The diagram shows different stages of pregnancy.

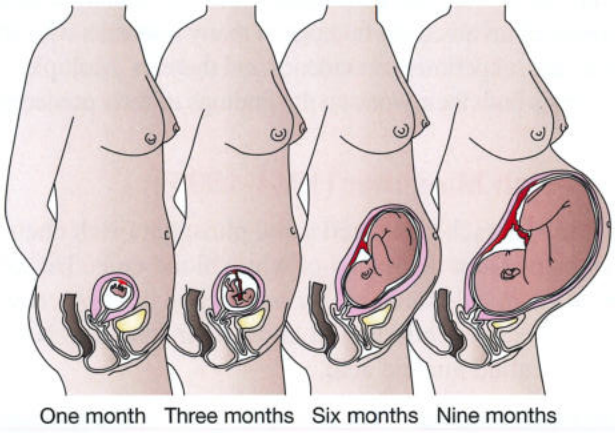


Figure 1.4 Different stages of pregnancy.

How long is human gestation?

17. What is parturition?
18. The diagram shows a type of human cell. What is the name of this cell and what is its function?

Figure 1.5 Type of human cell.

19. Distinguish between an embryo and a foetus.
20. Define a mutagen.
21. Why is Gregor Mendel often referred to as the 'father of genetics'?
22. Identify the factors that determine the features of an organism.
23. Use an example to show how environmental influences can affect the appearance of a person.
24. Use an example to show how genes determine the features of an organism.
25. What is the 'Watson-Crick' model of DNA?
26. Define genome.
27. What is a chromosome?
28. What is meant by genotype?
29. Define fertilisation.
30. Why is it important for gametes to have half the number of chromosomes of the species?
31. The diagram shows the structure of a female reproductive system.

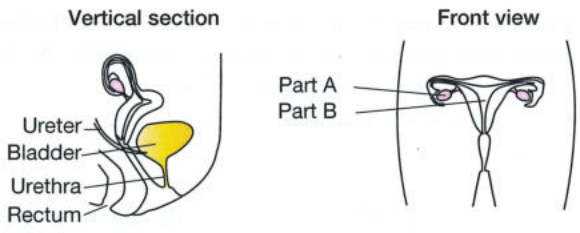


Figure 1.6 Female reproductive system.

State the function of part A and part B.

2 Discovering DNA

The discovery of DNA, its importance in the transmission of information from one generation to the next and its structural properties involved the findings of many scientists who shared their data, experimental evidence and theories. Multiple scientists built their work on the findings of their predecessors.

Friedrich Miescher (1844-1895)

In 1869 Miescher isolated some phosphate-rich chemicals that came from the nuclei of white blood cells. He called these chemicals **nuclein**. He researched the properties of these chemicals but did not know their function. Nuclein is now called nucleic acid.

Figure 2.1 Friedrich Miescher (left), Phoebus Levene (centre) and William Astbury (right).

Frederick Griffith (1879-1941)

In 1928 Griffith found that *Streptococcus pneumoniae* bacteria could change its form in a phenomenon known as **bacterial transformation**. He killed a pathogenic strain of bacteria with heat and then mixed the remains of the dead cells with living non-pathogenic bacteria. Some of the living cells transformed and became pathogenic and their offspring inherited the pathogenicity. The transforming factor that caused this heritable change was unknown.

Phoebus Levene (1869-1940)

In 1909 Levene discovered ribose and in 1929 discovered deoxyribose. He showed that DNA consisted of phosphate-sugar base units that he called **nucleotides** and in his **tetranucleotide hypothesis** he proposed that there were only four nucleotides per molecule of DNA. He also believed that the structure of DNA was too simple to hold the genetic code at a time when many believed that the protein component of chromosomes for inheritance. His work with nucleotides became the basis for the final determination of the structure of DNA.

William Astbury (1898-1961)

Astbury used X-ray diffraction to study the structure of many biological molecules. In 1938 he reported that the structure of DNA repeated every 2.7 nanometres and that the bases were arranged in flat stacks that were 0.34 nanometres apart.

Linus Pauling (1901-1994)

Pauling studied the structure of many biological molecules and found that many proteins had helical shapes. After reviewing the results obtained by Astbury he proposed that DNA was a triple helix. In 1949 he provided proof that a specific mutation in haemoglobin caused sickle cell anaemia and showed that diseases could be understood on a molecular level.

George Beadle (1903-1989) and Edward Tatum (1909-1975)

In 1940 Beadle and Tatum carried out an experiment using the bread mould *Neurospora crassa* showing there was a direct link between genes and enzymes with one gene responsible for the production of a single enzyme. They proposed the one gene-one enzyme hypothesis.

Figure 2.2 Linus Pauling (left),Oswald Avery (centre) and Colin McLeod (right).

Oswald Avery (1877-1955), Colin McLeod (1909-1972) and Maclyn McCarty (1911-2005)

In 1944 Avery-McLeod-McCarty reported that their experiments with pneumococcal bacteria showed that DNA was responsible for bacterial transformation. They had tested all the classes of molecules found in the remains of the dead pathogenic bacteria for transforming ability and found that only DNA could change non-pathogenic bacteria into pathogenic bacteria. Their discovery was met with interest and scepticism. At the time there was debate about the molecule that carried the genetic information with many believing that the protein was the heredity material.

Erwin Chargaff (1905-2002)

Chargaff analysed the base composition of DNA from a number of different species. In 1947 he showed that the number of units of adenine, thymine, cytosine and guanine units is different for different organisms. He also showed that in natural DNA the number of guanine units is the same as the number of cytosine units and the number of adenine units is the same as the number of thymine units. His findings became known as *Chargaff's rules*.

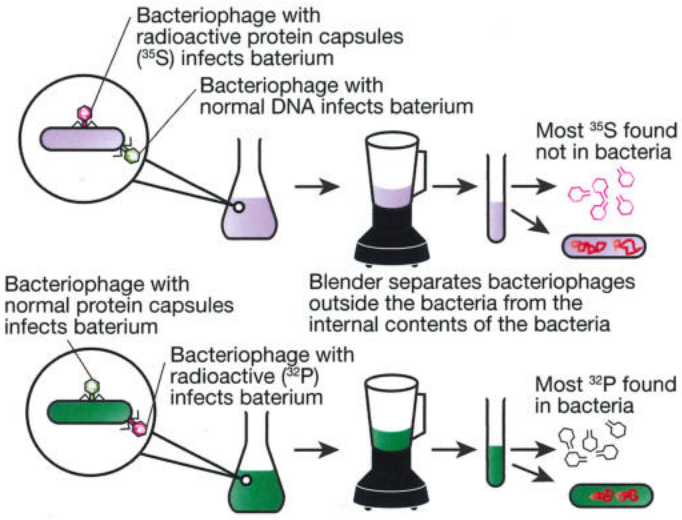


Figure 2.3 Hershey-Chase experiment.

Alfred Hershey (1908-1997) and Martha Chase (1927-2003)

In 1952 Hershey and Chase carried out experiments that confirmed DNA as the genetic material. **Bacteriophages** are viruses that can infect bacteria. Hershey and Chase used bacteriophages (phages) to infect *Escherichia coli* which is a bacteria that lives in the intestines of mammals. They used two sets of bacteriophages, one with proteins labelled with radioactive sulfur (using ^{35}S) and one with DNA radioactively labelled with phosphorus (using ^{32}P) and found that the radioactively labelled DNA entered the bacteria cells while the radioactively labelled proteins did not enter the bacteria cells. When the results of the Hershey-Chase experiment were published most of the scientific community accepted DNA as the heredity material.

Rosalind Franklin (1920-1958)

Franklin took X-ray diffraction images of DNA which showed the helical structure of DNA. Her photograph was used by Watson and Crick to determine the double helix structure of DNA.

(a) (b)

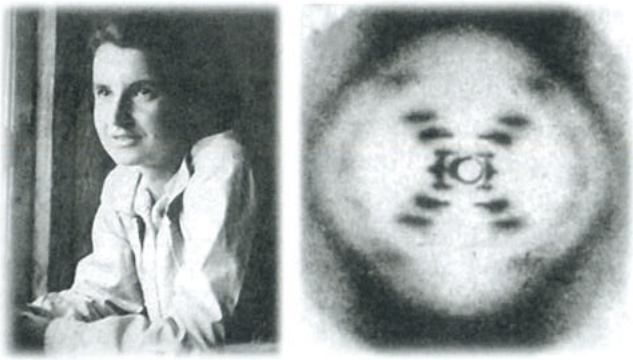


Figure 2.4 (a) Rosalind Franklin. (b) Franklin's X-ray diffraction pattern of DNA.

Maurice Wilkins (1916-2004)

Wilkins produced the first clear X-ray images of DNA with his work showing there was a repeating pattern in the DNA structure. As a friend of both Watson and Crick, Wilkins took a clear X-ray diffraction photograph of DNA taken by Rosalind Franklin and showed it to Watson who then shared the information shown in the photograph with Crick. This information was vital in the final determination of the structure of DNA. Wilkins shared the 1962 Nobel Prize for Physiology or Medicine with Watson and Crick.

James Watson (1928-) and Francis Crick (1916-2004)

In 1953, James Watson and Francis Crick suggested the double helix structure of DNA with two phosphate-sugar strands winding around the outside of nitrogenous bases with the pairs A-T and C-G. Watson and Crick based this structure on X-ray diffraction photographs of DNA taken by Rosalind Franklin. Maurice Wilkins was Franklin's colleague and showed her X-ray photographs to Watson. From the photographs, Watson and Crick were able to work out the distance between atoms, the angles between bonds and the size of atoms. They could then construct a three-dimensional model of DNA.

Figure 2.5 Watson and Crick.

Collaboration and competition

The accumulation of knowledge about the carrier of genetic information occurred over several decades and involved the cumulative work of multiple scientists. The final determination of the structure of DNA relied on experimental evidence and shows how collaboration and effective communication can lead to great scientific discoveries.

The collaboration between molecular biochemist Crick and biochemist Watson produced the final double helix structure. Wilkins collaborated when he showed the X-ray diffraction photographs to Watson. For this joint effort the three men were awarded the 1962 Nobel Prize for Physiology.

When Watson and Crick published their proposal for the double helix structure of DNA they did not credit Franklin with supplying the critical photograph or acknowledge that they did have permission from Franklin to use her work. This goes against scientific ethics and in later interviews Watson admitted that their work would not have been possible without the photograph taken by Franklin. There was also friction between Franklin and her colleague Wilkins. Franklin died of cancer in 1958 and could not be given credit posthumously for the Nobel Prize.

QUESTIONS

1. When were nucleic acids first identified and what were they called at this time?
2. Describe the bacterial transformation observed by Frederick Griffith.
3. (a) What is a nucleotide?
   (b) Who named nucleotides?
   (c) When nucleotides were first named, what was the common belief about the carrier of genetic information?
4. Explain why the work of William Astbury was important in determining the structure of DNA.
5. Outline how the work of Beadle and Tatum increased scientific understanding about genes and DNA.
6. Describe Chargaff's two rules for the composition of DNA.
7. Describe the Hershey-Chase experiment.
8. Outline the contribution made by Rosalind Franklin to our knowledge about the structure of DNA.
9. (a) Briefly describe how the work of James Watson, Francis Crick, Rosalind Franklin and Maurice Wilkins was interlinked and led to the final determination of the structure of DNA.
   (b) Using the discovery of the double helix structure, explain how the role of collaboration and effective communication is necessary in scientific research.
10. In the early 1950s Wilkins and Franklin were working at the University of London, Pauling was working at Caltech and Watson and Crick were working at Cambridge University. How does the work of these three groups show competition between scientists?
11. The diagram shows a bacteriophage.
    (a) What is a bacteriophage?
    (b) Explain how bacteriophages were used to help collect experimental data about the DNA molecule.

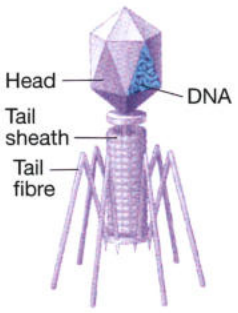


Figure 2.6 Bacteriophage.

12. The diagram shows the structure of a molecule found in living organisms.

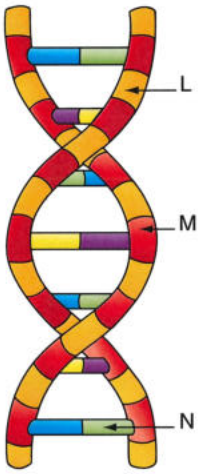


Figure 2.7 Molecule found in living things.

    (a) What is this molecule?
    (b) Identify the parts L, M and N.
    (c) Who discovered the structure of this molecule?
    (d) Discuss three aspects of the structure of this molecule that make it suitable for its function.
13. In 1868 Friedrich Miescher isolated a mixture of DNA and proteins from pus-soaked bandages of wounded soldiers. Later work proved the mixture contained the material of chromosomes. For many years scientists could not work out how information was coded for inheritance. Who proposed, in 1953, the double helix model of DNA?
    (A) Franklin and Wilkins.
    (B) Wilkins and Crick.
    (C) Watson and Crick.
    (D) Watson and Franklin.
14. What was Rosalind Franklin's major contribution to our understanding of the structure of DNA?
    (A) Took X-ray diffraction photographs of DNA.
    (B) Showed there were covalent bonds in nucleic acid polymers.
    (C) Suggested the double helix structure of DNA.
    (D) Deduced there is a purine-pyrimidine pair for nitrogenous bases.
15. Who determined that there were equal number of units of adenine and thymine and equal number of units of cytosine and guanine in DNA?
    (A) Levene.
    (B) Chargaff.
    (C) Avery.
    (D) Franklin.
16. Which of the following was named by Levene?
    (A) Nuclein.
    (B) Nucleotide.
    (C) Bacteriophage.
    (D) Bacterial transformation.

3 The Structure of DNA

Early in the 20th century scientists tried to determine if hereditary information was passed on through proteins or deoxyribose nucleic acid (DNA).

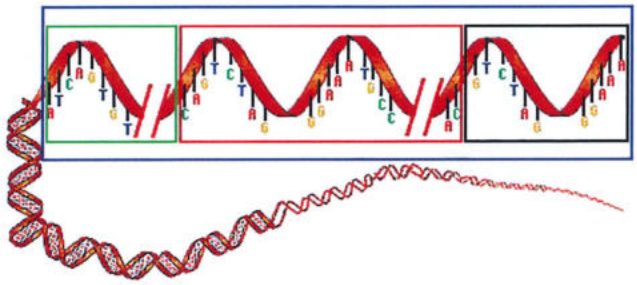


Figure 3.1 DNA.

Eventually enough evidence was gathered to show that DNA carried the genetic material from one generation to the next. In 1944 **Oswald Avery** demonstrated that genes were made of DNA. In 1952 **Rosalind Franklin**, with **Maurice Wilkins**, showed that the DNA molecule was helical using X-ray diffraction crystallography. In 1953 **James Watson** and **Francis Crick** proposed the double helix structure for DNA.

Each chromosome consists of many genes, i.e. each **gene** is a certain length of the DNA molecule. When working out the chemical structure of genes scientists knew that the substance would have to be able to encode a large amount of information, that it must be chemically stable, that it must be able to make an accurate replication of itself, that occasional errors (mutations) could occur and that the substance controls and directs protein synthesis.

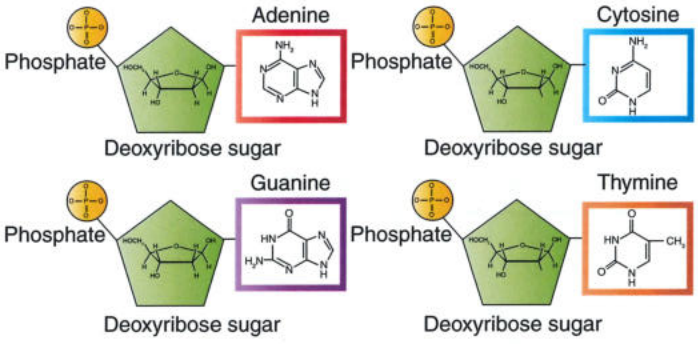


Figure 3.2 Four different nucleotides that make up DNA.

Nucleotides

The basic unit of DNA is the **nucleotide**. A nucleotide consists of a sugar, a phosphate and one of four nitrogenous bases – either adenine (A), guanine (G), cytosine (C) or thymine (T). To form the double helix, the sides of the ladder are made up of alternating sugar and phosphate molecules and the rungs consist of paired nitrogenous bases. Adenine always pairs with thymine and guanine always pairs with cytosine.

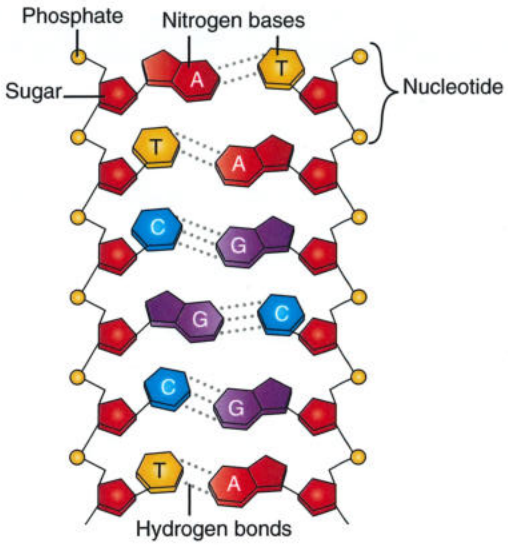


Figure 3.3 Base pairs of DNA.

The double helix is a structure where two spirals coil around each other keeping a constant diameter of the coil. It is a right-handed spiral polymer.

Bonding in DNA

The base pairs on the rungs of the ladder are held together by hydrogen bonds and by van der Waals interactions between the stacked bases. These hydrogen bonds break during DNA replication.

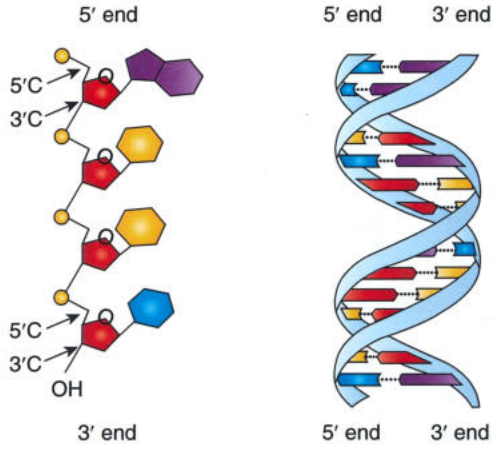


Figure 3.4 Four different nucleotides that make up DNA.

The nucleotides are joined together by covalent bonds between the phosphate group of one nucleotide and the sugar of the next. These bonds are called **phosphodiester linkages** and occur between the –OH group on the 3′ carbon of one nucleotide and the phosphate on the 5′ carbon of the next. This means that the free ends of the DNA molecule are different to each other with one side having a phosphate attached to a 5′ carbon and the other side having a hydroxyl (OH) group on a 3′ carbon. Since the two strands of the double helix run in opposite directions, they are said to be **antiparallel** to each other. The phosphate groups give the nucleic acids their acidic properties.

QUESTIONS

1. What is a gene?
2. What does 'DNA' stand for?
3. Use an example to show how major advances in scientific understanding have directed future biological research.
4. Describe the shape of DNA.
5. What is a nucleotide?
6. Which part of the nucleotide gives the nucleic acid its acidic properties?
7. Purines, e.g. adenine and guanine, are double-ringed nitrogenous bases and pyrimidines, e.g. cytosine and thymine, are single-ringed nitrogenous bases. How do these bases form complementary pairs?
8. List five features of DNA that make it a suitable genetic medium.
9. What type of bond holds the sugar of one nucleotide to the phosphate group of the next nucleotide?
10. What type of bond holds the nitrogenous bases on the 'rungs' of the ladder together?
11. Explain why the arrangement of the arrangement of the two strands of DNA are called antiparallel.
12. Which bonds break during DNA replication?
13. Outline the role of DNA in the transmission of genes from one generation to the next.
14. Explain the significance of DNA replication.
15. The diagram shows a short section of part of a DNA molecule.

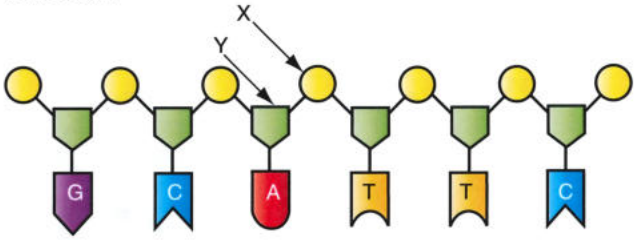


Figure 3.5 Part of a DNA molecule.

(a) Identify parts X and Y.
(b) Draw the complementary strand of DNA.
(c) Using the given section of DNA, describe the process by which this segment controls the production of proteins and/or polypeptides.

16. Since the 1940s it has been known that genes consist of which chemical?
(A) Ribose nucleic acid.
(B) Deoxyribose nucleic acid.
(C) Dinitroadenine acid.
(D) Adenosine triphosphate.

17. Which of the following identifies the basic components of a nucleotide?
(A) Sugar, phosphate, nitrogenous base.
(B) Adenine, thymine, cytosine, guanine.
(C) Glucose, nitrate, phosphate base.
(D) Sugar, amino acid, nitrogenous base.

18. What is shown by an analysis of the DNA molecule?
(A) The 'rungs' of the 'ladder' are made of phosphate-sugar bonds.
(B) The 'sides' of the 'ladder' are made of phosphate-sugar bonds.
(C) Nucleotides of one strand are identical to the nucleotides of the opposite strand.
(D) DNA with 40% thymine will have 20% guanine.

19. A certain chromosome has 20% guanine in its structure. What would you expect to be the amount of adenine present?
(A) 20%
(B) 30%
(C) 60%
(D) 80%

20. The diagram shows a short section of DNA.

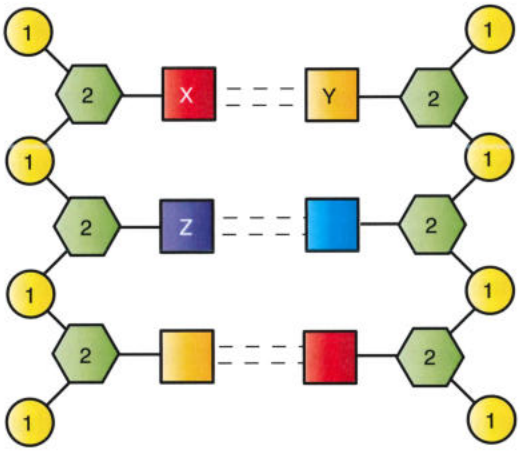


Figure 3.6 Short section of DNA.

If X represents adenine then what could Z represent?
(A) Thymine.
(B) Sugar.
(C) Phosphate.
(D) Cytosine.

21. The figure below represents a single strand of DNA. What is its complementary strand?

A A C T G G

(A) AACTGC
(B) GGTCAA
(C) TTGACC
(D) CCAGTT

22. What type of bonds would you expect to find between the nitrogenous bases on the rungs of the DNA ladder?
(A) Hydrogen bonds.
(B) Covalent bonds.
(C) Phosphodiester linkages.
(D) Peptide bonds.

4 Activity – Making a Model of DNA

In eukaryotic cells a chromosome is one extremely long linear DNA molecule. In prokaryotic cells there is a circular DNA molecule.

There are many ways you can construct a model to show the structure of DNA. Models can be made of paper, cardboard, pipecleaners, plastic sticks, coloured balls or created using multimedia graphics. The model needs to be able to show the triplet code of DNA which is the basic instruction of the genetic code.

(a) (b) (c)

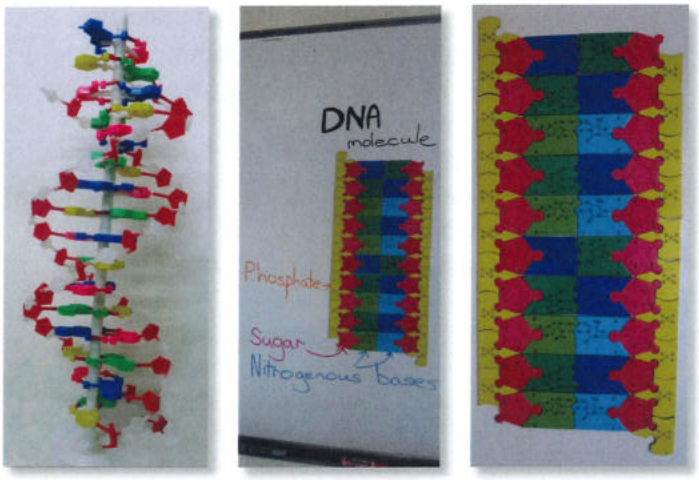


Figure 4.1 (a) Three-dimensional model of the DNA molecule. (b) Building a 2-D model of DNA on a classroom whiteboard using cardboard cutouts that have magnetic strips which stick to the whiteboard. (c) 2-D jigsaw model of DNA using a purchased DNA kit.

The **triplet code** is three bases next to each other in a molecule of DNA and specifies a particular amino acid. Reading the sequence of nucleotides in the triplet code in DNA means that you are reading the sequence of amino acids which will be joined together to form a polypeptide. There are three codes that are 'stop' signals which stop the process of gene expression and the translation of the code into a polypeptide. The DNA codes ATT, ATC and ACT signal 'stop'. The DNA code TAC is the code for the amino acid methionine but also acts as a 'start' signal for translation. Note that these start and stop codes are usually given as the mRNA code (e.g. stop codes UAA, UAG and UGA) as it is the **mRNA code** that is used in definitions and used to identify different amino acids. The mRNA code is complementary to the DNA code with uracil substituting for thymine. In gene expression the mRNA code is read in the cytoplasm when the code is translated to produce a polypeptide.

In gene expression when the DNA code is translated from the mRNA code into a polypeptide it is important to read the three nucleotide bases of an amino acid in the correct **reading frame**. Any alteration of the code will affect the ability to make a particular polypeptide.

Thus it is important when making a model of DNA to know what the code means when it is translated to produce a polypeptide.

3-D models

You can use molecular model kits with sticks and balls and make a model similar to the one built by James Watson and Francis Crick. A model using plastic sticks, coloured spheres and or pipecleaners has the benefit that it gives a three-dimensional vision of DNA to show the double helix structure.

Another way of showing this three-dimensional structure is to make an 'edible' DNA, e.g. using different coloured jellybeans to represent the nitrogenous bases, the phosphate and the sugar. The jellybeans can be joined with toothpicks to represent the bonds holding the nucleotides together.

2-D models

Many models of DNA use jigsaw cut-outs of coloured paper or cardboard and align the pieces to give a two-dimensional view of the ladder and rung structure of DNA. There are DNA jigsaw kits that can be purchased to use in class or you can create your own jigsaw. The jigsaw pieces can also be used to show the process of DNA replication and protein synthesis.

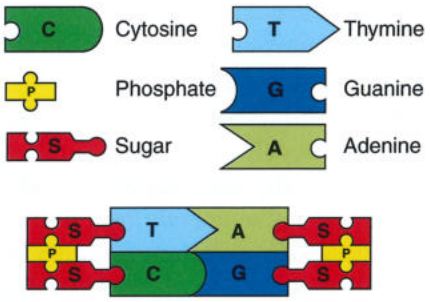


Figure 4.2 DNA jigsaw.

Evaluating a model of DNA

When evaluating a model the model must be consistent with known information. For example, one nucleotide of DNA must contain one phosphate, one deoxyribose sugar and one nitrogenous base, e.g. adenine (A), thymine (T), cytosine (C) or guanine (G). The nitrogenous bases must also be in complementary pairs, e.g. A-T and C-G.

An evaluation also needs to consider the advantages and the limitations of the model. Models of DNA are very useful in showing how a phosphate, a sugar and a nitrogenous base form a nucleotide. They also show how the nucleotides link together to form a ladder structure with the sugar-phosphates forming the sides of the ladder and the nitrogenous bases forming the rungs of the ladder. 3-D models have the added advantage that they can show how the ladder twists to form the double helix. 2-D cutout jigsaw models have the advantage that the two strands of the model can be easily separated to show the process of DNA replication.

Thus making a model of a section of DNA is an important tool in making the structure of DNA easier to understand giving a good visualisation of how each component fits into the molecule. The model can then be used as the bases for further knowledge about DNA, e.g. how DNA can replicate to form two identical double helix molecules and the model shows the genetic coding which is used to manufacture a particular polypeptide and protein.

Since proteins determine the different appearance and type of cells found in the body and also determine the reactions that occur in cells (enzymes are proteins), a section of DNA (gene) determines your appearance and phenotype.

QUESTIONS

1. Compare the structure of the DNA molecule in prokaryotic and eukaryotic cells.
2. Identify some ways in which DNA can be represented in models.
3. What are some features that must be shown in a model of DNA?
4. What is the triplet code?
5. Do all triplet codes specify a particular amino acid? Explain your reasoning.
6. (a) Which code is used in genetics in definitions and to identify amino acids?
   (b) What is the relationship between this code and the DNA code?
7. Draw a simple, fully labelled diagram of a part of a DNA molecule showing eight base pairs.
8. The table shows the percentage of adenine, guanine, cytosine and thymine in a series of DNA samples.

DNA source	Adenine	Guanine	Cytosine	Thymine
Yeast	33.2	16.1	16.9	33.8
Bacteria	26.9	22.7	23.0	27.4
Invertebrate	31.5	18.5	18.6	31.4

   Discuss how this data shows an important feature of the DNA molecule.
9. It has been found that cells from turtles contain, on average DNA with 30% adenine. From this information, how much guanine would you expect to find in the average turtle cell?
10. Give an advantage and a disadvantage of making 3-D models of DNA.
11. Explain why scientists construct models such as the model of DNA constructed by Watson and Crick.
12. DNA is a very regular polymer of nucleotides. How many different basic nucleotides are in DNA and what are they?
13. The diagram represents one nucleotide.
    (a) Identify the components X, Y and Z.
    (b) Draw a diagram to show how two more nucleotides would connect with this nucleotide.

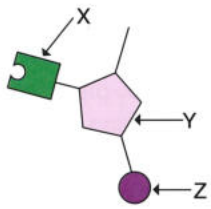


Figure 4.3 One nucleotide.

14. Describe a model you could construct to show the structure of the DNA molecule and assess the uses and limitations of this model.
15. What is a codon?
16. Explain why the coding of the base pairs in a section of DNA is important.
17. The diagram shows a section of DNA.

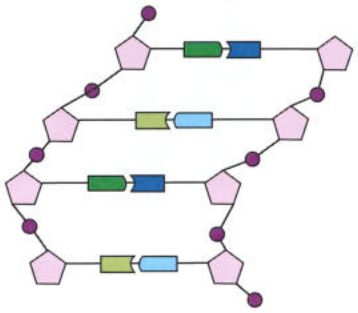


Figure 4.4 A section of DNA.

Copy this diagram and circle one nucleotide.

18. A biology student needed to evaluate a model of DNA made by another student. For the model to be consistent with known information, where should the phosphates be located in the model?
    (A) On the left-hand side of the ladder.
    (B) Alternating with sugars down both sides of the ladder.
    (C) Alternating with nitrogenous bases down both sides of the ladder.
    (D) On the rungs of the ladder.
19. Watson and Crick deduced the double helix structure of DNA and created a model of the DNA molecule. Many scientists went to see this model and their model is on display in the Science Museum, London. What is the benefit of creating models?
    (A) Models help visualisation of scientific concepts.
    (B) Models can be used to make further predictions.
    (C) Models aid communication.
    (D) All of the above.

5 DNA in the Nucleus and Organelles

DNA is deoxyribose nucleic acid and is a double helix molecule that carries genetic information. Some organelles in eukaryotes contain DNA. The genes in these organelles are called **extranuclear genes**. These organelles can reproduce themselves. In humans only mitochondria contain extracellular genes.

There are significant differences between the DNA in the nucleus and the DNA in cellular organelles, although both follow the **universality of the genetic code** with adenine pairing with thymine and cytosine pairing with guanine as complementary pairs. The triplet codes in DNA for amino acids are also the same in both types of DNA.

Nuclear DNA (nDNA)

Nuclear DNA has two sets of each chromosome – one paternal from the sperm and one maternal from the egg. The DNA is linear and has both exons and non-coding introns in its coding.

Chromatin is the name given to the complex of DNA and protein that makes up chromosomes in the nucleus of eukaryotic cells.

Nuclear DNA has a bead-like structure called **nucleosomes** that each consist of a protein core of histones with DNA wound twice around the group of histones. The histone tails extend outward from each nucleosome core.

Histones are important in the structure and packaging of chromatin in the nucleus. Without histones the DNA of human cells (eukaryotic cells) would be very long compared to the size of the cell and histones also aid the ability for certain genes to be expressed in the cell while other genes stay dormant. The histone tails are accessible to the enzymes that begin gene expression and transcription to start protein synthesis.

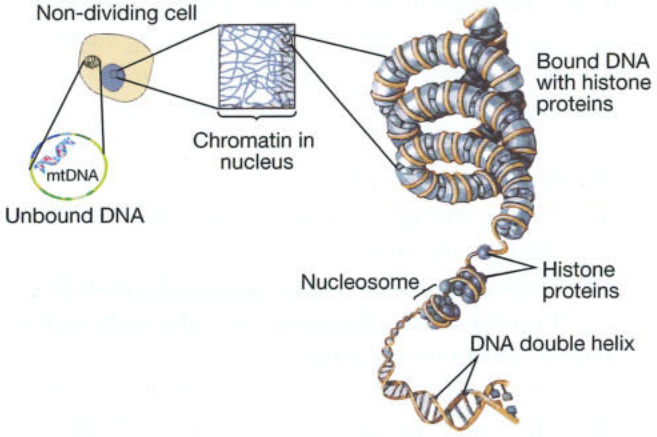


Figure 5.1 Bound DNA in the nucleus and unbound DNA in mitochondria.

Mitochondrial DNA (mtDNA)

Mitochondria are a cellular organelle that have their own DNA (mtDNA). mtDNA is circular, not bound to histone proteins and only encodes 37 genes. It is inherited with the egg which means it is only inherited from the mother.

Unlike nuclear DNA mtDNA does *not* have non-coding exons so that all the code is used in transcription and translation in protein synthesis.

Mitochondria are cylinder shaped organelles that vary in size from 2 to 8 micrometres in length. They are the site of aerobic respiration and generate most of the ATP in human cells. This means the number of mitochondria present in a cell depends on the energy needs of the cell.

Mitochondria have a double membrane with the inner membrane infolded to form **cristae**.

A mitochondrion can divide to form two mitochondria due to their circular DNA which resembles the DNA of prokaryotes. This means a mitochondrion can replicate itself even when the cell is not dividing. The dimensions of mitochondria are also similar to the size of prokaryotes.

Since the inheritance of mtDNA from one generation to the next does not involve both parents it has been used to track ancestry and evolutionary trees. In humans the analysis of mtDNA has been used to track the migration of different human populations following the female line back through many generations.

Endosymbiotic theory

The prokaryotic **endosymbiotic theory** suggests that mitochondria arose as a symbiotic relationship between a free living heterotrophic bacterium with aerobic respiration that entered a larger cell – either a larger prokaryote or an eukaryote and both then existed in a mutualistic partnership.

The endosymbiotic theory also explains the presence of unbound circular DNA in mitochondria in both plants and animals, and also the presence of DNA in plant plastids, e.g. chloroplasts.

The size of mitochondrial ribosomes is also the same size as ribosomes in prokaryotes rather than the same size as the ribosomes in the cytoplasm in human cells.

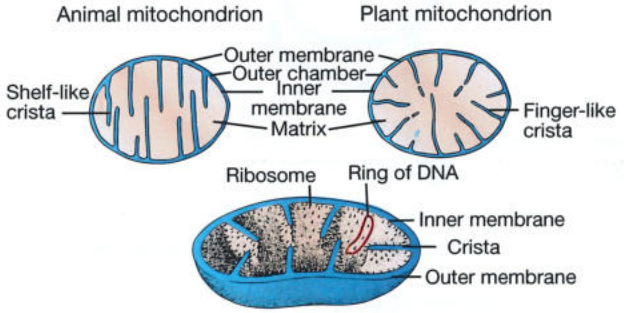


Figure 5.2 Plant and animal mitochondria.

QUESTIONS

1. What are extranuclear genes?
2. What is meant by the universality of the genetic code?
3. How is nuclear DNA inherited?
4. What is chromatin?
5. Describe a nucleosome.
6. Describe a benefit of DNA being bound to histones in nDNA.
7. Describe mitochondria.
8. Describe mitochondrial DNA.
9. How does the endosymbiotic theory apply to mitochondria?
10. Explain how human mtDNA has been used to trace human migration across the Earth.
11. Construct a table to compare mitochondrial DNA and nuclear DNA.
12. The diagram shows one proposal about the origin of mitochondria and chloroplasts.

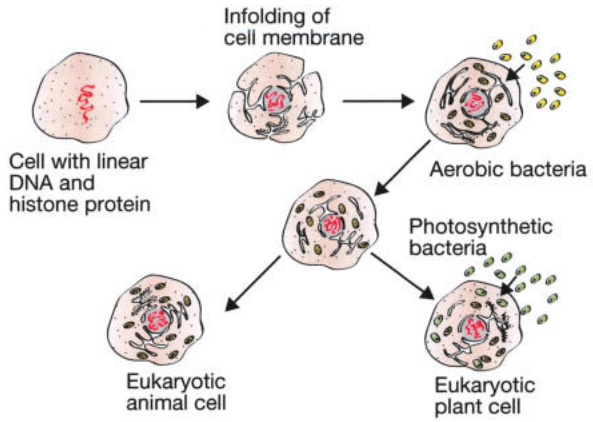


Figure 5.3 Origin of mitochondria and chloroplasts.

(a) Use this diagram to write out the sequence of events that has been proposed to explain the origin of human cells (eukaryotic animal cells).
(b) Explain how mitochondrial DNA and chloroplast DNA supports the endosymbiotic theory.

13. The diagram shows a DNA structure.

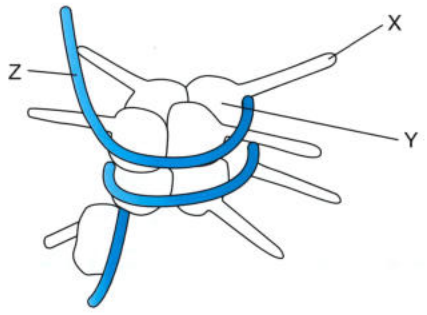


Figure 5.4 Origin of mitochondria and chloroplasts.

(a) Name the structure.
(b) Identify the parts X, Y and Z.

14. Which of the following has its own DNA?
    (A) Golgi apparatus.
    (B) Endoplasmic reticulum.
    (C) Mitochondria.
    (D) Lysosomes.
15. What is the structure of mitochondrial DNA?
    (A) Small circular ring.
    (B) Large circular ring with histone protein.
    (C) Linear chromatin.
    (D) Small circular ring linked with lipids.
16. Which of the following is the best description of chromatin?
    (A) Small protein with large number of negatively charged amino acids found in the nucleus.
    (B) Complex of DNA and protein that makes up chromosomes in the nucleus.
    (C) Circular DNA with few genes found in mitochondria.
    (D) One of two thread-like strands into which a chromosome divides longitudinally during cell division.

Use the diagram for the next TWO questions.

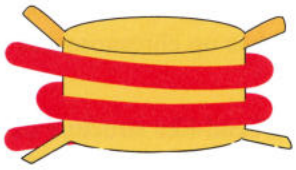


Figure 5.5 Biological structure.

17. What is the name of this structure?
    (A) Histone protein. (B) Chromatid.
    (C) Gene. (D) Nucleosome.
18. Where would you expect to find this structure in human cells?
    (A) Only in the nucleus.
    (B) Only in the mitochondria.
    (C) Only in Golgi apparatus.
    (D) In the nucleus and in mitochondria
19. In the cell cycle histones briefly leave DNA for a very short time. At which of the following steps would you expect this separation to occur?
    (A) During cytokinesis and division of the cytoplasm.
    (B) During DNA replication.
    (C) During metaphase when chromatids line up at the equator of the cell.
    (D) During anaphase when chromatids separate to become chromosomes going to separate poles.
20. What is meant by bound DNA?
    (A) DNA is restricted to being inside the nucleus.
    (B) DNA is in the process of protein synthesis.
    (C) DNA is wound around a protein core.
    (D) A protein is wound around a DNA core.

6 Genome, DNA and Bioinformatics

The genome can be considered to be the sum total of an organism's DNA measured in the number of base pairs contained in a haploid set of chromosomes.

DNA stands for deoxyribose nucleic acid.

The basic unit of DNA is the **nucleotide**. A nucleotide consists of a sugar, a phosphate and one of four nitrogenous bases – either adenine (A), guanine (G), cytosine (C) or thymine (T). To form the double helix, the sides of the ladder are made up of alternating sugar and phosphate molecules and the rungs consist of paired nitrogenous bases. Adenine always pairs with thymine and guanine always pairs with cytosine.

A **genome sequence** is the order of the As, Ts, Cs and Gs in the DNA code. The human genome has over 3 billion of these genetic letters in its coding. A **genome map** is less detailed than a genome sequence identifying short DNA sequences as markers in a genome.

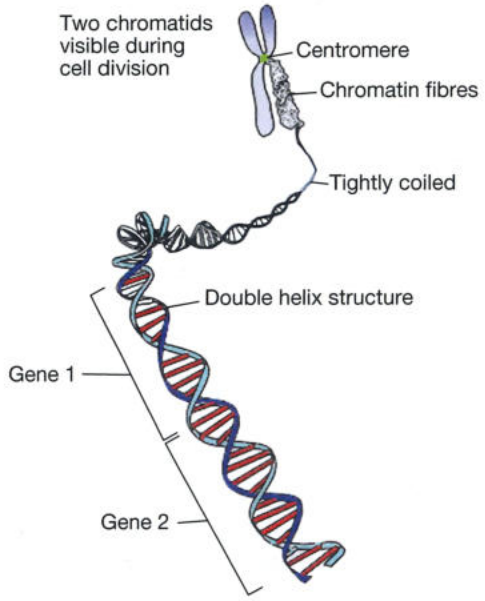


Figure 6.1 DNA.

DNA has a **double helix** structure where two spirals coil around each other keeping a constant diameter of the coil.

Bioinformatics

Bioinformatics uses computers, software and mathematical models to process and integrate biological information from large data sets. Bioinformatics is used to study and analyse genome sequences to help determine the genetics of disease, to understand gene expression and protein synthesis, to determine the effects of mutations, to understand the processes involved in the formation of tumours and cancers and to trace evolutionary changes in DNA and determine evolutionary relationships between different species.

QUESTIONS

1. What is the genome?
2. What does 'DNA' stand for?
3. Describe the shape of DNA.
4. What is a nucleotide?
5. The diagram shows a short section of part of a DNA molecule.

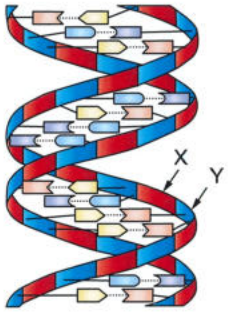


Figure 6.2 Part of a DNA molecule.

Identify parts X and Y.

6. Identify the base pairs of DNA.
7. Distinguish between a genome sequence and a genome map.
8. What is bioinformatics?
9. How is bioinformatics used with genome sdquences?
10. Which of the following identifies the basic components of a nucleotide?
    (A) DNA and protein.
    (B) Adenine, thymine, cytosine, guanine.
    (C) Glucose, nitrate, phosphate base.
    (D) Sugar, phosphate, nitrogenous base.
11. What forms the 'rungs' of the DNA molecule?
    (A) Phosphate and sugar molecules.
    (B) One phosphate and one nitrogenous base.
    (C) Paired nitrogenous bases.
    (D) One phosphate and one sugar.
12. The diagram shows a short section of DNA.

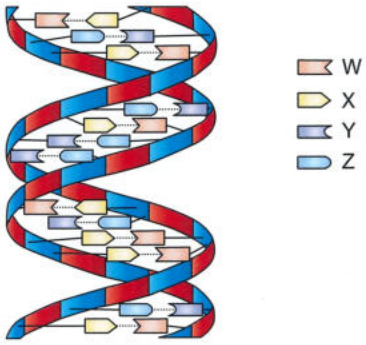


Figure 6.3 Short section of DNA.

What could be represented by W and X?
(A) Thymine and cytosine.
(B) Adenine and thymine.
(C) Phosphate and sugar.
(D) Cytosine and phosphate.

7 DNA Replication

Watson and Crick predicted that DNA replication involved the **semiconservative model** which involves each of the two daughter molecules having one old strand from the parental molecule and one newly made strand.

In the 1950s there were three models proposed for DNA replication, the conservative model where the parental strands reassemble after forming a template that forms a new double helix, the semiconservative model as proposed by Watson and Crick and the dispersive model where both the daughter molecules contain a mixture of old and newly synthesised DNA.

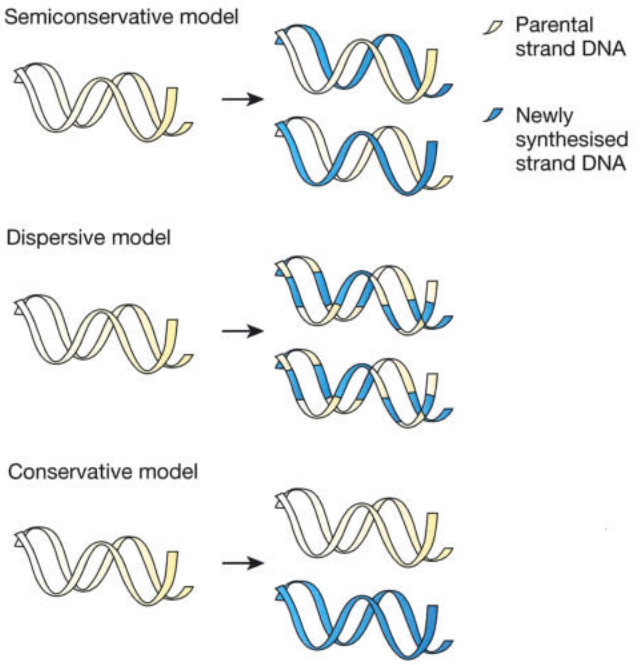


Figure 7.1 Three proposed models for DNA replication.

Meselson-Stahl experiment

In 1958 Matthew Meselson and Franklin Stahl carried out an experiment with the bacteria *Escherichia coli* labelled with radioactive nitrogen (^{15}N) to investigate the three models of DNA replication. They found that the amount of ^{15}N in the each generation after cell division and replication over several generations was consistent with the semiconservative model and confirmed the model of semiconservative replication proposed by Watson and Crick.

Process of replication

During DNA replication the DNA molecule unwinds and the double helix separates into two single strands. Each single strand is a template for the synthesis of a complementary strand so that adenine is opposite thymine and guanine is opposite cytosine.

The process of replication begins at **origin of replication** sites along a chromosome. These sites are sections that coded with a specific sequence of nucleotides. Proteins that start DNA replication detect and attach to these sections causing the two strands to separate to form a replication 'bubble'. There is a **replication fork** at the end of each bubble where the unwinding process begins and the bubble opens up further in both directions. **Helicase enzymes** untwist the double helix at the replication forks and break the bonds between the two strands of DNA. The separated sections in the bubble act as templates for new complementary strands.

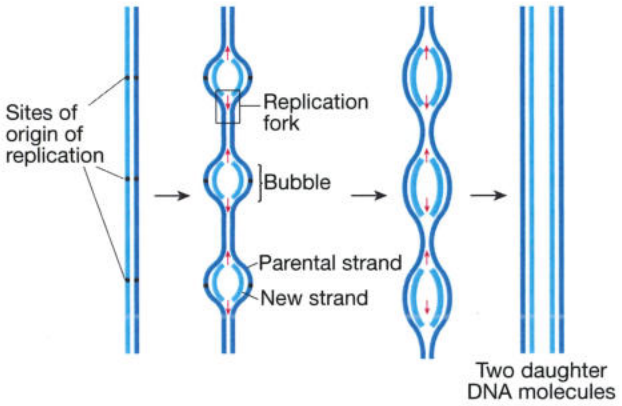


Figure 7.2 DNA replication begins at the sites of origin of replication.

The beginning nucleotide chain is a short RNA chain called a **primer** and is synthesised by the enzyme **RNA primase**. Once the primer has formed **DNA polymerase enzymes** add nucleotides to the end of the existing chain. Nucleotides can only be added to the 3′ end of the growing DNA strand. This means the new strand always grows in a 5′ → 3′ direction.

The **leading strand** is the new DNA strand that is **continuously** synthesised in the 5′ → 3′ direction. The other strand is called the **lagging strand** and is also synthesised in the 5′ → 3′ direction but grows **away** from the replication fork and forms **discontinuously** in a series of segments. These segments are called **Okazaki fragments**. DNA ligase enzymes join the sugar-phosphate backbones of the Okazaki fragments together to complete the new single DNA strand.

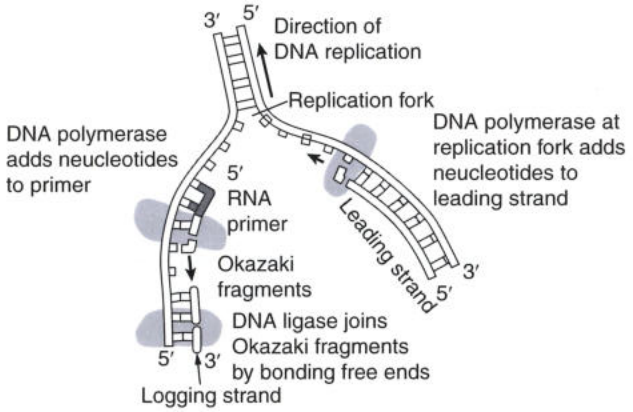


Figure 7.3 DNA replication occurs in 5′ → 3′ direction.

The cytoplasm of the cell contains many free nucleotides which are used during DNA replication. DNA polymerase proofreads each nucleotide against its template removing incorrect nucleotides or catalysing the formation of bonds to join the correct nucleotide to the DNA strand. If DNA polymerase while proofreading misses a set of mismatched nucleotides other repair enzymes can remove and replace the incorrect pair.

The significance of DNA replication is that large amounts of coded information can be copied and passed onto the next generation, providing continuity of a species. The method also allows for some changes and for mutations when the code is incorrectly copied.

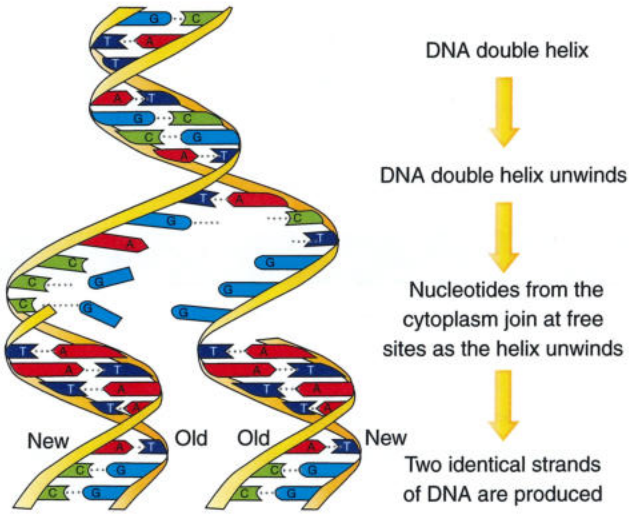


Figure 7.4 DNA replication.

QUESTIONS

1. Which model of DNA replication was proposed by Watson and Crick?
2. Construct a table to summarise the three models of DNA replication that were proposed in the early 1950s.
3. Describe the experiment that eventually showed the how DNA replicated.
4. The diagram shows a simple summary of the process of DNA replication.

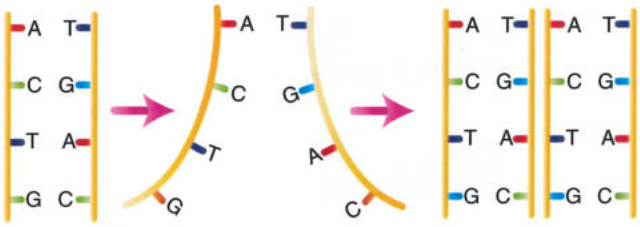


Figure 7.5 Simple summary of DNA replication.

Briefly explain what is happening in this diagram.

5. What is an origin of replication site?
6. What is a replication 'bubble'?
7. Outline the function of helicase enzymes.
8. (a) What is a primer?
   (b) What is the function of RNA primase?
9. Outline the function of DNA polymerase enzymes.
10. Distinguish between the leading strand and the lagging strand in DNA replication.
11. What are Okazaki fragments?
12. Outline the function of DNA ligase enzymes.
13. Explain the significance of DNA replication.
14. The diagram shows several stages of DNA replication.

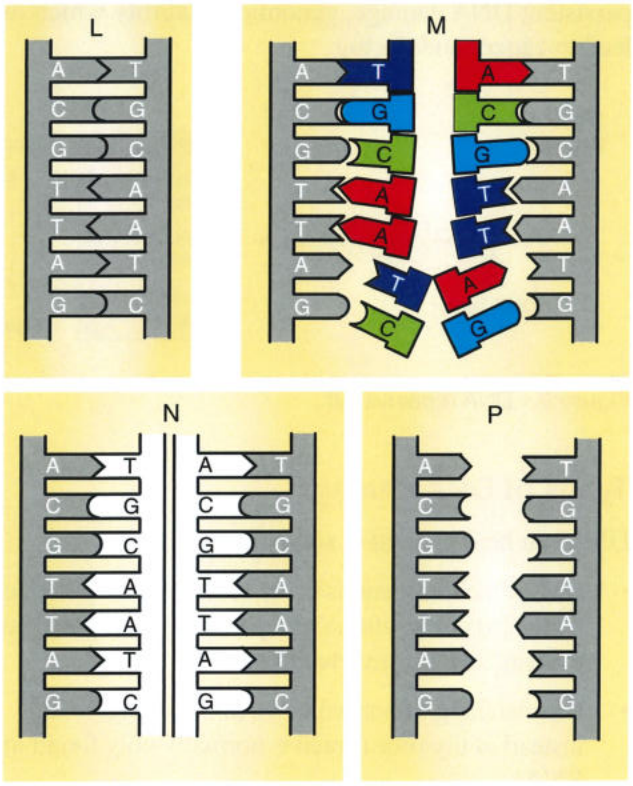


Figure 7.6 Stages of DNA replication.

Identify the correct sequential order.

(A) L, P, N, M
(B) L,M, N, P
(C) N, P, M, L
(D) L, P, M, N

15. The diagram shows a DNA replication bubble.

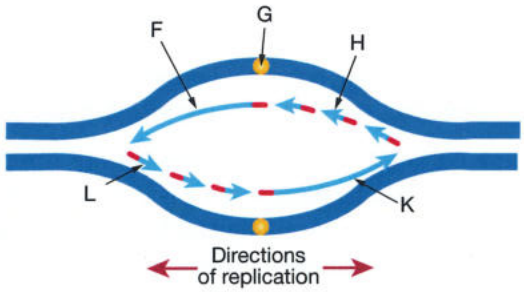


Figure 7.7 DNA replication bubble.

Which arrow(s) point to the leading strand in DNA replication?

(A) F only.
(B) H only.
(C) H and L.
(D) F and K.

8 DNA Repairs Itself

During DNA replication and other cellular metabolic activities DNA can be damaged and errors can occur. DNA that is exposed to chemical mutagens, radiation etc can become damaged and needs to be repaired.

There are several processes that correct damaged DNA. Repair enzymes move along the DNA checking the base pairs to make sure they are correctly matched. If the DNA is not repaired it can lead to mutations, replication errors, persistent DNA damage, genomic instability which in turn lead to cancer and ageing.

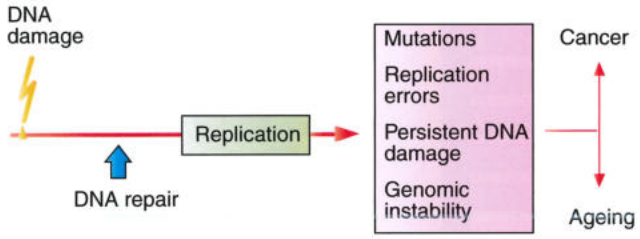


Figure 8.1 DNA repairs itself.

Types of DNA damage

DNA can be damaged in several ways.

- One of the nitrogenous bases in DNA is modified, e.g. by the loss of an amino group (deamination) such as cytosine being converted to uracil.
- Mismatching of normal bases in DNA, e.g. uracil instead of thymine (uracil is normally only found in RNA).
- Crosslinks between DNA bases either on the same DNA strand or between different DNA strands.
- Breaks in the backbone, e.g. X-rays and some chemicals such as peroxide cause breaks in the DNA backbone.

Types of damage repair systems

There are three basic types of damage repair systems: damage reversal, damage removal and damage tolerance.

Damage reversal – where enzymes restore structure without breaking the backbone or if it was a simple break in the backbone of one strand, DNA ligase can rapidly repair the damage. In many bacteria, insects and plants, the enzyme photolyase breaks the bond and reverses DNA damage in the presence of light.

Damage removal – where the damaged section of nucleotides is cut out, removed and replaced, e.g. glycosylase enzymes each remove a specific type of base damage by cutting the base-sugar bond. Uracil glycosylase will remove uracil from DNA. This is often called excision repair.

Damage tolerance – where a method is found to cope with the damage, e.g. if a thymine dimer forms, i.e. a covalent bond forms between two adjacent thymine bases, the thymine dimer will hinder DNA replication. During replication a 'gap' may be left on one strand, although this can be dangerous if the cell divides.

Rate of DNA repair

The rate of DNA repairs depends on several factors, e.g. the age of the cell, the type of cell and the extracellular environment.

Importance of DNA repair

In 2015 the Nobel Prize in Chemistry was awarded to Tomas Linmdahl, Paul Modrich and Aziz Sancar for their work on 'the mechanistic studies for DNA repair' that was carried out decades ago in the 1960s to 1980s. This shows the importance of understanding DNA repair mechanisms.

QUESTIONS

1. Why does DNA need to repair itself?
2. Identify two causes of DNA damage.
3. Outline some ways in which DNA can be damaged.
4. People with the hereditary disease xeroderma pigmentosum (XP) are sunlight sensitive. They get extensive freckle-like lesions on sun exposed skin and are highly susceptible to developing skin cancer. XP cells have defective DNA repair, especially excision repair. Explain how this defect affects DNA.
5. The diagram shows DNA repair caused by a covalent bond (a thymine dimer) forming between adjacent thymine bases.

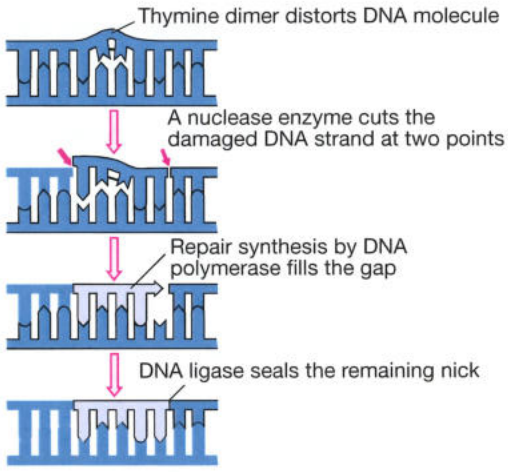


Figure 8.2 DNA repair.

   (a) How does the thymine dimer affect the DNA molecule?
   (b) Identify the enzymes that are involved in this DNA repair process.
6. Describe how DNA has the ability to repair itself.

9 The Beadle and Tatum Experiment

In 1941, George Beadle (1903-1989) and Edward Tatum (1909-1975) proposed the one gene–one enzyme hypothesis after conducting a series of experiments with the bread mould *Neurospora crassa*.

Figure 9.1 Edward Tatum and George Beadle.

Neurospora is a fungus with a haploid genome – only one copy of each gene. They exposed the spores of the fungi to X-rays or UV radiation to produce mutant varieties that had special nutritional needs. For example, one mutant would only grow if the agar plate contained the amino acid arginine. To clearly identify the defect Beadle and Tatum investigated the three steps in the synthesis of arginine.

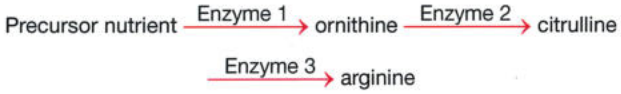


Figure 9.2 Steps in arginine synthesis.

They found three types of mutants for arginine when grown on a minimal medium and the chemical stages which led to arginine synthesis.

Bread mould	Minimal medium (MM)	MM + ornithine	MM + citrulline	MM + arginine
Normal type	Grew	Grew	Grew	Grew
Mutant 1	Did not grow	Grew	Grew	Grew
Mutant 2	Did not grow	Did not grow	Grew	Grew
Mutant 3	Did not grow	Did not grow	Did not grow	Grew

Some of the mutants could not live without a particular vitamin or amino acid in their food source, e.g. normal mould requires only the vitamin biotin, while some mutants also required thiamine or choline.

Investigations into proteins led to further refinement of Beadle and Tatum's theory. Since all genes do not code for enzymes, the one gene-one enzyme hypothesis has been refined to one gene-one polypeptide. This is needed as not all genes code for enzymes. Examples where the code is not for enzymes include some genes that have the code to build structural proteins, e.g. keratin and collagen. Many proteins have more than one polypeptide chain, e.g. haemoglobin has four polypeptide chains of two different types and each of the two chain types is controlled by a different gene.

In 1958 Beadle and Tatum received the Nobel Prize in Physiology or Medicine for their discovery that genes act by regulating definite chemical events. Joshua Lederberg also received the Nobel Prize that year for his work on genetic recombination and the organisation of the genetic material of bacteria.

QUESTIONS

1. What is *Neurospora*?
2. How did Beadle and Tatum produce mutants?
3. Describe the control in Beadle and Tatum's experiment.
4. Analyse the results of this experiment.
5. What conclusion could be drawn from this experiment?
6. How was the Beadle and Tatum hypothesis modified?
7. Explain why the hypothesis was modified.
8. How were Beadle and Tatum rewarded for their work on genes and enzymes?
9. Beadle and Tatum grew each culture on 'minimal medium', i.e. a medium with only those compounds needed for growth and reproduction of normal strains. Suggest why such substances as salts and a carbon source would be included in a minimal medium, but vitamins would not be included.
10. What was the evidence which led to Beadle and Tatum's 'one gene-one protein' hypothesis?
    (A) Interbreeding different pea plants to form hybrids.
    (B) Showing the bacteria, *Escherichia coli*, can be infected by viral DNA.
    (C) Growing bread mould, *Neurospora crassa*, on different agar growth mediums.
    (D) Splicing RNA of the the ciliated protozoan, *Tetrahymena*, using ribozymes.
11. Experimental work by Beadle and Tatum led to the 'one gene-one enzyme' hypothesis. This needed to be altered when it was realised that nearly all enzymes are proteins, but not all proteins are enzymes. It was first suggested that the hypothesis become 'one gene-one protein' – however, what has further evidence suggested that the statement should become?
    (A) One gene-one amino acid.
    (B) One gene-one polypeptide.
    (C) Two genes-one protein.
    (D) Two genes-one lipid.

10 Introns and Exons

Introns are non-coding segments of nucleic acid found in eukaryotic DNA.

Exons are coding segments of nucleic acid found in eukaryotic DNA that are involved in gene expression.

Gene expression

Gene expression is the process where DNA directs protein synthesis. A gene is a section of DNA that has the code for a particular polypeptide. When the gene is expressed the section is 'switched on' and the polypeptide is produced by the cell.

Gene expression and protein synthesis involves two steps – transcription and translation. In eukaryotic cells transcription occurs in the nucleus and translation occurs in the cytoplasm.

Triplet code

The smallest unit of DNA that passes information from gene to polypeptide is the **triplet code**. The triplet code is a sequence of three nucleotides. For example the base triplet TCG is the code for the amino acid cysteine and the position of this code along a DNA strand leads to cysteine being placed at the corresponding position in the polypeptide that is being produced.

In the transcription stage of gene expression the DNA triplet code is read to form a strand of messenger RNA (mRNA). RNA is ribonucleic acid. The base-pairing rules apply to transcription except that uracil (U) replace thymine (T) in RNA. mRNA takes the code from the nucleus to the cytoplasm.

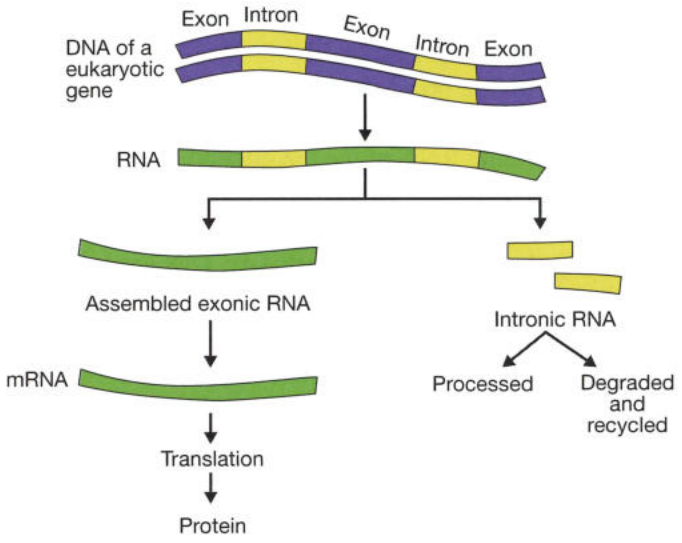


Figure 10.1 Exons and introns.

RNA processing

RNA processing is unique to eukaryotes and involves the modification of the RNA transcript taken from the DNA gene before it leaves the nucleus as mRNA. The initial transcript forms precursor-RNA or **pre-mRNA**. This primary transcript is modified in several ways. Any introns within the gene are removed and the exons are spliced together to form a continuous coding message for the polypeptide. Each end of pre-mRNA is also modified and the 5′ end is capped with the addition of a modified base.

The number of introns within a gene is variable with some genes having introns of greater combined length than the coding exon segments that are made into the functional message.

QUESTIONS

1. Distinguish between introns and exons.
2. What is meant by gene expression?
3. What is a gene?
4. (a) What are the two steps in protein synthesis and gene expression?
   (b) Where do these steps occur in eukaryotic cells?
5. What is the triplet code?
6. How do the base-pairing rules apply to RNA?
7. What is RNA processing?
8. Distinguish between pre-mRNA and mRNA.
9. Outline what happens in RNA processing.
10. The diagram shows a process that occurs in some types of cells.

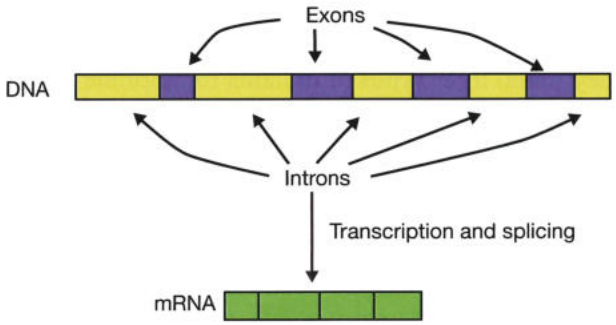


Figure 10.2 Exons and introns.

Where does this process occur?
(A) In prokaryote cells.
(B) In the nucleus of eukaryote cells.
(C) In the cytoplasm of eukaryote cells.
(D) In mitochondria.

11. What is the name of the sections of DNA that are not part of the functional code that is sent to the cytoplasm?
(A) Introns.
(B) Exons.
(C) Extrons.
(D) Intrudons.

11 Gene Expression

Gene expression is the process where DNA directs proteins synthesis. In gene expression a gene is 'turned on' to produce a polypeptide.

Gene expression is a process which converts the DNA information into the functions and structures of a cell. The genes in humans make around 30 000 proteins and each can be classified by their major function, e.g. structural proteins, regulatory proteins, immunological proteins, catalytic proteins (enzymes) and proteins that assist in the transportation of substances and body movements. At any one time only some genes are expressed. They can be switched on and off by various mechanisms.

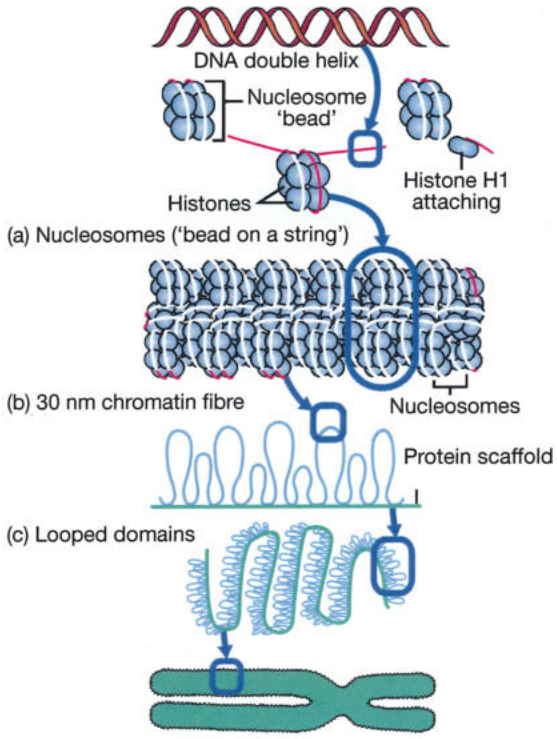


Figure 11.1 Packaging DNA into a nucleus.

Differential gene expression

Differential gene expression occurs when different genes are expressed in different sets of cells with the same genome. This leads to differentiation and cell specialisation.

Controls of gene expression

Each stage of protein synthesis from the DNA code, has some form of control which in turn means it controls gene expression. Most control occurs in the transcription of DNA into RNA. Other stages which control gene expression are the activation or deactivation of the chromosome, control of RNA transport, control of the initiation of translation and protein activity control.

DNA in the nucleus of eukaryotic cells is a tightly wound molecule which allows large amounts of information to be stored in a small area. The parts that are very highly condensed are the genes that are switched off, sometimes even permanently switched off.

Promoters are very important in controlling gene expression. A **promoter** is a regulatory area at the start of a gene that acts as the binding site for RNA polymerase. In eukaryotes, each gene has its own promoter and eukaryotes also have a separate RNA polymerase for each type of RNA, e.g. one enzyme for mRNA-coding genes such as structural proteins and another enzyme for larger rRNAs. If there is a problem with the promoter there will be a problem in starting transcription and thus the polypeptide will not be synthesised and the gene will not be expressed. If the binding strength of the promoter region is increased, there will be a greater quantity of the protein produced.

The presence of some chemical groups can block gene expression, e.g. methyl groups ($-CH_3$) block DNA from expression. While the presence of other groups can allow gene expression to occur more easily, e.g. acetyl groups (–COOH).

Application of gene expression

Because specialised cells, e.g. nerve cells and muscle cells, perform different specialised functions, we can use cDNA hybridisation to compare the gene expression in these cells. This information shows which genes are determining the specialised function and nature of these cells. For example, comparative hybridisation can be used to compare normal cells with cancerous cells to find out where the transcription change(s) occurred to change the cell from normal to cancerous.

Measurement of gene expression

Gene expression (expression profiling) can be indirectly measured using DNA microassay technology. This gives an approximate cellular concentration of the different mRNAs present.

QUESTIONS

1. What is meant by gene expression?
2. What is differential gene expression?
3. At which stage of protein synthesis is there most control of gene expression?
4. Why are some areas of DNA more tightly condensed and packed than other areas of DNA?
5. Define a promoter.
6. Use an example to show how the presence of certain chemicals can affect gene expression.
7. Explain how gene expression can be used to identify a disease.
8. Outline how gene expression can be measured.

12 Polypeptide Synthesis – Transcription

DNA stores genetic information. The sequence of bases determines the sequence of amino acids in protein molecules. The smallest unit that can code for amino acids is a **codon**, or triplet of bases. For example, the DNA code CCA codes for the amino acid glycine. Protein synthesis involves the **transcription** of the code from DNA to messenger ribonucleic acid (mRNA) and then the **translation** of the code from the mRNA to sequencing the amino acids to form a protein.

Scientific knowledge about the process of gene expression and polypeptide synthesis has rapidly increased as new research methods and developments in DNA technology have enabled biologists to investigate the structures and functions of molecules in cells.

The new technologies have shown how signals from outside the cell, e.g. hormones or other signalling molecules can start the process of polypeptide synthesis and gene expression.

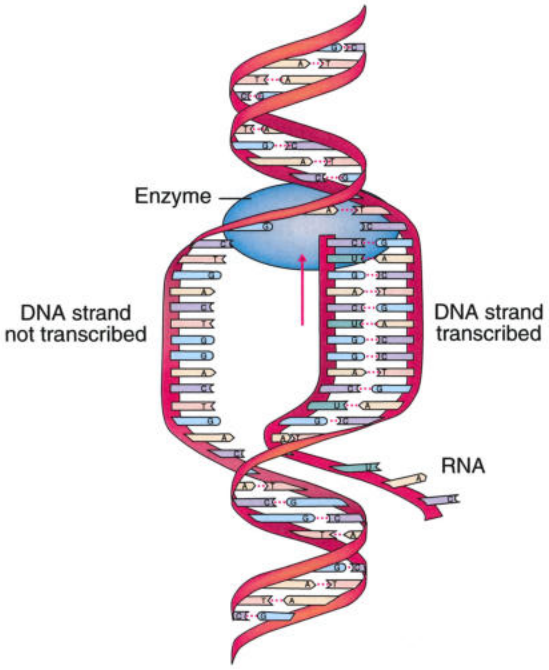


Figure 12.1 Transcription.

Pre-mRNA

The first step of polypeptide synthesis involves the unwinding of the DNA and one side, the **template strand**, is copied to form mRNA. RNA is similar to DNA except that it contains a **ribose** sugar instead of deoxyribose sugar, it is always single stranded and it is complementary with base-pairs to the template strand except that it contains the nitrogenous base **uracil** instead of thymine. The genetic code is just about universal for living organisms from unicellular bacteria to multicellular plants and animals.

In eukaryotic cells the RNA coding transcript from the DNA coding gene is modified in RNA processing before the functional code leaves the nucleus. The initial transcript forms **pre-mRNA** (precursor messenger ribonucleic acid).

Prokaryotic cells do not have a nucleus so when transcription begins translation will start before transcription is completed. There is no RNA processing of bacterial mRNA before translation begins.

Transcription occurs in the 5′ → 3′ direction from the template strand and starts at a specific **promoter** base sequence of nucleotides that signal the beginning of the gene. Different genes have slightly different promoter sequences which means that different genes can be transcribed at the same time. The enzyme **RNA polymerase** causes the two strands of DNA to separate and then links the RNA nucleotides together to form mRNA.

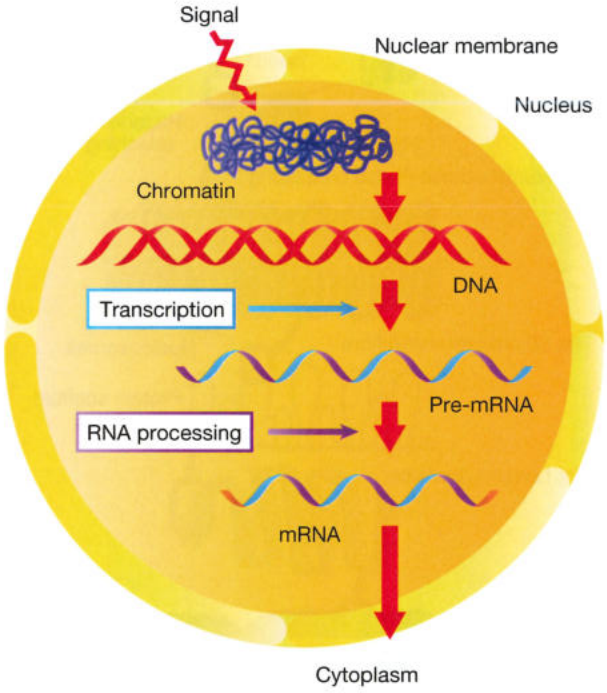


Figure 12.2 The first steps in gene expression.

RNA processing

RNA processing occurs in the nucleus of eukaryotes and involves several actions. Each end of the pre-mRNA is modified in a particular way to help protect the mRNA from being degraded by enzymes and to aid the movement of the functional mRNA to the cytoplasm.

The 5′ end of the mRNA chain is **capped** and at the 3′ end an enzyme adds a **poly-A tail** (polyadenylated tail). A poly-A tail is the addition of 50 to 250 adenine nucleotides to the end of the mRNA.

RNA processing also includes RNA splicing. During **RNA splicing** the noncoding introns are removed from the pre-mRNA and the exons are joined together.

Coding for polypeptides is an important function of DNA; however, less than 10% of the genome is involved in actual coding. Humans have about 3×10^9 base pairs in their genome, with the average gene being about 3000 bases long. The largest gene is over 2 million base pairs in length and the smallest gene is a few hundred base pairs long.

Once the mRNA has the code for the polypeptide, it moves from the nucleus into the cytoplasm. The mRNA is complementary to the DNA template, e.g. the DNA code CCA for glycine becomes the mRNA code GGU where uracil replaces thymine.

QUESTIONS

1. What is a codon?
2. Distinguish between transcription and translation.
3. What is the template strand?
4. What has enabled our knowledge of protein synthesis to be increased?
5. Construct a table to compare DNA and RNA.
6. Identify a difference between transcription in eukaryotes and prokaryotes.
7. What is the direction of transcription?
8. (a) Define the promoter.
   (b) Outline the significance of promoters having slightly different nucleotide sequences.
9. Outline the function of RNA polymerase.
10. Describe the relationship between base pairs and a gene.
11. What is the relationship between the coding in the human genome and protein synthesis?

The table gives the genetic code for triplets found in mRNA for some amino acids and is to be used for the next THREE questions.

Code	Amino acid	Code	Amino acid
CCC	Proline	ACC	Threonine
CCA	Proline	ACA	Threonine
CCG	Proline	ACG	Threonine
GCC	Alanine	CAC	Histidine
GCA	Alanine	CAA	Glutamine
AAC	Asparagine	AAA	Lysine
GAC	Aspartic acid	GAG	Glutamic acid

This code was found in mRNA.
CCCCAAAACGACACAACC

12. Identify what this code would produce.
   (A) Proline, glutamine, threonine, aspartic acid, threonine, threonine.
   (B) Proline, glutamine, asparagine, aspartic acid, threonine, threonine.
   (C) Proline, lysine, threonine, alanine, threonine, threonine.
   (D) Proline, glutamine, asparagine, alanine, lysine, threonine.
13. What is the DNA code for this mRNA code?
   (A) CCCCAAAACGACACAACC
   (B) GGGGTTTTGCTGTGTTGG
   (C) CCCCAAAACGACACAACC
   (D) TTTTGGGGTAGTGTGGTT
14. Which of the following would be the mRNA code for lysine, glutamine and alanine, in that order?
   (A) AAAGAACCA
   (B) GCCCAAGCA
   (C) GACAAAGCA
   (D) AAACAAGCA
15. The triplet code was first deciphered by Marshall Nirenberg in 1961 when he found the codon UUU specified the amino acid, phenylalanine. By the mid 1960s, all 64 codons were deciphered. The diagram shows a section of DNA, its transcription to mRNA and its translation into amino acids.

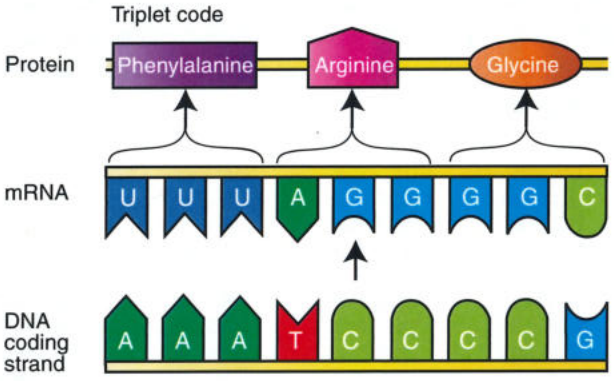


Figure 12.3 Triplet code.

Organisms share common codons, e.g. the codon CCG in mRNA translates to the amino acid proline in all organisms so far examined. What does this suggest?
   (A) The code was present in an ancestor common to all five kingdoms of life.
   (B) Convergent evolution occurred so that codons became similar.
   (C) Mutations have occurred over time, causing all codons to become the same.
   (D) The triplet code will further evolve into a quadruplet code for more complex organisms.
16. In polypeptide synthesis, what is the function of transcription?
   (A) Duplicating genetic information.
   (B) Using tRNA to produce the correct order of amino acids in a protein.
   (C) Rearranging parts of a chromosome.
   (D) Copying information from DNA onto mRNA.
17. Where does transcription occur in eukaryotic cells?
   (A) In the cytoplasm. (B) In the nucleus.
   (C) On a ribosome. (D) In a lysosome.
18. Which enzyme is involved in the synthesis of mRNA??
   (A) DNA polymerase. (B) RNA helicase.
   (C) RNA polymerase. (D) DNA helicase.

13 Polypeptide Synthesis – Translation

Translation is a cellular process that uses the genetic code on mRNA to synthesise a polypeptide. The sequence of nucleotides is translated into a sequence of amino acids which are joined together to form the polypeptide.

Translation occurs in the cytoplasm on ribosomes. The translator is **tRNA** (transfer ribonucleic acid).

Translation occurs in three main stages – initiation, elongation and termination. In **initiation** the mRNA, tRNA with the first amino acid of the polypeptide that is to be formed and the two subunits of a ribosome are brought together to form the initiation complex. In **elongation** the amino acids are brought to the mRNA and added to the growing polypeptide chain. In **termination** a stop codon (e.g. UAG, UAA, UGA) in the mRNA is recognised by a protein release factor causing the polypeptide to be completed and released and the translation assembly comes apart.

UUU Phe	UCU Ser	UAU Tyr	UGU Cys
UUC Phe	UCC Ser	UAC Tyr	UGC Cys
UUA Leu	UCA Ser	UAA stop	UGA stop
UUG Leu	UCG Ser	UAG stop	UGG Trp
CUU Leu	CCU Pro	CAU His	CGU Arg
CUC Leu	CCC Pro	CAC His	CGC Arg
CUA Leu	CCA Pro	CAA Gln	CGA Arg
CUG Leu	CCG Pro	CAG Gln	CGG Arg
AUU He	ACU Thr	AAU Asn	AGU Ser
AUC He	ACC Thr	AAC Asn	AGC Ser
AUA He	ACA Thr	AAA Lys	AGA Arg
AUG Met	ACG Thr	AAG Lys	AGG Arg
GUU Val	GCU Ala	GAU Asp	GGU Gly
GUC Val	GCC Ala	GAC Asp	GGC Gly
GUA Val	GCA Ala	GAA Glu	GGA Gly
GUG Val	GCG Ala	GAG Glu	GGG Gly

U – uracil (thymine)	Cys – cysteine	Met – methionine
C – cytosine	Gln – glutamine	Phe – phenylalanine
A – adenine	Glu – glutamine	Pro – proline
G – guanine	Gly – glycine	Ser – serine
Ala – alanine	His – histidine	Thr – threonine
Arg – arginine	He – isoleucine	Trp – tryptophan
Asn – asparagine	Leu – leucine	Tyr – tyrosine
Asp – aspartic acid	Lys – lysine	Val – valine

Figure 13.1 The triplet genetic code found in mRNA.

tRNA

Each tRNA can translate a particular mRNA codon into a particular amino acid. One end of the tRNA has a nucleotide triplet called the **anticodon** which is complementary to the mRNA code. The other end of the tRNA carries the amino acid and takes it to the ribosome for translation. This gives the tRNA its distinctive shape which looks a 3-leaf cloverleaf when flattened to draw in two dimensions. In eukaryotes tRNA is made in the nucleus and moves to the cytoplasm.

The cytoplasm of a cell contains a supply of amino acids that are either synthesised from other compounds or brought into the cell for protein synthesis. The 3′ end of tRNA is the attachment site for the amino acid. There are 20 different amino acids and there are 20 different **synthetase** enzymes that link the correct amino acid to the tRNA with the correct anticodon. Each tRNA can be used many times – to pick up an amino acid, to deposit the amino acid onto the polypeptide chain at the ribosome and then to move away to pick up another amino acid.

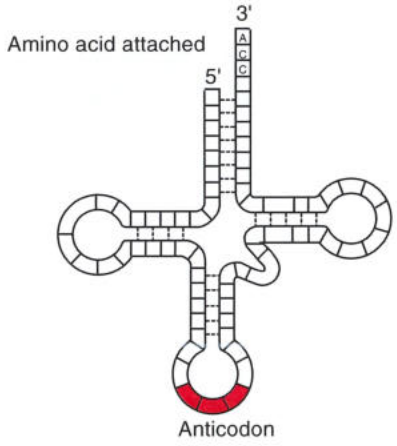


Figure 13.2 tRNA.

At the ribosome

In the cytoplasm the mRNA moves to a ribosome where it binds on to the ribosome at the end with the 'start' codon. The sequence of codons is translated into a sequence of amino acids that make up a particular polypeptide.

A transfer RNA (tRNA) in the cytoplasm moves to the codon on the mRNA which is at the ribosome. At one end of the tRNA there is an anticodon to complement the codon on the mRNA and on the other end of the tRNA is the amino acid.

Ribosomes are made up of two subunits which consist of proteins and ribosomal RNA (rRNA) with about two-thirds of the mass of a ribosome consisting of rRNA. The large subunit binds with tRNA and the smaller subunit binds with mRNA.

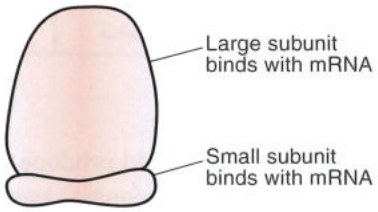


Figure 13.3 Structure of a ribosome.

A ribosome has *three* binding sites for tRNA. The **P site** holds the tRNA with the growing chain. The **A site** holds the tRNA carrying the amino acid about to be added to the chain. The **E site** has the exiting tRNA that has delivered its amino acid and is about to leave the ribosome.

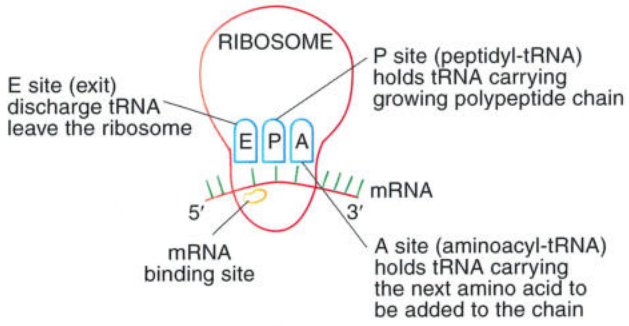


Figure 13.4 Binding sites on a ribosome.

Once the first amino acid is attached, the second tRNA brings in its amino acid to attach to the next codon on the mRNA. The ribosome then releases the first tRNA. The mRNA moves through the ribosome with the 5′ end first. Peptide bonds form between the amino acids and the sequence forms a polypeptide.

Stop codons, e.g. UAG, UAA, UGA stop translation and a **release factor** binds directly to the stop codon at the A site.

Polysomes

A polysome or polyribosome is a cluster of ribosomes bound to one mRNA. Once a ribosome has attached to a single mRNA and started to produced the polypeptide, another ribosome can attach and start to make the same polypeptide. Then another ribosome can attach. This means that multiple copies of a polypeptide can be quickly synthesised.

QUESTIONS

1. (a) Define translation.
   (b) Where does translation occur?
   (c) In translation, what is the 'translator'?
2. (a) Identify the main stages of translation.
   (b) Construct a table to summarise what happens during the main stages of translation.
3. Complete the following table to show the the mRNA codes and the corresponding amino acids for each of the listed DNA codes.

DNA code	mRNA code	Amino acid
GAC		
TCT		
GTA		
CCA		
TTA		
GTC		

4. (a) Describe the structure of tRNA.
   (b) Draw a diagram to show the shape of tRNA and label the location of the amino acid attachment and location of the anticodon.
5. In eukaryotes, where is tRNA formed?
6. Outline the function of synthetase enzymes in translation.
7. Construct a table to outline the function of each of the following in translation.
   (a) mRNA
   (b) tRNA
   (c) Ribosomes
8. Describe the structure of a ribosome.
9. How many subunits are in a ribosome and what are their function?
10. What binding sites are found on ribosomes?
11. What is the direction of translation?
12. State the codes for stop codons.
13. (a) What is a polysome?
    (b) Describe an advantage in the formation of a polysome.
14. Draw a brief flow chart to show the main sequence of events in the synthesis of a polypeptide.
15. The diagram shows a process that occurs in the body.

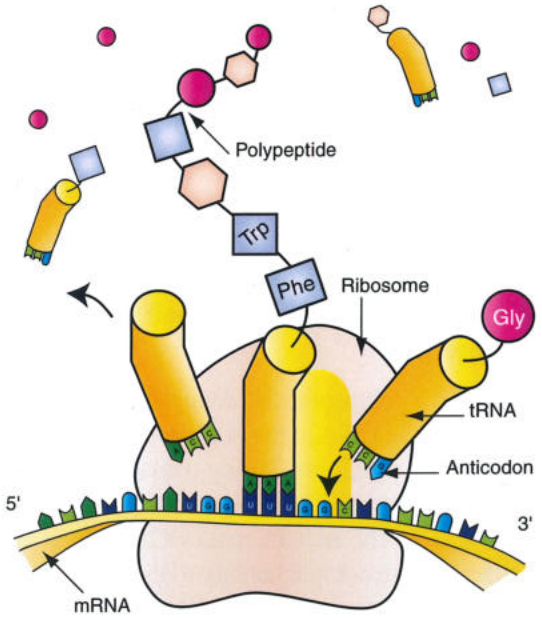


Figure 13.5 Body process.

What is the name of this process?
(A) Transcription. (B) Replication.
(C) Translation. (D) Transference.

16. What are the three main stages of translation?
    (A) Initiation → elongation → termination.
    (B) Elongation → initiation → termination.
    (C) Translocation → elongation → termination.
    (D) Initiation → elongation → transference.
17. Which of the following would cause termination and stop further synthesis of a polypeptide?
    (A) The arrival of the codon AUG at the A site.
    (B) The arrival of the codon UAG at the A site.
    (C) Synthetase acting on tRNA and an amino acid.
    (D) rRNA catalyses formation of a peptide bond between amino acids at A and P sites.

14 Activity – Making a Model of Protein Synthesis

Some Human Biology students were asked to form groups with each group to make a model of protein synthesis for eukaryotes. The models needed to include the stages of transcription and translation.

When presenting their work to the class each group needed to show how their model fully explained the process of protein synthesis and to explain why they chose their particular mode of presentation.

Table 14.1 Summary of student presentations.

Group	Mode of presentation
1	PowerPoint slideshow presentation with diagrams and animations.
2	Jigsaw cut-outs that were moved around a diagram of part of a cell.
3	Flow chart on coloured cardboard.
4	Series of diagrams in a word document.
5	PowerPoint slideshow presentation with diagrams and animations.

Background data given to students

The students were given the following background data which they needed to expand for their presentation by further research.

'In eukaryotes a polypeptide is produced when the coded information on the DNA in the nucleus is carried out of the nucleus into the cytoplasm. During **transcription** the DNA unzips and the enzyme RNA polymerase binds to one of the DNA strands. The enzyme makes a long chain of RNA nucleotides that are complementary to the DNA nucleotides. This forms the messenger RNA (mRNA) which is released from the enzyme and moves out of the nucleus into the cytoplasm. Then during **translation** the coding on the mRNA is used to make proteins on ribosomes in the cytoplasm when tRNA delivers amino acids in an order that matches up the complementary base-pairing between the mRNA and tRNA. The ribosome provides the structural site for the mRNA and tRNA. Sometimes there are clusters of ribosomes on an mRNA and since each one makes a polypeptide strand the final result is a more complex protein.'

QUESTIONS

1. (a) Define transcription.
   (b) Describe the steps that occur during transcription.
2. Identify what happens during translation.
3. Distinguish between the location of transcription and translation in eukaryotes.
4. When explaining why their group chose their particular mode of presentation all groups commented that they had considered constructing a three dimensional model, e.g. using plasticine, coloured balls, pipecleaners, toothpicks etc but had decided that it would be too hard to build and would not allow enough detail to be presented. Use your knowledge of the process of protein synthesis to explain why all groups would have reached the same conclusion about not building a three dimensional model.
5. Explain why a group activity to construct a group model of protein synthesis would assist student understanding of this biological process.
6. Give one advantage and one disadvantage for carrying out this activity as group work rather than each student individually making their own model.
7. (a) From Table 14.1 Summary of student presentations, what was the most favoured form of representing the process of protein synthesis?
   (b) Suggest a reason why students would have preferred this mode of presentation.
8. Name two types of RNA.
9. The table shows the mRNA codons for different amino acids as well as the stop codons.

Table 14.2 mRNA codons and amino acids.

First base	Second base: U	Second base: C	Second base: A	Second base: G	Third base
U	UUU Phe, UUC Phe	UCU Ser, UCC Ser, UCA Ser, UCG Ser	UAU Tyr, UAC Tyr	UGU Cys, UGC Cys	U, C
	UUA Leu, UUG Leu	CCU Pro, CCC Pro, CCA Pro, CCG Pro	UAA stop, UAG stop	UGA stop, UGG trp	A, G
C	CUU Leu, CUC Leu, CUA Leu, CUG Leu	CCU Pro, CCC Pro, CCA Pro, CCG Pro	CAU His, CAC His	CGU Arg, CGC Arg, CGA Arg, CGG Arg	U, C
			CAA Gln, CAG Gln		A, G
A	AUU Ile, AUC Ile, AUA Ile	ACU Thr, ACC Thr, ACA Thr, ACG Thr	AAU Asn, AAC Asn	AGU Ser, AGC Ser	U, C
	AUG Met		AAA Lys, AAG Lys	AGA Arg, AGG Arg	A, G
G	GUU Val, GUC Val, GUA Val, GUG Val	GCU Ala, GCC Ala, GAA Ala, GAG Ala	GAU Asp, GAC Asp	GGU Gly, GGC Gly, GGA Gly, GGG Gly	U, C, A, G
			GAA Glu, GAG Glu		

Amino acid	Abbreviation	Amino acid	Abbreviation	Amino acid	Abbreviation
Alanine	Ala	Glycine	Gly	Proline	Pro
Arginine	Arg	Histidine	His	Serine	Ser
Asparagine	Asn	Isolecine	Ile	Threonine	Thr
Aspartic acid	Asp	Leucine	Leu	Tryptophan	Trp
Cysteine	Cys	Lysine	Lys	Tyrosine	Tyr
Glutamine	Gln	Methionine	Met	Valine	Val
Glutamic acid	Glu	Phenylalanine	Phe		

The diagram shows a part of a mRNA molecule formed from a given length of DNA in the nucleus in a cell.

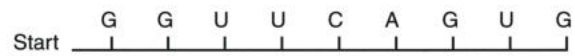


Figure 14.1 Part of mRNA.

(a) Identify the sequence of amino acids coded for by this length of mRNA.

(b) What codons identify the amino acid isoleucine?
(c) If the DNA which formed this mRNA had the third base from the right replaced by the base thymine, what amino acid sequence would be coded on the mRNA and what amino acid would this form?

10. Distinguish between a codon and an anticodon.
11. Discuss how the bases that make up RNA differ from the bases that make up DNA.
12. Compare mRNA and tRNA.
13. The following code is part of a single strand of DNA.

AAATAGTGGCCAAGC

Construct a table to show how this code would be found after transcription on the mRNA and translation on the anticodon of the tRNA with the corresponding amino acid that will be formed. (You will need to use the information in Table 14.2, mRNA codons and amino acids, to identify each amino acid.)

14. What happens at the beginning of transcription?
15. The following diagram shows a section of a polypeptide.

Figure 14.2 Part of a polypeptide.

Write down *two* DNA codes that could form this section of the polypeptide.

16. Insulin is a small simple protein consisting of 51 amino acids in two chains. What would be the minimum number of bases needed to code this protein? Why is it most likely that the number of bases in the gene is greater than this?
17. The diagram shows part of a PowerPoint slideshow from a student presentation on protein synthesis.

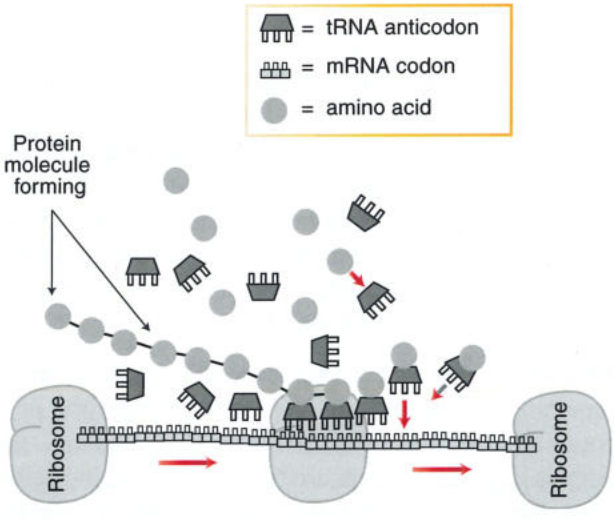


Figure 14.3 Part of student presentation on protein synthesis.

What is happening at this stage of protein synthesis?

18. One of the student presentations included an additional step that showed two possible pathways for the growing polypeptide.
'Protein synthesis begins on free ribosomes. Proteins made on free ribosomes are mainly used within the cell. If the polypeptide has a signal peptide the ribosome is moved to a particular part of the endomembrane system and becomes a bound ribosome, e.g. on rough endoplasmic reticulum. Proteins produced by bound ribosomes are mainly secreted from the cell.'
Suggest why this group felt it was necessary to add information about bound and free ribosomes.
19. The following diagram shows RNA and its corresponding DNA.

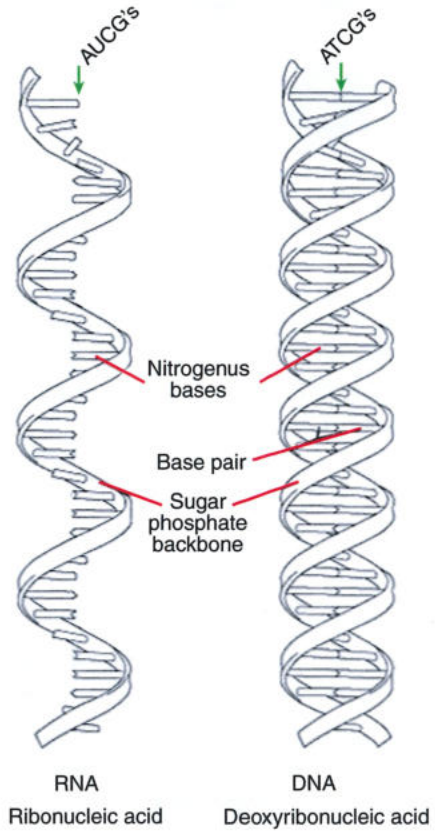


Figure 14.4 RNA and DNA.

Colour in a section of each of these using the colour code red = A, blue = C, green = G, black = U and yellow = T, to show RNA with a code for the three amino acids alanine, valine and glycine in a sequence.

20. In a student PowerPoint slideshow of the stages of protein synthesis, which of the following would be a slide before the small subunit of a ribosome binds to the 5′ end of mRNA?
(A) tRNA is drawn to a ribosome.
(B) tRNA codon matches an mRNA codon at A site.
(C) tRNA is released from E site.
(D) RNA polymerase links RNA nucleotides as they pair along the DNA template.

15 Protein Structure

Proteins make up more than 50% of dry mass of most cells and are involved in nearly every aspect of life. They are macromolecules that always contain the elements carbon, hydrogen, oxygen and nitrogen. Proteins are formed on ribosomes and are made from a linear sequence of amino acids.

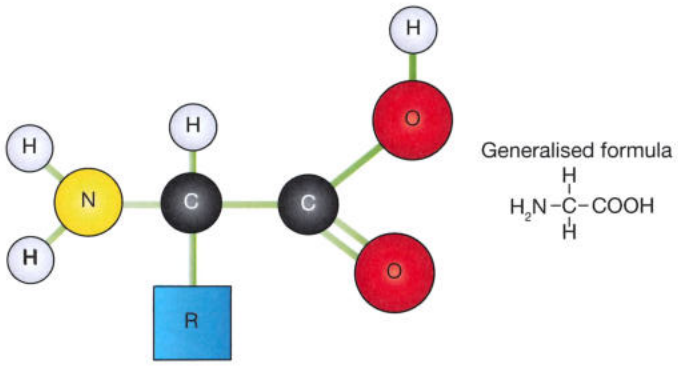


Figure 15.1 Generalised structure of an amino acid.

Amino acids

Amino acids are crystalline solids that dissolve in water but are insoluble in organic solvents. Each amino acid has an amino group (NH_2), a carboxyl group (COOH) and a side chain (R). The different chemical properties of different amino acids, e.g. polar (hydrophile) or non-polar (hyrodphobe) depend on the properties of R. The simplest amino acid is glycine where R = H.

There are over 150 types of amino acids that occur in cells but only 20 are used to make proteins.

Amino acids can be joined together by **peptide bonds**. Two amino acids joined together form a **dipeptide** while three amino acids joined together form a **tripeptide**. Hundreds or thousands of amino acids can be joined together to make a **polypeptide**.

Biuret solution (containing equal amounts of 1% copper sulfate solution and 5% potassium hydroxide solution) is used to test for peptide linkage. The biuret solution is a pale blue and a positive test for peptide linkage (protein) is a change to a purple colour.

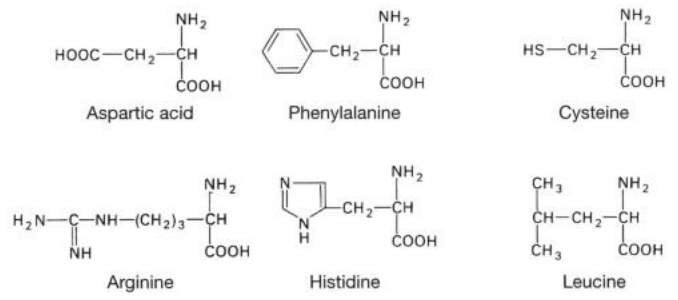


Figure 15.2 Structures of some amino acids.

Primary structure

The primary structure of a protein is the unique linear sequence of amino acids in its molecule and is determined by genetic information. The order of the amino acids is extremely important as it influences higher levels of organisation in the protein and its biological function.

Secondary structure

Secondary structures are coils or folds caused by hydrogen bonds between repeating CO and NH groups. Secondary structures include the helix and β pleated sheet. Some spirals are 'right-handed' (α-helix) and others are 'left-handed' (β-helix). Pleated sheets make up the basis of many globular proteins and fibrous proteins. If the coil of amino acids remains extended and threadlike it forms a **fibrous protein**. Keratin is a fibrous protein that mainly consists of α-helices and is found in hair, wool, claws, beaks, feather, horns and nails. Silk protein is a fibrous protein that mainly consists of β-sheet form.

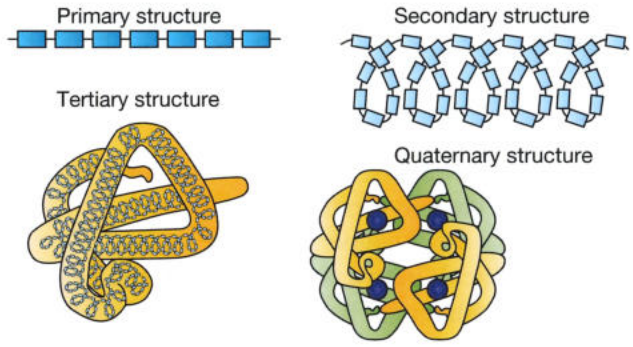


Figure 15.3 Protein structure.

Tertiary structure

Tertiary structure is formed by the folding of the secondary structure into a complex and compacted shape. The long coils of amino acids loop and twist and fold back on themselves to form **globular proteins**. The shape is determined by interactions between the R groups of the amino acids, hydrophobic interactions, hydrophilic interactions and hydrogen bonding. Three types of bonds hold the tertiary configuration in shape – hydrogen bonds, ionic bonds and disulfide bonds.

Disulfide bridges are strong covalent links between sulfur containing amino acids. Sulfur is often in the form of –SH groups and the most common amino acid with this –SH group is cysteine. If there are two cysteine amino acids distant from each other in the chain but folded to lie next to each other, their –SH groups with link to form a disulfide bridge –S–S– which will stabilise the structure. The folding of tertiary structures can provide structural strength, e.g. microtubules are globular proteins, or can produce an 'active site', e.g. enzymes are globular proteins.

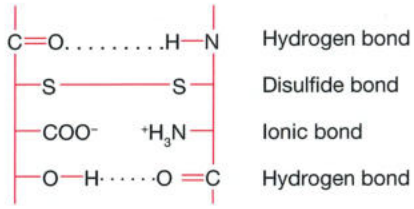


Figure 15.4 Bonds in proteins.

Quaternary structure

Quaternary structure involves the interaction between several polypeptide chains that aggregate into one functional macromolecule. The individual subunits are held together by hydrogen bonds rather than covalent linkages, though disulfide bonds occur. The hydrogen bonds are weak and can be easily broken by physical or chemical changes in the environment. If many bonds are broken, e.g. by excessive heat, pH, the globular structure cannot be maintained and the protein is **denatured**.

Conjugated proteins contain non-protein material and the non-protein part is called the prosthetic group. Many important proteins use quaternary structure to carry out a large variety of biological functions, e.g. collagen fibres and haemoglobin. Collagen is a fibrous protein with helical subunits twisted to form a triple helix which gives great strength to the long fibres. In the haemoglobin molecule of vertebrate blood, the protein component consists of four linked polypeptide chains – two α-chains and two β-chains. Haemoglobin also contains four haem groups which are the prosthetic group of the protein. Conjugated proteins include chlorophyll and many enzymes in the electron transport chain.

QUESTIONS

1. What is a protein?
2. (a) Name the main parts of an amino acid.
   (b) Draw the generalised structure of an amino acid.
3. What determines the different properties of an amino acid?
4. Identify the simplest amino acid.
5. Distinguish between a dipeptide, a tripeptide and a polypeptide.
6. (a) What is the distinguishing test for protein?
   (b) What does this test actually detect?
7. What is meant by:
   (a) The primary structure of a protein?
   (b) The secondary structure of a protein?
   (c) The tertiary structure of a protein?
   (d) The quaternary structure of a protein?
8. Describe the types of coils/folds of secondary structures of proteins.
9. Construct a table to distinguish between a fibrous protein and a globular protein.
10. Identify the three types of bonds that hold the tertiary structure of proteins in shape.
11. Outline the significance of disulfide bonds in protein structure.
12. (a) What happens when a protein is denatured?
    (b) List some factors that can cause denaturation of a protein.
13. The diagram shows one polypeptide chain of a particular protein.

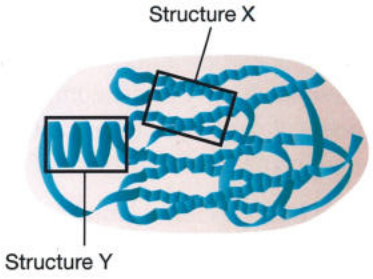


Figure 15.5 One polypeptide chain of a particular protein.

(a) Discuss how this diagram shows the primary, secondary and tertiary structure of this protein.
(b) Identify structure X and structure Y on the diagram.
(c) Is this a fibrous or globular protein? Explain your reasoning.
(d) If this protein consists of two of these polypeptide subunits, draw a possible appearance of this protein and identify the level of structure that has been formed.

14. The diagram shows what happened to a particular protein when it was heated to a high temperature.

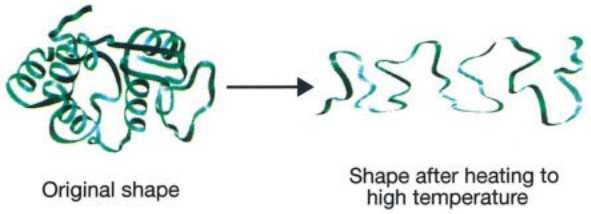


Figure 15.6 A protein is heated.

What has happened to this protein?
(A) Renaturation. (B) Denaturation.
(C) Melting. (D) Mutation.

15. What are the main components of all amino acids?
    (A) A phosphate group, a pentose sugar and a nitrogen base.
    (B) A carboxyl group (COOH) and an unbranched hydrocarbon chain($CH_3(CH_2)_n$–)
    (C) An amine group (NH_3), a carboxyl group (COO) and a side chain (R).
    (D) An amine group (NH_2), a carboxyl group (COOH) and a side chain (R).
16. What is the simplest amino acid?
    (A) Glycine (B) Leucine
    (C) Cysteine (D) Histidine

16 Polar and Non-Polar Amino Acids

Amino acids consist of an amino group ($-NH_2$), a carboxyl group (–COOH) and a side chain (–R). The amino group is basic and the carboxyl group is acidic. An **amphoteric** substance is both acidic and basic.

The two functional groups of the amino acid (the amino group and the carboxyl group) in most amino acids are attached to the same carbon atom which is called the α-carbon. Amino acids vary in the composition of the side chain. The chemical and physical properties of the side chain determine the particular features of each amino acid,

If the amino acid has one amino group and one carboxyl group it will be **neutral**. If the amino acid has additional amino groups it will be a **basic amino acid** while additional carboxyl groups will make it an **acidic amino acid**.

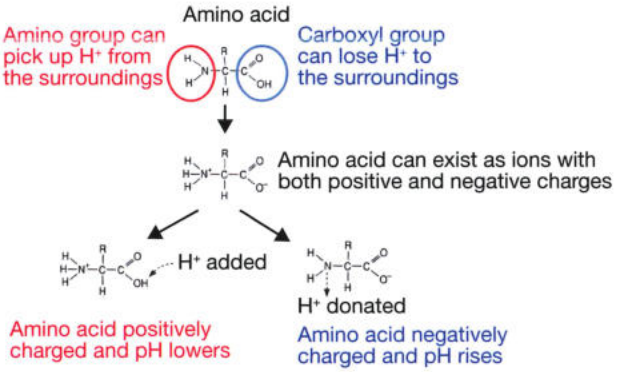


Figure 16.1 Amino acids can be amphoteric.

In an acid solution the amino acid picks up H^+ ions and becomes positively charged. In an alkaline solution the amino acid donates H^+ ions to the surroundings and becomes negatively charged. This means that the presence of amino acids in a solution stabilises the pH of the solution and amino acids act as **buffers**. Proteins are important in stabilising the pH of cells and body fluids.

Polar amino acids

Amino acids with a polar side chain (R) are hydrophilic. Hydrophilic molecules have an affinity for water and are usually ionic or polar substances. Soluble proteins have a large number of polar amino acids.

Non-polar amino acids

Amino acids with a non-polar side chain (R) are hydrophobic. Hydrophobic substances do not have an affinity for water and can even seem to repel water. Hydrophobic substances are nonionic and nonpolar substances. Integral proteins in the lipid bilayer of the cell membrane have an arrangement of polar and non-polar amino acids to anchor them in the membrane.

The distribution of polar, non-polar and electrically charged amino acids in a protein determines the tertiary structure and the quaternary structure of the protein. This in turn determines the properties of the protein and its biological function.

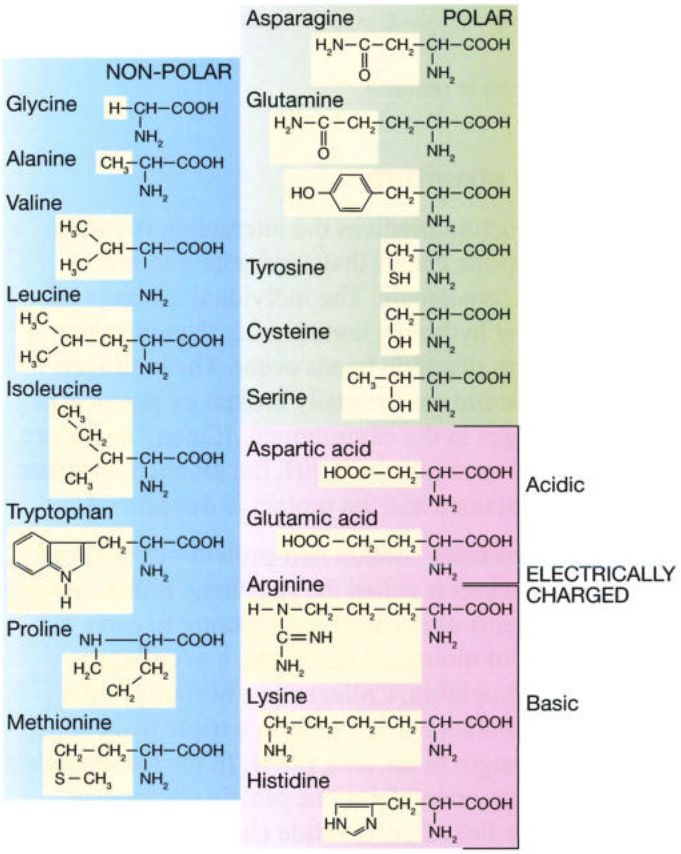


Figure 16.2 Polar, non-polar and electrically charged amino acids.

QUESTIONS

1. What are the components of an amino acid?
2. Which part of the amino acid is:
   (a) Basic? (b) Acidic?
3. What is meant by an amphoteric substance?
4. When describing amino acids, organic chemists talk about the functional groups present in the molecule. What are the functional groups of amino acids?
5. All amino acids ,except for proline and hydroxyproline, have an α-carbon atom. What is an α-carbon atom?
6. What molecule composition leads to a neutral amino acid, a basic amino acid and an acidic amino acid?
7. Explain why proteins can act as buffers and stabilise the pH of cells and body fluids.
8. Which amino acids are likely to be hydrophilic and which are likely to be hydrophobic?
9. Which proteins are most likely to be soluble?
   (A) Those with a large number of polar amino acids.
   (B) Those with a large number of non-polar amino acids.
   (C) Those containing over 150 amino acids.
   (D) Those with a high number of nucleotides containing adenine.

17 The Role of Proteins in Human Biology

Proteins are essential to cell structure and functioning. The amount and type of protein in a cell determines the appearance of the cell and how it functions, e.g. muscle cells have large amounts of the proteins myosin and actin which give the cell its ability to contract and give muscle cells their distinctive appearance.

There are several different types of proteins.

Enzymes

Enzymes are proteins that catalyse reactions. They change the rate of a reaction without being consumed by the reaction. Enzymes control specific reactions with many working in 'teams' in a series of reactions, e.g. cellular respiration involves specific enzymes at each step of the reaction pathway. Some enzymes only catalyse one particular reaction, e.g. sucrase only splits sucrose and urease only decomposes urea to ammonia and carbon dioxide. While other enzymes catalyse a type of reaction, e.g. peroxidase catalyses several different peroxides including hydrogen peroxide and pancreatic lipase splits the ester bonds connecting the glycerol and fatty acids of a large range of fats.

Most enzymes are globular proteins.

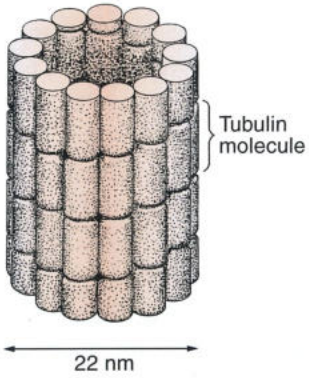


Figure 17.1 Tubulin, a structural protein, forms microtubules. Microtubules are in cilia, flagella, centrioles and the spindles formed in cell division.

Structural proteins

Structural proteins provide support. Many are fibrous proteins, e.g. collagen and elastin which are part of connective tissue, e.g. in cartilage and keratin which is highly important in providing tough support, e.g. in nails. Humans use structural proteins of other animals, e.g. silk fibres made by insects and spiders for cocoons and webs and keratin in feathers. Some structural proteins are globular proteins, e.g. actin and tubulin which are part of microfilaments and microtubules in the cytoskeleton of cells. The cytoskeleton controls the shape and movements of the cell.

Membrane proteins are a range of different proteins embedded in the lipid bilayer of the cell membrane. Different types of cells have different sets of membrane proteins with some membrane proteins held in place by connections to the internal cytoskeleton and some membrane proteins attached to the extracellular matrix. In animal cells these attachments to the membrane proteins provide support and give the cell a stronger framework.

Transport proteins

Transport proteins are involved in the movement of other substances, e.g. haemoglobin in vertebrate red blood cells transports oxygen from the lungs/gills to other parts of the body.

Some membrane proteins in the lipid bilayer of the cell membrane are transport proteins and are involved in assisting the movement of molecules across the cell membrane. **Channel proteins** form a hydrophilic channel that allows certain molecules or ions to pass through the cell membrane, e.g. aquaporins channel water molecules. **Carrier proteins** shuttle molecules across the membrane by changing shape, e.g. glucose carrier proteins.

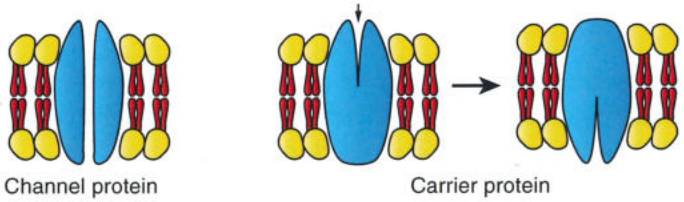


Figure 17.2 Transport proteins in the cell membrane.

Storage proteins

Storage proteins store amino acids, e.g. ovalbumin is the protein in egg white which provides amino acids for the developing embryo and casein is the protein in milk which provides amino acids for a breast-fed baby. Some storage proteins store metals, e.g. ferritin is a storage protein that stores iron which can then used to make haemoglobin for new red blood cells. The seeds of plants contain storage proteins, e.g. the seeds of many cereal grains store prolamines – a group of storage proteins with a high amount of the amino acid proline, e.g. wheat stores gliadin which is a part of gluten and barley stores the glycoprotein hordein. These are important in dietary requirements.

Hormones

Hormones are formed in specialised cells in multicellular organisms and are chemical signals that travel in body fluids to act on specific target cells to change the functioning of the target cells. Hormones are involved in the coordination of different activities in an organism.

Since hormones are defined by function there is a range of chemical structures in the group, e.g. peptide hormones such as insulin, growth hormone and ADH are chains of amino acids, steroid hormones such as testosterone and cortisol which come from cholesterol and glycoprotein hormones such as luteinising hormone and thyroid stimulating hormone which are chains of amino acids with carbohydrate side chains. Insulin is a hormone secreted by the pancreas and regulates the concentration of sugar in the blood of vertebrates by causing cells in fat tissue and skeletal muscle to absorb glucose from the blood.

Receptor proteins

Receptor proteins are involved in the response of a cell to chemical stimuli. They are usually found on the surface of a cell or inside the cell. Some membrane proteins act as receptors, e.g. G protein linked receptor proteins are used in embryonic development and sensory perception. Intracellular receptor proteins are found in either the cytoplasm or the nucleus of target cells. Receptors in the dendrites of neurons detect the neurotransmitter released by other neurons and respond by continuing the nervous impulse.

Contractile and motor proteins

Contractile and motor proteins are involved in movement. Motor proteins enable cytoplasmic streaming and the active transport of proteins and vesicles in the cytoplasm The proteins actin and myosin cause the movement of muscles.

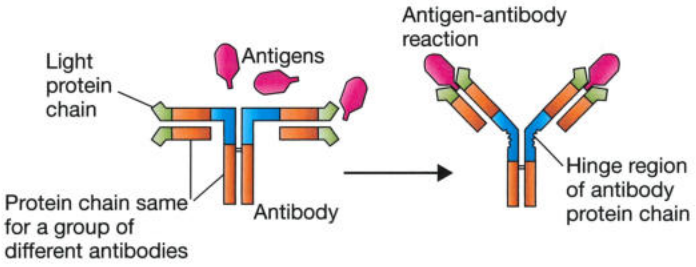


Figure 17.3 Function of antibodies.

Defence proteins

Defence proteins aid protection against disease. The skin and respiratory tract secrete antibacterial peptides such as beta defensins. The **complement system** is a number of small blood proteins that are part of the innate immune system that help remove pathogens from the body. **Interferons** are proteins made and released by host cells in response to a pathogen, e.g. virus, bacteria, parasite or tumour cell. Interferons enable communications between cells to trigger the immune response, e.g. interferon alpha and interferon beta are secreted by virus-infected cells and help nearby cells to resist viral infection. Viral particles produced in cells affected by interferon are not very effective at infecting other host cells.

Antibodies (immunoglobulins – Ig) are large Y shaped glycoproteins produced by plasma cells in response to a particular antigen and defend against bacteria and viruses forming an antigen-antibody complex. Antibodies are highly specific and consist of four polypeptide chains – two identical long chains called heavy chains and two identical short chains called light chains linked by disulfide bridges.

QUESTIONS

1. Define an enzyme.
2. Describe the specificity of enzymes.
3. What is the structure of most enzymes?
4. Describe the function of three named structural proteins.
5. Explain how membrane proteins are involved in support.
6. Explain why haemoglobin is classified as a transport protein.
7. Explain why some membrane proteins can be classified as transport proteins.
8. (a) What is a hormone?
   (b) Explain why you cannot give an exact description for the chemical structure of hormones.
   (c) Give some examples of hormones that are proteins.
9. Describe the function of insulin.
10. Name some storage proteins.
11. What are the prolamines and why are they important?
12. (a) Where do you find receptor proteins?
    (b) How are receptor proteins involved in the transmission of a nervous impulse?
13. Name some proteins involved in movement.
14. Construct a table to summarise the roles of at least three types of defence proteins.
15. Draw a diagram of an antibody and label the light and heavy protein chains and the disulfide bridges.
16. Construct a table to summarise the different roles of proteins in living things.
17. A single protein molecule can contain thousands of atoms and scientists are determining the amino acid sequence and three-dimensional shapes of known proteins. The diagram shows how X-ray crystallography is used to determine the three-dimensional shape of a crystallised protein.

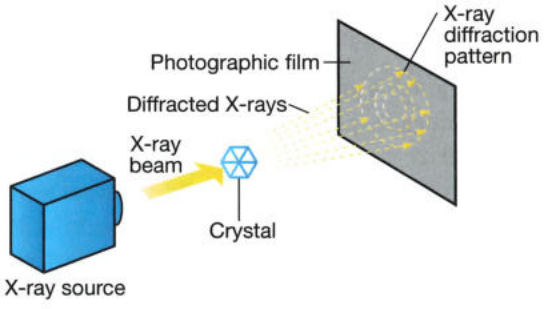


Figure 17.4 X-ray crystallography.

Use the diagram to explain X-ray crystallography.

18 Phenotype and Epigenetics

Epigenetic inheritance refers to the inheritance of traits in ways that do not directly involve the nucleotide sequence.

Phenotype refers to the actual appearance of an organism. It includes all aspects of external and internal anatomy, physiology and behaviour. The **genotype** refers to the genetic constitution of the organism. Genotype can also refer to a particular set of alleles in an organism. The **genome** is the complete complement of an organism's genetic information.

The phenotype of an individual does not always show the genotype. For example, a plant may have the genotype TT and should grow to a tall height but if the person has a poor diet through childhood and growth they may not reach their potential height (phenotype).

The range of phenotypic appearances for a trait is called the **norm of reaction** for a genotype. The norm of reaction can range from no breadth, e.g. for traits that have a specific genotype/phenotype such as ABO blood group or there can be a very broad norm of reaction, e.g. for human skin colour. Traits that are polygenic with two or more genes determining a single phenotypic feature usually have broad norm of reaction.

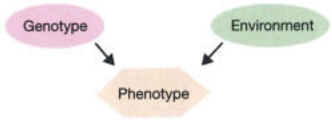


Figure 18.1 Phenotype.

Pleiotropy occurs when a single gene has multiple effects, e.g. many genetic diseases caused by a single pair of alleles such as cystic fibrosis and sickle cell anaemia have multiple symptoms. Albinism causes a lack of pigmentation in a number of characteristics, e.g. hair, skin and eyes.

Phenotype depends upon:

- Factors controlling transcription and translation during protein synthesis.
- Products of other genes.
- The environment.

When many factors determine the final appearance of a particular feature the trait is called **multifactorial**. The development of cancer in a person is multifactorial. The fundamental cause of many cancers is DNA damage that accumulates over a person's lifetime. Contributing factors include age (the older you are the more likely you are to develop a cancer), lifestyle (e.g. diet, exercise, alcohol intake, smoking cigarettes), DNA damage (e.g. by carcinogens in the environment), inheritance of a fault in one of the genes, infection by some types of virus (e.g. infection by human papilloma virus – HPV – is linked to cervical cancer) and problems with the immune system.

Factors controlling protein synthesis

The **genotype** of an individual determines the particular polypeptides that will be produced by cells. A gene is a section of DNA that has the code for a particular polypeptide and when the gene is expressed that section of the DNA molecule is 'switched on' and the polypeptide is produced by the cell. All organisms control which genes are expressed at any one time.

Gene expression and protein synthesis involves two steps – transcription and translation. In **transcription** the DNA code for a gene is used as a template to form mRNA. In **translation** a polypeptide is synthesised under the direction of the mRNA at a ribosome. Gene expression can be measured by the amount of functional protein produced by the cell.

The beginning of gene expression requires a section of DNA to be modified so that transcription can occur. **Chromatin-modifying enzymes** act on the DNA and have the initial control of gene expression. The initiation of transcription is regulated by specific **transcription factors** – activators or repressors which bind with multiple DNA **control elements** called **enhancers**. In eukaryotes **RNA processing** in the nucleus controls the final form of the mRNA that leaves the nucleus and moves to the cytoplasm. The initiation of translation is controlled by **initiation factors**. After translation the protein can be processed and modified and these steps control the functionality of the final product. Each protein is functional for a specific length of time and **proteases** regulate protein degradation.

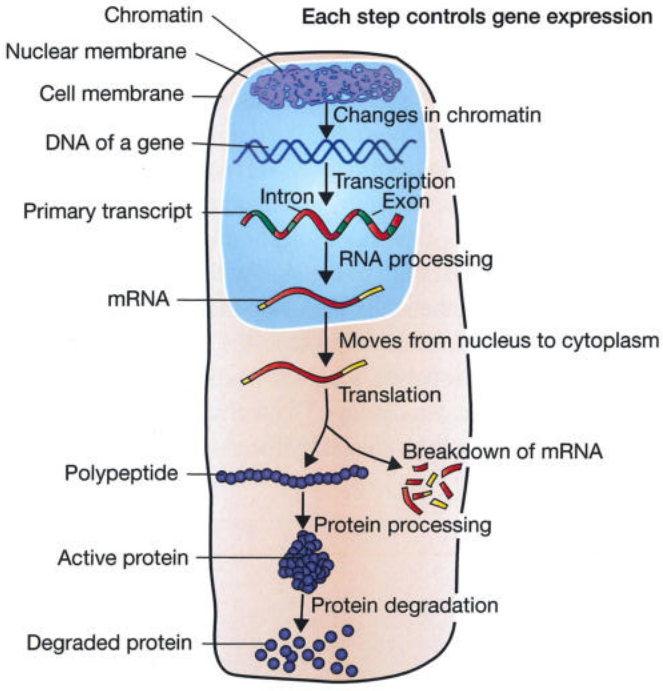


Figure 18.2 Gene expression is controlled at each step of protein synthesis and degradation.

Products of other genes

Many metabolic pathways involve a series of reactions with each step catalysed by a different enzyme. Each enzyme is coded by a particular gene. This means that several genes need to be expressed for the complete metabolic reaction to occur. The appearance of a particular trait can thus depend on the products of several genes.

For human genes it has been found that some coexpressed genes are clustered near each other on the same chromosome while other coexpressed genes are scattered over different chromosomes, e.g. for enzymes on a particular metabolic pathway. Genes with the same control elements will be activated by the same chemical signal.

Epistasis is a gene interaction with the presence of a particular allele of one gene pair determining whether certain alleles of another gene are expressed.

Environment

The environment is all the living and non-living surroundings. The conditions under which a person lives can affect their appearance, e.g. the quality and quantity of nutrients in the diet will affect growth, development and final body size and the amount of exposure to sunlight will affect skin tone and colour with sun tanning and potential skin cancers with overexposure to UV rays.

Melanocytes are pigment producing cells that are found in abundance between the dermis and the epidermis. The number of melanocytes is genetically determined and varies between different parts of the body, between different members of the same race and between different races. When exposed to UV light melanocytes produce increased amounts of melanin which is a black pigment. Activation of secretion of melanocyte stimulating hormone (MSH) causes secretion of melanin which is absorbed by surrounding epithelial cells increasing darkening of the skin. Injection of MSH into a person will cause darkening of the skin with a greater effect in people who have a genetically dark skin than in people with genetically light skin. Melanin protects the skin from harmful effects of sunlight. However, long term exposure to large amounts of UV light can damage DNA changing the amounts and kinds of chemicals produced by the skin.

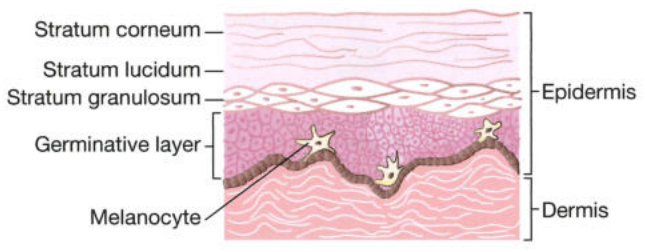


Figure 18.3 Surface skin layers.

QUESTIONS

1. Define epigenetic inheritance.
2. Define phenotype.
3. Distinguish between genotype and genome.
4. Use an example to show how the phenotype of an individual does not necessarily reveal the genotype of that individual.
5. (a) What is meant by the norm of reaction for a particular genotype?
   (b) Give an example of a trait with a narrow norm of reaction and a trait with a broad norm of reaction.
   (c) What is the likely norm of reaction for polygenic traits?
6. (a) Define pleiotropy.
   (b) Give an example of pleiotropy.
7. What is meant by a multifactorial trait?
8. List some of the factors that lead to the development of a cancer.
9. What is a gene?
10. Outline what happens when a gene is expressed.
11. Identify the two steps in gene expression.
12. Construct a table to summarise what happens in gene expression and where each step occurs in human cells.
13. How can gene expression be measured?
14. Draw a flow chart to show the steps in protein synthesis and degradation in a human cell and identify how this determines phenotypic expression of the gene.
15. Explain why the appearance of a particular trait can depend on the products of several genes.
16. What is meant by coexpressed genes?
17. Define epistasis.
18. Define the environment.
19. Use an example to show how the environment can affect the phenotype of an organism.
20. Repeated exposure to the poison ivy toxin can cause an allergic reaction with activated helper and cytotoxic T cells moving into skin tissues causing red streaks and small hive bumps to appear. Severe reactions can cause lung oedema and asthmatic attacks. Explain how this is an example of epigenetic inheritance.
21. The list shows several steps in protein synthesis.
    (i) Polypeptide is formed.
    (ii) RNA processing occurs.
    (iii) Transcription.
    (iv) Translation.
    (v) Chromatin changes.
    What is the correct order of these steps?
    (A) ii, v, iii, iv, i
    (B) ii, v, iv, iii, i
    (C) v, iii, ii, iv, i
    (D) v, ii, iii iv, i

19 Genetic Profiling and Screening

The completion of the Human Genome Project identified many features of DNA, genes and chromosomes and opened up new areas for further research. Genomic research has led to the development of diagnostic genetic tests that aid the early detection and diagnosis of human diseases.

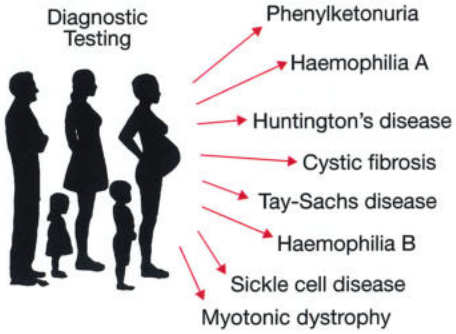


Figure 19.1 Diagnostic testing for diseases.

Prenatal testing

Prenatal tests involve tests, e.g. using placental tissue and/or amniotic fluid to provide information about the baby before birth. Prenatal screening tests are carried out, e.g. using ultrasound to check the developmental stage of the embryo or foetus and these tests can indicate if further diagnostic testing needs to be carried out. For example, an ultrasound test carried out at 8 to 10 weeks will confirm the stage of pregnancy, identify the number of embryos present in the uterus and show how the embryo is developing. A chorionic villus sampling (CVS) test can be carried out at 11 to 13 weeks to detect a range of physical and intellectual problems.

Chromosome microarray testing

A chromosome microarray (CMA) test is used to check for extra or missing DNA segments. A small blood sample is taken and tested for the number of copies of particular DNA segments. **Copy number variations** (CNVs) are changes in the genome due to a gene or segments of a gene being duplicated or deleted. There are some copy number variations that have a normal variation in the number of copies of one or more sections of the DNA. The CNV can be one gene or a set of genes.

If the diagnostic test finds a copy number variant further testing is carried out to determine which particular genes are involved. Knowledge of the particular genes will then enable the medical practitioners to determine the health risk, if symptoms will develop over time and if offspring will also be affected. Some copy number variants are benign and do not decrease the health of the person or affect development.

During DNA replication duplication of DNA segments can occur so that one person may have cells with different numbers of CNVs. CNVs are associated with cancer cells, autism, learning disabilities and schizophrenia.

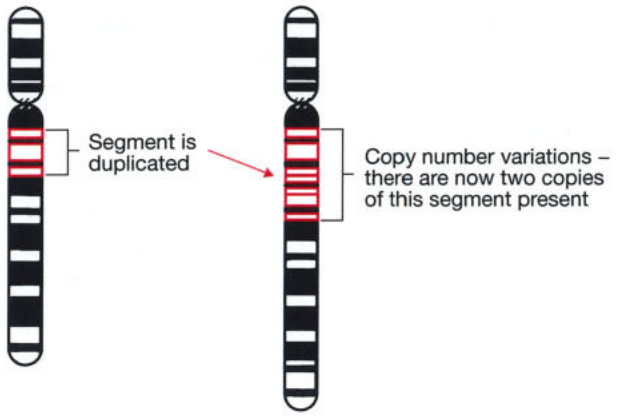


Figure 19.2 Copy number variations can occur when segments of DNA are duplicated.

DNA fingerprints and DNA profiles

Genetic testing is used to identify particular individuals. DNA profiling, also called DNA fingerprinting or genetic fingerprinting identifies particular DNA sequences and the resulting pattern can be used to identify individuals. Each person has a unique DNA profile, except for identical twins. The tests looks at the information in the non-coding DNA in the chromosome, especially where there are large numbers of repeated letter sequences. **Short tandem repeats** (STRs) are short microsatellites that consist of a unit of two to thirteen nucleotides repeated hundreds of times in a row on the chromosome and the pattern of the STRs at particular sites on a chromosome is unique to each person.

DNA profiles are used in forensic science to solve crimes and in paternity cases.

QUESTIONS

1. What is meant by a prenatal test?
2. Describe an example of prenatal testing and discuss why it is important.
3. (a) What is a CMA test?
   (b) What are CNVs?
   (c) Outline what happens if diagnostic tests find CNVs.
4. (a) What are STRs?
   (b) Identify a use of STRs.
5. What is the best description of a short tandem repeat?
   (A) Gene segments that are duplicated in a chromosome.
   (B) Insertion of an additional A-T 'rung' in a double helix code after another A-T 'rung'.
   (C) Duplication of a short chromosome.
   (D) Unit of 2 to 13 nucleotides repeated hundreds of times in a row in the DNA molecule.

20 The Cell Cycle

The **cell cycle** is a series of events that occur in the life of eukaryotic cells. It consists of mitosis, cytokinesis and the stages of interphase which are G_1 stage (growth stage 1), S stage (synthesis stage) and G_2 stage (growth stage 2). Mitosis is the shortest phase of the cell cycle taking less than 10% of the time and interphase taking 90% of the time of the cycle.

Mitosis is the division of the nucleus in cell division. **Cytokinesis** is the division of the cytoplasm and follows mitosis. In growing cells most of the time is interphase when DNA, other molecules and cell components are synthesised.

The timing of the cell cycle varies for different types of cells and for different times in the life of a particular species.

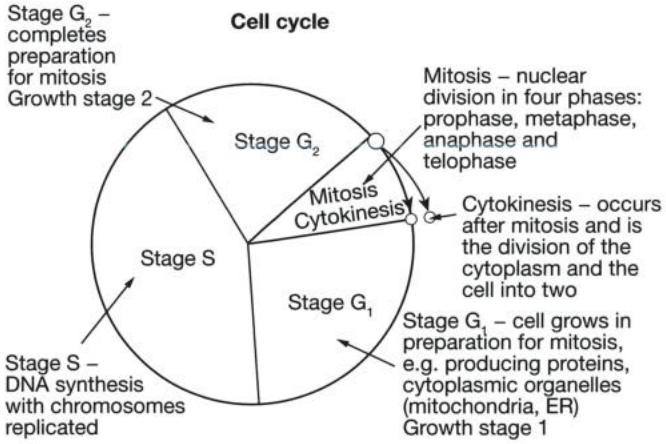


Figure 20.1 The cell cycle.

M phase

The M phase is when mitosis occurs, followed by cytokinesis and the eukaryotic cell divides to form two identical daughter cells.

G_1 phase

The G_1 phase is the 'first gap' when normal cellular activity resumes after the cell division and mitosis. The G_1 phase can be a long phase and is where cells that become differentiated leave the cell cycle and stop dividing. The cells grow by producing proteins and cytoplasmic organelles.

S phase

The S phase is a synthesis phase when DNA is replicated. The amount of DNA in the nucleus doubles in this phase. The chromosomes are not visible with a light microscope during this phase. The chromosomes exist as very long thin threads of **chromatin** which is a three-dimensional network of DNA and associated protein molecules. The associated proteins help keep the structure of the chromosome and help control the activity of the genes.

G_2 phase

The G_2 phase is the 'second gap' when there is synthesis of other cellular molecules in preparation for cell division. The G_2 phase usually involves rapid cell growth.

Cell cycle checkpoints

A checkpoint is a control point in the cell cycle where stop and go-ahead signals regulate the cycle. Three checkpoints are the G_1 checkpoint, G_2 checkpoint and the metaphase checkpoint.

The **G_1 checkpoint** is the restriction point and if it receives the go-ahead the cell will usually complete the cell cycle and divide. In mammalian cells the G_1 checkpoint determines if the rest of the cycle occurs or if the cell goes into a non-dividing state called the **G_0 phase**. The G_0 phase is a resting state and can be the state when some types of cells reach their mature form, e.g. nerve and heart muscle cells move to the G_0 phase when differentiation is complete. Some cells stay at the G_0 phase for some time and will begin to divide if needed, e.g. parenchymal cells in the liver and kidney. While other cells rarely go into the G_0 phase, e.g. epithelial cells.

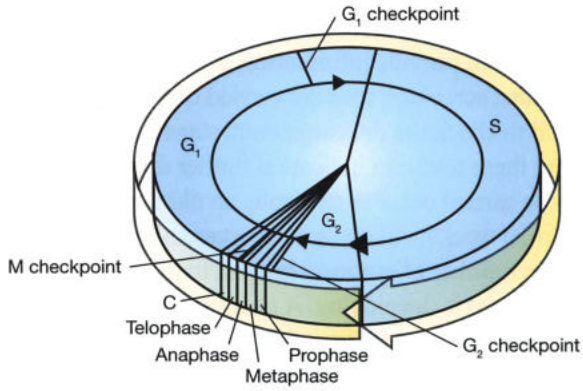


Figure 20.2 Checkpoints in the cell cycle.

The **G_2 checkpoint** triggers mitosis and makes sure chromosomes have been replicated and DNA is not damaged before mitosis. If DNA damage is detected at the G_2 phase checkpoint the cell cycle will halt until the damage is repaired.

The **M checkpoint** is the metaphase checkpoint in mitosis and is the DNA damage point to check if sister chromatids are correctly attached before anaphase. The mitotic spindle needs to be aligned correctly at this checkpoint for the rest of mitosis to occur.

Checkpoint proteins

At certain times of the cell cycle there are checkpoint proteins that keep the steps of the cycle in order and keep the cycle operating normally, e.g. the S phase promoting factor (spf) and the mitosis promoting factor (mpf). These regulatory proteins are mainly **kinases** and **cyclins** and are part of the **cell cycle control system**.

Disruption of the cell cycle

The cell cycle can be disrupted due to genetic predisposition or the action of mutagens that gives rise to uncontrolled cell division including cancer and abnormal embryonic development.

Genetic predisposition means that the genome of the person contains specific genetic variations, often inherited from a parent that gives the person an increased likelihood of developing a particular disease or disorder.

A **mutagen** is a chemical or physical agent that causes a mutation changing DNA and the genetic information.

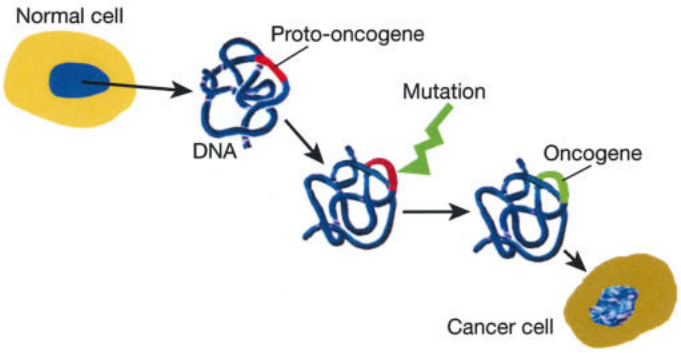


Figure 20.3 Proto-oncogenes and oncogenes.

Proto-oncogenes are a group of normal cellular genes that have the potential to become oncogenes. Proto-oncogenes code for proteins that stimulate cell growth and division, inhibit cell differentiation and cause cell death. An **oncogene** is a gene that is part of the normal genome or a gene found in viruses that triggers cancerous characteristics. Mutations can cause a proto-oncogene to become an oncogene. This means the oncogene will cause increased cell division, decreased cell differentiation and will inhibit cell death. The uncontrolled cell division will form a mass of cells called a **tumour** (neoplasm). Tumours can occur in any organ or tissue. A benign tumour is a mass of cells that will stay at the original site and have a slower growth rate than a malignant tumour. The cells of a malignant tumour detach and spread to other organs and tissues.

QUESTIONS

1. (a) What is the cell cycle?
   (b) What events make up the cell cycle?
   (c) What is the shortest phase of the cell cycle?
2. What proportion of the cell cycle is usually interphase?
3. What is mitosis?
4. What is cytokinesis?
5. Construct a table to summarise what happens during each of the phases of interphase.
6. What is a checkpoint?
7. What is the function of the G_1 checkpoint?
8. What is the function of the G_2 checkpoint?
9. What is the function of the M checkpoint?
10. If cells are given radioactively-labelled thymine the processes of the cell cycle can be investigated. Thymine is part of DNA but not part of RNA. Explain why the radioactive thymine is able to show the timing of the different phases of the cell cycle.
11. What is involved in the cell cycle control system?
12. Identify the regulatory proteins involved in the cell cycle control system.
13. What is the G_0 phase and when does it occur?
14. What is chromatin?
15. In active mammalian tissue the cell cycle can take 12 to 20 hours. The diagram shows the cell cycle for an active skin cell.

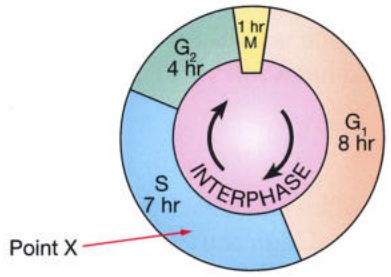


Figure 20.4 The cell cycle in an active mammalian tissue.

   (a) How long is the cell cycle for this cell?
   (b) What is the longest phase of this cycle?
   (c) What would be happening in the cell at point X of the cell cycle?
16. What can disrupt the regulation of the cell cycle?
17. What is meant by genetic predisposition?
18. What is a mutagen?
19. Define proto-oncogene.
20. Outline the function of most proto-oncogenes.
21. Define an oncogene.
22. What can cause a proto-oncogene to become an oncogene?
23. Outline what happens when a proto-oncogene becomes an oncogene.
24. What is a tumour?
25. Differentiate between a benign tumour and a malignant tumour.
26. Which of the following occurs when a proto-oncogene is mutated to become an oncogene?
   (A) Cells become specialised.
   (B) Uncontrolled cell division.
   (C) Increased cell death.
   (D) Increased cell differentiation.
27. Which of the following is least likely to disrupt the cell cycle?
   (A) Ionising radiation.
   (B) Genetic predisposition.
   (C) Chemical mutagens.
   (D) Cells entering G_2 phase.

21 Mitosis and Cytokinesis

Mitosis is a process during cell division in which the cell nucleus divides into two. Mitosis is needed to create new cells for growth, repair and reproduction. Since all organisms begin life as one cell, a fertilised egg, mitosis is essential to become a multicellular organism.

Mitosis increases the number of cells. Even when the organism has ceased 'growing', mitosis is still needed. Older cells need replacement as they are damaged or worn out and some parts, even in adults, are continually growing, e.g. hair and fingernails in humans.

Genetic information is transferred to new cells as DNA and DNA is present in nuclei, mitochondria and chloroplasts. Each new daughter cell is identical to the parent cell and has a full set of chromosomes.

Interphase

Interphase is often called the 'resting' stage but it is actually a time of high activity between cell divisions, when the DNA is replicating and when other organic molecules and cell components are being synthesised. The nucleus appears as a dark staining material called chromatin and the nucleolus is visible. Chromosomes are *not* visible. The cell cycle consists of mitosis and interphase.

The stages of mitosis

Mitosis is a continuous process that can take from fifteen minutes to several hours to complete. In mitosis a cell splits into two with each new daughter cell having the same genetic information as the parent. For convenience, mitosis has been divided into stages: prophase, metaphase, anaphase and telophase.

Prophase. The chromosomes become visible as long, thin threads which shorten and thicken. The nucleolus and then the nuclear membrane disappear. The chromosomes appear as two chromatids joined together by a centromere. In animal cells the centriole divides into two and each moves to opposite ends of the cell. Plant cells do not have centrioles.

Metaphase. The chromatids line up along the central plane of the cell and the spindle forms. Each pair of chromatids are attached to the spindle at the centromere. At this time the chromosomes have maximum shortening and thickening with individual chromatids and the centromere clearly visible.

Anaphase. Anaphase is the shortest stage in mitosis. The centromere divides and the pair of chromatids separate as each one is pulled to opposite ends of the cell by the spindle fibres as the fibres shorten.

Telophase. The spindle disappears, the sets of chromosomes at the poles condense, the nucleoli reform, and a nucleus forms around each group. At this time cytokinesis occurs and the cells split into two distinct groups. Two new cells are formed.

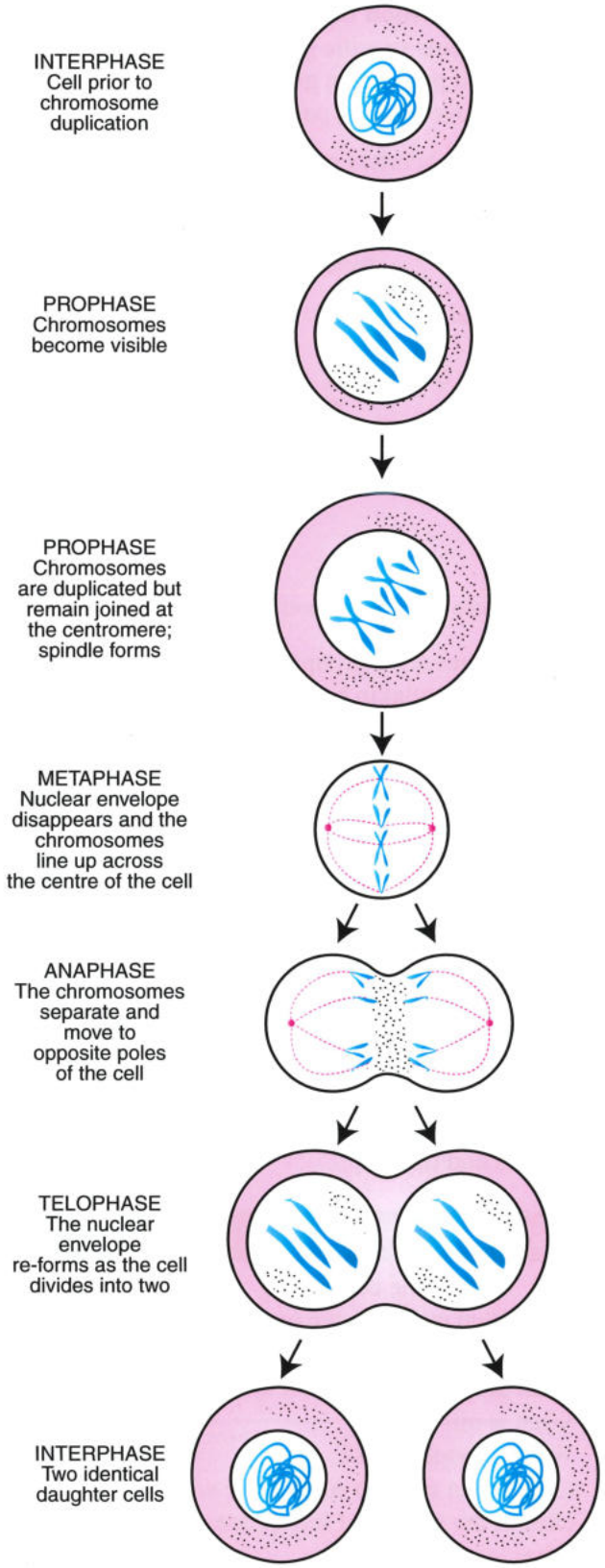


Figure 21.1 Stages of mitosis.

Cytokinesis

Cytokinesis is the division of the cell's cytoplasm following the division of the nucleus. Cytokinesis is important because it stabilises the internal concentration of materials in the two new cells.

The cell organelles are evenly divided and distributed between the two daughter cells. Each new cell needs sufficient organelles such as mitochondria, ribosomes and endoplasmic reticulum so that it can grow and carry out the processes of living.

In cytokinesis a cleavage furrow begins at the centre of the cell. The cell membrane constricts with the help of a contractile ring of microtubules and microfilaments which appear near the cell surface and contract. Cleavage continues as the cell membrane continues to be pinched in until they are separated into two segments.

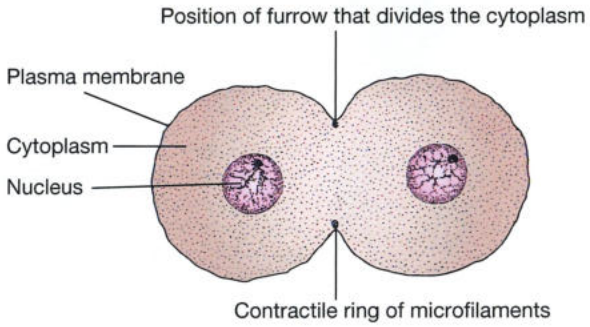


Figure 21.2 Cytokinesis.

Occurrence of mitosis

Once the egg is fertilised to form a zygote cell division by mitosis increases the number of cells. The embryo and then the foetus grow due to mitosis. Most individual cells have a limited life span. They are replaced by identical daughter cells formed by mitosis. The frequency of division varies for different types of cells. Cells that are continuously being replicated include epithelial cells, e.g. in the skin, mucous membranes, secretory glands, ducts and uterus lining. Stable cells that retain the ability to replicate include liver, kidney and pancreatic cells, smooth muscle cells, fibroblast and osteoblasts. Cells that are permanent and do not replicate after completion of normal growth include skeletal and cardiac muscle and most neurons.

QUESTIONS

1. (a) Define mitosis.
   (b) Explain why mitosis is vital for multicellular organisms.
2. (a) Define cytokinesis.
   (b) Explain why cytokinesis is important.
3. Outline the role of mitosis.
4. Which organelles contain DNA?
5. During the 'resting' phase of the cell cycle, is the cell really 'resting'?
6. Describe how cleavage occurs in cytokinesis.
7. How are the cell organelles divided between the two new daughter cells?
8. Describe the appearance of a cell at the beginning of mitosis.
9. Briefly outline the process of mitosis.
10. Describe spindle fibres and their function.
11. When will cells divide?
    (A) The surface area to volume ratio becomes too large.
    (B) Nutrients cannot efficiently reach all parts of the cell by diffusion.
    (C) Membrane surface area is too large to efficiently support cellular metabolism.
    (D) Wastes leave the cell too quickly.
12. A cell with 14 chromosomes undergoes mitosis. How many chromosomes will be in each daughter cell?
    (A) 7 (B) 14 (C) 21 (D) 28
13. The diagram shows mitosis and interphase.

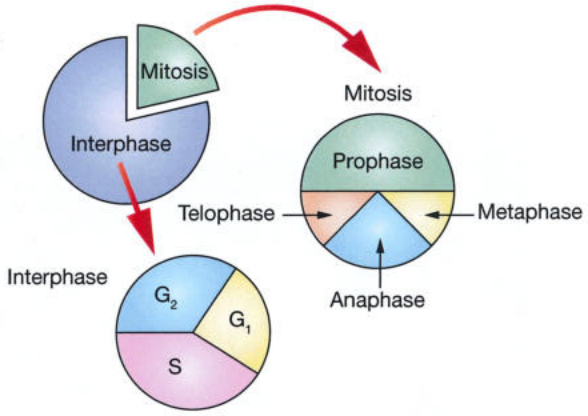


Figure 21.3 Interphase and mitosis.

From this diagram what is the longest phase of mitosis?
(A) S phase
(B) Interphase
(C) Prophase
(D) Anaphase

14. The diagram shows different stages of mitosis.

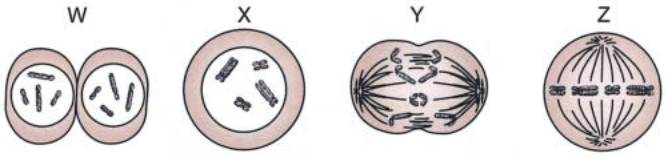


Figure 21.4 Stages of mitosis.

What is the correct order of these phases?
(A) W, Y, Z, X
(B) X, Y, Z, W
(C) Z, Y, X, W
(D) X, Z, Y, W

22 Experiment – Investigating Mitosis

A Human Biology class was given the task of investigating mitosis. Some students decided that they would use a light microscope to view dividing cells. They collected prepared stained slides of dividing epithelial cells.

The students studied the slides and took photos with their cameras of cells at different stages of mitosis.

Once the students had found cells at different stages of mitosis they used secondary sources found in school textbooks and reliable websites to find out what was happening during each phase of mitosis.

The students collated the data and put it into a table.

Other students decided to construct models to show the stages of mitosis.

Student results

The photograph shows the model of mitosis made by one of the students who decided to show the phases of cell division in another way.

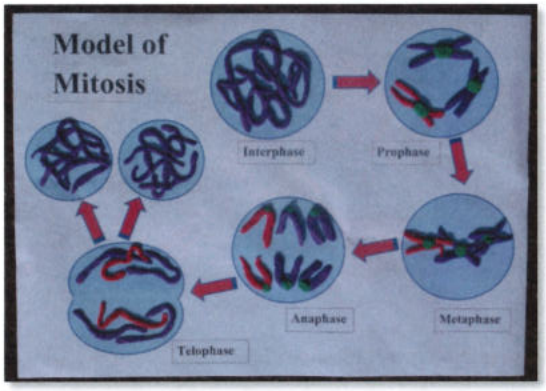


Figure 22.1 Student model of mitosis using plasticine.

The diagram shows a student drawing of the observed prepared slide.

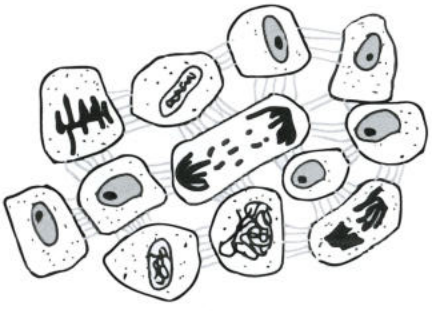


Figure 22.2 Student drawing of human epithelial cells, high power (400x).

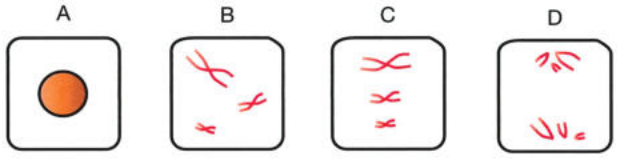
The table shows the results of the work collected by the students who used secondary sources to collect data.

Table 22.1 Stages of mitosis.

Stage of mitosis	Diagram of stage	What is happening
Interphase		Resting phase. Individual chromosomes not visible. Chromatin is replicating.
Prophase		The chromatin fibres of the chromosomes becomes more tightly coiled and the chromosomes become visible as long thin threads, then become thicker and shorter to become two chromatids joined at the centromere. Nuclear membrane disappears, the nucleoli disappear and a spindle forms.
Metaphase		Chromatids line up at the equator of the cell and centre of spindle. Centromere is attached to spindle. The centrosomes at the opposite ends of the cell, at the spindle pole.
Anaphase		Spindle fibres shorten pulling the two chromatids apart taking each to opposite poles of the cell. When the sister chromatids part they each become a full-fledged chromosome. At the end of anaphase each end of the cell has a full set of chromosomes.
Telophase		Chromosomes at each pole. Spindle fibres disappear and two daughter nuclei begin to form. The nuclear membrane forms around each cluster and the chromosomes become less condensed. At the end of telophase cytokinesis occurs and two smaller daughter cells are formed.

QUESTIONS

1. Explain why the students using a light microscope to investigate mitosis chose prepared slides of epithelial cells.
2. What safety precautions would the students need to take when studying prepared slides with a light microscope?
3. Draw a flow diagram to show the process of mitosis.
4. When studying cells under a microscope at what stage are most body cells?
5. The diagram represents different stages of mitosis.



At which stage does the amount of DNA double?
(A) A (B) B (C) C (D) D

23 The Need for Mitosis

Mitosis is the division of the nucleus in cell division and is needed for growth, repair, maintenance and replacement of tissues within the body. There are many sites of mitosis and most unspecialised cells keep the ability to divide. Once many cells become specialised they lose the ability to divide, e.g. many cells of the nervous system.

The frequency of cell division for cells that retain the ability to divide depends on the need for the type of cell being produced. Cells in bone marrow may divide every 8 to 10 hours to produce new blood cells. Other factors that can influence the rate of cell division are the supply of nutrients to the cell and hormones.

A **stem cell** is a relatively unspecialised cell that can divide by cell division to produce identical daughter cells and to differentiate to form different specialised cells. Mitosis occurs at sites where there are stem cells.

Sites of mitosis

In humans, cells are replaced over a period of time but some areas have mitosis occurring most of the time, e.g. skin is constantly replacing cells that are rubbed off, the cell lining of the digestive system is being replaced and bone marrow is constantly making new blood cells.

The growth pattern in humans is direct development from juvenile to adult with a change in body proportions and an increase in size. Mitosis is involved in producing new cells for growth of the individual organism from juvenile to adult.

Table 23.1 Location of mammalian stem cells.

Part of body	Cell types produced at that site
Brain	Oligodendrocyte Astrocyte Neuron
Skin	Epidermis Hair follicle
Gut	Secretory cells
Bone marrow Blood stem cell	Red blood cells White blood cells Platelets
Bone marrow Blood vessel stem cell	Arteries Veins Capillaries
Bone marrow Mesenchymal stem cell	Muscle cell Bone cell Cartilage cell Fat cell

In humans **adult stem cells** are found in several locations. These cells generate specific cell types and can self-renew. During mitosis in **asymmetric division** a stem cell will produce one identical daughter cell and a more specialised daughter cell that will become further specialised.

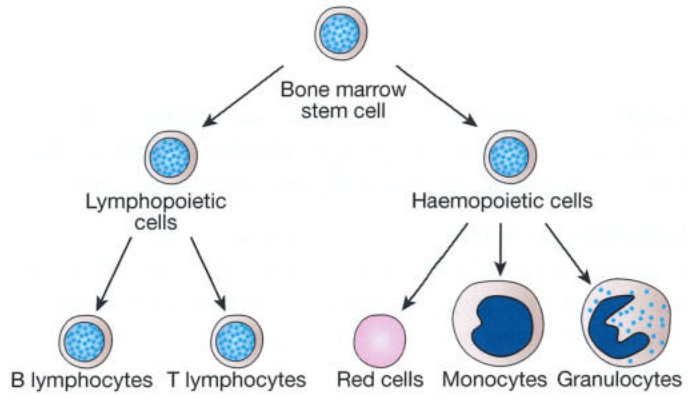


Figure 23.1 Cell types from bone marrow.

Summary of mitosis

Mitosis involves a series of stages. At the beginning of mitosis the nucleus is surrounded by the nuclear membrane. The genetic material has already duplicated, during the S phase, but cannot be seen as the chromosomes have not condensed. During mitosis the chromosomes become visible as two sister chromatids, the chromatids line up at the equator of the cell, the two sister chromatids separate to become chromosomes. The spindle fibres are microtubules with associated proteins which attach to the centromere of the chromatids. During anaphase the microtubules shorten and the chromosome attached to the centromere moves to the pole. Daughter nuclei form at the poles.

Control of mitosis and the cell cycle

The cell cycle involves mitosis and interphase. Interphase consists of three stages – G_1, S and G_2. A group of molecules act in a coordinated manner forming a cell cycle control system. Near the end of the G_1 stage of the cell cycle there is an important checkpoint for human cells. If this checkpoint receives a go-ahead signal the rest of the cycle will continue and the cell will divide. If the checkpoint does not receive the signal the cell switches to a non-dividing state, G_0 phase. Most body cells are in G_0 phase.

QUESTIONS

1. Distinguish between mitosis and cytokinesis.
2. Explain the importance of mitosis.
3. What is the relationship between specialised cells, unspecialised cells and the ability to divide?
4. Identify some types of cells that lose the ability to divide.
5. (a) What is a stem cell?
   (b) What is the relationship between stem cells and sites of mitosis?
6. Mitosis is part of the cell cycle. How is the cell cycle controlled?

24 Stem Cells

A **stem** cell is a relatively unspecialised cell that can divide by cell division to produce identical daughter cells and to differentiate to form different specialised cells.

Totipotent stem cells can give rise to any type of differentiated cells and form a foetus. The union of sperm and ovum forms a zygote and the cells formed from the first few divisions before the formation of the blastocyst are totipotent.

Pluripotent stem cells can form into any of the embryonic germ layers and give rise to all cell types of the body, e.g. embryonic stem cells from the inner cell mass of the blastocyst are pluripotent. They cannot give rise to a placenta and umbilical cord. In mammals pluripotent stem cells are found in the embryo blastocyst – the early stage of development when the cells form a hollow ball. In humans the blastocyst forms about one week after fertilisation.

Multipotent stem cells can develop into more than one cell type but not all cell types, e.g. adult stem cells found in specific tissues are multipotent such as bone marrow stem cells which give rise to the different types of blood cells and brain stem cells which give rise to nerve cells. Multipotent stem cells function to replenish cells lost from normal turnover or disease.

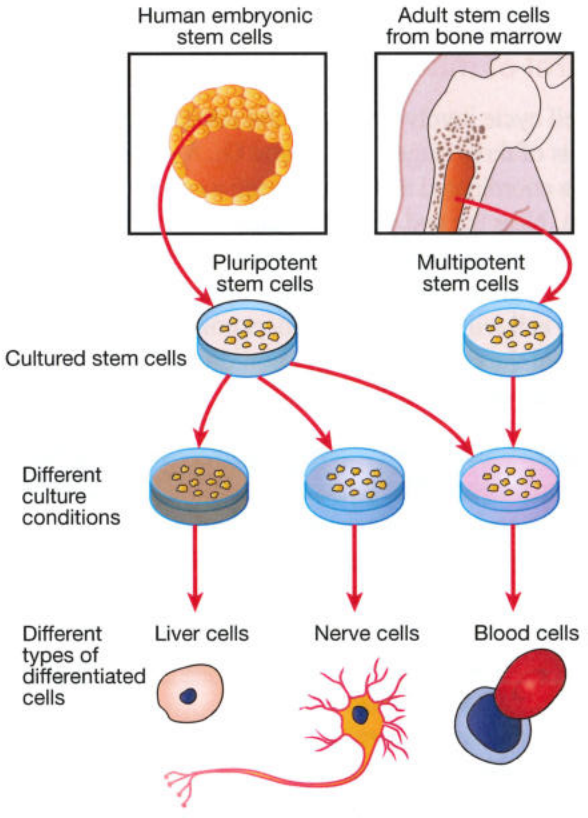


Figure 24.1 Stem cells.

Induced pluripotent stem cells (iPS) have been created in laboratories, e.g. the first iPS were formed using viruses to insert extra copies of genes into tissue specific cells. They can rise to all cell types but there are some differences to naturally occurring pluripotent stem cells.

Embryonic stem cells

If given the correct environmental conditions, embryonic stem cells can keep dividing and propagating themselves indefinitely.

Embryonic stem cells that are pluripotent come from the inner cell mass of a blastocyst. The **blastocyst** is an embryonic stage in mammals that is a hollow ball of cells produced about one week after fertilisation in humans. The blastocyst consists of over 100 cells with a blastocoel which is a fluid-filled cavity and an inner cell mass (ICM). The blastocyst implants in the endometrium, e.g. in humans implantation begins around day 7 and is usually completed by day 14.

Some of the **trophoblast**, the outer layer of cells of the blastocyst develops into the chorion which is an extraembryonic membrane that encloses the entire embryo and forms the placenta. The inner cells of the blastocyst will develop into the embryo and other extraembryonic membranes. These membranes include the amnion which secretes amniotic fluid in which the embryo floats. The inner cell mass will give rise to specialised cells, e.g. muscles, nerves, heart, lung, liver, kidneys. In humans the heart begins beating by the fourth week and can be detected with a stethoscope by the end of the first trimester.

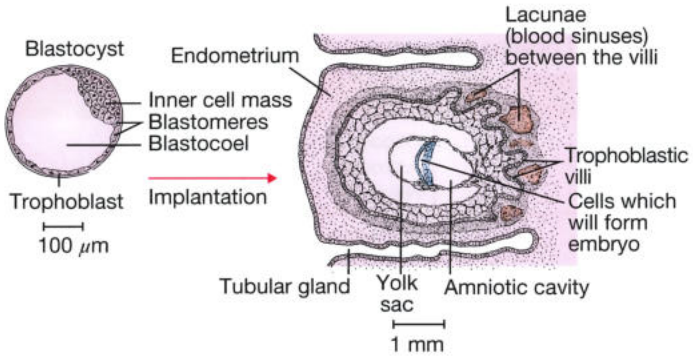


Figure 24.2 Blastocyst and implantation.

The trophoblast cells of the blastocyst cause chemical changes in the endometrium as the embryo penetrates into the endometrial tissues during implantation. As it grows out it begins to form the placenta.

At eight weeks the embryo is called a **foetus**.

Humans, like other vertebrates are triploblastic with three germ layers. The three germ layers are the ectoderm, the mesoderm and the endoderm. Each layer gives rise to particular body parts. **Organogenesis** is the development of body organs from the three germ layers.

The table shows some of the body parts associated with each germ layer.

Table 24.1 Body development from three embryonic germ layers.

Ectoderm	Mesoderm	Endoderm
Skin epidermis Sensory receptors in skin Epithelial lining of mouth and rectum Nervous system	Notochord Skeletal and muscular system Excretory system Circulatory and lymphatic systems Dermis of skin	Epithelial lining of digestive tract and respiratory system Liver and pancreas Thyroid and parathyroid glands Lining of urethra, urinary bladder and reproductive system

Adult stem cells

Adult stem cells or tissue-specific stem cells are unspecialised and undifferentiated and found in particular locations in the body. They divide and produce new cells for repair and maintenance of body parts. The daughter cells will differentiate and become specialised with specific structures and functions. There are several types of adult stem cells, e.g. haematopoietic stem cells in the bone marrow, mammary stem cells, intestinal stem cells, mesenchymal stem cells, endothelial stem cells, neural stem cells, testicular stem cells, olfactory adult stem cells and neural crest stem cells.

Neurospheres are free-floating clusters of neural stem cells cultured in vivo. Investigations into neurospheres aim to isolate neural stem cells which could be used in treating brain diseases.

QUESTIONS

1. Define a stem cell.
2. What are totipotent stem cells and where are they found in mammals?
3. What are pluripotent stem cells and where are they found in mammals?
4. What are multipotent stem cells and where are they found in mammals?
5. What is a blastocyst?
6. In humans what happens to the blastocyst?
7. What is the trophoblast?
8. What happens to the trophoblast?
9. Which cells of the blastocyst become the embryo?
10. In humans when does an embryo become a foetus?
11. Define organogenesis.
12. Jellyfish and corals are diploblastic with an ectoderm and an endoderm. How does this compare with humans and other vertebrates?
13. What is cell differentiation?
14. Describe adult stem cells.
15. What is the function of adult stem cells?
16. In humans identify the body parts that develop from:
    (a) The ectoderm.
    (b) The mesoderm.
    (c) The endoderm.
17. The diagram shows the location of stem cells in an adult human.

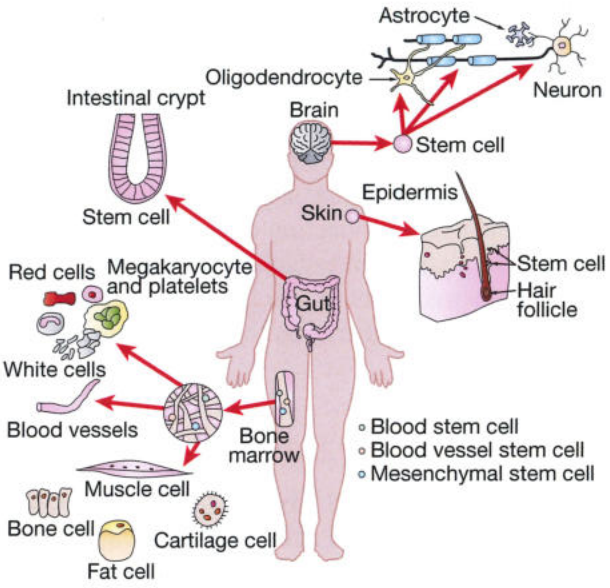


Figure 24.3 Location of stem in in an adult human.

Construct a table to show the location of stem cells in the human body and the cell types that are produced at that site.

18. What are induced pluripotent stem cells?
19. Which cells are likely to form from haematopoietic stem cells?
    (A) Lining of blood vessels.
    (B) Mammary gland cells.
    (C) Lining of the lung.
    (D) Blood cells.
20. The diagram shows a blastocyst.

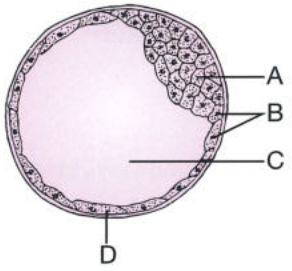


Figure 24.4 Blastocyst.

Which part of the blastocyst has the stem cells that will develop into the embryo and give rise to all cells of the body?
(A) Part A
(B) Part B
(C) Part C
(D) Part D

25 Stem Cell Research

Unlike typical somatic cells of the body which become specialised to perform specific functions in particular parts of the body, stem cells are unspecialised and can keep dividing, renewing themselves even if they have been inactive for a relatively long period of time. Stem cells also produce daughter cells that can be induced under specific conditions to differentiate into many different cell types. This makes stem cells an important resource for investigating the processes involved in normal cell development, the signals that control the genes that lead to the production of the chemicals that control differentiation, investigating the effect of new drugs to aid the treatment of specific diseases and for testing new cell-based therapies for the treatment of disease and injury.

Embryonic stem cell lines

Since embryonic stem cells have the potential to be grown into any type of specialised cells, research scientists are interested in embryonic stem cells as these cells could be used to repair or replace ailing organs or tissues.

In the 1970s researchers started in vitro fertilisation (IVF) procedures fertilising human eggs in test tubes for couples who had difficulty conceiving by other methods. This led to a number of leftover embryos and an ethical problem of the status and what to do with the human test tube babies.

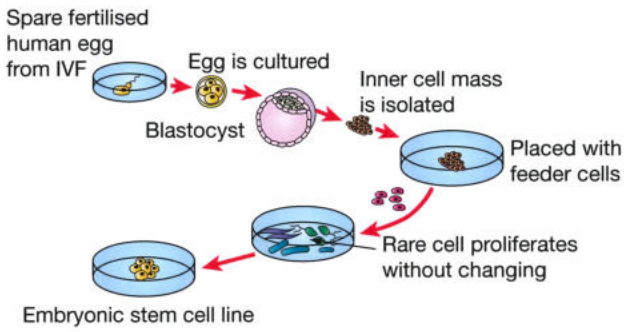


Figure 25.1 Creating an embryonic stem cell line.

In the 1990s, stem cells were successfully removed from available embryos at fertility clinics. This created the world's first **embryonic stem cell line**. If the plated cells survived, divided and filled the plate, individual cells were removed from the plate becoming a source of embryonic stem cells for subculturing. These established stem cell lines can provide millions of embryonic stem cells which can be frozen and sent to other laboratories for subculturing and other research investigations.

Stem cell debate

The international scientific community was excited about the development of embryonic stem cell lines as it has the potential to dramatically change the way diseases and impairments are treated saving lives and improving the quality of life for many people. Treatments were proposed for degenerative conditions, e.g. Parkinson's disease and osteoarthritis, spinal cord injuries and genetic diseases using stem cells in combination with gene therapy.

The use of embryonic stem cells also caused concern in religious and political circles about the potential for unethical use and experimentation. There is the moral dilemma between the need to stop or reduce the amount of suffering of a person with a degenerative condition, impairment or disease and the moral status of the human embryo. Early use of embryonic stem cells required the destruction of the embryo.

Some governments have prohibited some types of stem cell research and other governments have limited funding or placed restrictions on scientific investigations.

In 2007 a new stem cell line was created without destroying the parent embryo. Also in 2007 a method was found to reprogram skin cells to behave like embryonic stem cells, first using mouse skin cells and then using human skin cells. Retroviruses were used to induce extra cloned copies of four 'stem cell' master regulatory genes. These **induced pluripotent stem** (iPS) **cells** behave like embryonic stem cells in many ways though there are differences in the expression of some genes and in cell division. Further research needs to find out more about the actions for iPS cells and if the disrupted DNA in these cells may form damaged and cancerous tissue.

In 2010 the first human was injected with cells from human embryonic stem cells in a clinical trial sponsored by Geron Corp. The cells were injected into people with severe spinal cord injury. There were no signs that the treatment helped the patients.

In 2012 a patient's own stem cells were used to regenerate heart tissue to repair tissue damaged in a heart attack.

International groups

There are several groups that are concerned about stem cell research.

The **International Society for Stem Cell Research** (ISSCR) is an independent non-profit organisation that was established in 2002 to improve the international exchange of information on the developments in stem cell research. ISSCR has experts in science, ethics and law and its members represent the academic community, industry and governments from around the world. ISSCR has developed guidelines for human embryonic stem cell research and promotes a transparent and uniform practice worldwide.

Cell based therapies

Current treatments for many diseases or injuries often involve organ or tissue transplants. The supply of these tissues and organs does not meet the need. Researchers are investigating the use of stem cells for cell based therapies as a source of replacement tissues, e.g. for macular generation, spinal injury, heart defects, diabetes, osteoarthritis and rheumatoid arthritis.

Clinical trials investigate specific new treatments or new ways of using a current treatment and find out if a proposed new therapy is effective and safe. Clinical trials usually start with a small number of people checking for potential risks and patient safety following strict ethical practices governed by regulatory agencies.

Multiple sclerosis (MS) is an autoimmune disease that affects the specialised cells of the central nervous system that produce the myelin which forms a protective sheath around neurons. The damage to the myelin sheath interferes with the proper function of the neurons and there is no current known cure for MS. Researchers are using stem cells to investigate the causes of MS, a way to limit the attack of the immune system on the myelin-generating cells and ways to replace lost or damaged neurons.

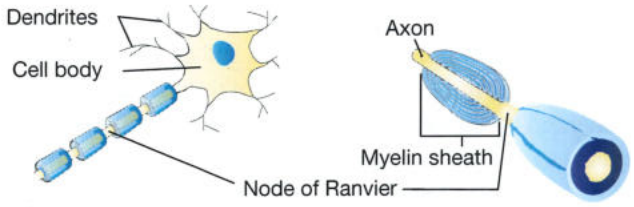


Figure 25.2 Neuron with myelin sheath.

Cardiovascular disease refers to a range of diseases of the heart and blood vessels, e.g. coronary heart disease, heart failure, stroke, hypertension and congestive heart failure. Cell types being investigated in stem cell research into cardiovascular disease include the cardiac muscle cells of the heart, the cardiac pacemaker cells and the endothelial cells that line blood vessels. iPS have been used for screening drugs and plated cells have been created that can beat like a heart in a culture dish. Research is investigating how to repair and replace damaged cells and tissues.

QUESTIONS

1. In what ways are stem cells different to normal somatic body cells?
2. Why are scientists interested in stem cells for research and investigations?
3. Why are embryonic stem cells important for research?
4. What is IVF?
5. Mesenchymal stem cells (MSC) give rise to a variety of cell types, e.g. osteocytes (bone cells), chondrocytes (cartilage cells) and adipocytes (fat cells). Suggest why there is research in mesenchymal stem cells.
6. Discuss why there is a debate about using embryonic stem cells.
7. Outline the importance of international groups in the use of stem cells and stem cell research.
8. Explain why multiple sclerosis is called a demyelinating disease.
9. What is cardiovascular disease?
10. What are the three types of cells being investigated in stem cell research for cardiovascular diseases?
11. In macular degeneration the retinal pigment epithelium (RPE) that lies behind the photoreceptors – the rods and cones of the retina, stops its support function delivering oxygen and nutrients from the blood and removing waste products causing the rods and cones to die. The waste products can build up underneath the RPE.

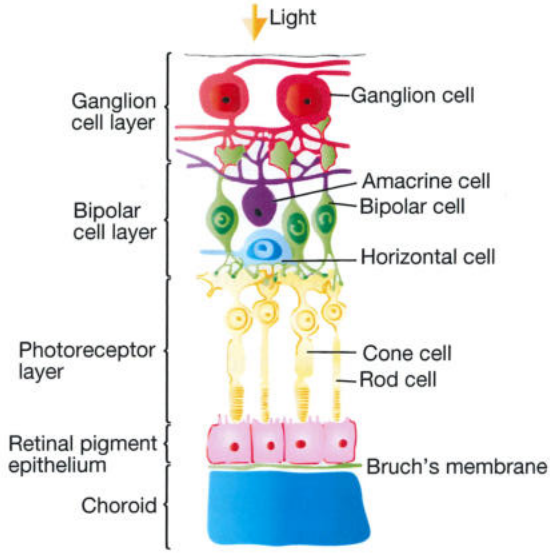


Figure 25.3 Layers in the eye.

Distinguish between each of these research investigations and outline how they could aid treatments for macular degeneration –
Method I – Using induced pluripotent stem (iPS) cells to grow rods and cones or RPE cells.
Method II – Using RPE-specific stem cells from adult RPE from donated eyes to generate new RPE cells.

12. In type I diabetes the immune system attacks the beta cells of the pancreas. Beta cells produce insulin which is a hormone that causes signals to other cells to take up glucose from the blood. High blood glucose levels can cause damage to kidneys, eyes and the nervous system. Which of the following is the most suitable investigation using stem cells to determine a new treatment for type I diabetes?
    (A) Making induced pluripotent cells from pancreatic cells.
    (B) Producing human insulin from transgenic bacteria.
    (C) Creating new exocrine pancreatic cells from adult stem cells.
    (D) Making beta cells from embryonic stem cells.

26 Tumours and Cancers

A **tumour** is an abnormal growth of tissue of a part of the body. This abnormal mass of cells forms a lesion or a lump which can interfere with processes not of the tissue in which it is found but also the processes occurring in adjacent areas. Not all tumours are life threatening.

The tumour cells **stop responding** to normal control mechanisms and ignore the normal checkpoints in the cell cycle continuing to divide and divide.

Metastasis is the spread of the abnormal tumorous cell from its point of origin to another location in the body.

Types of tumours

There are two types of tumours – benign and malignant.

A **benign tumour** stays at the original site and usually does not cause serious problems. The cells of a benign tumour usually appear like relatively normal cells with features of the specialised cells in the tissue where the tumour is located. Most benign tumours can be removed by localised surgery and in many cases a new growth does not reappear.

A **malignant tumour** can spread to new tissues and affect the functions of one or more organs. The genetic and cellular changes of the abnormal cells of a malignant tumour allow the cells to survive in different tissues and organs with the cells of a malignant tumour tending to be less differentiated and have less specialised features. A malignant tumour becomes a **cancer**, though not all cancers have a tumour growth, e.g. blood cancer. Malignant tumours have a **primary growth area** that spreads to other areas called **secondary growths**. These areas can be removed by surgery which is usually followed with chemotherapy and radiotherapy. **Chemotherapy** uses chemicals to kill the cancer cells and **radiation therapy** uses high energy radiation to shrink the tumour and kill cancer cells by damaging their DNA. The high energy radiation damages DNA in cancer cells more than the surrounding cells.

The benign tumour usually has a slower growth rate than a malignant tumour.

Cause of tumours

Tumours and cancers are the result of uncontrolled division of cells. They can be caused by the following factors.

- **Genetic mutations** – the changes in the DNA can be naturally occurring or due to outside agents, e.g. tobacco smoke, ionising radiation, and environmental pollutants. Translocations, deletions and other chromosome abnormalities can be seen in cancer cells.
- **Inherited genetics** – many cancers require two or more mutations before the cancer develops and some families have a hereditary tendency to cancer with one or more mutated genes already in the inherited genome.
- **Lifestyle and behavioural factors**, e.g. overweight, lack of exercise and stress factors. Some physical irritants, e.g. some types of food scar the lining of the intestinal tract causing rapid mitosis to replace damaged cells.

  In many instances the tumour or cancer is due to several factors and not just one issue.

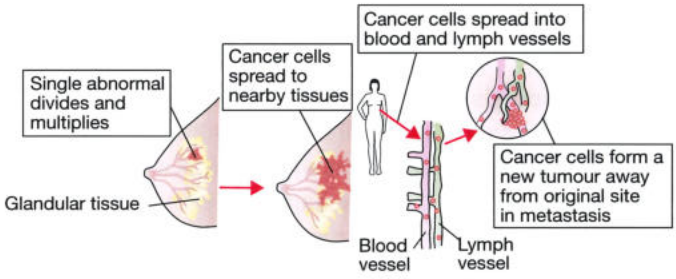


Figure 26.1 Breast cancer.

Different cancers

Somatic mutations occur in body cells and do not have the potential to alter the genetic make-up of the species as they are not inherited by the offspring. Cancers begin when one body cell starts to reproduce uncontrollably and can develop from most types of cells. Each cancer has its own set of causes, is associated with different parts of the body and will spread in different ways and at a different rate. Different cancers include non-melanoma skin cancer, breast cancer, kidney cancer, prostate cancer, pancreatic cancer, thyroid cancer, lung cancer.

Non-melanoma skin cancer develops on sun-exposed areas especially the head and neck. There are two types of non-melanoma skin cancer: **Basal cell carcinoma** begins in the lowest layer of the epidermis and is the most common form. **Squamous cell carcinoma** begins in cells called keratinocytes in the upper level of the epidermis. It can develop in scars or skin ulcers. It is more aggressive and can invade fatty tissues beneath the skin spreading to lymph nodes.

QUESTIONS

1. What is a tumour?
2. Why do tumours form lumps?
3. Construct a table to compare benign and malignant tumours.
4. Outline some causes of uncontrolled cell division and cancer.
5. Discuss how cancers can differ.
6. Distinguish between basal cell carcinoma and squamous cell carcinoma.

27 Meiosis

Meiosis is a two stage type of cell division to produce gametes – eggs and sperm. The four daughter cells produced in meiosis are haploid and not identical to the parent cell. Meiosis only occurs in the gonads – testes or ovaries. Meiosis involves two divisions which are usually called meiosis I and meiosis II.

During meiosis the homologous chromosomes loosely pair along their length aligning gene by gene. **Homologous chromosomes** are pairs of chromosomes of the same length and centromere position with one being inherited from the father and one from the mother. When the homologous chromosomes are aligned during prophase I, they form a **tetrad**.

Crossing over

When the homologous chromosomes are aligned forming a tetrad in prophase I, crossing over can occur. Crossing over involves the exchange of genetic information between two adjacent chromatids. This occurs at the chiasmata and provides new genetic combinations and variation.

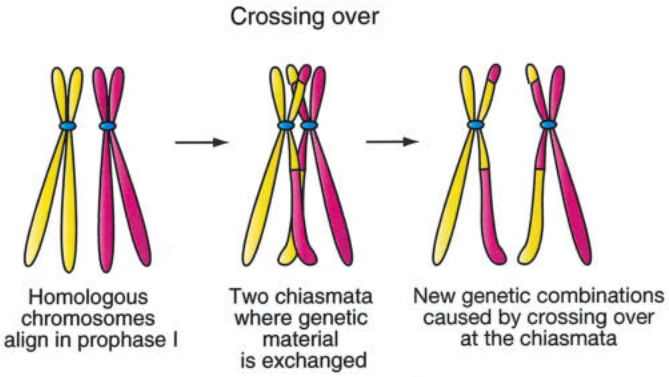
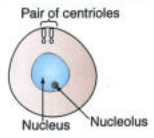
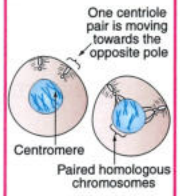
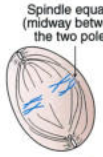
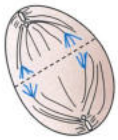
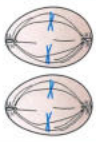
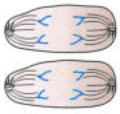
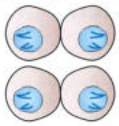


Figure 27.1 Crossing over.

At metaphase I the tetrads line up at the equator of the cell. The separation of the homologous pair with one pair moving to either pole accounts for the law of independent assortment. The **law of independent assortment** states that alleles of genes on non-homologous chromosomes sort independently during gamete formation. Linked genes, on the same chromosome will sort together – unless crossing over occurs.

A gamete has only one allele for each gene. During meiosis the two alleles separate. This is the **law of segregation**. Cytokinesis occurs after meiosis to form the four daughter cells. Fertilisation involves the union of two haploid gametes. This means the zygote is diploid. Thus meiosis is an essential process to enable sexual reproduction to occur, provide variation and maintain the diploid number of the species.

Table 27.1 Stages of meiosis.

Stage of meiosis	Diagram of stage	What is happening
Interphase		Cell growth. Chromosomes replicate forming identical sister chromatids joined at centromere. Centrosome replicates forming two centrosomes.
Prophase I		Chromosomes condense. Diploid number of chromosomes appear as long, thin threads, they shorten, thicken and homologous chromosomes pair with each other gene by gene along their length and are seen as a tetrad of four chromatids. Crossing over can occur at chiasmata. Nuclear membrane disappears. Centrosomes move and spindle forms from microtubules.
Metaphase I		Pairs of homologous chromosomes in the form of tetrads line at the equator with one chromatid of each pair facing each pole. Assortment is random. One homologue of tetrad is attached by fibres to one pole and the other homologue is attached to the opposite pole.
Anaphase I		Chromosomes move towards the poles pulled by spindle fibres as microtubules contract. Sister chromatids stay attached at centromere and move as one unit.
Telophase I and cytokinesis		Each half of cell has complete haploid set with two sister chromatids. Spindle disappears. Cytokinesis and division of cytoplasm occurs at the same time. Two haploid cells form.
Prophase II		Spindle apparatus forms at right angles to previous spindle. Due to earlier crossing over, the two sister chromatids may not be genetically identical.
Metaphase II		Sister chromatids line up at equator and attach to spindle fibres.
Anaphase II		Centromeres separate and sister chromatids are pulled apart and move to opposite poles. Chromatids now act as individual chromosomes.
Telophase II and cytokinesis		Chromosomes reach poles and spindle disappears. Nuclei form, chromosomes begin condensing, nuclear membrane reappears, nucleolus becomes visible and cytokinesis occurs. Four daughter cells form which can each be genetically unique.

QUESTIONS

1. Define meiosis.
2. How many daughter cells are produced in meiosis?
3. Where does meiosis occur?
4. Describe the daughter cells produced by meiosis.
5. What are homologous chromosomes?
6. State the law of independent assortment.
7. What are linked genes?
8. Identify how linked genes can be inherited independently of each other and become part of different gametes.
9. State the law of segregation.
10. Draw a flow chart to show the process of meiosis.
11. The diagram shows a cross-section of human testis using low power of a light microscope.

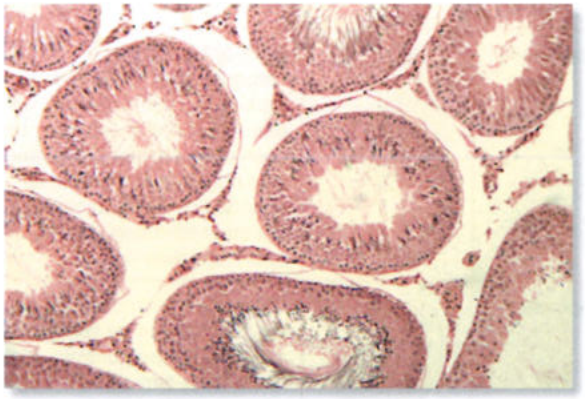
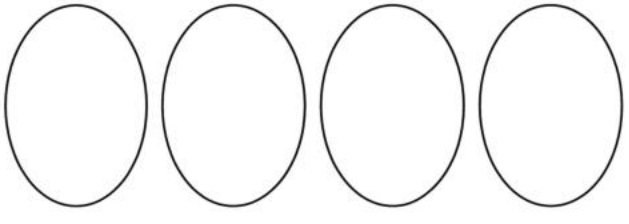


Figure 27.2 LP human testis showing several seminiferous tubules.

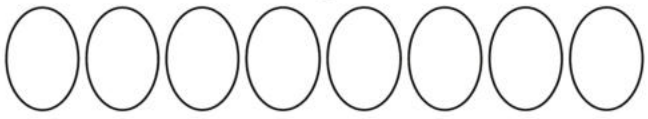
Explain why a testis was chosen to study the stages of meiosis.

12. The diagram shows a cell with $2n = 6$. Copy the diagram and fill in the possible cell alignments for metaphase I and the possible gametes (no crossing over).

Possible alignments in metaphase I



Possible gametes



13. The diagrams show the stages of spermatogenesis in a testis and how it relates to meiosis.

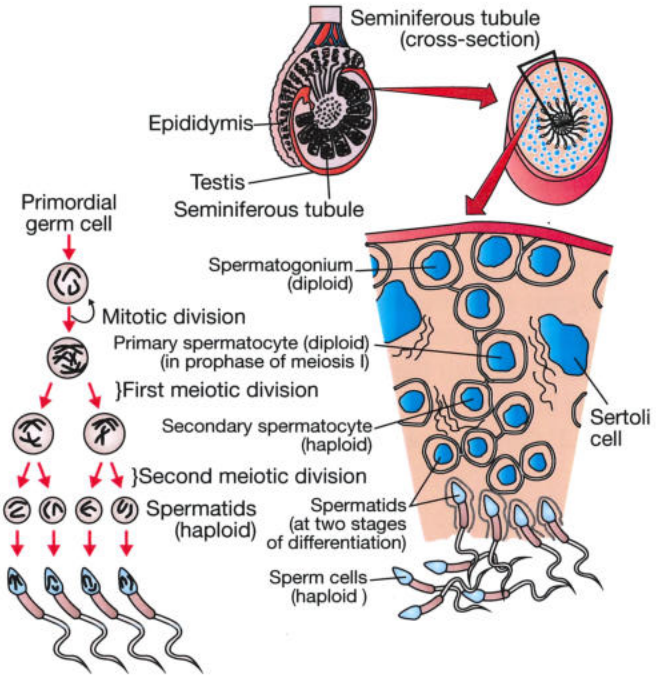
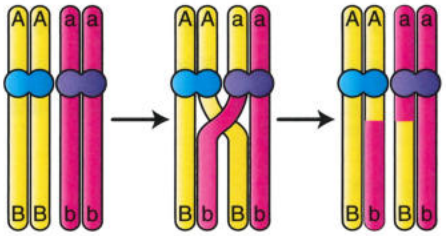


Figure 27.3 Spermatogenesis and meiosis.

(a) Define spermatogenesis.
(b) Distinguish between a primary spermatocyte and a secondary spermatocyte.

14. The diagram shows crossing over with a pair of homologous chromosomes. Alleles A, a B and b are indicated on each chromosome.



What are the genotypes of the daughter cells?
(A) AA, aa, BB, bb (B) AB, Ab, aB, ab
(C) AB, AB, ab, ab (D) AA, AB, bb, ab

15. In which stage of meiosis would you expect to see the homologous chromosomes separated and moving to opposite poles?
(A) Prophase I (B) Anaphase I
(C) Prophase II (D) Anaphase II

16. Which of the following occurs in meiosis but not in mitosis?
(A) Homologous chromosomes align.
(B) Chromatids line up at the equator.
(C) Spindle fibres move chromosomes to the poles.
(D) Chromatids joined at centromere.

28 Processes in Meiosis

Meiosis was first discovered and described in the 19th century. In the 20th century new staining techniques allowed biologists to better observe cell division in meiosis.

Genetics is the study of heredity and hereditary variation. Meiosis is very important in providing a means for variation to occur from one generation to the next.

Meiosis is cell division to produce haploid daughter cells. It is **a reduction division**, coming from the Greek word meaning 'make smaller' and only occurs in sexually reproducing organisms.

The 'normal' number of chromosomes for a species is the **diploid number**, or $2n$, and is found in body cells such as muscle cells, nerve cells and epithelial cells. The **haploid number**, or 'n', is half the diploid number and is the number of chromosomes found in a gamete, or sex cell (sperm and ovum). For humans the diploid number is 46 which means that a normal body cell has 46 chromosomes.

During interphase before meiosis the chromosomes replicate. When meiosis begins the chromosomes appear as long threads that shorten and thicken to become visible as two chromatids joined together at the centromere. In meiosis the homologous chromosomes pair up to form a tetrad or **bivalent** in prophase I. The paired chromosomes in the bivalent are joined by an extremely thin protein framework and crossing over is likely to occur at this time. The bivalents then line up at the equator in metaphase I and separate during the first meiotic division. **Homologous chromosomes** are the corresponding pairs that look the same, except for the sex chromosomes X and Y. Each gamete formed by meiosis will contain only one of each homologous pair.

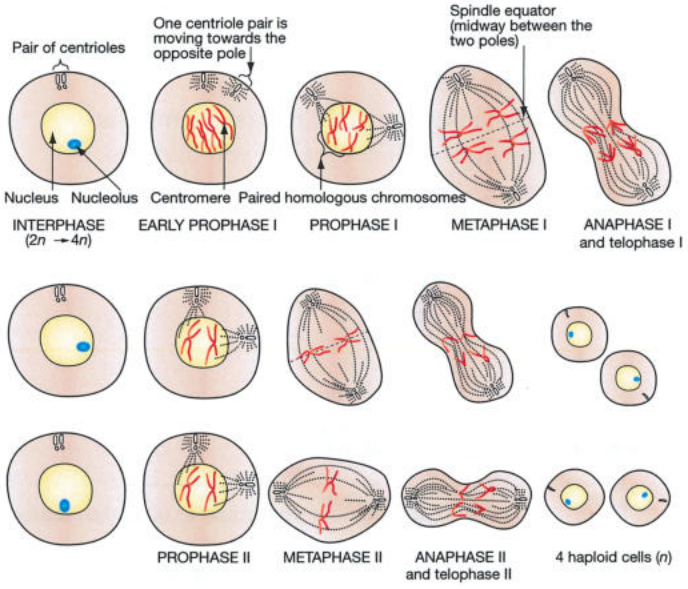


Figure 28.1 Stages of meiosis.

During meiosis, when the homologous chromosomes are paired in the tetrad, crossing over can occur. **Crossing over** is the exchange of certain sections of chromosomes, producing new linkage groups. The point at which the chromosomes cross and break is called the **chiasma** (plural chiasmata).

Cytokinesis follows the division of the nucleus and the cell organelles are distributed.

Meiosis produces haploid gametes. The diploid number is restored when fertilisation occurs. Random fertilisation also increases genetic variation.

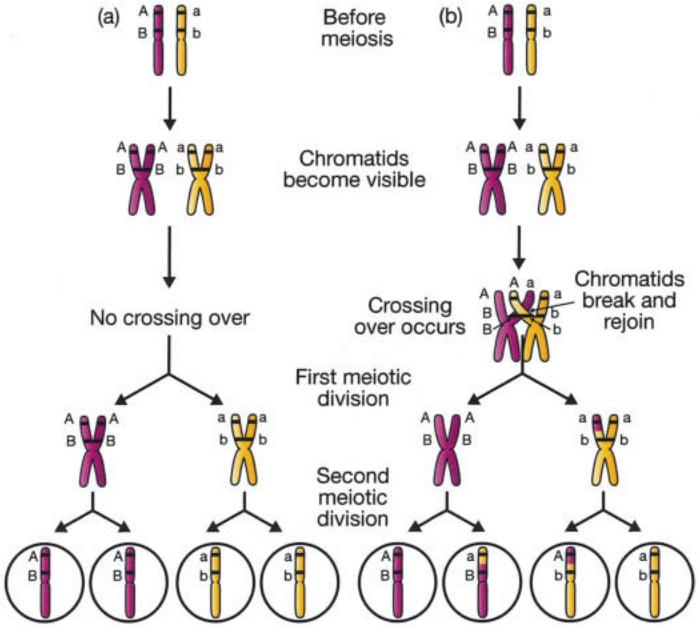


Figure 28.2 Crossing over.

QUESTIONS

1. Define meiosis.
2. Define genetics.
3. Explain why the process of meiosis is studied as part of a genetics study.
4. What is the difference between the diploid number and the haploid number?
5. What is the diploid and haploid number for humans?
6. What are homologous chromosomes?
7. Define crossing over.
8. What is the chiasma?
9. Explain why crossing over is important.
10. At which stage of meiosis is crossing over likely to occur?
11. Explain why meiosis is called the 'reduction division'.
12. In humans the diploid number is 46. What would be the number of chromosomes after:
    (a) Meiosis? (b) Mitosis?
13. When does the DNA replicate for meiosis?
14. Name three different factors that can lead to variation in a population.

15. (a) Draw a diagram of a cell undergoing cell division to show crossing over between linked genes G and H. The parent is heterozygous for both linked genes with genotype GgHh. Label the crossing over point A and each allele.
(b) Identify the type of cell division.
(c) Write the genotypes of the possible daughter cells from this division.
(d) State the biological significance of this event.
(e) Identify a part of a human where this process occurs.

16. The diagram shows a cell which will divide by meiosis.

Figure 28.3 Cell about to divide by meiosis.

Which of the following shows the result of this division?

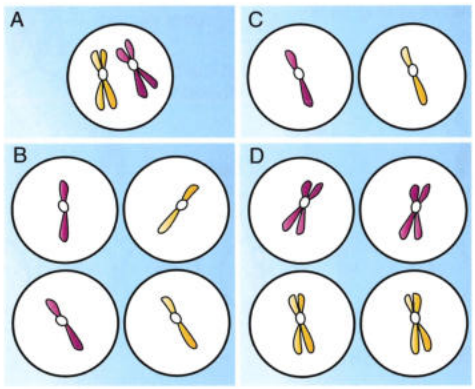


Figure 28.4 Results of cell division.

17. The diagram shows the parent cell and two daughter cells which formed during cell division.

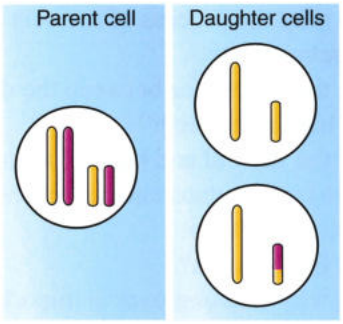


Figure 28.5 Parent cell and two daughter cells.

What can you conclude about the cell division?
(A) Mitosis and crossing over occurred.
(B) Meiosis and all chromosomes experienced crossing over.
(C) Meiosis and crossing over occurred in one analogous pair.
(D) Meiosis and crossing over occurred in one homologous pair.

18. The diagram shows a cell.

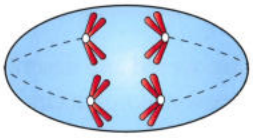


Figure 28.6 Cell.

Which of the following is correct for this cell?
(A) It is found in a mammalian ovary.
(B) It is found in the dermis layer of the skin.
(C) It is a female gamete.
(D) It is a zygote.

19. The diagram shows a map of some of the genes on human chromosome 19.

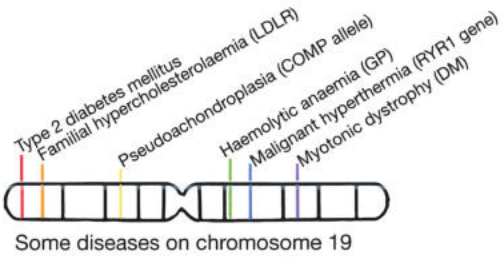


Figure 28.7 Chromosome 19.

Between which would have the greatest frequency of crossing over?
(A) Malignant hyperthermia and myotonic dystrophy.
(B) Haemolytic anaemia and pseudoachondroplasia.
(C) Type 2 diabetes and familial hypercholesterolaemia.
(D) Type 2 diabetes and myotonic dystrophy.

20. Why does gamete formation and sexual reproduction produce more variation of phenotypes in the offspring?
(A) Meiosis involves two divisions rather than one division.
(B) There is random assortment of homologous chromosomes in meiosis.
(C) Mutation is more frequent in meiosis than all other processes.
(D) Increased rate of cell division in meiosis compared to other types of cell division.

21. The diagram shows four cells P, Q, R and S at different stages of meiotic division.

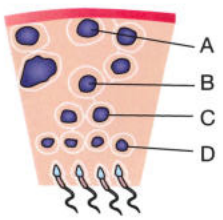


Figure 28.8 Part of a seminiferous tubule in a human testis.

At which location is crossing over most likely taking place in a cell?
(A) A (B) B (C) C (D) D

29 Non-Disjunction

Non-disjunction is an error in either mitosis or meiosis where both members of an homologous pair or both sister chromatids fail to properly separate. Non-disjunction is a cause of aneuploidy. **Aneuploidy** is a genetic condition where there is an abnormal chromosome number, e.g. there are one or more additional copies of chromosomes or there is a deficiency in one or more chromosomes.

Non-disjunction in meiosis

If an homologous pair fails to separate properly during meiosis one of the daughter cells which will become a gamete will not have a particular chromosome while another daughter cell, another gamete, will have two copies of a particular chromosome.

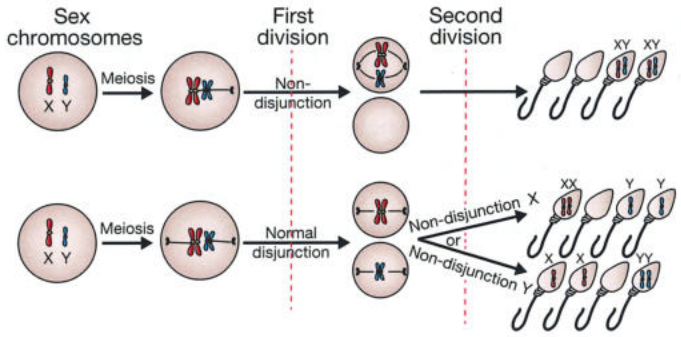


Figure 29.1 Non-disjunction of sex chromosomes at meiosis I or at meiosis II.

Non-disjunction in the first meiotic division is due to the homologous chromosomes failing to separate leading to the formation of two gametes with two of the same type of chromosome and two gametes with no copy of that chromosome. While non-disjunction in the second meiotic division is due to sister chromatids failing to separate.

When a gamete with an abnormal number of chromosomes unites with a normal gamete during **fertilisation** the offspring will have an abnormal chromosome number. The health of the offspring will be affected depending on which chromosome is missing/duplicated.

Down syndrome or trisomy 21 is caused by non-disjunction and an extra copy of chromosome 21. Trisomy means that the individual is diploid but has three copies of one chromosome. A karyogram will show Down syndrome with three sets of chromosome 21.

Autosomal monosomy where there is only one member of a homologous pair seems to be incompatible with life and is not seen in live births. However, there are several **sex chromosome abnormalities**, e.g. a male with an extra X has sex chromosomes XXY which is Klinefelter syndrome and a person with only one sex chromosome (XO) develops as a female with Turner syndrome.

QUESTIONS

1. What is meant by non-disjunction?
2. Define aneuploidy.
3. What happens if an homologous pair fails to separate properly in meiosis?
4. The diagram shows several steps in meiosis.

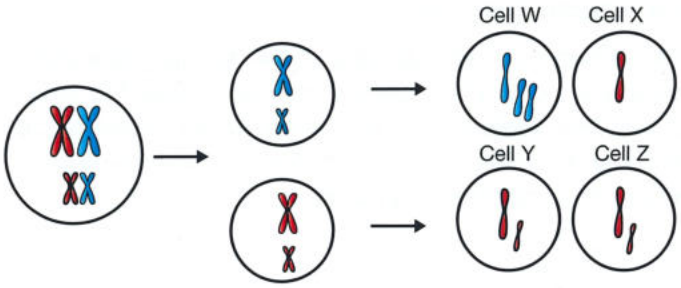


Figure 29.2 Steps in meiosis.

Explain what has happened in this cell division.

5. The microphotograph shows a section of testis seen under high power light microscope.

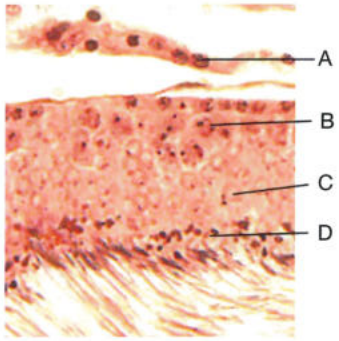


Figure 29.3 Part of a prepared slide of a seminiferous tubule, testis, HP light microscope.

At which point would you expect non-disjunction in meiosis I to occur in the testis?

(A) A (B) B (C) C (D) D

6. The diagram shows two chromosomes in a cell at the beginning of meiosis. If non-disjunction occurs at meiosis I to chromosome 1 which of the four gametes is a possible abnormal result of this non-disjunction?

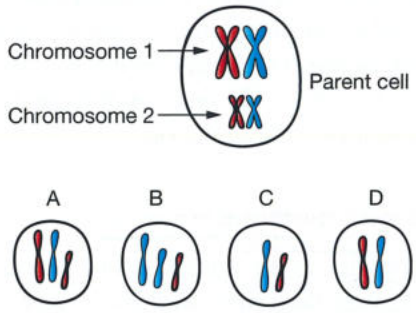


Figure 29.4 Cell beginning meiosis.

(A) A (B) B (C) C (D) D

30 Case Study – Down Syndrome

Down syndrome (trisomy 21) was first identified by Langdon Down in 1866 and in 1959 Jerome Lejeune identified the cause as an extra copy of chromosome 21. It is a congenital condition.

The karyotype of an individual shows the chromosomes present in a cell. The chromosomes are arranged in order of size from longest to shortest with the sex chromosomes last. Chromosome 21 is the smallest chromosome with a relatively small number of genes.

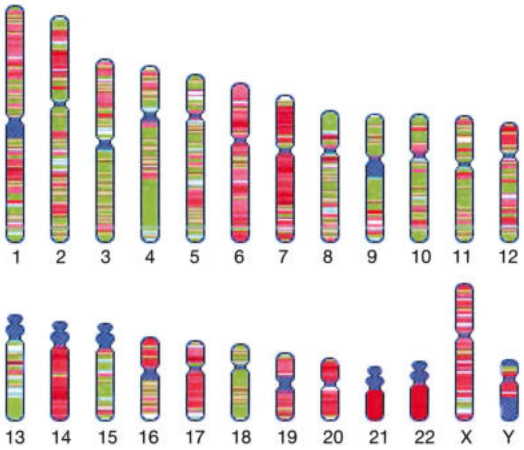


Figure 30.1 Human chromosomes.

Genetic cause of trisomy 21

Most cases are caused by **non-disjunction** of chromosome 21. For example, non-disjunction during gamete formation causes either the egg or the sperm to have two copies of chromosome 21. About 95% of people with Down syndrome have an extra copy of chromosome 21. The other 5% of people with Down syndrome are either mosaics or have translocation Down syndrome. Non-disjunction can also occur after fertilisation during mitosis in the embryo so that some cells have 47 chromosomes and some cells have 46 chromosomes, making these people **mosaics**. **Translocation** occurs when part or all of chromosome 21 attaches onto another chromosome. Translocation usually involves chromosomes 13, 14, 15, 21 or 22 so that the fused chromosome is now a combination, e.g. 13/21 or 14/21.

Incidence of trisomy 21

Non-disjunction is more common in older mothers, so that women having their first child over 36 years are far more likely to have a trisomy 21 child than a 20 year old mother. For this reason, women in the 'at risk' group often have genetic testing of the embryo.

The prospective parents can then use genetic counselling to decide their options. If a baby is found to have mosaicism, 20 different cells, e.g. blood, skin, muscle, bone marrow are analysed and the mosaicism is expressed as a percentage of cells. For example, if five cells have 46 chromosomes and 15 cells have 47 chromosomes, it is called a mosaicism level of 75%.

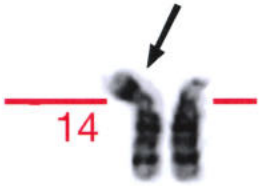


Figure 30.2 Translocation 14/21.

Testing for trisomy 21

There are several tests that can be carried out to detect Down syndrome. Ultrasound screening and blood tests can be used, e.g. ultrasound imaging looks for a small or no nasal bone, large ventricles and measures the nuchal fold thickness and blood tests test the mother's blood for foetal DNA. An amniocentesis or chorionic villus sampling is needed to confirm the diagnosis.

Effects of trisomy 21

There are many symptoms of trisomy 21 and if the person is a mosaic, they will show reduced symptoms depending on which cells and the number of cells affected. Symptoms of Down syndrome include wide-set eyes; flattened face, short, flat-bridged nose; spinal weakness; heart defects (major risk of heart malformations); intestinal defects (risk of duodenal atresia where part of the small intestines is not developed); susceptibility to infection; reduced life expectancy; mental retardation and a significant risk of acute leukaemia. Down syndrome can cause Alzheimer's disease by age 40.

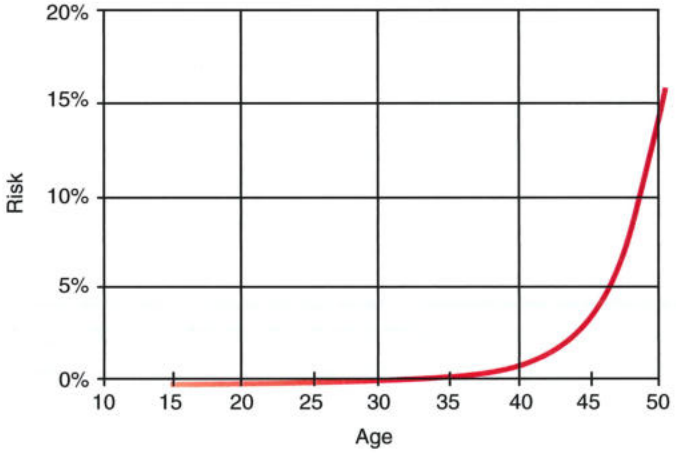


Figure 30.3 Relationship between mother's age and risk of trisomy 21.

Because there are three copies of chromosome 21, Down syndrome causes an overexpression of genes found on chromosome 21. Some studies show that trisomy 21 can cause increased activity in the superoxide dismutase (SOD) gene. SOD converts oxygen radicals to hydrogen peroxide and water. If these oxygen radicals are not converted they can damage cellular structures. Some believe that increase SOD activity causes a build-up of hydrogen peroxide which also damages cells. Oxidative damage to neurons can cause rapid brain ageing similar to Alzheimer's disease. Chromosome 21 also has the gene for the precursor of amyloid protein. Amyloid plaques damage nerve cells causing a decrease in the number of nerve cells in the cortex and are found in both people with Alzheimer's disease and Down syndrome. However, people with Down syndrome also have a very low rate of breast, lung, and gastrointestinal cancers.

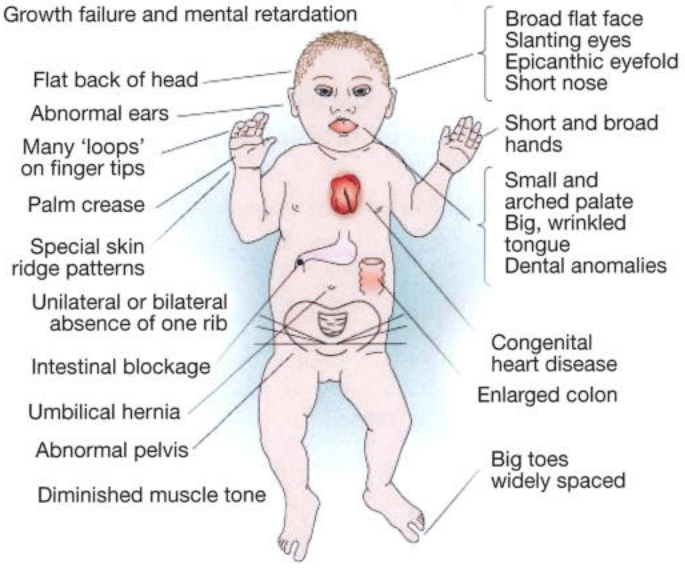


Figure 30.4 Effects of trisomy 21.

Treatment

Children with Down syndrome need help developing language skills and coordinated movement. It is important they have physical therapy, speech therapy and occupational therapy to help develop skills and abilities. For example, they need assistance with gross motor skills for walking, stair-stepping and being aware of body location in space and speech skills for enunciation and articulation.

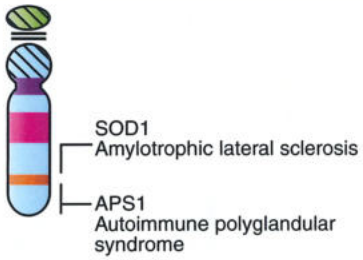


Figure 30.5 Chromosome 21.

QUESTIONS

1. Define trisomy.
2. Down syndrome was the first disease to be recognised as a chromosomal mutation. Who discovered the cause of this disease?
3. What is meant by a congenital condition?
4. Study the karyotype shown below.

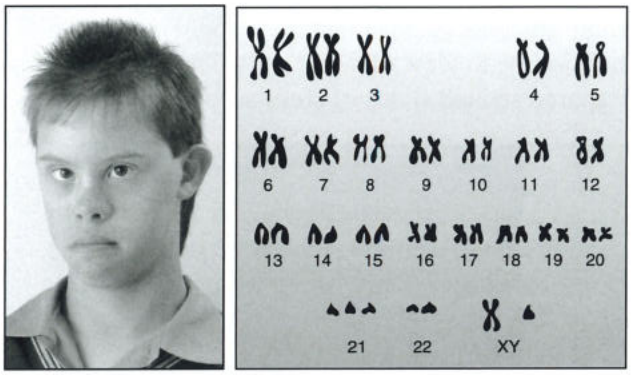


Figure 30.6 Down syndrome.

Suggest the cause of this case of trisomy 21 and give a reason for your choice.

5. Describe some symptoms of trisomy 21.
6. For trisomy 21, distinguish between the effects of a germ line mutation and a somatic mutation.
7. Explain mosaicism in Down syndrome.
8. Explain how an extra copy of chromosome 21 can cause cell impairment and brain damage.
9. Suggest why people with mosaic Down syndrome show a wide range and variety of symptoms.
10. Outline some procedures that can assist people with Down syndrome.
11. Genome and gene research often focuses on particular diseases, e.g. the gene which causes Alzheimer's disease is on chromosome 21. People with Alzheimer's disease have senile plaque caused by an amyloid protein. To study this gene, outline the processes involved in cloning the gene.
12. A research project is investigating the centromere region of chromosome 21. Chromosome 21 has unusual alphoid DNA in the centromere region. Suggest why this research may provide information about the cause of trisomy 21.
13. It has been found that parents with a Down syndrome child who both have normal karyotypes have about a 1% chance of having a second child with Down syndrome. What are the risk factors for Down syndrome?
14. What is the genetics of Down syndrome?
    (A) Three copies of chromosome 14.
    (B) Three copies of chromosome 21.
    (C) Multiple copies of 3 chromosomes.
    (D) Base-pair substation on chromosome 3.

31 Experiment – Investigating Meiosis

A Human Biology class was given the task of investigating meiosis. Each student had to present their work to the class and they were given a marking criteria which they would use to grade each other's work.

Some students decided that they would use a light microscope to view dividing cells. They collected a set of prepared stained slides of cross-sections of testis.

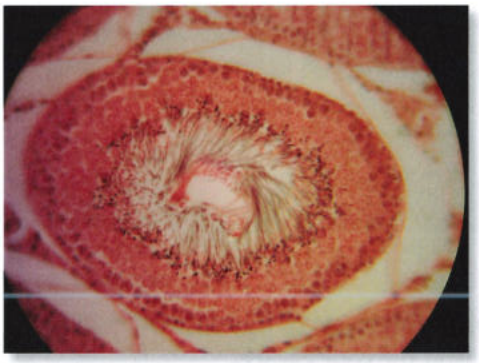


Figure 31.1 Cross-section of testis, low power.

The students took photos of the slides with different magnifications and drew diagrams of the cells they found at the different stages of meiosis. Using secondary sources of textbooks and reliable websites they collected data about what was happening at each stage of meiosis and annotated their diagrams.

Other students decided to use plasticine or other materials to construct models to show the sequence of events in meiosis. They also used secondary sources of textbooks and reliable websites to collect the data about what was happening at each stage of meiosis and used this data to make their models.

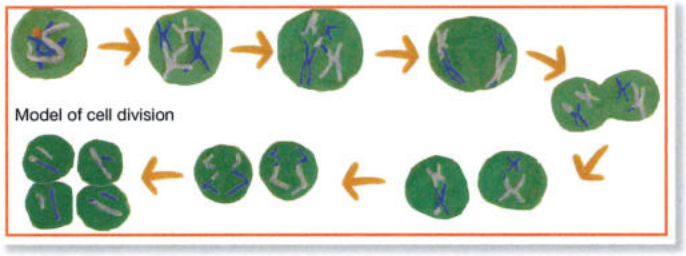


Figure 31.2 Student model of meiosis using plasticine.

After gathering the first-hand and secondary data, the students displayed their work around the classroom. They then graded each other's work assessing the 'correctness' and 'completeness' of information and evaluating each presentation for its appearance, its ability to be easily understood and its use as a model to convey scientific knowledge.

QUESTIONS

1. Explain why the students who used prepared slides and a light microscope to investigate meiosis chose to use slides of cross-section of testis.
2. In this task, what is the first-hand data collection and what is the secondary data collection?
3. Why do scientists make models?
4. When the students were using secondary sources to collect information, how could they determine if the secondary source was reliable?
5. When the students were using secondary sources to collect information, how could they determine if the source was accurate?
6. When the students were using secondary sources to collect information, how could they determine if the source was valid?
7. Discuss why the grading criteria assessed the 'completeness and 'correctness' of each student's presentation.
8. Explain why some would have chosen to make a model of meiosis for their investigation and presentation.
9. The diagram shows several steps in meiosis.

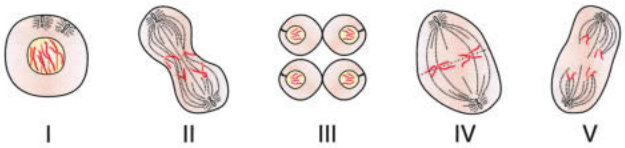


Figure 31.3 Steps in meiosis.

What is the correct order for these steps in meiosis?

(A) I, IV, V, II, III (B) I, IV, II, V, III
(C) III, V, II, IV, I (D) II, V, IV, III, I

10. The diagram shows changes in ovarian cells during the production of gametes.

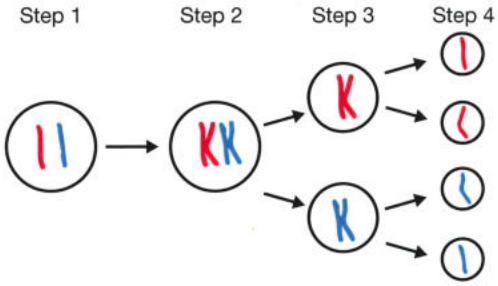
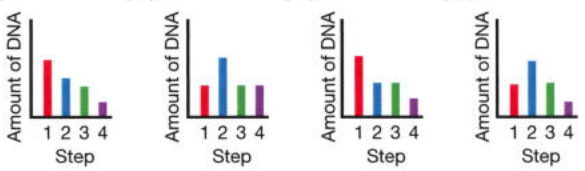


Figure 31.4 Steps in meiosis.

Which of the following graphs shows the relative amount of DNA in a cell at each of the stages 1 to 4?

(A) (B) (C) (D)



32 Comparing Meiosis and Mitosis

Meiosis is a type of cell division that produces gametes. **Gametes** are sex cells and have half the normal number of chromosomes. **Mitosis** is the process in which the cell nucleus divides into two.

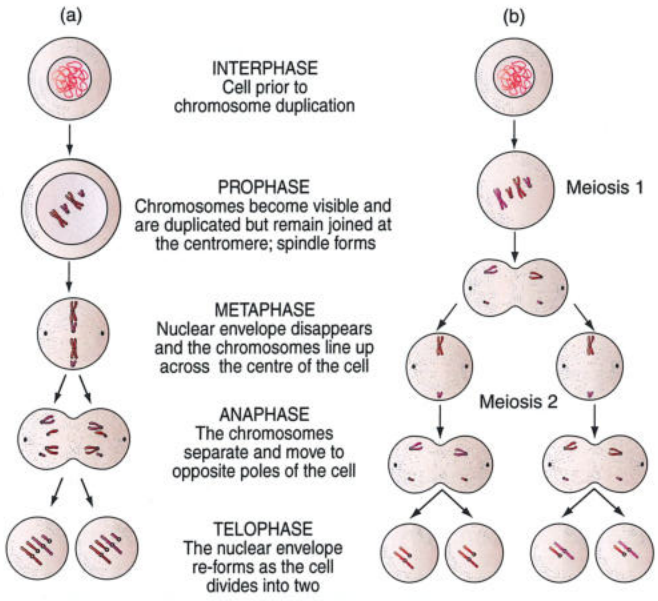


Figure 32.1 Meiosis and mitosis cell division. (a) Mitosis produces cells genetically the same as the parent. (b) Meiosis produces gametes with half the number of chromosomes.

Stages in mitosis

Mitosis begins with the chromosomes becoming visible as two identical chromatids. The nuclear membrane disappears and the chromatids line up at the equator. The chromatids separate and move to the poles carried by spindle fibres. A nuclear membrane forms around each new nucleus and cytokinesis occurs forming two identical daughter cells.

Stages in meiosis

Like mitosis, the cell division begins with the chromosomes becoming visible as long thin threads which condense until they are double strands called chromatids joined at the centromere. The nuclear membrane disappears. However, in the first division, the chromosomes line up in homologous pairs and crossing over can occur increasing the variation in genetic material in the gamete unlike mitosis where the chromosomes line up singly. The spindle apparatus forms and the sister chromatids remain attached and are pulled as a unit towards opposite ends of the cell. The spindle disappears and the cell divides into two. In the second division, the chromosomes do not duplicate. A spindle forms and the sister chromatids line up along the equator. The sister chromatids separate and each moves to an opposite pole. The spindle disappears, the nuclear membrane re-forms and there are now four daughter cells.

There are some similarities between meiosis and mitosis. However, meiosis involves two cell divisions and produces four haploid daughter cells while mitosis involves one cell division and produces two identical daughter cells. Mitosis occurs in all living things while meiosis only occurs in organisms that sexually reproduce. In multicellular organisms mitosis occurs in body cells (somatic cells) and is needed for growth, repair and maintenance while meiosis occurs in the gonads (testes and ovaries) and is needed for sexual reproduction.

QUESTIONS

1. Define meiosis, gamete and mitosis.
2. Construct a table to show the differences between meiosis and mitosis.
3. Explain why meiosis is often called the 'reduction division' while mitosis is called the 'equating division'.
4. What is meant by the diploid number and haploid number of chromosomes?
5. Explain what a homologous pair is using the chromosomes found in humans as an example.
6. Random assortment during meiosis means many different chromosome combinations are possible. With a diploid number of 46, how many different possible combinations could one person produce?
   (A) 46^2 (B) 2^{46} (C) 2^{23} (D) 23^2
7. How is mitosis different to meiosis?
   (A) Mitosis involves two divisions.
   (B) Mitosis forms four daughter cells.
   (C) Meiosis has daughter cells identical to the parent cell.
   (D) Meiosis has the chromosomes line up in homologous pairs.
8. Where does meiosis occur?
   (A) In any somatic cell.
   (B) In the gonads.
   (C) In sperm and ova.
   (D) In any living cell.
9. When can non-disjunction occur?
   (A) Only in meiosis.
   (B) In mitosis and meiosis.
   (C) Only in mitosis.
   (D) Only in the first division of meiosis.
10. What type of cell division forms sperm?
    (A) Meiosis. (B) Mitosis.
    (C) Parthenogenesis. (D) Differentiation.

33 Fertilisation

Fertilisation is the union of two gametes. **Gametes** are sex cells – eggs and sperm are haploid – they contain only one set of chromosomes. When the two gametes fuse they form a fertilised egg called a **zygote**. Fertilisation maintains the diploid number of a species. The contact of the sperm with the egg also triggers the start of the development of the embryo. Human reproduction involves **internal fertilisation** when the sperm and ovum unite inside the female's body.

Gametes are produced in **gonads**, which are the sex organs in males and females. The female gonads are the **ovaries** and the male gonads are the **testes**.

Fertilisation and variation

In fertilisation males release millions of sperm at a time and it is random which sperm fertilises which egg. Fertilisation means offspring have a different combination of characteristics to either parent.

Sexual reproduction is very important as a way to produce **variation** in a population. Variation in a population in sexually reproducing organisms occurs due to fertilisation and two processes that occur during meiosis – random segregation and crossing over. Variation has been important in human evolution with natural selection leading to the current distinctive features of *Homo sapiens*.

Human sperm are microscopic cells that swim using a relatively long tail. The nucleus in the sperm is in a small head piece with an acrosome at the tip which contains enzymes that help the sperm penetrate the egg. Behind the head the sperm has a middle piece with large numbers of mitochondria which provide the energy needed for the movements of the tail.

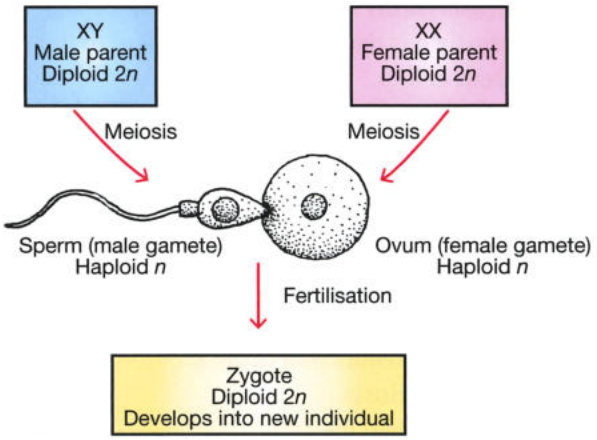


Figure 33.1 Fertilisation in humans.

Acrosomal reaction

The **acrosome** is a vesicle at the tip of a sperm cell that helps the sperm penetrates the egg. The **acrosomal reaction** is a process that occurs when the sperm approach the zona pellucida of the egg. The **zona pellucida** is the extracellular matrix of the egg.

The acrosome releases hydrolytic enzymes that digest the jelly coat making a pathway through the zona pellucida so that the sperm head can fuse with the plasma membrane of the egg. The sperm head is engulfed and there is exocytosis of cortical granules that alter the zona pellucida to stop other sperm entering the egg. In the cortical reaction the zona pellucida hardens to form a block to polyspermy.

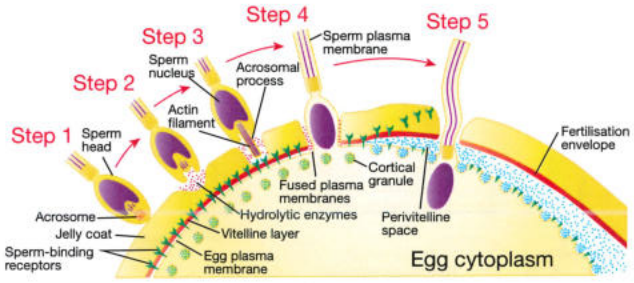


Figure 33.2 Acrosomal reaction.

There is a 'lock and key' recognition of the protein.

QUESTIONS

1. Define fertilisation.
2. What is a gamete?
3. Define zygote.
4. Explain why fertilisation maintains the diploid number of a species.
5. Explain why fertilisation produces variation.
6. Define copulation.
7. (a) What is the acrosome?
   (b) What is the acrosomal reaction?
8. What are the gonads?
9. Describe an ovum
10. Describe sperm.
11. Identify some advantages of sexual reproduction.
12. The diagram shows sexual reproduction.
    Identify process X.
    (A) Mitosis.
    (B) Meiosis.
    (C) Spermatogenesis.
    (D) Fertilisation.

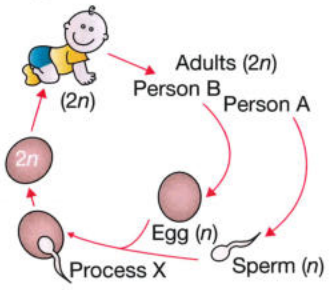


Figure 33.3 Sexual reproduction in humans.

34 The Male Reproductive System

The structure and function of the male reproductive system facilitates the production of offspring. The primary male gonad is the testes (singular testis) and in the male embryo the testes develop in the abdominal cavity and descend into the scrotum usually before birth. Sperm are not produced in undescended testes.

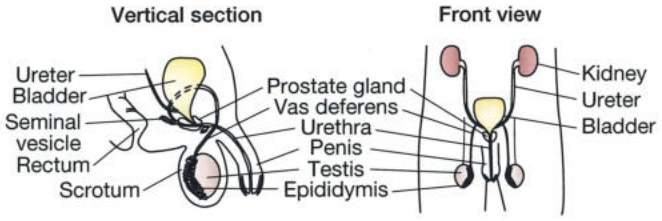


Figure 34.1 Structure of male reproductive system.

Role of the male reproductive system

The male reproductive system:

- Produces the male gamete (sperm) in the testes.
- Delivers the sperm in a liquid medium (semen) to the vagina during copulation.
- Produces the main male sex hormone (testosterone).

Structure of the male reproductive system

The male reproductive system consists of two **testes** with each subdivided into around 250 **lobules** each containing one to four tightly packed **seminiferous tubules**. Between the seminiferous tubules there are interstitial cells (**Leydig cells**) that secrete **testosterone** after puberty. The tubules contain sperm-producing cells (spermatogenic cells) and **Sertoli cells**. The testes are an **endocrine gland** as they synthesise testosterone from cholesterol. **Testosterone** begins the formation and the maintenance of secondary sexual characteristics, e.g. growth of gonads, enlargement of larynx, muscular development and body hair.

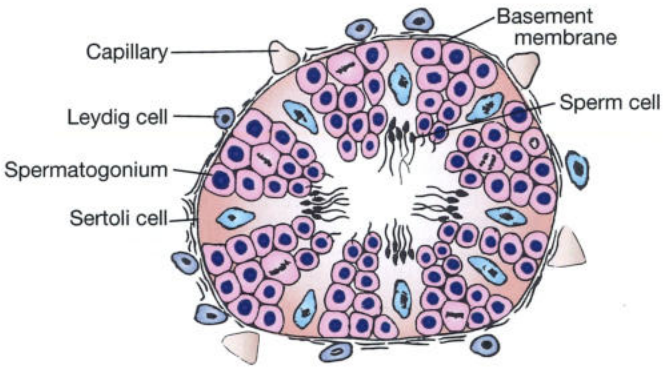


Figure 34.2 Cross-section of a seminiferous tubule.

The testes are held in the **scrotum** which is a pouch made of pigmented skin, fibrous tissue, connective tissue and smooth muscle. The scrotum is divided into two sections, which each hold one testis, one epididymis and the testicular end of the spermatic cord. The scrotum allows the testes to be kept about 2°C below body temperature which is needed for sperm production. Sperm move from the site of formation in the seminiferous tubule into the epididymis and stay there for two to three weeks becoming more motile. Those that are not ejaculated will be reabsorbed into the body usually within four weeks.

In an ejaculation sperm make up about 10% of the semen. There are about 40 to 100 million sperm per mL of semen and a normal ejaculate contains around 2 to 5 mL of semen.

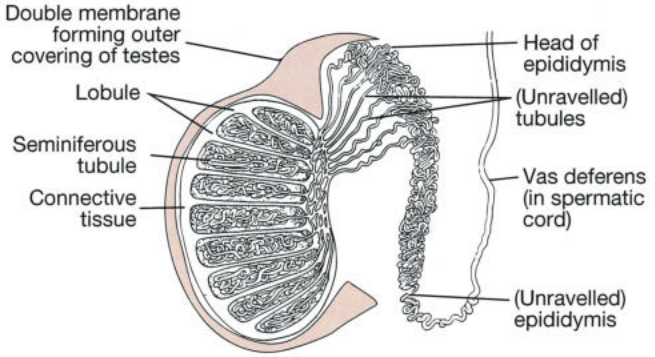


Figure 34.3 Structure of testes.

The **spermatic cords** suspend the testes in the scrotum with each containing vascular tissue (artery, vein and lymph vessels) and the vas deferens. In a **vasectomy** the sperm ducts are cut and each duct is tightly tied in two places and a short length removed so sperm cannot reach the urethra. This allows the male to still have sexual intercourse but is a permanent method of birth control causing sterility.

The **seminal vesicles** produce seminal fluid which makes up about 60% of the fluid being ejaculated as semen. This fluid is rich in sugars to nourish the sperm.

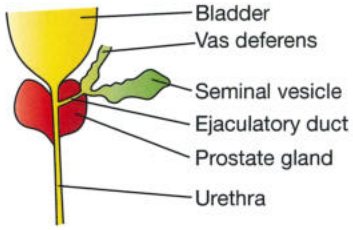


Figure 34.4 Ejaculatory duct and urethra.

The **prostate gland** secretes a thin, milky fluid that makes up about 30% of semen. The fluid contains a clotting enzyme which thickens the semen in the vagina and it is also slightly alkaline to protect the sperm in the acidic vagina. Enlargement of the prostate gland is common in men over 60 years and can cause problems urinating.

Cowper's gland is beneath the prostate gland and secretes a lubricant that precedes emission of semen.

The **urethra** is a common pathway in males for both urine from the bladder and semen. During ejaculation the sphincter muscle at the base of the bladder is closed so sperm/semen cannot travel to the bladder and urine cannot enter the semen.

The **penis** is a copulatory organ to transfer sperm to the female vagina. At its tip there is an expanded sensitive region call the **glans** which is covered by retractable skin (**foreskin**). When a male is circumcised the foreskin is removed. This can be done for religious or health reasons.

QUESTIONS

1. Name the main male gonad.
2. What is a consequence of the testes descending into the scrotum before and just after birth?
3. Outline the roles of the male reproductive system.
4. Describe the structure of a testis.
5. Explain why a testis is an endocrine gland.
6. Identify some of the functions of testosterone.
7. (a) Which cells in the testes secrete testosterone?
   (b) Where are these cells located?
   (c) Suggest why these cells are active in the foetus but then become inactive until puberty when they become active again.
8. Study the diagram and identify the parts of the male reproductive system.

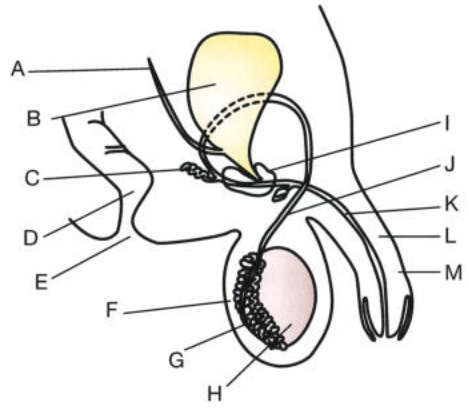


Figure 34.5 Identifying the components of the male reproductive system.

9. (a) Describe the scrotum.
   (b) Outline the function of the scrotum.
10. What happens to sperm once they are formed in the seminiferous tubules?
11. In an ejaculation what are the components of semen?
12. Outline the structure and function of the spermatic cords.
13. (a) What procedure is carried out when a man has a vasectomy?
    (b) Explain why many men choose to have a vasectomy.
14. Outline the function of the fluid from Cowper's gland.
15. The diagram shows a cross-section of a prepared slide of testis with LP, light microscope.

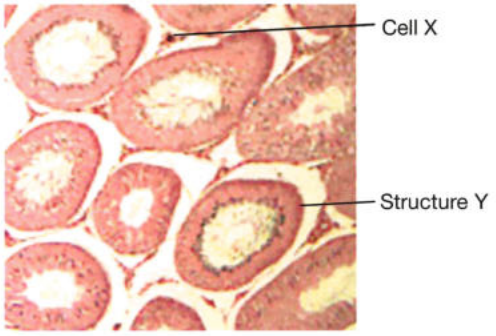


Figure 34.6 Prepared slide of testis, LP light microscope.

   (a) What is the most likely identity of cell X?
   (b) What is structure Y?
16. Which of the following correctly matches structure with function?

	Scrotum	Epididymis	Vas deferens
(A)	Pouch to lower temperature of testes	Stores sperm till mature	Secretes seminal fluid
(B)	Stores sperm till mature	Pouch to lower temperature of testes	Secretes lubricant
(C)	Sperm duct from testes	Stores sperm till mature	Pouch to lower temperature of testes
(D)	Pouch to lower temperature of testes	Stores sperm till mature	Sperm duct from testes

17. Which cells produce testosterone?
    (A) Sertoli cells.
    (B) Leydig cells.
    (C) Spermatogenic cells.
    (D) Spermatogonium.
18. Which procedure involves the removal of the foreskin from the glans?
    (A) Circumcision. (B) Vasectomy.
    (C) Spermatogenesis. (D) Circumlocution.
19. How many sperm would you expect to find in a normal ejaculate?
    (A) Approximately 70 sperm per mL.
    (B) Approximately 7×10^3 sperm per mL.
    (C) Approximately 7×10^7 sperm per mL.
    (D) Approximately 7×10^{12} sperm per mL.
20. What is the function of the urethra in males?
    (A) Allow semen only to pass to the exterior.
    (B) Allow urine only to pass to the exterior.
    (C) Allow semen or urine at the same time to pass to the exterior.
    (D) Allow semen or urine at different times to pass to the exterior.

35 The Female Reproductive System

The structure and function of the female reproductive system facilitates the production of offspring. The primary female gonads are the ovaries (singular ovary) and the genitalia can be divided into external and internal organs.

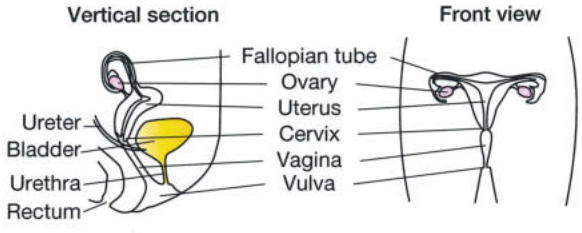


Figure 35.1 Structure of female reproductive system.

Role of the female reproductive system

The female reproductive system:

- Produces the female gamete (ova) in the ovaries.
- Delivers the ovum to the oviduct where fertilisation occurs.
- Prepares the lining of the uterus for implantation and development of an embryo.
- Produces the main female sex hormones (oestrogens).

Structure of the female reproductive system

The external genitalia are collectively known as the **vulva**. The vulva consists of the labia majora, labia minora, clitoris, the vaginal orifice, the vestibule and hymen.

The internal genitalia consists of the ovaries, the fallopian tubes (oviducts), the uterus and vagina.

The **ovaries** are whitish oval bodies about 3 to 4 cm long that lie either side of the uterus. They are held in place by ovarian ligaments and covered by a layer of **germinal epithelium**. Beneath the epithelium the ovary can be divided into an inner **medulla** in the centre consisting of fibrous tissue, blood vessels and nerves and a **cortex** around the medulla consisting of connective tissue or stroma.

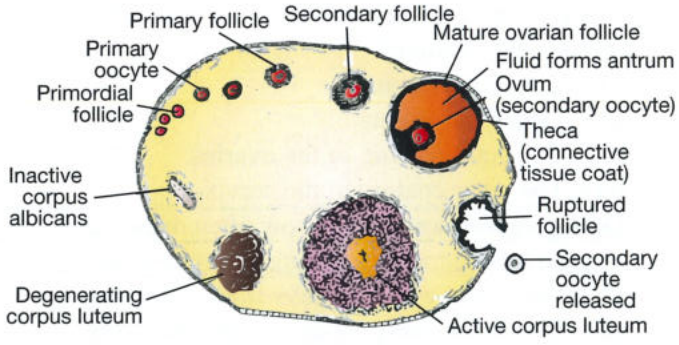


Figure 35.2 Structure of an ovary with stages of development of follicle and corpus luteum.

In a female embryo potential eggs move from the yolk sac to the germinal epithelium where they stop development at prophase I. Some degenerate so that at birth there are around 1 million **primary oocytes** in each ovary with around 200 000 remaining till puberty.

When an oocyte moves from the germinal epithelium into the cortex of the ovary it is surrounded by **follicle cells** to become a **primary follicle**. At puberty a few follicles begin to mature each month. A primary follicle develops into a **secondary follicle** which is a fully grown oocyte surrounded by a zona pellucida, about nine layers of granulosa cells and **theca cells** which secrete hormones. During **ovulation** the secondary oocyte is released from the follicle and the remaining follicular material – the granulosa and theca cells form a solid mass called the **corpus luteum**. During child-bearing years, one ovarian follicle matures and ruptures to release an ovum around every 28 days.

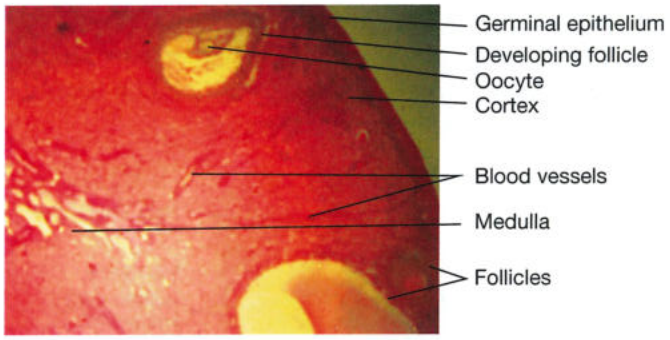


Figure 35.3 Prepared slide cross-section ovary, LP light microscope.

The **oviducts** are around 10 cm long and have a funnel shaped opening with fingerlike projections called **fimbriae** near each ovary that collect an egg when it is released from the ovary. The oviducts are *not* directly connected to the ovaries. Each oviduct is a narrow tube that connects to the uterus and cilia on the epithelial lining of the oviduct and help move the egg to the uterus. **Fertilisation** usually occurs in the oviduct.

The **uterus** (womb) is a thick muscular organ that is pear shaped and is about 80 mm long. During pregnancy it can expand to hold the embryo/foetus. The walls of the uterus consist of three layers. The inner layer is a mucous membrane called the **endometrium** that consists of columnar epithelial cells and mucus-secreting glandular cells in two layers – the functional layer and the basal layer. The surface section of the endometrium is the **functional layer** that thickens and becomes rich in blood vessels in the first half of the menstrual cycle and is shed in **menstruation** (monthly period) if the egg is not fertilised. The **basal layer** is not shed in menstruation and will regenerate the outer functional layer for the next menstrual cycle. Most of the uterine wall is smooth muscle. If fertilisation occurs the embryo will embed in the uterine wall.

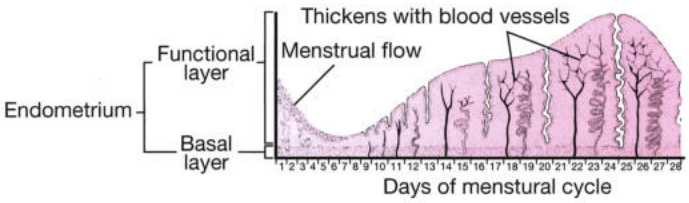


Figure 35.4 Functional layer of endometrium changes size during menstrual cycle.

The **cervix** is the neck of the uterus and opens into the vagina.

The **vagina** is a chamber with a wall of elastic and muscle fibres that is a passageway for menstrual flow, the site for insertion of the penis and sperm deposits and acts as the birth canal. It opens to the outside at the vulva.

The **mammary glands** (breasts) are exocrine sweat glands over the pectoral muscles. Each breast contains around 15 to 20 lobes each consisting of many lobules of glandular tissue with fibrous tissue and fatty tissue. Breast development begins at puberty due to higher oestrogen and progesterone levels. Milk ducts grow and fat deposits are laid down behind the glandular tissue.

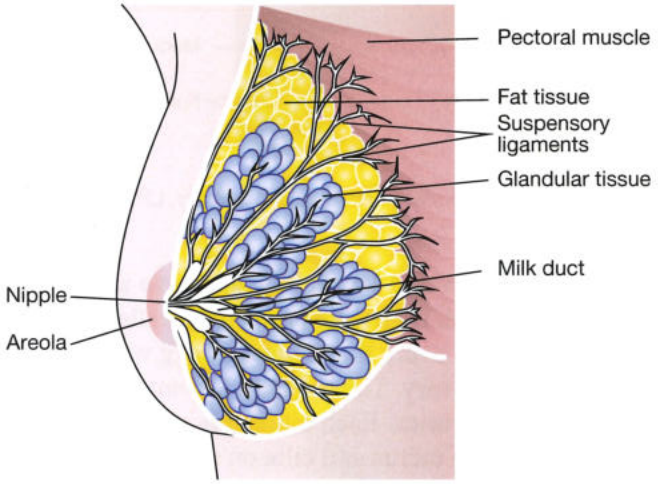


Figure 35.5 Mammary gland.

The mammary glands become active in late pregnancy and after the birth of a baby milk is produced in lactation.

QUESTIONS

1. What is the primary female gonad?
2. Outline the main roles of the female reproductive system.
3. What is the vulva?
4. Describe the ovaries.
5. What holds the ovaries in place?
6. What is the germinal epithelium?
7. How many primary oocytes does a female have per ovary:
   (a) At birth? (b) At puberty?
8. What happens during ovulation?
9. The diagram shows the female reproductive system.

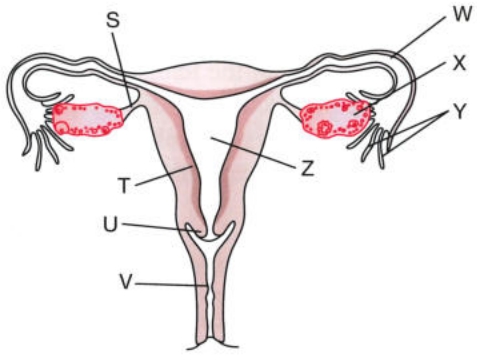


Figure 35.6 Female reproductive system.

Identify each part of the female reproductive system.

10. Where does fertilisation usually occur?
11. Outline a function of the fimbriae.
12. Describe the uterus.
13. Outline the structure and the function of the endometrium?
14. What is the cervix?
15. List some of the functions of the vagina.
16. Describe the structure of female mammary glands.
17. What causes breast development in females at puberty?
18. Define lactation.
19. The diagram shows four structures that would be seen under a light microscope when looking at a cross-section of an ovary.

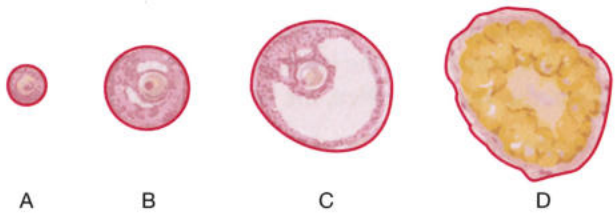


Figure 35.7 Structures found in an ovary.

Which of these shows a primary follicle?
(A) A (B) B (C) C (D) D

20. What is the name of the inner layer of the uterus?
    (A) Vulva (B) Endometrium
    (C) Fimbriae (D) Corpus albicans
21. Where is the germinal epithelium in the female reproductive system?
    (A) Lining the uterus.
    (B) Lining the vagina.
    (C) The outer coating of the ovaries.
    (D) The outer coating of the cervix.
22. During child-bearing years how often does one ovarian follicle mature, rupture and release an ovum?
    (A) One about every 28 days from one ovary.
    (B) One about every 14 days from one ovary.
    (C) One about every 28 days per ovary.
    (D) One about every 14 days per ovary.

36 Male and Female Hormones

Hormones are produced in specialised cells in endocrine glands and travel in body fluids to act on specific target cells to cause specific responses. They are very important in long distance signalling.

Steroid hormones

The sex hormones of males and females are steroid hormones that contain four fused carbon rings and are formed from cholesterol. The steroid hormones are lipid soluble and can easily pass through cell membranes. The receptors for steroid hormones can be in the cytoplasm or in the nucleus. The pathways of hormonal control of reproduction is similar in both sexes with **gonadotropin releasing hormone** (GnRH) from the hypothalamus controlling the secretion of **luteinising hormone** (LH) and **follicle stimulating hormone** (FSH) from the anterior pituitary gland.

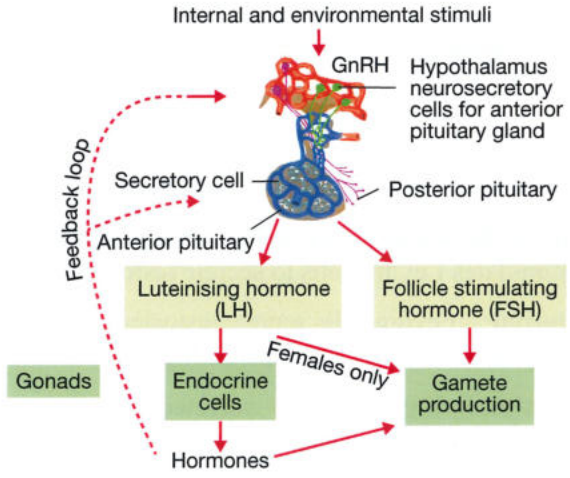


Figure 36.1 Reproduction hormones.

Male hormones

The **androgens** are any steroid hormone that stimulates the development and maintenance of the male reproductive system and male secondary characteristics. Androgens can be natural or synthetic compounds. The main androgen is testosterone. GnRH from the hypothalamus controls the release of LH and FSH from the anterior pituitary gland and these hormones control the release of testosterone from the testes.

Testosterone is secreted by the testes with some being produced by the adrenal glands and the ovaries in females. In males testosterone is produced by the interstitial cells (Leydig cells) of the testes. Certain levels of testosterone are needed for sperm production and also cause the voice to deepen, body hair, e.g. facial and pubic hair to grow, the building of muscles, the strengthening of bones and is important in developing the prostate gland and male reproductive tissues, e.g. testes and penis.

Female hormones

Female hormones include the oestrogens and the progestins.

Oestrogens – Oestrogens are responsible for the development of secondary female characteristics stimulating breast growth, causing deposits of fatty tissue, thickening of the vagina and causing it to secrete mucus. The follicular cells and then the corpus luteum of the ovary produce oestrogen and in the menstrual cycle cause the development of the endometrium (lining of the uterus). Oestrogen levels can also affect skin appearance, bone strength and protect against heart disease. FSH stimulates the ovaries to produce oestrogen.

Progesterone – Progesterone is important in maintaining pregnancy. The corpus luteum of the ovary produces progesterone which helps to maintain the endometrium. A reduction in the levels of oestrogen and progesterone leads to a decrease in the thickness of the uterine lining. Some research links symptoms of premenstrual syndrome (PMS) to high levels of progesterone with bloating, mood swings and breast tenderness.

Pregnancy and female hormones

If pregnancy occurs then oestrogen and progesterone levels stay high and the endometrium is maintained until the placenta develops and takes over this role. Menstruation does not occur if oestrogen and progesterone levels are high. It has been found that there is a decreased level of progesterone in women about to miscarry.

QUESTIONS

1. What produces hormones?
2. How do hormones travel in the body?
3. Describe steroid hormones.
4. What is GnRH?
5. Suggest why LH and FSH are sometimes called the gonadotrophic hormones.
6. Define androgen.
7. What are the main actions of testosterone?
8. What are the two main types of female hormones?
9. What are the main actions of the oestrogens?
10. What are the main actions of progesterone?
11. The diagram shows the hormone pathway in females. Identify hormone X and hormone Y.

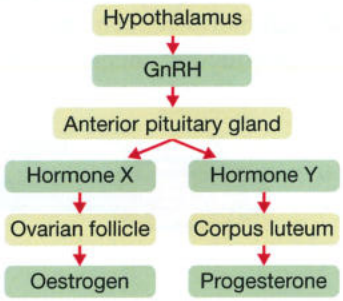


Figure 36.2 Hormone pathway in females.

37 Spermatogenesis

Spermatogenesis is the production of sperm, and occurs in all seminiferous tubules during the active sexual life of the male.

Spermatogonia

Spermatogonia (singular spermatogonium) are germinal epithelial cells that line the outside section of the seminiferous tubules. They usually form two or three layers and continually divide and maintain the layer.

In spermatogenesis a spermatogonium migrates to a Sertoli cell moving towards the lumen and centre of the seminiferous tubule. As it moves the spermatogonium gets larger to become a primary spermatocyte.

Meiosis

The **primary spermatocyte** divides to form two secondary spermatocytes. Each of these then divides to form **spermatids**.

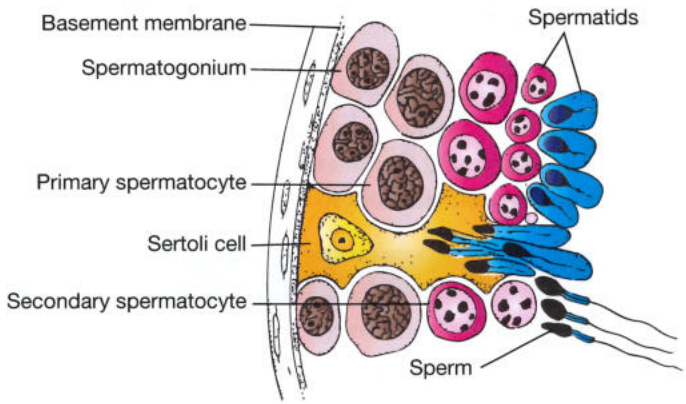


Figure 37.1 Spermatogonia to spermatids in seminiferous tubule.

Spermatid to spermatozoa

Newly formed spermatids have many features of an epithelial cell. The cells stay embedded at the lumen end of the Sertoli cell as they elongate, lose most of their cytoplasm and microtubules extend to form the tail. Mitochondria concentrate in the neck region of the sperm and provide the energy for movement.

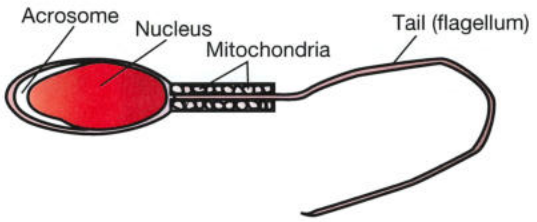


Figure 37.2 Structure of sperm.

Spermatogenesis takes round 64 days from spermatogonium division until the sperm are released into the tubule. When the sperm are released from the Sertoli cell they are not fully mature and able to swim by themselves. Bulk flow pushes the sperm into the **epididymis** where they complete the process and are stored.

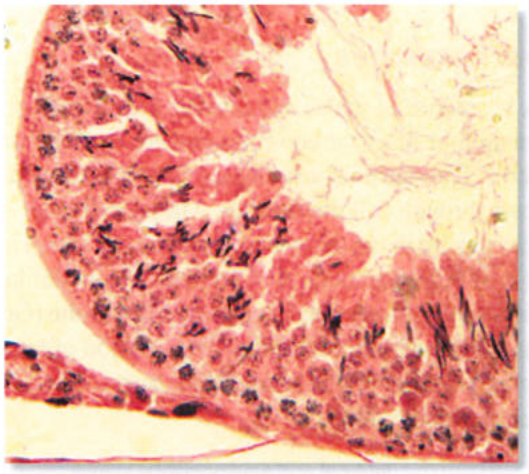


Figure 37.3 Cross-section of testes seminiferous tubule.

Hormones and spermatogenesis

Several hormones are needed for spermatogenesis.

Luteinising hormone – secreted by anterior pituitary gland stimulates Leydig cells to secrete testosterone.

Testosterone – Leydig cells secrete testosterone needed for growth and division of the spermatogonia.

Follicle stimulating hormone – secreted by anterior pituitary gland stimulates Sertoli cells to convert spermatids to sperm.

QUESTIONS

1. Define spermatogenesis.
2. What are spermatogonia?
3. What is the first step in spermatogenesis in the development of a spermatogonium into sperm?
4. Draw a flow chart to show the steps from spermatogonium to sperm.
5. Relate the steps in meiosis to spermatogenesis.
6. What hormones are involved in spermatogenesis?
7. Where do sperm become fully mature and capable of swimming?
8. During spermatogenesis what is the function of luteinising hormone?
   (A) Stimulate the Sertoli cells to produce testosterone.
   (B) Stimulate the Leydig cells to change primary spermatocytes into secondary spermatocytes.
   (C) Stimulate the Sertoli cells to change primary spermatocytes into secondary spermatocytes.
   (D) Stimulate the Leydig cells to produce testosterone.

38 Oogenesis

Oogenesis is the production of ova.

Oogonia

Oogonia (singular oogonium) are stem cells in the germinal epithelium of the ovaries of a female foetus. Oogonia divide in the ovary many times by mitosis during prenatal development and they migrate to the connective tissue and increase in size to become **primary oocytes**. They become surrounded by a layer of follicle cells to become a **primary follicle**. This has usually occurred by the fifth month of embryonic development.

Meiosis

When a female child is **born** the primary oocytes in the primary follicle are up to **prophase I** in meiosis and go into a resting phase. At puberty further development occurs and meiosis restarts. Several oocytes may start to develop each month but usually only one matures and the rest degenerate.

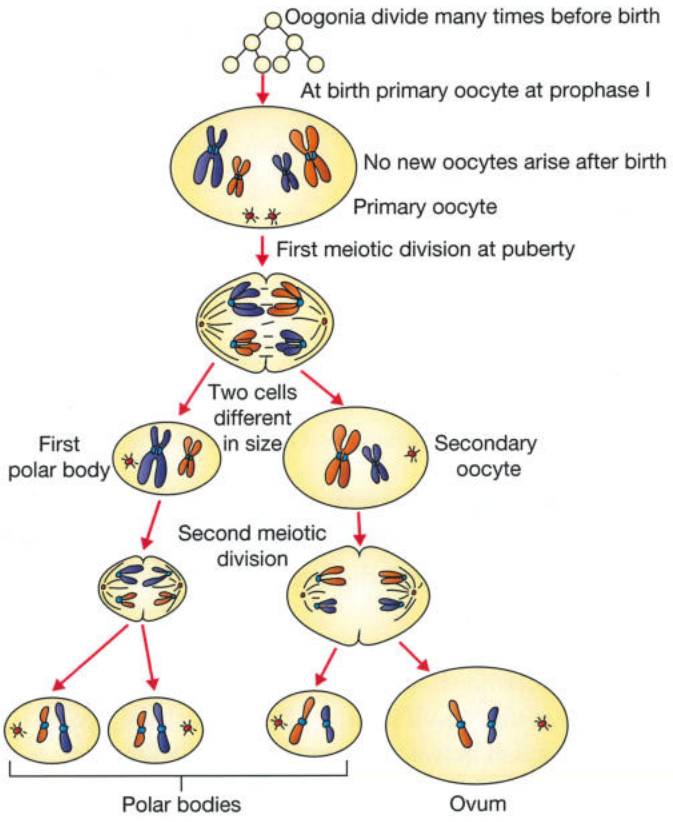


Figure 38.1 Oogenesis.

First division of meiosis – produces two cells of different size with the larger being the **secondary oocyte** and the very small other cell being the **first polar body**. Each cell has 23 duplicated chromosomes.

Second division of meiosis – the secondary oocyte begins the second division but halts at metaphase II unless it is fertilised. The secondary oocyte at metaphase II is released from the ovary in **ovulation**.

The first polar body can divide to form two polar bodies but all polar bodies will eventually break down. The formation of polar bodies is a way to remove unneeded chromosomes without complete formation and then needing to dispose of a large amount of cytoplasm.

Follicle development

During the development of the oocyte the **zona pellucida**, a thick membrane, surrounds the oocyte separating it from the follicle cells. The follicle cells then secrete fluid which collects in the space. Glandular cells secrete oestrogen and the follicle moves closer to the surface of the ovary.

After ovulation the ruptured ovarian follicle develops into the **corpus luteum** (yellow body) which will become the endocrine gland of pregnancy if fertilisation occurs. If fertilisation does not occur the corpus luteum degenerates becoming a white scar, the **corpus albicans** (white body).

Hormones and oogenesis

Several hormones are needed for oogenesis.

Luteinising hormone – secreted by anterior pituitary gland stimulates ovulation, the development of the corpus luteum and secretion of progesterone.

Follicle stimulating hormone – secreted by anterior pituitary gland stimulates maturation of ovarian follicles and the secretion of oestrogen for ovulation.

Oestrogen – secreted by follicular cells and then corpus luteum cause development of the endometrium.

QUESTIONS

1. Define oogenesis.
2. What are oogonia?
3. Describe a primary oocyte.
4. Draw a flow chart to show the steps from oogonium to ovum.
5. What is a primary follicle?
6. At what stage of meiotic development are oocytes during female childhood?
7. When does the first division of meiosis occur in females?
8. Outline a benefit of the formation of polar bodies in oogenesis.
9. At what stage of meiosis is the ovum at ovulation?
10. What is the corpus luteum?
11. Which hormones help regulate oogenesis?
12. Compare spermatogenesis and oogenesis.

39 Menstrual and Ovarian Cycles

Ovarian cycle refers to the monthly cycle of egg development in the ovary with follicular phase, ovulation and the luteal phase regulated by hormones.

Menstrual cycle refers to the cyclic preparation of the endometrium of the uterus for pregnancy in females with discharge of blood and tissue through the cervix if implantation does not occur.

The ovarian cycle

There are three main phases in the ovarian cycle.

Follicular phase – follicles take between 10 to 21 days to develop in the ovary.

Ovulation – a functional ovum is released from the ovary.

Luteal phase – the ruptured follicle develops into the corpus luteum which secretes hormones to prepare the endometrium of the uterus for pregnancy. If pregnancy does not occur the corpus luteum breaks down for the start of a new cycle.

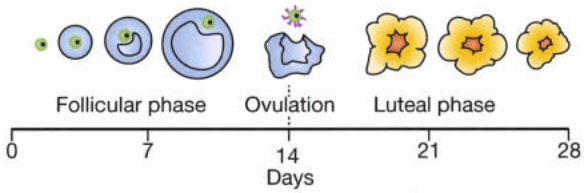


Figure 39.1 Ovarian cycle.

The menstrual cycle

At puberty gonadotropic releasing hormone (GnRH) from the hypothalamus stimulates the release of follicle stimulating hormone (FSH) and luteinising hormone (LH) from the anterior pituitary gland. FSH and LH stimulate the ovary to become active and with oestrogens and progesterones from the ovary they regulate the menstrual cycle.

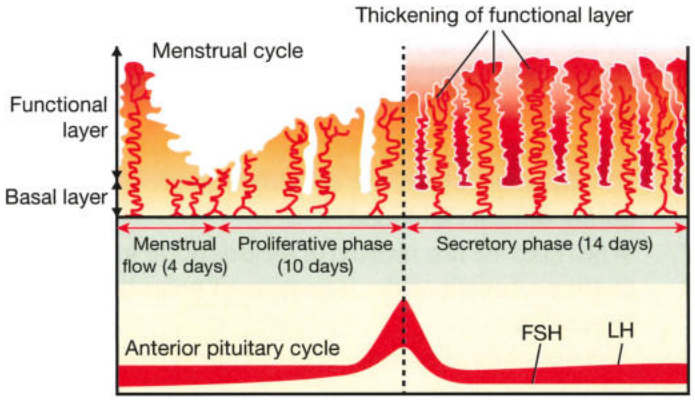


Figure 39.2 Menstrual cycle and anterior pituitary cycle.

The menstrual cycle is usually between 25 and 35 days in length, with 28 days taken as the average.

Menarche is the beginning of menstruation at puberty and **menopause** is when the menstrual cycle stops and the ovaries do not release any more ova. **Menstruation** is the shedding of the blood and tissues from the endometrium of the uterus when an ovum is not fertilised.

The menstrual cycle consists of three main phases.

Menstrual flow – is menstruation with the shedding of blood and tissue from the endometrium.

Proliferative phase – involves the thickening of the endometrium caused by increasing levels of oestrogen secreted by follicle cells and is coordinated with the follicular phase of the ovarian cycle.

Secretory phase – begins after ovulation with oestrogens and progesterone secreted by the corpus luteum stimulating the continued development and maintenance of the endometrium. If fertilisation does not occur the corpus luteum breakdown causes a rapid drop in ovarian hormones and the endometrium disintegrates leading to the menstrual flow.

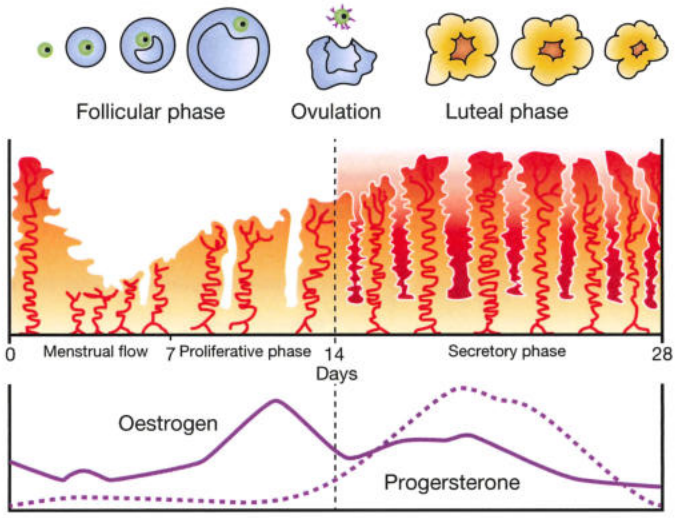


Figure 39.3 Linking the ovarian and menstrual cycles.

QUESTIONS

1. Distinguish between the ovarian cycle and the menstrual cycle.
2. Outline the three phases of the ovarian cycle.
3. Which hormones stimulate the ovary to become active?
4. What is the length in time of the menstrual cycle?
5. Distinguish between menarche and menopause.
6. Define menstruation.
7. Describe the three phases of the menstrual cycle.
8. How are oestrogen and progesterone levels linked to the menstrual cycle?
9. About 7% of women suffer from endometriosis when cells from the endometrium begin to grow outside the uterus, e.g. on ovaries, fallopian tubes or bowel. Explain why ovulation causes pain for females with endometriosis.

40 Conception and Pregnancy

Conception is the process of becoming pregnant with the union of viable sperm and ovum at the optimal time in the ovarian cycle. The fertilised egg is called a **zygote**. After conception the zygote will divide by mitosis in a process called **cleavage** that converts the zygote to a small mass of cells. After around five days the embryonic mass of cells is called a **blastocyst.** The blastocyst consists of a ball of cells around an inner cavity and implants in the endometrium of the uterine wall. Implantation causes the endometrium to be maintained and stops the next menstrual flow.

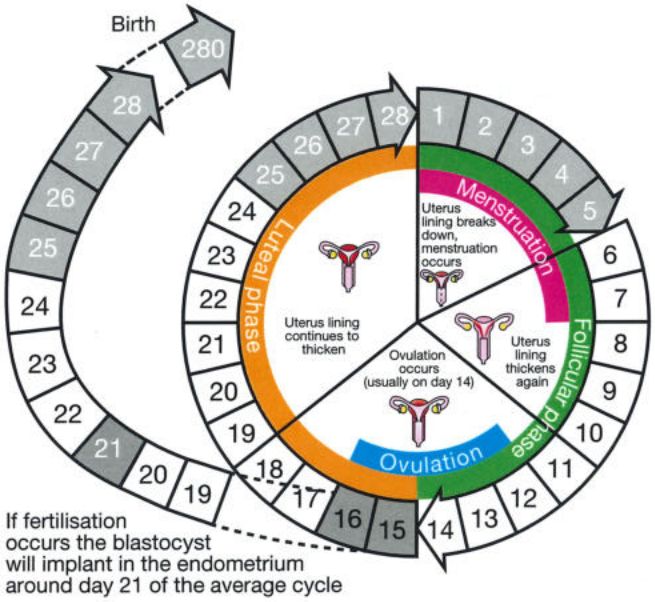


Figure 40.1 Change in the menstrual cycle with fertilisation.

Hormones

Human chorionic gonadotropin (hCG) is an embryonic hormone produced by some of the cells in the blastocyst/ embryo when it implants in the endometrium. hCG maintains the corpus luteum in the mother stimulating it to continue to produce oestrogen and progesterone through the first few months of pregnancy. As the embryo develops these cells become part of the placenta and it is the placenta which continues to produce hCG. If hCG is not present or levels are low the corpus luteum of the mother will break down, progesterone levels will drop and the mother will menstruate and the embryo will be lost in a **miscarriage**. Since high levels of hCG are needed to maintain oestrogen and progesterone secretion, some hCG is excreted in the mother's urine and many early **pregnancy tests** involve checking for hCG.

Oestrogen and **progesterone** – around the third month of pregnancy oestrogen and progesterone begin to be produced by the placenta so that only small amounts of hCG are needed for the rest of the pregnancy.

Hormone levels can be monitored during pregnancy by urine analysis.

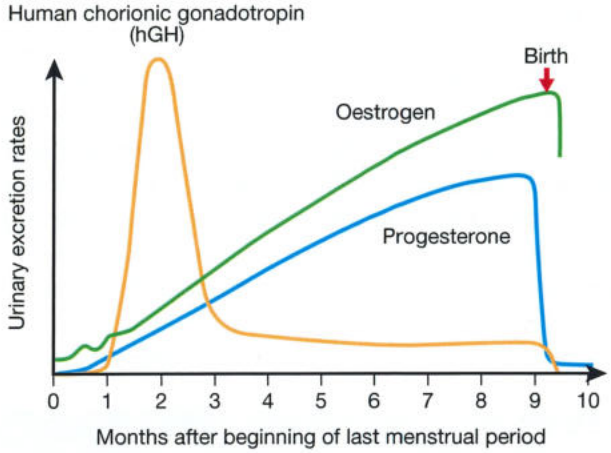


Figure 40.2 Hormone levels as measured in urine analysis during pregnancy.

After implantation the corpus luteum does not degenerate and menstruation does not occur. The corpus luteum remains for approximately 20 weeks and secretes progesterone and oestrogens to maintain the uterine wall in a suitable form for the developing embryo and then foetus. As the placenta develops it also secretes oestrogens and progesterone. The oestrogens produced during pregnancy suppress the secretion of FSH which means no follicles develop and there is no ovulation. Oestrogen and progesterone also suppress the secretion of luteinising hormone from the pituitary gland. During pregnancy the mother's **thyroid gland** can enlarge up to 50% due to hCG and a specific thyroid stimulating hormone called **human chorionic thyrotropin**.

Pregnancy

Human gestation or pregnancy is on average 266 days (approximately 38 weeks). Since timing is related to the menstrual cycle it is usually counted as 40 weeks since the start of the last menstrual cycle. Not all fertilised eggs develop and some pregnancies spontaneously terminate, e.g. due to chromosome abnormalities or development problems.

Multiple births can occur and the occurrence of multiple births is increasing mainly due to the increase in the use of fertility drugs. Monozygotic (MZ) twins come from a single fertilised egg while dizygotic (DZ) twins develop from two separate eggs which are each fertilised by different sperm. In Australia MZ twins occur in about 4 per 1000 and about 1 in 80 births in Australia is a twin of which about 30% are MZ twins. Siamese twins occur when the cells incompletely separate so that the twins share some organs. Depending on which organs are being shared some Siamese twins can be surgically parted after birth.

An **ectopic pregnancy** occurs when the embryo develops outside the womb, e.g. in the fallopian tube. In most cases the embryo dies as a suitable placenta cannot form and provide sufficient nutrients. If the fallopian tube ruptures immediate surgery and a blood transfusion is needed.

Infertility is the inability of a couple to achieve conception after at least one year of unprotected sex (and six months if a woman is over the age of 35). Infertility can be due to male related problems, e.g. blocked sperm ducts, too few numbers of sperm (e.g. < 20 million sperm per cm^3 semen) or defective sperm (e.g. > 50% immobile sperm in semen or > 35% sperm have abnormal morphology) or chemical components such as enzymes in incorrect proportions or high pH of semen; or due to female related problems, e.g. blocked oviducts; and in some cases the reason is a combination of male/female problems or unexplained.

QUESTIONS

1. What is conception?
2. Define a zygote.
3. What is meant by cleavage?
4. Describe a blastocyst.
5. If fertilisation occurs when is the blastocyst likely to start implantation in the endometrium?
6. The diagram shows a human egg and sperm.

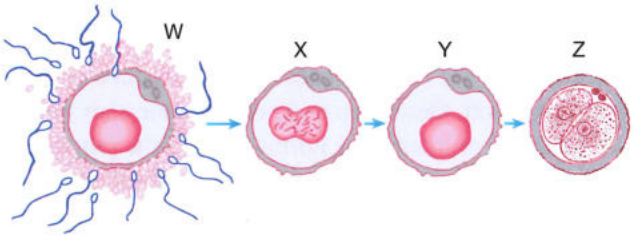


Figure 40.3 Human egg and sperm.

Describe what is happening at steps W, X, Y and Z.

7. The diagram shows the stages of conception and early pregnancy.

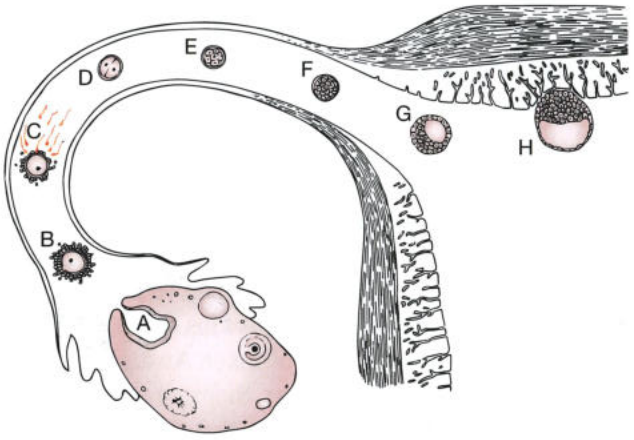


Figure 40.4 Conception and early pregnancy.

(a) What has happened at point A?
(b) Which of the steps shows fertilisation?
(c) What is the name of the embryonic stage at point G?
(d) Describe what is happening at point H.

8. (a) Explain how an embryonic hormone causes the endometrium to be maintained and prevents a menstrual flow.
   (b) Explain why an embryonic hormone can be used as a pregnancy test.
9. Explain why only small amounts of hCG are needed after the first three months of pregnancy.
10. Explain why no follicles develop during pregnancy.
11. Explain why the mother's thyroid gland can enlarge up to 50% during pregnancy.
12. (a) How long is human gestation ?
    (b) How is human gestation usually timed?
13. Distinguish between monozygotic twins and dizygotic twins.
14. Suggest why the occurrence of multiple births is increasing.
15. What are Siamese twins?
16. (a) What is an ectopic pregnancy?
    (b) What happens in most instances of an ectopic pregnancy?
17. Define infertility.
18. What are the main reasons for infertility?
19. The diagram shows the levels of three hormones found in the urine of a pregnant woman.

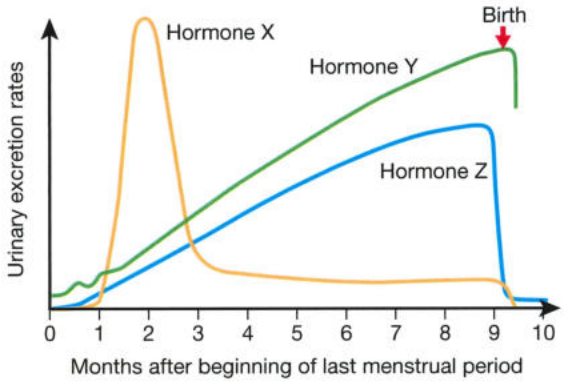


Figure 40.5 Hormone during pregnancy.

What is the correct identity of these hormones?

	Hormone X	Hormone Y	Hormone Z
(A)	Oestrogen	FSH	Progesterone
(B)	Progesterone	hCG	Oestrogen
(C)	Oestrogen	Progesterone	FSH
(D)	hCG	Oestrogen	Progesterone

20. Which hormone is suitable to use as a pregnancy test?
    (A) Human chorionic gonadotropin
    (B) Oestrogen
    (C) Progesterone
    (D) Follicle stimulating hormone
21. When does implantation occur in the average cycle?
    (A) Day 7 (B) Day 14
    (C) Day 21 (D) Day 28

41 Play – Cell Productions

Dramatic personae:

Nelly – nucleus
Myra – gene for the protein myosin
Colin – gene that leads to acetylcholine production
Neil – nuclear membrane
Ian – gene that leads to insulin production
Gary – a gamma radiation photon
Dion – a DNA repair molecule
Bernie – the blastocyst
Minnie – a mitochondrion
Dennis – a DNA repair molecule
Daphne – a DNA repair molecule

SCENE 1 – THE GAMETE

(Three genes wake up in the nucleus of a gamete)

Nelly Welcome to my home.
Ian Is this a gamete?
Nelly I'm the nucleus of an ovum to be precise.
Ian How egg-cellent.
Nelly Egg jokes are not very funny.
Ian Egg-actley.
(Nelly glares at Ian with an angry look)
Myra It's very cosy in here and a bit squishy.
Nelly The amount of DNA in this nucleus will double when a sperm unites with the egg.
Colin That's ridiculous. There's no room in here for more DNA.
Ian We want to stay egg-clusive. We don't want more chromosomes in here!
Nelly At the moment the nucleus is haploid.
Myra What do you mean by haploid?
Nelly Haploid means there is only one set of chromosomes present in the nucleus.
Colin So why can't we stay haploid? I like the nucleus the way it is.
Nelly Gametes with their haploid nucleus only have a limited lifespan. This cell is destined to be fertilised and we will grow and develop into a complex multicellular organism.
Ian If a gamete has a limited lifespan, I don't mind joining with the sperm nucleus. I don't want to become egg-tinct.
Nelly I told you to stop the egg jokes.
(Nelly glares at Ian again)
Neil We've got company. Watch out everyone inside.
Colin Who are you?
Neil I'm the nuclear membrane that surrounds the nucleus.
Myra What company do we have?
Neil A sperm made contact with the jelly coat of this egg triggering the acrosomal reaction. This caused fusion of the sperm and egg membranes.
Ian Our egg-terior has been breached!
Colin This sounds dangerous. What happens next?
Nelly The acrosome of the sperm releases enzymes that lets the sperm move through the outer jelly coat and our egg has released cortical granules in the cortical reaction to stop polyspermy.
Myra What's polyspermy?
Neil Polyspermy means the egg is fertilised by more than one sperm which means there would be three or more copies of each chromosome.
Colin It's starting to get even more squishy in here.
Nelly The sperm nucleus has entered the egg and is heading our way.
Ian This is egg-citing.
Myra Oh no! Here comes another nucleus!
Nelly It's going to fuse with us and I'm going to become a diploid nucleus.
(The two nuclei fuse to form the zygote – the fertilised egg)
Colin That wasn't too bad. But there seems to be two of me.
Myra And there's two of me.
Nelly A diploid cell has two sets of chromosomes, one set inherited from each parent. There are two alleles for each gene.

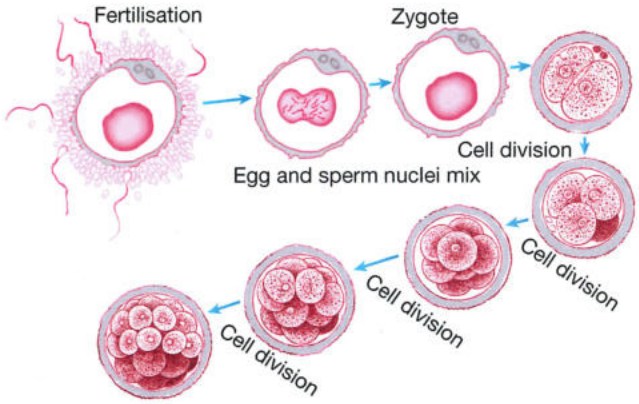


Figure 41.1 Fertilisation and early embryonic development.

Colin What's an allele?
Neil An allele is the alternative version of a gene that produces distinguishable phenotypes.
Ian I don't mind egg-isting as one of two alleles for a gene.
Myra What happens now?
Nelly We divide and grow into a human being.
Colin If there are two sets of chromosomes in this fertilised egg, how many chromosomes are actually here?
Nelly There are now forty-six chromosomes in this nucleus. That's twenty-three pairs of chromosomes.

Questions Scene 1

1. What is meant by haploid?
2. Outline what happens when a sperm contacts an egg?
3. What is polyspermy?
4. What is meant by diploid?
5. What is an allele?
6. What is the diploid number for humans?

SCENE 2 – THE BLASTOCYST

(The friends are shaken by an unexpected burst of energy)

Colin What was that?

Bernie My whole structure has been shaken.

Myra Who are you?

Bernie I'm Bernie the blastocyst.

Colin How did we get here?

Bernie After fertilisation the zygote begins dividing in a process called cleavage and after about five days in humans the embryonic stage is called the blastocyst.

Myra So this means we are now part of a blastocyst?

Bernie Correct. The blastocyst is a very important stage. You are part of the inner mass of cells that forms the embryoblast.

Colin Are we in stem cells?

Bernie Correct. The inner mass of cells of the blastocyst is the source of embryonic stem cells.

Nelly What exactly is a stem cell?

Bernie A stem cell is any relatively unspecialised cell that can produce, during a single division, one identical daughter cell and one more specialised cell that can undergo further differentiation.

Colin This is very interesting but what was that burst of energy?

Myra it was very powerful, whatever it was!

Neil It passed right through me as though I wasn't there. The nuclear membrane is meant to control what enters and leaves the nucleus.

Bernie I think this blastocyst has just been exposed to some dangerous radiation.

(Gary, a gamma radiation photon enters the nucleus)

Gary That's right. I'm dangerous and I'm a type of electromagnetic radiation.

Colin Who are you?

Gary I'm Gary a gamma radiation photon.

Myra What's gamma radiation?

Gary Gamma radiation is an extremely high frequency radiation with frequencies over 10^{19} Hertz. I'm high energy and my kind have wavelengths less than 10 picometres which is less than the diameter of an atom.

Colin Wow! Where did you come from?

Gary There are several natural sources of gamma rays such as from naturally occurring radioisotopes and from cosmic rays.

Myra So gamma rays only come from natural sources?

Nelly Gamma rays can also be released from artificial sources such as nuclear reactors and high energy physics experiments.

Neil You haven't answered our question Gary. Where did you come from?

Gary I was released from a nuclear reactor. An earthquake caused a meltdown at the plant. The walls came down and we escaped.

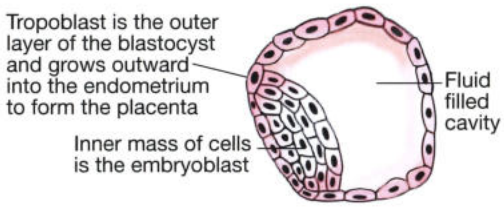


Figure 41.2 The blastocyst.

Myra Were you a prisoner?

Ian You're an egg-scaped convict!

Gary The people keep us prisoners in the reactor so they can convert my energy into heat.

Colin What do they do with your energy?

Gary They heat water to make steam which is used to drive turbines to make electricity.

Bernie So you lived in a big kettle?

Gary It was a big nuclear reaction driven kettle. It's much nicer out here. Water is very boring. Two hydrogens and one oxygen and their molecule is a little bent.

Colin This physics and chemistry is all very interesting but what are you doing here in this cell with us?

Neil I don't trust this particular gamma ray. Gamma rays can damage the DNA in the cell and change the coding.

Ian Sounds egg-cruciating.

Nelly You can stop the egg jokes. The egg was fertilised and became a zygote which divided by mitosis to become many cells. And then differentiation occurred to make specialised cells. And we are now part of a young embryo. So no more egg jokes.

Ian I was egg-joying myself, but I can stop.

(Ian hangs his head in shame)

Myra What do you mean by 'damage the DNA in the cell'? I'm a gene for the protein myosin. I am very important in muscle contraction.

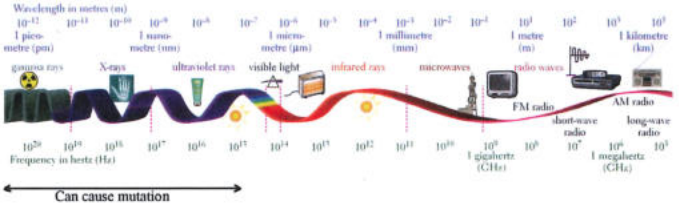


Figure 41.3 Electromagnetic spectrum.

Gary I thought I would visit this nucleus and have a bit of fun.

Colin What do mean by having fun? I'm the gene for choline acetyltransferase, which the scientists call ChAT.

Myra I didn't know that. What does ChAT do?

Colin The enzyme that is made by my code is responsible for the synthesis of the neurotransmitter acetylcholine.

Ian This sounds very technical. Could you egg-splain in more simple terms?

Colin My code leads to the production of acetylcholine which is the chemical released by neurons, nerve cells so the nervous message can cross the synapse from one nerve cell to the next.

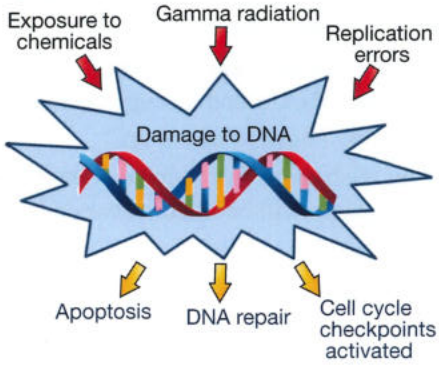


Figure 41.4 Damage to DNA and DNA repair.

Ian I'm more interesting than that! My code is the *INS* gene and I make the preproinsulin precursor of insulin.

Myra What does insulin do?

Ian Insulin controls the levels of glucose in the blood.

Bernie What is Gary going to do here in the blastocyst?

Gary I thought I would mutate everyone. A deletion here and a substitution there. Gamma rays like to change the coding in genes.

Myra Oh no! This is terrible. We're all being mutated!

Gary That was fun, but now I'm moving on. I think I'll go and kill some bacteria. I'm free at last to do as I wish!

Questions Scene 2

1. What is the function of the nuclear membrane?
2. What is a blastocyst?
3. What genetic codes are in the nucleus of a cell?
4. What is an embryonic stem cell?
5. Describe gamma radiation.
6. Identify some natural and artificial sources of gamma radiation.
7. How is electricity generated in nuclear reactors?
8. Which parts of the electromagnetic spectrum can cause mutations?
9. Why is the protein myosin needed by the body?
10. Why is the gene for ChAT needed by the body?
11. Why is the *INS* gene needed by the body?

SCENE 3 – THE GROWING EMBRYO

(The friends are activated by gene expression)

Colin I've been activated and the chromatin in my section of the chromosome is changing so my gene can be expressed.

Myra I've also been activated. Is this gene expression?

Neil What's gene expression?

Nelly Gene expression, means that information coded in the DNA directs the production of proteins or in some cases the production of RNA.

Neil Is that why this play is called 'Cell Productions'?

Nelly Correct. My nucleus has the code for all of the proteins produced by this body.

Minnie No you don't!

Colin Who are you?

Minnie I'm Minnie, a mitochondrion.

Myra What's a mitochondrion?

Minnie A mitochondrion is an organelle found in eukaryotic cells that carries out cellular respiration. I'm the powerhouse of the cell. I'm very important.

Myra Why did you contradict Nelly Nucleus and say she is does not have the code for all of the proteins the proteins that are produced by the cell?

Minnie I have my own DNA and I have the code to produce several proteins.

Nelly You don't have as many codes as I do!

(Minnie sighs)

Minnie She is quite correct. In humans mtDNA codes for 37 genes.

Nelly I am far more important.

Minnie No you're not! My DNA sequences have been used to work out the evolutionary relationships between many species. I'm more important.

Ian The two of you can stop bickering and be quiet. I've also been activated and my gene is being egg-spressed.

Nelly I'm tired of your egg jokes.

Colin The jokes aren't funny.

Ian I think I'm egg-ceptionally funny. My jokes are better than fighting over who has the most important DNA code.

(Colin hiccups and starts to change shape)

Myra What's the matter with Colin? He's looking strange.

Neil I think he was mutated by Gary and it's affecting his ability to produce his protein.

Dion Out of my way. Can everyone move back and let me get to my patient.

Nelly Good to see you Dion. I was wondering when you would show up.

Colin Who's this? I don't want any more strangers changing my code.

Dion Settle down. I'm Dion, a DNA repair molecule.

Colin What are you doing?

Dion I'm going to use the your sister chromatid as a template to recover the original information in your code.

Colin You can do that?

Dion There are several ways we can repair damaged genetic information. In your case the damage changed the spatial structure of the double helix and it was easy for me to see the problem.

Colin I'm starting to feel better already.

Dion I bind near the site of the damage and bring in other molecules to actually repair the damage.

(Colin is repaired and becomes a fully functional gene)

Colin This is fantastic. My code is correct and I will produce choline acetyltransferase. This body can now make acetylcholine and messages can pass from one nerve cell to the next.

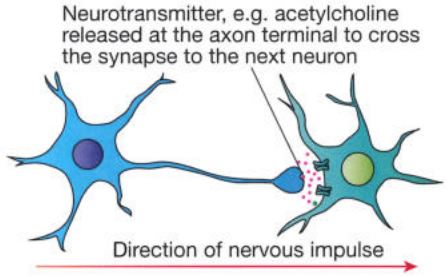


Figure 41.5 Acetylcholine, a neurotransmitter is produced at the axon terminal of neurons and is released into the synapse. It diffuses across the synapse to the next neuron and stimulates the next neuron thus passing the nervous message from one nerve cell to the next.

Myra I'm starting to feel funny. What's the matter with me?

Nelly You were mutated by the gamma radiation and are having problems with gene expression.

Myra Dion! Come over here and repair my DNA. You repaired the damage to Colin's code and you need to repair my DNA.

Dion I'm very sorry Myra. I can't fix your type of problem.

Dennis Out of my way. I'm here to fix the Myosin gene.

Myra You can repair the damage in my code?

Dennis Of course I can. There's a range of enzymes that detect and repair mutated sections of DNA.

Myra There are a lot of DNA damage checkpoints.

Dennis Correct. We repair damage and the cell cycle can be stopped at multiple points to make sure there is normal cell division and growth.

Myra I feel much better. This cell can now produce myosin. Cell productions are under way.

Dennis Myosin is a motor protein and is the driving force for muscle contraction. This person would have had a lot of problems if that mutation had stayed in the cell.

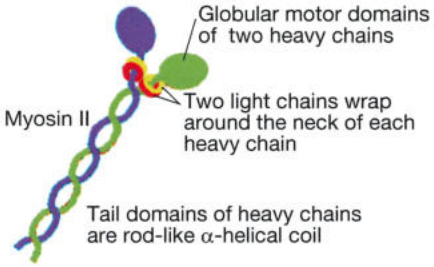


Figure 41.6 Myosin.

Ian What about me? I feel peculiar and need help. Who's going to help me?

Dion I can't fix that type of mutation.

Dennis I can't fix that type of mutation.

Ian What's going to happen to me?

Daphne Out of my way, I'm here to repair the gene.

Myra Are you a DNA repair molecule?

Daphne Correct. I remove damaged bases by cutting them out of the DNA strand through cleavage.

Ian Oh no. Do I have to be cut up?

Daphne If you wish to survive to have to be cut up and the resulting gap will be filled and sealed.

Ian This is not fair.

Nelly At least you're not being egg-terminated.

Myra Or made egg-tinct.

Colin Or be egg-liminated.

Ian I don't think I like egg jokes anymore. I don't want to be cleaved and filled in.

Daphne Too late.

(The damaged section of the gene is removed)

Nelly Well you need to say goodbye to each other now.

Myra Are we being separated?

Neil Each cell will have a copy of each gene but you will all soon be active in different cells.

Colin What do you mean?

Nelly Differentiation is occurring and the cells in this embryo are becoming specialised. Each of you will be active in different cells.

Ian I am pleased we talked and I have learnt a very important lesson.

Nelly No more egg jokes?

Ian Even more important than that! Stay out of the path of gamma radiation!

Questions Scene 3

1. What is the first step in gene expression?
2. What is meant by gene expression?
3. Does the nucleus have the codes for all the proteins produced by an eukaryotic cell? Explain your reasoning.
4. In what way has a study of mtDNA advanced scientific knowledge?
5. What is the function of DNA repair molecules?
6. Outline the function of acetylcholine.
7. What happens during differentiation?

42 Germ Layers and Embryonic Development

Developmental biology is the study of the process by which organisms grow and develop. It includes the study of the processes by which organisms grow and develop from a zygote to a fully grown adult, e.g. genetic control of cell growth, differentiation and morphogenesis. Developmental biology includes the study of classical embryology with molecular genetics.

Embryology is the study of the development of an embryo from fertilisation to the foetus stage.

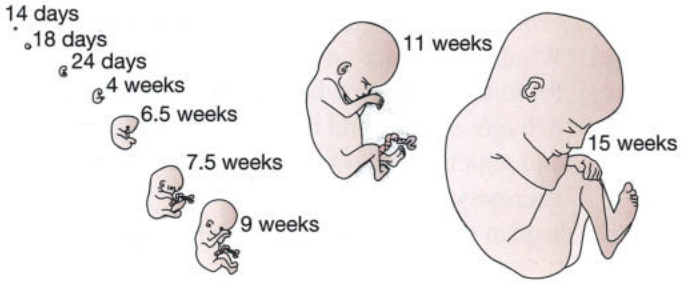


Figure 42.1 Development of human embryo.

Human development occurs at different points, e.g. cleavage of the zygote to form a blastula and gastrulation. The basic mechanisms of development and the regulatory genes that control these stages are common to many animals.

In humans the first two weeks is called the **pre-embryonic period** and includes the stages of cleavage (the zygote divides by mitosis to become a hollow ball called a blastula), implantation (the blastula embeds in the wall of the uterus) and gastrulation (the formation of the three germ layers – the ectoderm, endoderm and mesoderm from which all tissues and organs form). In humans the third to eighth week are called the **embryonic period** and the third to ninth month is called the **foetal period**.

Development of the placenta

After fertilisation the zygote divides to become a two-cell embryo and **cleavage** continues as it travels down the oviduct moved by the actions of cilia and muscle contraction.

When it reaches the uterus the **zona pellucida** has dissolved and it has formed a **blastocyst** with an outer layer of cells called the **trophoblast** and an inner mass of cells. The trophoblast cells that contact the endometrium of the mother secrete enzymes that allow the embryo to invade the lining of the uterus. The **chorion** forms from the trophoblast and the chorion secretes chorionic gonadotropin hormone. The trophoblast cells send out projections that become placental villi and then develop into foetal capillaries of the placenta.

A specialised region of the chorion will become the **placenta**. The placenta will cover around 20% of the uterus and its major function is to provide food and oxygen to the embryo/foetus and remove excretory wastes from the embryo/foetus.

Gastrulation with the formation of the three-layered embryo (ectoderm, mesoderm, endoderm) occurs during the second and third weeks of development.

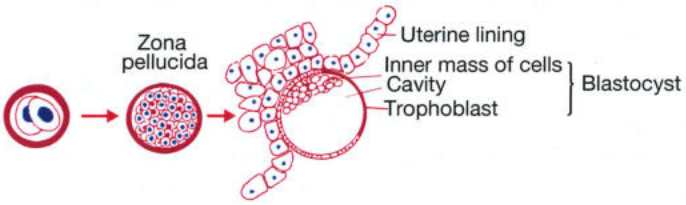


Figure 42.2 Development of the blastocyst.

Germ layers

The germ layers are the group of cells in an embryo at the stage called the **gastrula** that will form the various tissues and organs of an animal body. **Diploblastic** animals have two germ layers, e.g. sponges, jellyfish and corals while **triploblastic** animals have three germ layers, e.g. vertebrates.

In humans and other triploblastic animals after fertilisation of an egg development follows the pathway: zygote → cleavage and mitotic division → morula → blastula → gastrula (with three germ layers).

The **ectoderm** develops from the outer layer of germ cells and is the most exterior of the germ layers. The ectoderm gives rise to skin epidermis, sensory receptors in skin, epithelial lining of mouth, nostrils, and rectum and the nervous system.

The **mesoderm** is the middle germ layer and gives rise to the notochord, skeletal and muscular system, the excretory system, the circulatory and lymphatic systems and the dermis of skin.

The **endoderm** is the innermost layer of cells and gives rise to the epithelial lining of digestive tract and respiratory system, the liver and pancreas, the thyroid and parathyroid glands and the lining of urethra, urinary bladder and reproductive system.

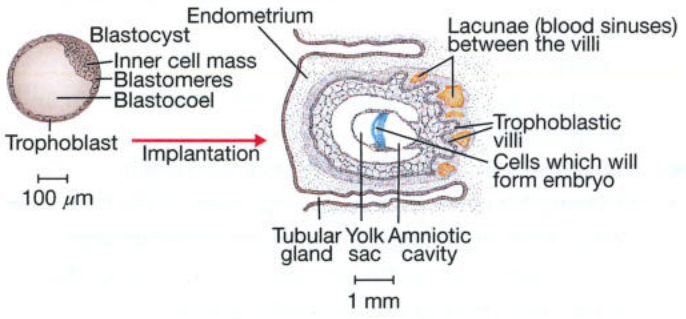


Figure 42.3 Implantation and the germ layers.

Patterning

Patterning is a process where genes cause embryonic cells to differentiate into body regions. Cells become specialised to perform particular functions in particular organs. When genes become 'switched on' the cell will produce particular proteins. These proteins are often enzymes which control specific reactions which in turn alter the cell's metabolism and structure. Thus the 'switching on' of particular genes will lead to the formation of particular cell types, tissues and organs.

Pattern formation is controlled by **positional information** which is molecular cues and signals that regulate the genes controlling development and differentiation. The development of a cell is determined by its location with respect to the position of other cells in the developing embryo.

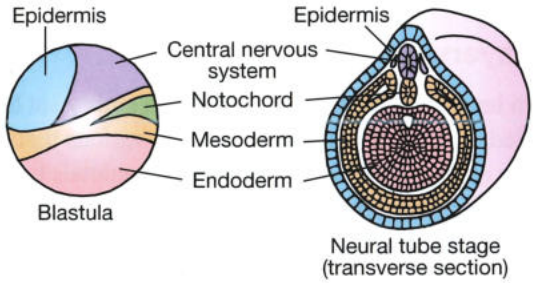


Figure 42.4 Development of organs.

In embryonic development the formation of some tissues requires programmed cell death (PCD). **Apoptosis** is one of the main types of PCD by deliberate suicide of an unwanted cell. For example, the separation of human fingers needs the cells in between the fingers to start apoptosis. Apoptotic cells and their nuclei shrink and are removed by phagocytosis. Selective cell proliferation and selective apoptosis causes tissues to form their definitive shapes. Genetic research has identified genes that code for apoptosis-inducing proteins and also genes that have the opposite effect, i.e. they protect the cells from programmed cell death.

QUESTIONS

1. What is developmental biology?
2. What is a germ layer?
3. Distinguish between diploblastic and triploblastic animals.
4. Define zygote.
5. Explain what is meant by blastula, implantation and gastrulation.
6. Discuss why gastrulation is very important.
7. Construct a table to summarise the parts of the body that come from the ectoderm, mesoderm and endoderm.
8. In humans when is the pre-embryonic period, the embryonic period and the foetal period?
9. Outline what happens in 'patterning'.
10. Describe what happens when particular genes become 'switched on' in the embryo.
11. The diagram shows an embryo and the structure of the placenta.

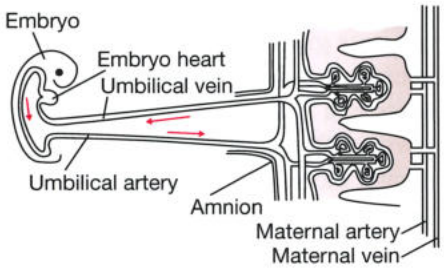


Figure 42.5 Embryo and placenta.

(a) From which cells does the placenta develop?
(b) What is the function of the placenta?
(c) From the diagram which embryo blood vessel delivers oxygen and nutrients to the embryo and which embryo blood vessel removes excretory wastes?

12. The diagram shows a summary of the three germ layers.

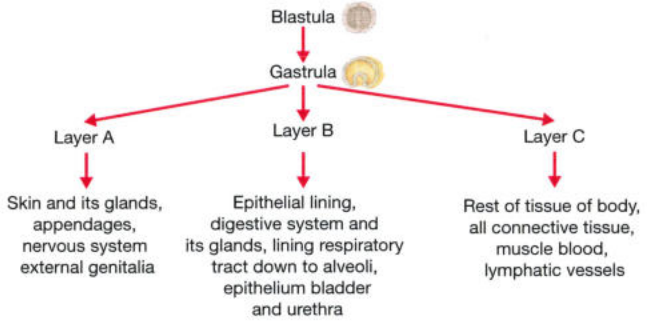


Figure 42.6 Three germ layers.

Identify layers A, B and C.

13. What is positional information?
14. What is apoptosis and how does it influence embryonic development?
15. Identify the role of genes in apoptosis.
16. Discuss why the last five weeks of embryonic period are highly critical.
17. During the first month of development the heart begins to take shape and beat. An average heart rate at this stage is around 60 beats/minute and will increase to around 140 beats/minute before birth. A congenital anomaly is a defect that occurs during foetal development. If a heart defect was due to a mistake in the development of a germ layer, which layer was affected to cause a congenital heart problem?
18. What body structures form from the ectoderm?
    (A) Skin and nervous system.
    (B) Skeletal tissue and muscles.
    (C) Liver and pancreas.
    (D) Renal system and lungs.

43 Birth

Parturition means the birth of a baby.

Labour is a series of strong contractions of the uterus to push the baby out of the uterus, through the vagina for birth. Labour is regulated by hormones, e.g. oestrogen and oxytocin and local regulators, e.g. prostaglandins.

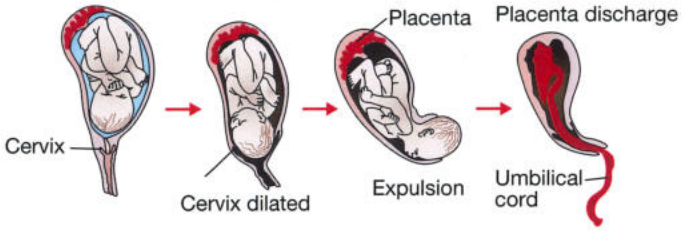


Figure 43.1 Stages of birth.

In the late stages of pregnancy the foetus increases greatly in size and weight, e.g. doubling in size in the last two months. The increased size of the foetus causes a stretching of the walls of the uterus and cervix which are mechanical factors that help initiate labour.

Lanugo is downy hair produced by the foetus from about five months and is usually shed before birth. Some babies are born with lanugo and shed this hair within the first few days or weeks. In the last months the foetus is covered in a waxy/greasy substance called **vernix** which covers the skin and acts to help moisturise and protect the skin and conserve heat.

Stages of labour

Labour is divided into three stages – dilation, expulsion and placental stages.

Stage 1 dilation – the dilation of the cervix is usually the longest stage, e.g. usually from 2 to 16 hours. It usually begins with mild contractions of the uterus usually 15 to 30 minutes apart and at the end of the stage the contractions are more intense and can be 1 to 2 minutes apart. The cervix needs to open to around 10 centimetres to fit the head of the foetus. This stage is usually shorter for subsequent pregnancies.

Stage 2 expulsion – when dilation is complete the amniotic membrane breaks and the amniotic fluid flows out. The head of the foetus is seen in 'crowning'. The baby is born with the most powerful contractions. This stage is usually about 30 minutes long in a first pregnancy and shorter, even just a few minutes in subsequent pregnancies.

Stage 3 placenta expulsion – more uterine contractions expel the placenta as the 'afterbirth'. As the placenta separates the blood vessels in the endometrium close off to minimise blood loss.

Breech births occur when the foetus is presenting its bottom rather than the head.

Hormones and labour

After the sixteenth week of pregnancy the placenta releases increasing amounts of **oestrogen** and **progesterone**. Immediately before birth there is a rapid drop in the level of progesterone. **Oxytocin** from the mother's posterior pituitary gland is released and helps to regulate birth by stimulating rhythmic contractions of the muscles in the wall of the uterus and stimulating the placenta to produce **prostaglandins**.

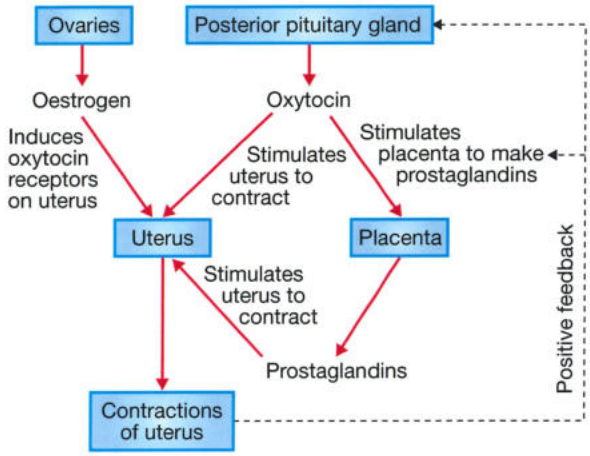


Figure 43.2 Birth flow chart.

Changes in circulatory system at birth

At birth the circulatory system of the foetus has to swap from using the placenta as the source of oxygen and nutrients to using its own circulatory system to collect oxygen and remove carbon dioxide in the lungs in gas exchange. Before birth most of the blood bypasses the lungs with blood from the right atrium flowing directly into the left atrium via an opening called the **foramen ovale**. After birth, the foramen ovale should close due to pressure of blood in the left atrium from the pulmonary circulation. If it does not close, insufficient blood goes to the lungs for oxygenation and deoxygenated blood mixes with the oxygenated blood and insufficient oxygen is supplied to the tissues, giving the baby a bluish tinge.

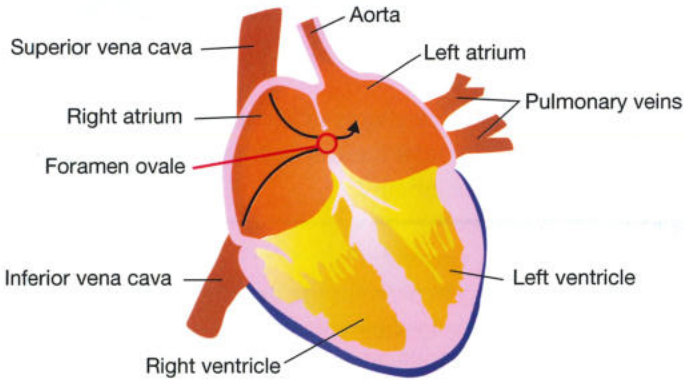


Figure 43.3 Foramen ovale closes at birth.

Ductus venosus is a foetal blood vessel that carries blood from the umbilical vein to the inferior vena cava. Most foetal blood bypasses the liver as the foetus obtains its oxygen and nutrients from the mother. **Ductus arteriosus** is a foetal blood vessel that connects the left pulmonary artery with the aorta. Most foetal blood bypasses the lungs as they are not functional and oxygen is obtained from the mother. After birth the ductus arteriosus closes to become the ligametum arteriosum.

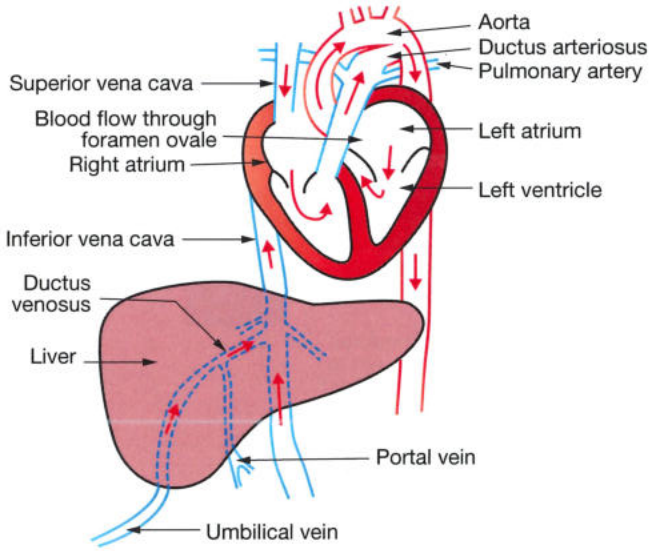


Figure 43.4 Foetal blood flow.

When the placenta stops removing carbon dioxide the rise in carbon dioxide in the baby's blood stimulates the respiratory centre and triggers the **first breath**.

Head changes

The body plates of a baby's skull are soft and flexible and have sutures which allow the bones to move during the birth process. While in the birth canal the pliable bones of the baby's skull can overlap causing the baby's head to change shape. The **fontanelles** are the suture spaces that allow the change of head shape to allow passage through the birth canal. The fontanelles are covered by tough membranes that protect the brain and they close up after birth.

QUESTIONS

1. Define parturition.
2. Define labour.
3. What are the three stages of labour?
4. Identify mechanical factors that help initiate labour.
5. What regulates labour?
6. What is languo?
7. (a) What is vernix?
   (b) Suggest a benefit for the development of vernix.
8. Construct a table to summarise the three stages of labour.
9. What is a breech birth?
10. At what stage of pregnancy does the placenta produce increasing amounts of progesterone and oestrogen?
11. (a) Where is oxytocin produced?
    (b) Outline the function of oxytocin.
12. The diagram shows a feedback mechanism that occurs during labour.

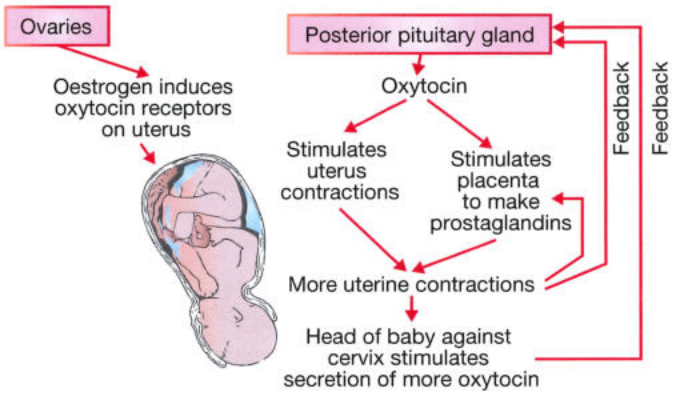


Figure 43.5 Feedback mechanism during birth.

Explain why this is an example of positive feedback.

13. (a) What is the foramen ovale?
    (b) Outline the function of the foramen ovale.
    (c) Explain why the foramen ovale needs to close at birth.
14. What triggers the baby's first breath?
15. The table summarises developmental stages during pregnancy.

Trimester 1	Trimester 2	Trimester 3
Zygote divides forming blastocyst which implants in the endometrium. Rapid growth of vital organs and limbs and outer cell layers blastocyst form the placenta.	Heart, kidneys and limbs further develop and arms flex, legs kick, can suck its thumb. Eyes sensitive to light. Can swallow, hear, move, gets hiccups.	Fat stores and muscle mass developing. Can open/close eyes, bones fully developed. Movements strong and fingernails grow.

What is the approximate number of weeks for each of these trimesters?

16. Explain the significance of foetal ductus venosus.
17. Explain the significance of foetal ductus arteriosus
18. In a foetus what is the blood flow and blood vessels connected by the ductus venosus?
    (A) Umbilical vein to inferior vena cava.
    (B) Vena cava to umbilical artery.
    (C) Left pulmonary artery to aorta.
    (D) Aorta to right pulmonary artery.
19. What is the function of the foetal foramen ovale?
    (A) Allow blood flow from the left atrium to the right atrium.
    (B) Allow blood flow from the right atrium to the left atrium.
    (C) Allow blood flow from the right atrium to the left ventricle.
    (D) Allow blood flow from the left atrium to the right ventricle.

44 After the Birth

A **neonate** is a newborn child. When a child is born it is assessed on an APGAR score which looks at appearance, pulse, grimace, activity and respiratory effort.

Table 44.1 APGAR scale.

Feature	Rating		
	0	1	2
Appearance (skin)	Pale or blue	Body pink, limbs blue	Whole body pink
Pulse (heart rate)	None	Less than 100	More than 100
Grimace (reflex to sole of foot stimulation)	No response	Facial grimace and some movement	Facial grimace and extensive movement
Activity (muscle tone)	Flaccid	Extremities flexed	Good muscle tone and active movement
Respiratory effort	None	Irregular, slow, no crying	Rapid, regular, strong cry

Lactation

During pregnancy hormones prepare the breasts for lactation. **Lactation** is the production of milk from the mammary glands and begins 2 to 3 days after giving birth. The sucking of the neonate is the main stimulus for the release of prolactin which is the main hormone involved in starting milk secretion.

Colostrum is the first milk released after birth. Unlike mature milk colostrum contains large amounts of white blood cells and antibodies, especially immunoglobulin A (IgA). These components are important in preventing pathogens invading the baby's circulatory system through its immature intestinal walls. IgA also helps prevent food allergies. Colostrum also contains a mild laxative which helps the baby pass the first faeces called the meconium. Mature milk has higher levels of carbohydrates, lipids and potassium compared to colostrum.

Newborn reflexes

Newborn babies are born with unlearned or automatic actions known as reflexes. For example, sucking and swallowing are necessary for feeding; crying is needed so that needs can be met and blinking protects the eyes.

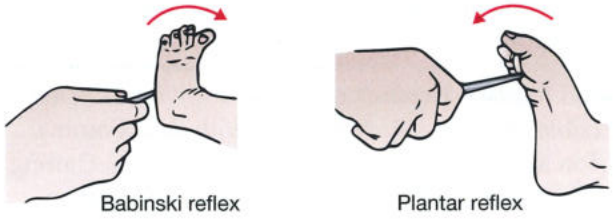


Figure 44.1 Foot reflex changes after about 2 years.

Table 44.2 Developmental milestones.

Developmental milestone	Age when occurs
Raises head	3 months
Sits up	6 to 9 months
Crawls	6 to 9 months
Walks	12 to 18 months
Jumps and hops	2 to 3 years
Throws objects for amusement	6 to 9 months
Places objects in mouth	3 to 6 months
Repeats sounds	9 to 12 months
Asks lots of questions	3 to 4 years
Plays with other children	2 to 3 years

QUESTIONS

1. Define a neonate.
2. What is an APGAR score and give an example.
3. What is lactation?
4. (a) What is colostrum?
   (b) Outline the importance of a neonate drinking the colostrum.
   (c) How is colostrum different to mature milk?
5. Identify some innate reflexes of newborn babies and suggest why the reflex is needed.
6. The diagram shows stages of embryo development.

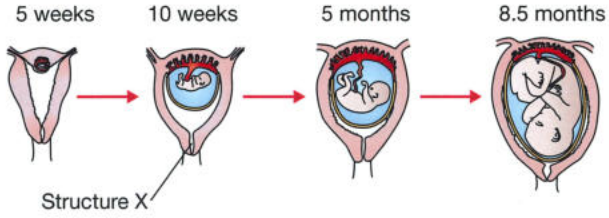


Figure 44.2 Embryo development.

   (a) Identify structure X.
   (b) How is this structure involved in the birth process?
   (c) During the birth process the baby's head can often change shape to some degree. Explain how this can occur.
   (d) The last diagram shows the most suitable presentation for birth. Explain what this means.
7. Why is colostrum beneficial for a neonate?
   (A) Has a laxative effect to pass early stools.
   (B) Maternal antibodies reduce risk of pathogen infection.
   (C) Helps prevent food allergies.
   (D) All of the above.

45 Contraception

Contraception is the deliberate use of methods to prevent pregnancy. Many methods of contraception do not protect against sexually transmitted diseases (STD).

Methods of contraception can be divided into five main areas. **Natural family planning** (also called rhythm method) involves periodic abstinence with intercourse only during 'safe' times of the cycle when conception is unlikely. The pregnancy rate for couples using natural family planning is around 10% to 20%. This means that 10 to 20 females out of 100 per year will become pregnant.

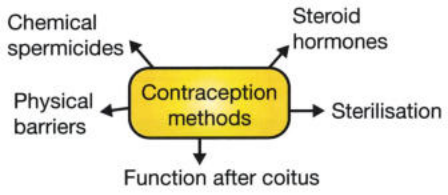


Figure 45.1 Main methods of contraception.

Steroid hormones

Female contraceptive pills use steroid hormones, e.g. synthetic forms of oestrogen and/or progesterone. A **combined female contraceptive pill** has both of these hormones and is taken daily for 21 days of the cycle at the same time of day. The hormones cause negative feedback in the ovarian cycle stopping the release of gonadotropin releasing hormone (GnRH) from the hypothalamus. This stops the release of LH and FSH from the pituitary gland so follicles do not develop and ovulation does not occur. These hormones of the combined pill can be injected, or inserted in a ring into the vagina or worn as a skin patch. Pregnancy rate is around 1% and the pill is also used to reduce period pain, make periods regular and reduce menstrual flow. The pill can have side effects such as increased incidence of blood clots, heart attack, hypertension, weight gain, mood swings, headaches and regular smokers taking the pill have a greater risk of cardiovascular disease. There is also a possible increased risk of infertility.

The **mini pill** only contains progesterone which is suitable for women who cannot take oestrogens. The hormone causes an increase in the viscosity of cervical mucus which hinders sperm penetration, makes the endometrium unsuitable for implantation and decreases the frequency of ovulation. It must be taken within a couple of hours the same time each day or lose its effectiveness.

Male contraceptive pills are being researched, e.g. hormone treatments that will supress the release of gonadotropin to stop spermatogenesis. Hormone combinations including testosterone cause inhibition of the hypothalamus and pituitary hormones, which reduces testosterone production to levels below that necessary for sperm production.

Physical barriers

Some methods of contraception involve a physical barrier separating the male and female gametes.

The **male condom** is a thin sheath of rubber/plastic that slides over the erect penis before intercourse preventing sperm entering the vagina. The male condom also prevents the transmission of STDs, e.g. HIV. Disadvantages include a slightly decreased sensation for the male, foreplay is interrupted to put it on and a tear or slip during or after ejaculation stops protection.

The **female condom** is a lubricated pouch that lines the vagina with a closed end at the cervix. This prevents sperm entering the vagina and also prevents the transmission of STDs. Used correctly is has a pregnancy rate of 5%.

A **cervical cap** is a small thin rubber cap that fits directly over the cervix and a **diaphragm** is a larger dome of soft rubber or silicone that covers the cervix and upper part of the vagina. They must be prescribed (and fitted) by doctor. Both devices can be inserted ahead of time to prevent sperm reaching the uterus. They are effective, relatively cheap and must remain in place for 6 hours after intercourse. Diaphragms must be used with a spermicide cream or gel and have a rate of 10% to 39%.

Chemical spermicides

Chemical methods and spermicides include sponges, foams, jellies, aerosols and creams that immobilise sperm. Some react with moisture in the vagina to form bubbles of gas that are a physical barrier to sperm. They are relatively cheap, simple to use and most are effective in the vagina for up to 24 hours after insertion. They are unreliable on their own and are usually used with mechanical barriers.

Surgical methods

Surgery can involve **sterilisation** to prevent sperm reaching the egg, e.g. **tubal ligation** for females where the fallopian tubes are cut and tied and **vasectomy** for males where a small part of the vas deferens is removed and the ends tied off. New microsurgery procedures mean that these procedures can be reversed in some cases.

A **vasectomy** is a simple procedure that is a permanent solution that can be done in a doctor's surgery under local anaesthetic. The vas deferens is cut and tied on each side so that sperm cannot become part of the semen released in an ejaculation.

A **tubal ligation** requires a stay in hospital and is not reversible. It is usually performed with an abdominal incision and is done using general anaesthesia. Cutting the oviduct means the ovum cannot reach the uterus but the ovary can still secrete hormones and there is no change in the menstrual cycle.

Function after coitus

Function after coitus involves the use of emergency contraceptive pills and **intrauterine devices** (IUDs). IUDs are plastic or metal objects inserted into the lumen of the uterus by a doctor. Many are a small loop or coil and can be left in place until the woman wishes to have a child. They cause an inflammatory reaction in the endometrium so white blood cells engulf sperm or ovum and they prevent implantation of the embryo. Some contain chemicals, e.g. copper or hormones that also assist contraception. IUDs are 98% effective and are usually only recommended for women who have had children and completed their childbearing. They can have side effects such as bleeding and pain, cause pelvic inflammatory disease (PID), be expelled from uterus and cause an increased risk of tubal pregnancy.

The **morning after pill** is an emergency contraception that involves taking 2 pills up to 72 hours after intercourse and 2 more 12 hours later after having unprotected sex. Many morning after pills contain progestogen, a synthetic version of progesterone which will prevent ovulation or implantation. Some contain up to 50 times the normal dose of oestrogen. The morning after pill is reliable with about 85% effectiveness, though a pregnancy test is also advisable, especially if the next period is more than 7 days late.

QUESTIONS

1. Define contraception.
2. Is there a relationship between contraception and sexually transmitted diseases (STDs)?
3. (a) What is the rhythm method?
   (b) What is the pregnancy rate for couples using the rhythm method?
4. The diagram shows methods used in natural family planning methods.

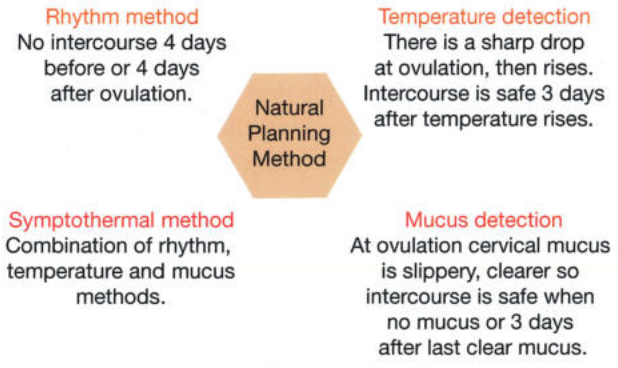


Figure 45.2 Different methods used in natural family planning.

Construct a table to summarise different ways to implement natural planning.

5. List the main methods of contraception.
6. Which steroid hormones are used as a method of contraception?
7. Outline the actions that need to be taken for the contraceptive pill to be effective.
8. How does the contraceptive pill prevent pregnancy?
9. Describe some of the problems related to the use of the female contraceptive pill.
10. Discuss why some women need to take the mini pill.
11. What is the male contraceptive pill?
12. Construct a table to summarise different types of physical barriers that are used as methods of contraception.
13. What are chemical spermicides?
14. The diagram shows two surgical procedures.

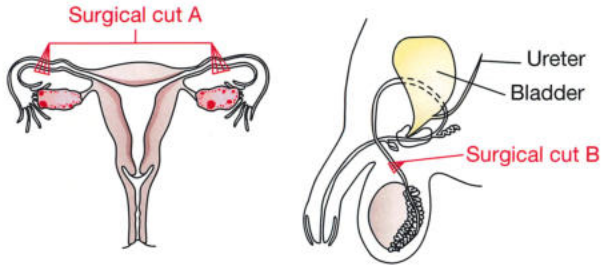


Figure 45.3 Main methods of contraception.

Identify and explain what is happening in surgical cut A and surgical cut B.

15. Describe how IUDs prevent pregnancy.
16. What is the morning after pill?
17. Coitus interruptus is from the Latin *coitus* meaning sexual intercourse and *interruptus* meaning cut short or interrupted. Explain why this method is not very reliable as a contraceptive method.
18. Explain why a vasectomy and sterility does not affect masculinity or the ability to have sexual intercourse.
19. The diagram shows the hormone levels of a woman who uses natural family planning methods of contraception.

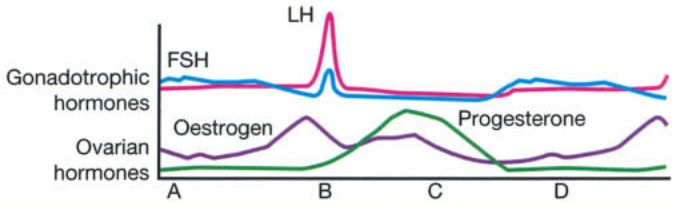


Figure 45.4 Chart of female hormone levels.

At which point in the cycle would sexual intercourse lead to a pregnancy and failure of this method of contraception?

(A) Point A (B) Point B
(C) Point C (D) Point D

46 Lifestyle Choices and Foetal Development

Lifestyle choices, e.g. diet, illicit drugs, alcohol and nicotine may affect foetal development. External factors have the greatest impact on embryonic development in the first two months of life. In the fourth and fifth weeks of development the limbs are rudimentary and exposure to several substances at this stage can lead to limb abnormalities, e.g. thalidomide babies.

Mutagens are chemicals or substances that cause mutations. **Teratogens** are chemicals or agents that cause birth malformations and defects. Teratogens can be any substance, chemical, physical or biological that exposure to during pregnancy can cause birth defects in the developing foetus.

Table 46.1 Teratogens.

Type of teratogen	Example
Medication or drug	Medications can be prescription, over the counter, recreational or illegal drugs. Examples include alcohol, tobacco, aminopterin which is used to treat cancerous tumours and isotretinoin (accutane) which is used to treat severe acne. Thalidomide was given to pregnant women to control nausea of early pregnancy and it prevented normal formation of foetal limbs with the arms which develop first more affected than legs.
Infections	Some maternal infections during pregnancy can affect foetal development, e.g. cytomegalovirus, herpes simplex, chickenpox, zika virus and rubella (German measles). If the mother contracts rubella during the fourth to twelfth weeks of pregnancy the virus can affect embryonic formation of the heart, the eye lens, the inner ear and brain depending upon when she contracted the disease. About 90% of infected babies have major congenital abnormalities.
Maternal disease	Maternal diseases that affect foetal development include diabetes, maternal PKU. Maternal diabetes can involve hyperglycaemia (excessive glucose in the bloodstream) which can affect foetal development causing cardiovascular and neural tube (brain, spine or spinal cord) defects, e.g. spina bifida.
Environmental chemicals and heavy metals	Organic mercury compounds, polychlorinated biphenyls (PCBs), herbicides, lead, industrial solvents, mercury, chemicals in cigarette smoke.
Ionising radiation	Exposure to X-rays during the early stages of pregnancy can produce permanent abnormalities.

TORCH syndrome

TORCH syndrome refers to the infection of a developing foetus or newborn by any of a group of infectious agents. TORCH stands for T (toxoplasma), O (other, e.g. chickenpox), R (rubella), C (cytomegalovirus), H (herpes simplex). TORCH symptoms include small birth weight, petechial rash, enlarged liver and spleen, jaundice, hearing impairment, vision impairment, mental retardation, autism and even death.

Figure 46.1 Teratogens cause birth defects.

Nicotine

There are many chemicals in cigarettes and cigarette smoke. Nicotine whether from actually smoking, inhaling cigarette smoke or wearing nicotine patches can lead to many birth defects. Nicotine is linked with lower birth weight, increased risk of abortion, gastrointestinal problems, respiratory problems and the SIDS risk increases.

Alcohol

Maternal alcohol consumption is linked with several birth defects including foetal alcohol syndrome which causes lower than normal birth weight, slow growth after birth, small head, irregularities of the face, narrow eye slits, sunken nasal bridge, defects of the heart, malformations of arms and legs, mental retardation, hyperactivity, extreme nervousness and poor attention span. The four main problems caused by alcohol are mental retardation, impaired growth, facial malformations and organ defects, e.g. heart, kidneys, muscles and joints.

Heroin

Heroin is derived from morphine and is a highly addictive opiate drug that can cross the placenta and affect embryonic development. It is linked with premature delivery and when the baby is born it has withdrawal symptoms such as irritability, convulsions, diarrhoea, fever, sleep abnormalities, and joint stiffness. Birth defects can include low birth weight, breathing difficulties, hypoglycaemia and bleeding in the brain.

Marijuana

Marijuana contains the chemical delta-9-tetrahydrocannabinol (THC) which is a mind altering chemical that acts on specific brain cell receptors. Maternal intake of THC leads to an increased chance of miscarriage, low birth weight, premature birth, developmental delays in the infant and behavioural and learning problems as the child gets older. THC can also be passed to breastfeeding babies.

Risk possibilities

The **absolute risk** is the rate of occurrence of the abnormal phenotype for those exposed to the agent, e.g. a mother who is a chronic alcoholic has a 45% chance of having an infant with foetal alcohol syndrome. The **relative risk** is the ratio of the condition appearing among those exposed compared to those not exposed. The risk of a smoker having a low birth weight child is twice that of non-smokers and the relative risk is 2.

Data collection

The **Australian Bureau of Statistics** (ABS) carries out many studies looking at selected characteristics of babies, infant mortality and illness, immunisation statistics and facts about mothers. According to ABS there were 259 800 births in 2005, 247 300 births in 1985 and 299 700 births in 2014 with an increase in the average age of mothers from 27.3 years in 1985 to 30.7 in 2005 to 30.9 in 2014.

Infant mortality refers to the death of the child before their first birthday and the rate is calculated at deaths per 1000 live births. The ABS recorded an infant mortality rate of 9.9 in 1985 and a rate of 5.0 in 2005. The infant mortality rate is linked to low birth weight and the ABS data confirms this link with low birth weight infants having longer hospitalisation and an increased risk of problems and disabilities. The change in the infant mortality rate from 1985 to 2005 is related to improvements in medical care and technology and a reduction in the number of deaths from sudden infant death syndrome (SIDS).

QUESTIONS

1. List some lifestyle choices that can affect foetal development.
2. What is a mutagen?
3. Outline the difference between a mutagen and a teratogen.
4. Explain why it is important to limit exposure to teratogens especially in the first two months of embryonic development.
5. Use some examples to show how the use of medications can affect foetal development.
6. The zika virus was first isolated in 1947 and comes from forests in Uganda. It is spread by the *Aedes* mosquitoes from person to person and by unprotected sexual intercourse. If a pregnant woman becomes infected the virus can cross the placenta to the developing foetus causing microcephaly, brain defects, eye defects, hearing defects and impaired growth. Suggest some preventive measures a pregnant woman can take if she is in a zika virus area.
7. The Immunise Australia Program of the Australian government includes rubella in its vaccination program with injections of the MMR (measles-mumps-rubella) vaccine given at 12 months of age and the MMRV (measles-mumps-rubella-varicella) given at 18 months of age. Discuss why the Australian government has seen the need to include rubella in its national immunisation program.
8. There is some evidence that maternal high persistent fever during pregnancy especially during the first trimester is associated with multiple congenital abnormalities (MCAs), e.g. heart defects and cleft palate. Suggest why pregnant women are advised to avoid hot tubs and saunas.
9. (a) What is TORCH syndrome?
   (b) Identify some TORCH syndrome symptoms.
10. List some of the problems that can occur if the mother smokes while pregnant.
11. Identify the four main problems caused by alcohol consumption during pregnancy.
12. (a) According to the Australian Bureau of Statistics what is the change in the number of births and the age of mothers from 1985 to 2005 to 2014?
    (b) In Western Australia in 2004 there were 25 295 births and in 2014 there were 35 403 births. How does this trend compare with the national figures?
13. (a) Define infant mortality.
    (b) Outline the change recorded by the Australian Bureau of Statistics in the infant mortality rate from 1985 to 2005.
    (c) Suggest a reason for the change in the infant mortality rate.
14. According to data from the Australian government around 16% of Australian women smoked during their pregnancy in 2008, 12.7% smoked during their pregnancy in 2011 and 11.2% smoked in the first 20 weeks of their pregnancy in 2013. What trend is shown in this data and suggest a possible effect this trend will have on the rate of birth defects.
15. What is the best term for a chemical or agent that causes birth malformations and defects?
    (A) Mutagen (B) Teratogen
    (C) Amphetamine (D) Abnormalagen

47 Mutagens

A **mutagen** is a chemical or physical agent that causes a mutation changing DNA and the genetic information. Mutagens can be environmental factors such as radiation (e.g. X-rays, cosmic rays, UV rays and alpha, beta and gamma radiation from radioisotopes) or chemical agents (e.g. benzene, formalin or Agent Orange) or biological agents, e.g. viruses and bacteria.

Table 47.1 Mutagens.

Chemical mutagens	Radiation	Biological agents
Disrupt DNA replication, e.g. deaminate nitrogenous bases, e.g. nitrous oxide. Mutagens can resemble base pairs leading to insertion or deletion of an extra base pair, e.g. ethidium bromide.	Ionising radiation, e.g. X-rays, gamma rays and alpha particles break down DNA sequences leading to chromosome rearrangement. Lower energy radiation, e.g. UV rays penetrates cellular and nuclear membranes which can crosslink two bases together.	Viral DNA can insert into the genome and disrupt metabolic activities and correct protein synthesis. Bacteria can produce toxins that cause DNA damage and affect DNA repair systems.

Mutation

A **mutation** is a change in the DNA genetic make-up of a cell. Mutations can cause a change in the structure of DNA, e.g. insertion, deletion, duplication or translocation of genes on the chromosomes changing the arrangement or quantity of DNA on the chromosomes. Mutations can also involve a change in the coding of a particular gene.

Mutations can occur spontaneously and randomly, e.g. due to errors in DNA replication or cell division, or they can be induced by mutagens.

Somatic mutations

If the mutation occurs in a body cell it is a somatic mutation, e.g. skin cell mutation that causes a melanoma and skin cancer. Somatic cell mutations affect the individual and cannot be inherited by offspring and passed on to the next generation.

Germ cell mutations

If the mutation is in a gonad cell in the reproductive organs, e.g. in testes or ovaries, the change can be inherited by offspring and passed on to the next generation. It may be inherited with the sperm or ova.

Teratogens are any substance, e.g. chemical, physical or biological that can cause a birth defect affecting the developing embryo/foetus during the pregnancy.

Exposure to teratogens during the embryonic period when cell differentiation occurs and the major organs and systems are forming can lead to health problems in the child.

In the 1950s and 1960s thalidomide was a drug used to treat morning sickness in pregnant women. It was also used as a sedative to help them sleep. After investigations into children with birth defects it was realised that **thalidomide** affects limb development in the growing embryo. At the time it had not been universally accepted that there were environmental factors that were non-toxic to the mother that could affect the developing embryo. Many had believed that the embryo was totally protected from harm while in the womb. Thalidomide was then acknowledged to be a **teratogen** with the sensitive period from day 35 to day 49, e.g. if thalidomide was taken by the mother on days 35 to 37 the child is born without ears and is deaf, if taken from day 39 to day 41, the child has no arms.

Mutations can lead to cancer

Cancers and other genetic disorders can occur when the mutation occurs in the genes involved in DNA repair. *BRCA1* and *BRCA2* (genes named for **BR**east **CA**ncer) are regulator genes of the cell cycle and the proteins that are coded by the *BRCA1* and *BRCA2* genes play an important role in embryo development. Mutations of the *BRCA1* and *BRCA2* genes are associated with breast cancer and less DNA repair. When defective *BRCA1* proteins do not fix damaged DNA, mutations in other genes can allow uncontrollable cell division to form a tumour.

QUESTIONS

1. What is a mutagen?
2. Define mutation.
3. What type of agents can be mutagens?
4. How can chemical agents cause mutations?
5. Distinguish between spontaneous and induced mutation.
6. Distinguish between a somatic mutation and a germ cell mutation.
7. What is a teratogen?
8. Explain why it is beneficial that a women is not exposed to teratogens in the first trimester of pregnancy.
9. What is thalidomide and explain why the discovery that thalidomide is a teratogen was important in the scientific understanding of prenatal development.
10. What are *BRCA1* and *BRCA2*?
11. What is the name of the process when sequences of bases in DNA are changed due to exposure to radiation such as X-rays?
    (A) Crossing over.
    (B) Mutation.
    (C) Random segregation.
    (D) Replication.

48 Sexually Transmitted Diseases

Sexually transmitted infections (STIs) (also called venereal disease – VD) are diseases that are transmitted through unprotected sex or genital contact. Early detection and treatment of the infection are important and reduce the likelihood of serious health consequences. STIs are caused by organisms that thrive in the warm, moist environment of reproductive systems.

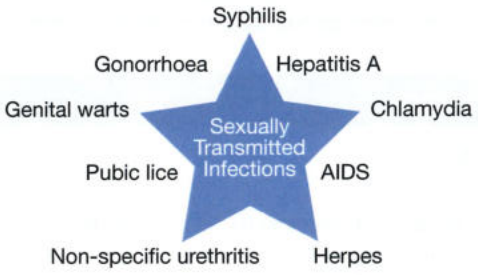


Figure 48.1 Different STIs.

Syphilis

Syphilis is caused by the spirochete bacteria *Treponema pallidum*. Syphilis is a progressive disorder that can affect all parts of the body, e.g. brain, bones, spinal cord, heart and if left untreated can cause blindness, brain damage, heart disease and death.

In the **primary stage** small, painless open sores called **chancres** appear around 10 to 90 days after infection usually on the genital organs, rectum, cervix, fingers, lips or anywhere where there was direct contact.

In the **secondary stage** which occurs within 6 weeks to 6 months of initial contact there are flu-like symptoms with fever and sore throat then skin rashes, mouth ulcers, mild fevers, disorders of bones or eyes. This stage lasts up to 2 years.

In the **tertiary stage** the disease is less contagious and there is damage to many body organs with syphilitic heart disease, brain damage and insanity, blindness, physical incapacity. This stage occurs 5 to 40 years after initial contact. The bacteria can be detected in blood tests and treated with antibiotics in all stages. Surgery in later stages can remove damaged areas and it is best prevented by maintaining safe sex practices. The bacteria can cross the placenta and infect a foetus.

Syphilis is rare in Australia. The Australian Bureau of Statistics (ABS) reported that in 2011 there were 12 men/100 000 men diagnosed compared to 1.7 women/100 000 women diagnosed with the disease and there were differences in the age group affected compared to other STIs, e.g. middle aged men (40 to 44 years) most likely to be infected.

Gonorrhoea

Gonorrhoea is caused by the bacteria *Neisseria gonorrhoeae* and is one of the most frequently reported STIs. It predominantly affects the penis in men, the vagina in women and the mouth and anus in both sexes and if left untreated becomes a blood infection causing heart problems, sterility and arthritis. Males experience burning sensations in the urethra, pain in urination and yellow discharge from the penis usually within 2 to 10 days of infection. The bacteria can spread to the prostate gland and male duct system causing inflammation. Females may show no symptoms with no pain, little or no discharge though some females have painful urination with cloudy urine, vomiting and diarrhoea. It can lead to pelvic inflammatory disease (PID) which can lead to infections of ovaries, oviducts and sterility. During birth passage through the birth canal can infect the child causing gonococcal conjunctivitis. Microscopic examination of a sample of the discharge will identify the bacteria and the infection is treated with antibiotics, e.g. penicillin. Prevention requires the use of safe sex practices, e.g. using condoms.

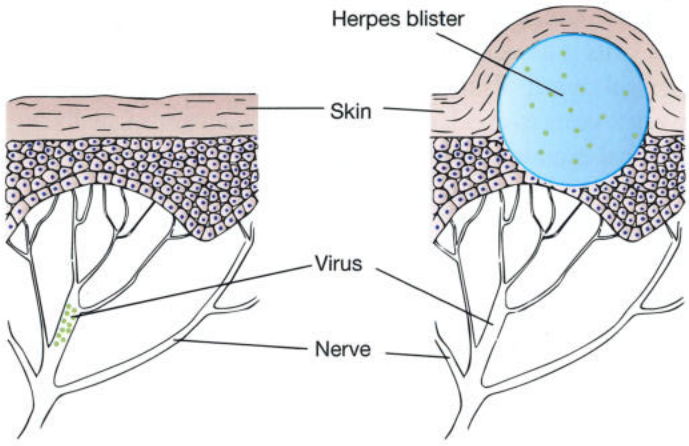


Figure 48.2 Herpes infection.

Herpes

Herpes is caused by the virus herpes simplex type 2 and is a widespread STI. It can be treated but not cured and is only contagious during active periods, e.g. times of stress or illness. Symptoms appear, e.g. itching blisters filled with fluid on the genital organs, then disappear with the virus lodged permanently in the nervous system in nerve cells. During an active period there are flu-like symptoms, e.g. fever and fatigue. It can pass across the placenta causing serious malformations of the foetus and transmitted during childbirth causing meningitis, brain damage in the newborn. Diagnosis involves microscopic examination of a culture of the fluid contained in the blisters. Treatment involves antiviral drugs which reduce the length and severity of attack but do not the cure the disease. Prevention needs the use of safe sex practices, e.g. using condoms.

Chlamydia

Chlamydia is caused by the bacteria *Chlamydia trachomatis*. Many people do not show immediate signs of infection. In males it causes a burning in the urethra, pain in urination and a yellow discharge from the penis. In some cases it can lead to epididymitis and occasionally infertility. In females there are often no symptoms. Infection can cause bleeding between periods or after sex, pain when urinating and can lead to infertility (56% of females) if untreated, eye infections, arthritis, pelvic inflammatory disease and increase risk of premature or stillborn child. Treatment involves antibiotics and prevention requires use of a condom as a preventive barrier.

Genital warts

Genital warts are caused by the human papillomavirus (HPV) and are one of the most common STIs. There are many strains of HPV and only some cause the formation of visible warts and affect the genitals. Symptoms are warts on the genital area that appear about 6 to 8 months after infection and if untreated the warts can grow quite large. The warts are soft, moist pink and can appear alone or in clusters. The virus is associated with some types of cancer especially cervical cancer. Women are advised to have a **Pap smear** every 2 years to check cervical cells. Diagnosis is usually by physical examination as the wart has a distinct, easily visible appearance. Tissue samples are also usually tested, e.g. for cancer cells and to rule out other STIs. Treatment involves antiviral preparations and wart removal. The removal of the warts does not remove the virus so warts can return. The virus is contagious and transmission can occur even if a wart is not visible. In Australia free HPV vaccination is offered for males and females. It is best completed before a person becomes sexually active.

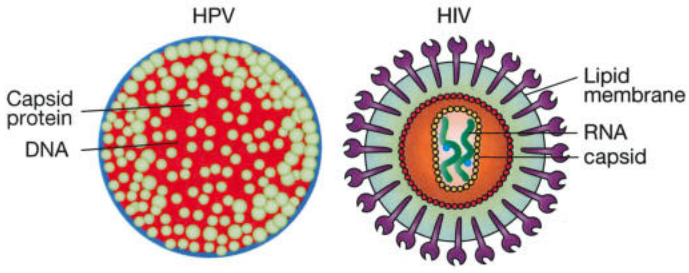


Figure 48.3 Human papillomavirus and human immunodeficiency virus.

AIDS

Acquired immunodeficiency syndrome (AIDS) is caused by human immunodeficiency virus (HIV) and in Australia is most commonly spread by unprotected sexual intercourse. HIV weakens the immune system making people more susceptible to other diseases and infections. It is a progressive disease with vulnerability to other diseases increasing over time.

Category A has symptoms similar to glandular fever with swollen lymph nodes, headaches, fever and dry cough. **Category B** has persistent generalised lymphadenopathy with swelling of lymph nodes, weight loss, diarrhoea, fatigue, fever, unexplained bleeding and memory loss. **Category C** is AIDS with the development of serious diseases such as pneumonia, cancer, brain disorders. Diagnosis involves HIV testing, e.g. ELISA (enzyme linked immunosorbent assay) test. Treatment involves anti-HIV drugs, e.g. drugs that interfere with the replication cycle of the virus. New medications and types of medication are constantly being developed. At present there is no cure. Prevention involves safe sex practices.

QUESTIONS

1. Define a STI.
2. List some STIs.
3. Why is early detection of STIs important?
4. What is a chancre?
5. How can you detect the progression of syphilis infection from secondary stage to tertiary stage?
6. (a) What causes gonorrhoea?
   (b) Which areas are usually most affected in gonorrhoea?
7. According to the Australian Bureau of Statistics (ABS) notification rates of gonorrhoea increased in Australia from 40 per 100 000 people aged over 15 years to 65 per 100 000 people aged over 15 years. In 2011 more men were diagnosed (8056 men and 3789 women) with men aged 20 to 24 having the highest diagnosis rate followed by men aged 25 to 29 years while women aged 15 to 19 then 20 to 24 had the highest diagnosis rate. How does this compare with the diagnosis rate of syphilis in 2011?
8. (a) What causes herpes?
   (b) Discuss why herpes can be a threat to a foetus.
9. Describe the cause and symptoms of chlamydia.
10. According to the Australian Bureau of Statistics (ABS) chlamydia was the most frequently reported notifiable condition in Australia in 2011 with 79 833 notifications for people aged 15 years and over and in 2012 there were 82 707 new cases with remote areas reporting 81% of cases who were 15 to 24 year olds. What treatment and preventive measures are recommended to reduce the spread of chlamydia?
11. Explain why the national immunisation program includes the HPV vaccination.
12. Discuss the statement – 'HIV infection is not AIDS'.
13. How is HIV infection diagnosed?
14. What is the cause of syphilis?
    (A) *Pallidia syphillosis*
    (B) *Treponema pallidum*
    (C) *Neisseria gonorrhoeae*
    (D) *Chlamydia trachomatis*

49 Reproductive Technologies

There is a variety of **assisted reproductive technologies** (ART) that are medical interventions with the use of any reproductive technology to help overcome infertility problems, though each has its limitations, risks and benefits. They include in vitro fertilisation, artificial insemination, embryo transfer, hormone treatment, cryopreservation of sperm/eggs, embryo and fertility medications, mapping of a woman's ovarian reserve with the ability of the ovary to produce viable egg cells, follicular dynamics and associated biomarkers and semen analysis.

In Australia in 2004 approximately 2.5% of all births were the result of ART treatments showing about a 74% increase since 1989 with mothers using ART having an average age of 34.5 years compared to an average age of 29.7 years for all mothers in 2004.

Infertility

Infertility is defined as the failure to conceive a child after one year of regular sexual intercourse without birth control. Infertility is not sterility. **Sterility** is the permanent, irreversible inability to have children.

One of the main causes of **male infertility** is low sperm count measured as the number of active sperm per millilitre of semen. This can be caused by low levels of testosterone, exposure to chemicals, pesticides, heat, radiation, very frequent intercourse that depletes the sperm supply or tube blockage, e.g. caused by varicose veins in the scrotum. Male infertility can also be caused by irregular shaped sperm without the oval shaped head needed to fertilise an egg or sperm unable to swim properly.

Female infertility can be due to infrequent ovulation, tube blockage, pelvic inflammatory disease (PID), a sexually transmitted infection, endometriosis, hormonal imbalance or cervical problems, e.g. a weak cervix due to a previous abortion or surgery so that it cannot hold the weight of a growing embryo.

A **miscarriage** is the natural or spontaneous end of a pregnancy where the embryo/foetus is unable to survive. It usually refers to events that happen in early pregnancy. It is estimated that miscarriages occur in the first 20 weeks of pregnancy to about 20% of women who know they are pregnant.

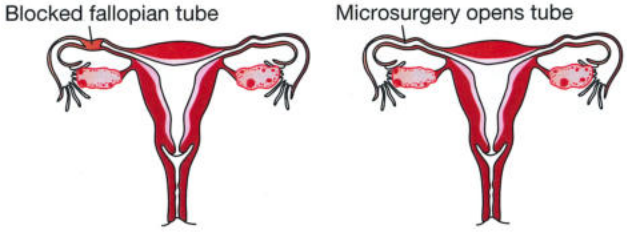


Figure 49.1 Microsurgery can open blocked fallopian tubes.

Pregnancy testing

Common methods for pregnancy testing involve testing the urine or blood of the female for human chorionic gonadotrophin (hCG) hormone. By around the sixth day the embryo (at trophoblast stage) contacts the endometrium for implantation and releases hCG which will stay present in urine until around week 8. Many home pregnancy test kits tests for hCG are used within two weeks of conception when the female has a 'missed period'. Blood tests are more sensitive and can confirm pregnancy.

In vitro fertilisation

There are several methods of in vitro fertilisation (IVF) used to aid couples to have children. In vitro fertilisation involves mixing oocytes with sperm in culture dishes and incubating for several days to ensure the beginning of gastrulation. The embryo needs to be at least 8 cells in size to be implanted into the uterus.

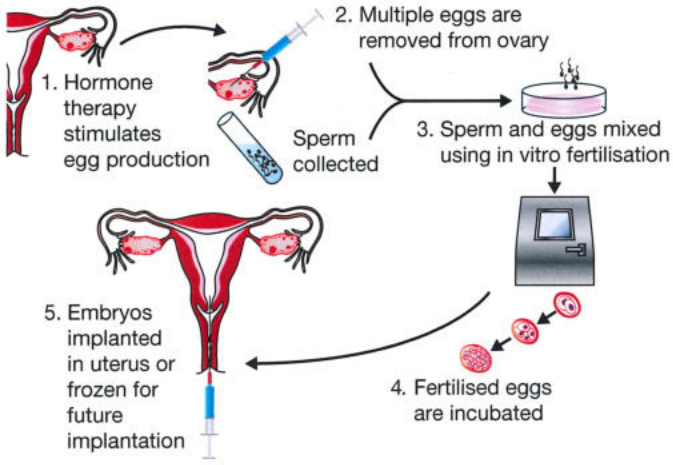


Figure 49.2 In vitro fertilisation.

ZIFT (zygote intrafallopian transfer) involves fertilising the eggs in vitro and then immediately transferring the zygote to the oviducts. **GIFT** (gamete intrafallopian transfer) involves placing eggs and sperm in the oviducts hoping fertilisation will occur.

There are several risks involved with IVF. The use of fertility drugs, e.g. hCG can cause ovarian hyperstimulation syndrome which causes the ovaries to become swollen and sore, along with nausea, vomiting and diarrhoea. If more than one embryo is implanted at a time there is a risk of multiple births which has a risk of early labour and low birth weight and in a small percentage of cases IVF can lead to an ectopic pregnancy with the fertilised egg implanting in the fallopian tube.

Ovarian reserve testing

The **ovarian reserve** is the current supply of eggs in the ovary and associated with reproductive potential. There is a finite number of eggs and the supply of eggs slowly decreases over time until menopause when no eggs remain in the ovary. This means that age is the most important factor in determining the number of remaining eggs in the ovary. Fertility treatments often involve boosting egg production and it is important to know the size of the ovarian reserve. The decrease in the number of follicles becomes critical after the age of 38 years.

There are several ovarian reserve tests (ORTs). Assessment methods include measuring the levels of FSH, LH, oestrogen, progesterone, inhibin-B and anti-mullerian hormone (AMH) which is a hormone produced by granulosa cells in ovarian follicles. Ultrasound is also often used, e.g. to determine if ovarian cysts are present and if present they need to be aspirated (collapsed) before starting IVF treatment.

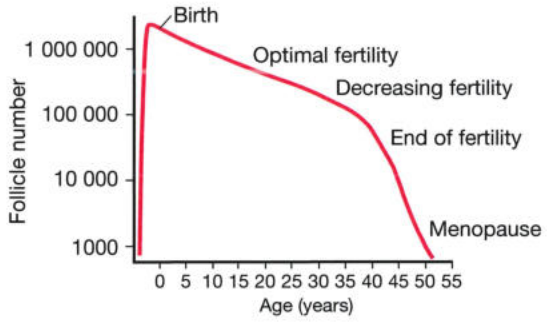


Figure 49.3 Generalised ovarian reserve.

Cryopreservation

Cryopreservation is a process where biological material, e.g. cells, tissues, organs, embryos are preserved by cooling to very low temperatures, e.g. −80°C to maintain their viability. Eggs, sperm and embryos can be cryopreserved and used in fertility programs. There have been pregnancies from embryos stored for 16 years and data indicates that there is no increase in birth defects for children who were frozen embryos. The survival rate of frozen embryos is around 90% to 95%. Donor sperm is quarantined for 6 months to confirm donor does not carry any viral diseases. There are two main methods for cryopreservation – slow freezing and vitrification (ultrarapid freezing).

There are several risks associated with cryopreservation. There is a potential loss and damage to stored material, e.g. with equipment or power failures causing premature sample thaw. For example, in September 2016 fierce storms that caused power blackouts across the entire state of South Australia led to the destruction of embryos at Flinders Fertility in the Flinders Medical Centre. About 12 patients were affected with about 25 embryos lost per patient. Another risk for cryopreservation is that there is a possibility of transmission of infectious disease between samples and there are safety issues for the people working in these facilities.

QUESTIONS

1. Define assisted reproductive technology.
2. List some types of assisted technology activities.
3. Define infertility.
4. Explain why infertility does not mean sterility.
5. What is a miscarriage?
6. There are several types of miscarriage, e.g. incomplete miscarriage where some pregnancy tissue remains in the uterus, a spontaneous miscarriage which is unplanned and there is complete loss and an induced miscarriage which is a planned and voluntary termination of the pregnancy. Explain why it is important to have different treatments for different types of miscarriage.
7. What technologies are used in pregnancy testing?
8. Define in vitro fertilisation.
9. Distinguish between GIFT and ZIFT.
10. List the main steps in the in vitro fertilisation process.
11. (a) Define the ovarian reserve.
    (b) What is the most important factor in determining the size of the ovarian reserve?
12. Identify some risks associated with in vitro fertilisation.
13. (a) Define cryopreservation.
    (b) What can be cryopreserved in fertility programs?
14. Use an example to show a risk and limitations of using cryopreservation.
15. Explain why donor sperm is usually kept quarantined for 6 months after being donated.
16. An ovarian cyst is a fluid filled sac in an ovary. Many ovarian cysts are related to ovulation and can be detected by ultrasound.
    (a) Suggest why women with persistent ovarian cysts may need assisted reproductive technologies when trying to become pregnant.
    (b) How are ovarian cysts detected and suggest how they can be aspirated.
17. Suggest why testing for levels of follicle stimulating hormone (FSH) is one of the most widely used ovarian reserve tests.
18. Biomarkers are naturally occurring molecules, genes or characteristics by which a particular body process or condition can be identified. Which of the following would act as a biomarker for a low ovarian reserve?
    (A) High human chorionic gonadotropin levels.
    (B) Low human chorionic gonadotropin levels.
    (C) High follicle stimulating hormone levels.
    (D) Low follicle stimulating hormone levels.
19. Which condition involves the permanent and irreversible inability to have children?
    (A) Sterility
    (B) Infertility
    (C) Impotency
    (D) Incontinence

50 Genetic Screening

Genetic screening is done for a particular condition in individuals, groups or populations without family history of the condition.

Carrier testing is a type of screening and is usually given to those known to be at risk.

There are several types of screening carried out in Western Australia. These include the following.

Maternal serum screening (MSS) is a blood test for pregnant women for the detection of Down syndrome and neural tube defects, e.g. spina bifida or trisomy 18.

Newborn screening is a public health program for all newborn babies and tests for several metabolic disorders including phenylketonuria (PKU), hypothyroidism and cystic fibrosis. These conditions are treatable but not always clinically evident at birth.

Carrier screening is used to find out if a person has an allele for a genetic disorder, e.g. screening for cystic fibrosis (CF) can detect the presence of the abnormal CFTR allele. It is also used to test for fragile X symdrome and spinal muscular atrophy.

Thalassemia screening is usually for couples in antenatal care and it screens for haemoglobinopathies which have defects in the structure of the haemoglobin molecule in red blood cells.

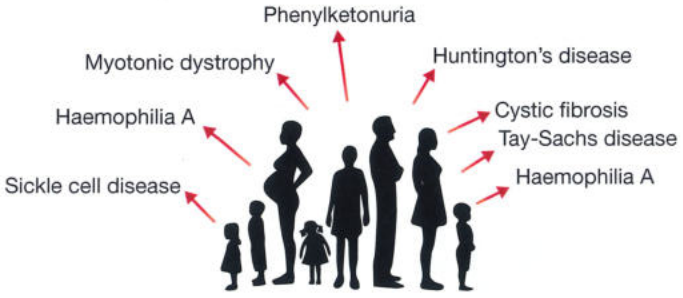


Figure 50.1 Genetic screening.

Process involved

Gene tests or DNA based tests are tests that check for a particular code sequence which indicates a genetic disorder or biochemical tests that check for gene products such as enzymes and other proteins.

Examples

The Western Australian state government provides many types of health services including population screening of newborns and second trimester foetuses for selected genetic conditions, diagnostic, predictive testing and counselling services for people affected by, or at risk of birth defects, genetic disorders of childhood or later onset genetic disorder such as familial cancer.

Gene tests currently available include tests for haemophilia A and haemophilia B, tests for Huntington's disease, cystic fibrosis, Tay-Sachs disease, phenylketonuria and sickle cell disease. Some susceptibility genetic tests currently available include tests for *BRCA1* and *BRCA2* (breast cancer genes 1 and 2) and Alzheimer's disease. Susceptibility tests can show the presence of a gene, but the development of the disease is a probability not a certainty.

Gene testing is used for carrier screening and for people with family histories of genetic disorders, testing of in vitro fertilisation embryos, prenatal diagnostic testing, newborn screening, to confirm a suspected disease for an individual exhibiting symptoms, and for forensic and paternity cases.

Gene test results can be used to assist diagnosis and direct the medical practitioner towards the most suitable treatment. Tests can also identify people with a high risk factor for particular diseases that may be preventable. Test results can be used to inform couples about the likelihood of offspring inheriting a disease.

Genetic screening can identify if a person has a particular genetic disorder. Experimental treatments that remove defective genes, e.g. in patients with cystic fibrosis, bone marrow disorders and Huntington's disease show that there can be large impacts on the wellbeing of patients improving their quality of life and reducing their dependency on drugs and constant medical attention. This has an economic effect on the health care system and the time and effort of care staff, family and community. However, issues can arise over confidentiality of information about the results of genetic screening impacting on a person's right to privacy and how this data is used, e.g. by insurance companies and employers. Knowledge about the specific genes involved in genetic diseases has led to the development of specific drugs that can better target a disease. This aids patients but commercialisation of the drugs with associated costs and market distribution may mean only some people are able to access the new drugs. This can cause ethical issues of inequality.

QUESTIONS

1. What is genetic screening?
2. Construct a table to summarise some different types of genetic screening.
3. Identify some tests provided by the Western Australian state government.
4. Explain some issues that arise due to genetic screening.
5. Genetic Services Western Australia (GSWA) provides genetic counsellors to support genetic testing. Explain why many people may need this service in conjunction with genetic testing.

51 Screening Embryos

There are a range of techniques available to genetically screen embryos before implantation or during early development, including blood tests, amniocentesis and chorionic villus sampling.

Screening tests look for the increased risk of a birth defect and look at data from a blood test and an ultrasound. **Diagnostic tests** identify particular conditions and use data from chorionic villus sampling, amniocentesis and ultrasound, e.g. testing for chromosome conditions, e.g. trisomy 13, trisomy 18, testing for conditions such as neural tube defects, congenital heart conditions and congenital kidney conditions and checking for sickle cell and thalassemia for at risk women.

Blood tests

In Western Australia there is a range of blood tests that are offered during pregnancy. These tests can check for infections in the mother and identify specific problems of the growing embryo/foetus. Some of the blood tests carried out during a pregnancy include:

- Full blood count of mother in first trimester.
- Mother's glucose levels for maternal diabetes.
- Mother's blood group and antibodies present.
- Check for infectious diseases of mother, e.g. syphilis, rubella, hepatitis B, C, HIV antibodies, chlamydia screening.

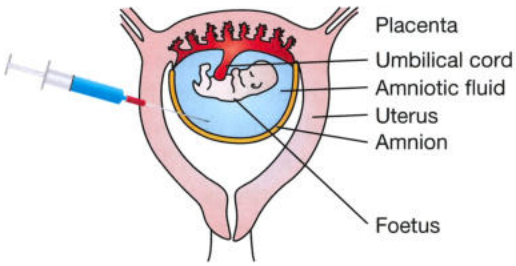


Figure 51.1 Amniocentesis.

Amniocentesis

Genetic amniocentesis is a procedure of prenatal diagnosis which removes about 10 to 20 mL of amniotic fluid from the womb of a pregnant woman by aspiration from a needle inserted into the uterus. The procedure is usually carried out in conjunction with ultrasound to determine the exact location and position of the foetus.

The fluid contains living foetal cells which are cultured then analysed to check for chromosome abnormalities, e.g. determine the number and structure of chromosomes in the foetus and to check for biochemical defects.

When the karyotype is studied any chromosomal anomaly can be identified, e.g. the number of chromosomes 13, 18, 21 and X or Y is assessed. The test is usually carried out during weeks 16 to 20 of pregnancy and recommended for women over 35 years of age having their first child and those with a family history of a genetic disorder.

Risks in the procedure can involve foetal injury if carried out too early, potential miscarriage, potential Rh sensitisation if Rh positive foetal blood contacts the mother's Rh negative blood, potential infection transmission, e.g. HIV, hepatitis C from mother to foetus, potential leakage of amniotic fluid and there is a risk of the needle contacting the foetus causing needle injury. There are also ethical issues as most conditions which are often incurable are not determined until the second trimester when abortion is difficult medically and psychologically. Combined with changing attitudes in society and new technology advances the results of the diagnostic testing can require the couple to look at many ethical issues.

Chorionic villus sampling

Chorionic villus sampling (CVS) is a procedure of prenatal diagnosis that involves using a long narrow tube through the cervix or through the abdominal wall to take foetal cells from the chorionic membrane. The sample from the placenta is cultured and tested for genetic and biochemical disorders. It is considered to be an invasive technique as it involves entry into the placenta by the insertion of an instrument.

The procedure is usually carried out during weeks 9 to 13 of pregnancy which is earlier than an amniocentesis and more likely to be used if there is a family genetic disorder.

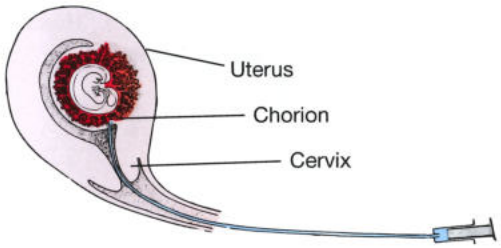


Figure 51.2 Chorionic villus sampling.

Chorionic villus sampling has a higher risk of miscarriage and a higher risk of infection than an amniocentesis. It can cause leakage of amniotic fluid which can affect foetal development and there is a risk of Rh sensitisation.

Preimplantation genetic diagnosis

Preimplantation genetic diagnosis (PGD) is a technique used to screen embryos before pregnancy. A cell from the embryo or sometimes an oocyte that has been created in a laboratory is tested for chromosomal or genetic abnormalities.

If testing an embryo a cell is removed when the embryo is 6 to 8 cells in size and all cells are totipotent. The removed cell is cultured and used in genetic testing to check for chromosomal abnormalities for biochemical anomalies. This is important for couples that have family histories of genetic disorders as an embryo that is not carrying the gene for the disorder can be chosen for implantation.

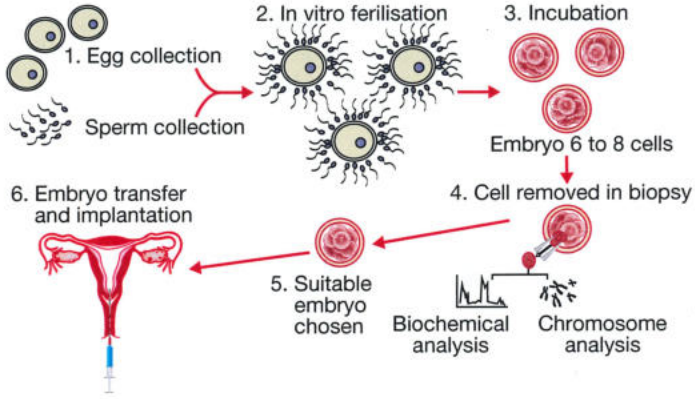


Figure 51.3 Preimplantation genetic diagnosis.

Ethical concerns

The use of assisted reproductive technologies raises several areas of ethical concern. Diagnostic testing can identify specific human genetic profiles raising issues that can be legal, social, moral and/or economic.

1. Potential parents may face many ethical issues when embryo screening provides data about an embryo and the data may affect their decision about having children, terminating a pregnancy or to having a child with a genetic disorder.
2. In the process of applying assisted reproductive technologies a certain amount of data about an embryo can be compiled before implantation or birth. Ethical issues arise when potential parents may wish to use the data to choose the sex of their child or choose other features that are not related to avoiding their offspring inheriting a particular genetic disease.
3. Equity issues arise due to the cost of assisted reproductive technologies. Most procedures are expensive. Who should pay for the procedures? Should ART be available to all couples in all socioeconomic brackets?
4. Privacy issues arise with specific data being compiled about a person's genome. General access to this information could affect the ability of a person to get insurance for health care, get a job or get suitable life insurance.
5. There are several questions that need to be answered – What happens to unused frozen embryos, sperm, eggs? Who has legal rights to unused frozen embryos, sperm, eggs?

QUESTIONS

1. Distinguish between screening testing and diagnostic testing.
2. List some diagnostic tests that can be carried out during a pregnancy.
3. List some blood tests that can be carried out during a pregnancy.
4. (a) What is an amniocentesis?
   (b) At which stage of a pregnancy is an amniocentesis usually carried out?
   (c) For which group of women is an amniocentesis recommended?
5. List some risks that are associated with an amniocentesis.
6. (a) Define chorionic villus sampling.
   (b) At which stage of a pregnancy is a chorionic villus sample usually taken?
   (c) Explain why chorionic villus sampling is considered to be an invasive technique.
7. Compare the risks of amniocentesis to chorionic villus sampling.
8. What is preimplantation genetic diagnosis?
9. Give a reason why preimplantation diagnosis is carried out for a couple planning a family.
10. The Australian government and the National Health and Medical Research Council (NHMRC) have strict guidelines on the use of assisted reproductive technologies in clinical practice and in research. Suggest why the ART guidelines restrict the use of PGD for sex selection in Australia.
11. Describe some ethical concerns about the use of ART.
12. The diagram shows two assisted reproductive technologies.

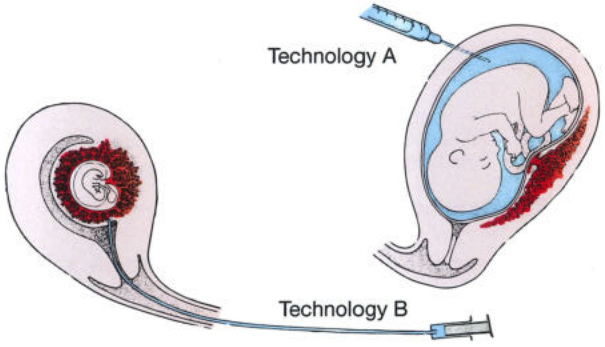


Figure 51.4 Two technologies.

Identify these technologies.

	Technology A	Technology B
(A)	Amniocentesis	Karyotyping
(B)	Preimplantation genetic diagnosis	Chorionic villus sampling
(C)	Karyotyping	Amniocentesis
(D)	Amniocentesis	Chorionic villus sampling

52 Pap Smears and Breast Screening

New technologies, e.g. Pap smear and breast screening have made early detection of cancers possible. If cancers are detected in the early stages then treatment has a greater chance of being successful and reducing the number of early deaths due to cancer.

Pap smear

A **Pap smear** is a screening test that detects the early signs of cervical cancer and has saved the lives of millions of women. Cells are collected from the outer opening of the cervix and examined under the microscope for abnormalities. If abnormal cells are detected they can be removed in one of the most treatable forms of cancer.

In 1928 **Dr Georgios Papanicolaou** discovered that there is a change in the appearance of cervical cells before they become cancerous. He developed the Papanicolaou stain which has 5 dyes in 3 solutions and colours cells across the spectrum ROYGBV of white light. The stain shows up cytoplasmic keratinisation and is still the basic stain in use today. Dr Papanicolaou published his book '*Diagnosis of Uterine Cancer by the Vaginal Smear*' in 1943 and the Pap smear has been widely used since the end of World War II. In Western Australia the result of a Pap smear test is sent to the **Cervical Screening Registry** (CSR) of Western Australia with the results given as negative or normal, unsatisfactory or abnormal.

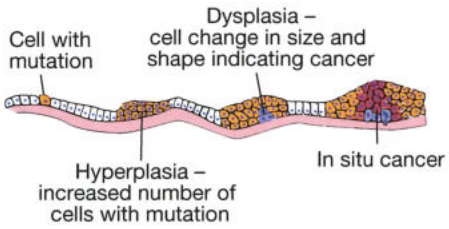


Figure 52.1 Development of cervical cancer.

In a **Pap smear procedure** cervical cells are swabbed from the cervix at the neck of the uterus and then the cells are smeared onto a glass slide. The slide is sent to a pathology laboratory which analyses the cells looking for a change in the size and shape of the cells (**dysplasia**). Cancer cells have a large nucleus surrounded by a relatively small amount of cytoplasm and some have multiple nuclei. The development of computer programs that can scan slides to analyse results and to store data have greatly assisted the early detection of abnormal cancerous cells.

New testing procedures have improved the accuracy of diagnosis, e.g. in **ThinPrep collection technique** and **liquid based cytology** the cells are rinsed into a preservative solution. The slides produced from these techniques are higher quality with cells well spread for easier analysis.

A **negative result** occurs for about 9 out of 10 Pap smear results and another Pap smear is not needed for another 2 years. An unsatisfactory result occurs for around 2 out of 100 Pap smears and means there was a problem collecting cells, e.g. mucus or blood obscured the cervical cells on the slide or not enough cells were obtained, or the smear was not properly prepared or broken in transport. Another Pap smear is required within 3 months. Abnormal results means there were different cervical cells present which could be low grade abnormality, e.g. cells with HPV infection, high grade abnormality with highly altered cervical cells.

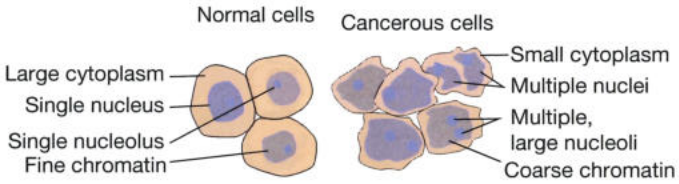


Figure 52.2 Normal cells and cancerous cells.

Other tests are also carried out in conjunction with Pap smears, e.g. a test for HPV (**human papillomavirus**) and optoelectronic screening. HPV causes many cases of cervical cancer and optoelectronic screening uses a device that picks up electrical and light signals from the tissue beneath the cervix and compares the cells with a databank of normal and abnormal cells. Once abnormal cells are detected more sensitive diagnostic techniques, e.g. using a **colposcope** (instrument used to examine the vagina and the cervix of the womb) are used to determine the situation and if a cervical biopsy is needed.

Treatment for abnormal cells can involve using a wire loop to remove abnormal cells from the cervix, using a laser under a local anaesthetic or a short stay in hospital under a general anaesthetic, a diathermy using heat to destroy the cells, a cone biopsy to remove a small piece of cervical tissue or a hysterectomy which removes the uterus.

Breast screening

BreastScreen Australia is a national program that promotes free two-yearly breast screening to women aged 50 to 74. Screening involves taking an X-ray of each breast. **Mammography** is a radiography screening method that uses low energy X-rays to check breasts for tumours and the X-ray picture is a **mammogram**. Mammography screening can identify tumours, cysts, calcifications and early breast cancer.

In Australia around 1 in 8 women develop breast cancer with the risk increasing with age with around 75% of breast cancers in women aged over 50 years. The cancer cannot be prevented but early diagnosis gives a better chance of successful treatment.

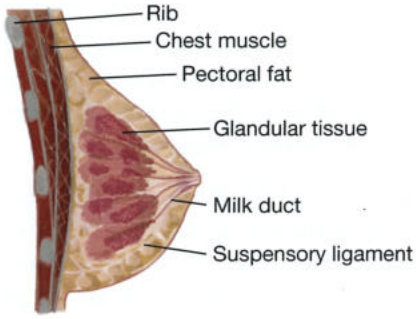


Figure 52.3 Breast structure.

The construction of X-ray machines for mammography was an important technological development that aided the early detection of breast cancer. The screening process involves taking two X-rays of each breast one from the side and one from the top.

BreastScreen Western Australia has recognised that **high breast density** which is due to white fibrous supportive tissue can obscure breast lesions and notified GPs recommending that women with very dense breasts have regular breast examinations and additional non-mammographic screening, e.g. ultrasound and/or magnetic resonance imaging (MRI).

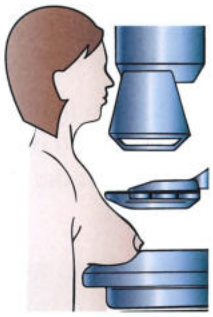


Figure 52.4 Breast screening.

Further developments in technologies have aided detection and diagnosis of breast cancer, e.g. digital mammograms, nuclear medicine techniques such as molecular breast imaging, ultrasonography and magnetic resonance imaging.

There are several risk factors associated with breast cancer.

- Genetic predisposition and family history, e.g. mutations in the *BRCA1* or *BRCA2* genes (*BR*east *CA*ncer genes).
- Exposure to ionising radiation.
- Being childless or giving birth after 30 years of age.
- Hormone replacement therapy during menopause.
- Gaining a lot of weight in adulthood especially after menopause.
- Drinking alcohol, e.g. more than 2 standard drinks a day.
- Early age at first menstruation, e.g. before 12 years.
- ***BRCA1* and *BRCA2*** are human genes that produce tumour suppressor proteins that repair damaged DNA. Mutated forms of these genes are associated with several types of cancer, e.g. breast and ovarian cancers. **Genetic testing** for *BRCA1* and *BRCA2* provides information about the possibility of breast cancer developing in a person and is carried out in family cancer clinics. In a DNA microarray (DNA chip) assay a small amount of a large number of single stranded DNA fragments of a specific gene are fixed to a glass slide in a DNA chip which is a tightly spaced grid. The DNA is then treated with labelled fluorescent dyes to test for gene expression. The DNA chip can test thousands of genes simultaneously and comparisons of gene expression of normal and cancerous breast tissue can rapidly identify mutant forms of *BRCA1* and *BRCA2*. Accurate diagnosis then aids in determining appropriate treatment measures and strategies.

QUESTIONS

1. Define a Pap smear.
2. How was the Pap smear named?
3. What is the name of the book that provided the information about detecting cervical cancer?
4. Name some new techniques that are used in diagnosing cervical cancer.
5. Describe what happens when a Pap smear is taken.
6. What happens if the result of a Pap smear is negative?
7. (a) Why can a Pap smear result be unsatisfactory?
   (b) What happens if a Pap smear result is unsatisfactory?
8. What is a colposcope?
9. What other tests are often carried out in conjunction with Pap smears?
10. List ways that remove abnormal cervical cells.
11. Define mammography.
12. What is BreastScreen Australia?
13. What is involved in having a mammogram?
14. Why do women with high density breasts need to be particularly mindful of breast cancer?
15. List some technologies that have further developed detection and diagnosis of breast cancer.
16. Identify some risk factors for breast cancer.
17. (a) What are *BRCA1* and *BRCA2*?
    (b) Why are these genes studied in relation to breast cancer?
18. (a) What is a DNA chip?
    (b) How is a DNA chip used in genetic testing?
19. Which pathogen is most closely linked to cervical cancer?
    (A) *Treponema pallidum*
    (B) *Neisseria gonorrhoeae*
    (C) Human papillomavirus
    (D) Human immunodeficiency virus

53 Prostate Cancer

New technologies, e.g. blood tests for prostate cancer have made early detection of cancers possible.

The prostate gland is situated in the pelvic cavity in front of the rectum. It consists of an outer fibrous covering, a layer of smooth muscle and glandular tissue composed of columnar epithelial cells.

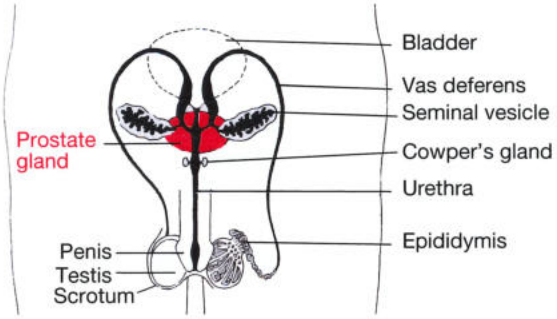


Figure 53.1 Prostate gland in the male reproductive system.

Function of the prostate gland

The prostate gland secretes a thin, milky fluid that makes up about 30% of semen. It contains calcium, citrate ion, phosphate ions, a clotting enzyme, profibrinolysin and is slightly alkaline. This fluid provides a protective local environment for the sperm especially against the acidic female vaginal secretions which has a pH of 3.5 to 4.0. The sperm become motile when the pH of the environment around them drops to around 6.5 to 6.0. The clotting agents help thicken the semen in the vagina increasing the probability they will remain near the cervix.

Prostate cancer

Prostate cancer is the development of a tumour in the prostate gland of the male reproductive system. The prostate normally slowly gets bigger with age. Many of the symptoms of prostate cancer relate to urination, e.g. there is a need to urinate frequently, especially at night and difficulty beginning to urinate, incontinence, a slow, interrupted flow of urine, blood in urine or semen, lower back or pelvic pain and painful ejaculation.

The first check for prostate cancer is usually a **digital rectal examination** (DRE test) in which the doctor inserts a gloved finger into the rectum to check the size of the prostate gland and any other changes that may be present.

Technology and prostate cancer

A new technology to test for prostate cancer is the prostate specific antigen (PSA) blood test. The **prostate specific antigen** test measures the amount of PSA in a man's blood in ng/mL.

This protein helps nourish sperm and there is normally very little PSA in the bloodstream, though the amount does increase with age. A man with prostate cancer has a high blood level of PSA as the cancer cells interfere with cell functioning and normal metabolism.

After diagnosis regular PSA blood tests are carried out to monitor the development of the cancer. This helps to determine if raised PSA levels are due to an infection of the prostate or the enlargement is a benign prostatic hyperplasia (BPH).

If a man has a high PSA level and/or a digital rectal examination (DRE) performed by the doctor to check the feel of the prostate gland indicates there is a possible cancer in the prostate gland, a **prostate biopsy** is performed. A **transrectal ultrasound probe** gives a picture inside the rectum to make the biopsy easier and more efficient. In a prostate biopsy tissue samples are taken from the prostate gland and the cells examined to check for abnormalities.

Other tests can be used to check on the progression of the cancer. A **magnetic resonance imaging** (MRI) scan is often done before a biopsy and a **bone scan** checks if the cancer cells have moved into bone tissue.

Early diagnosis means treatment can begin and the condition effectively managed. The development of **keyhole surgery** with small telescopes on cameras means surgery to cut away and remove the prostate is less invasive and recovery is faster.

If surgery is carried out then in many instances a **pelvic lymph node dissection** is carried out as well removing nearby lymph nodes to check if the cancer cells have entered the lymphatic system.

QUESTIONS

1. Where is the prostate gland?
2. Describe the structure of the prostate gland.
3. Identify the components in the fluid from the prostate gland.
4. Discuss why it is beneficial for the fluid from the prostate gland to be slightly alkaline.
5. Why is a clotting agent beneficial in the fluid from the prostate gland?
6. Define prostate cancer.
7. (a) What is a DRE?
   (b) How is it carried out?
8. (a) What is PSA?
   (b) What is the PSA test?
9. What are the main indications that a man has prostate cancer?
10. Why is PSA monitored for a while after high PSA levels are detected?
11. How is prostate cancer usually treated?
12. Why are pelvic lymph node dissections often carried out in conjunction with surgery?

54 Genome, Gene and Allele

The **genome** is the complete genetic information of an organism. It usually refers to the haploid genome of an organism. **Genomics** is the study of whole sets of genes and their interactions, e.g. it includes how particular genes are expressed, how they are regulated, their interactions with other genes and how gene expression affects growth, development and evolution.

The genome of humans and other eukaryotes consists of a number of DNA molecules which are found in the nucleus which is surrounded by the nuclear membrane while the genome of prokaryotes, e.g. bacteria and cyanobacteria is often a single DNA molecule which is found in the nucleoid region of the cell (nucleoid region is *not* surrounded by a membrane). Human (eukaryotic) DNA is linear and precisely packed with a large amount of protein while the bacterial chromosome is usually a circular DNA molecules with some associated proteins.

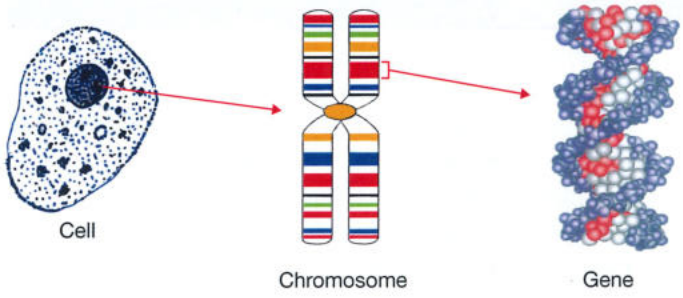


Figure 54.1 Cell, chromosome and gene.

A **chromosome** is a cellular structure that holds genetic information in the coding of the DNA molecule. In eukaryotes chromosomes consist of a very long DNA molecule with associated proteins. The proteins help control the activity of the genes. Each eukaryotic species has a distinct number of chromosomes – the diploid number ($2n$), e.g. humans *Homo sapiens* have $2n = 46$, and chimpanzee *Pan troglodytes* $2n = 48$.

A **gene** is a heritable factor that determines a specific characteristic, e.g. gene for stem height in pea plants. It is a section of DNA coding for proteins that expresses itself as the phenotype for that trait.

An **allele** is an alternative for a particular trait, e.g. there are two alleles for height in pea plants – tall (T) or short (t). Alleles are the alternative forms of a gene and differ from each other by only one or a few bases. If two alleles at a locus differ then one will be the **dominant allele** and will show in the phenotype and the other will be the **recessive allele** and will only show in the phenotype if there is no dominant allele present.

Case study – PKU gene

Phenylketonuria (PKU) is an autosomal recessive genetic disorder caused by a defective gene on chromosome 12 for the enzyme phenylalanine hydroxylase (the official name of the gene and given the symbol *PAH*). There are over 400 different mutations of the *PAH* gene that cause the disease. A defective gene prevents the breakdown of the amino acid phenylalanine into the amino acid tyrosine as the defective gene does not produce a functional enzyme. The non-functional enzyme stops the reaction causing a build-up of phenylalanine in the body which in turn causes problems with brain development leading to mental retardation, stunted growth, hypersensitivity and limb and hand tremors.

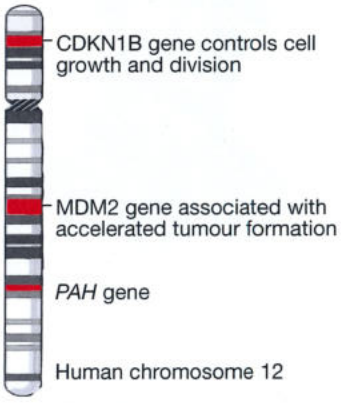


Figure 54.2 *PAH* gene on chromosome 12.

PKU is detected by screening newborns 2 to 7 days after birth with tests for PKU. A child diagnosed with PKU is prescribed a diet that is low in phenylalanine and high in tyrosine, e.g. diet low in meat, chicken, nuts, cheese and potatoes with tyrosine supplements.

QUESTIONS

1. Define a genome.
2. What is genomics?
3. How is the genome of a prokaryote different to the genome of a eukaryote?
4. Define a chromosome.
5. Define a gene.
6. What is an allele?
7. Distinguish between a dominant allele and a recessive allele.
8. (a) What is PKU?
   (b) Identify a gene that is involved with PKU.
   (c) How is PKU detected?
9. Where would you find a chromosome in a eukaryotic cell?
   (A) Nucleoid region (B) Nucleus
   (C) Cytoplasm (D) Ribosome
10. What chemical contains the genetic code?
    (A) Polypeptides (B) Carbohydrates
    (C) Nucleic acids (D) Lipids

55 Gregor Mendel and His Experiments

Gregor Johann Mendel (1822-1884) was an Augustinian monk who began to study the inheritance of different characteristics in pea plants in 1856. He started with many strains and bred them for several years to find easily recognisable traits that bred true. He published his results in 1866.

Figure 55.1 Gregor Mendel.

Mendel crossed purebreeding round seeds with purebreeding wrinkled seeds and found that the first generation (F_1) were all round seeds. When he crossed two of the offspring he found that the ratio was 3 round : 1 wrinkled in the second generation (F_2). Mendel concluded that there were two factors for a character and one (e.g. round seed shape) was dominant over the other (e.g. wrinkled seed shape). He studied seven characteristics in pea plants.

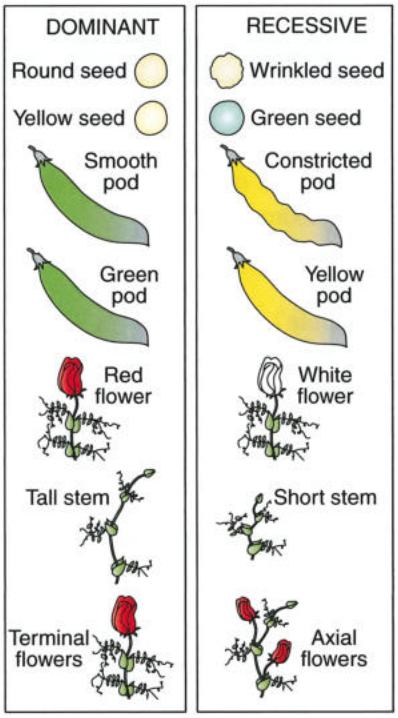


Figure 55.2 Seven characteristics in pea plants.

Mendel repeated his experiments many times with a large number of crosses which gave his results statistical verification. He was meticulous in making the crosses using purebreeding lines and transferring pollen from one plant to another using an artist's brush, preventing random pollination. In each experiment he investigated a limited number of characteristics, e.g. round/wrinkled seed shape; or red flower/white flower.

Mendel worked out the law of segregation, which states that for each characteristic there are two factors which segregate, with one factor in each reproductive cell, and then recombine at fertilisation. The characteristics do not blend; one dominates over the other. He also worked out the law of independent assortment, which states that pairs of factors segregate independently of other pairs of factors when forming reproductive cells. Mendel's work was presented to a small group of scientists. Since he was an Augustinian monk and did not move in scientific circles, the significance of his work was not recognised at that time.

QUESTIONS

1. Around 1856, Gregor Mendel began his experiments. Outline one of his experiments and what this showed.
2. Outline three aspects of Mendel's work that showed his excellent experimental technique.
3. Explain Mendel's two laws.
4. Suggest one reason why Mendel's work was not recognised for many years.
5. Distinguish between homozygous and heterozygous genotypes in monohybrid crosses.
6. Use examples to distinguish between the terms allele and gene.
7. Use examples to explain the difference between dominant and recessive genes.
8. Distinguish between genotype and phenotype, using an example.
9. Who is considered to be the 'father of genetics'?
   (A) Gregor Mendel.
   (B) Walter Sutton.
   (C) Theodor Boveri.
   (D) Thomas Hunt Morgan.
10. The work of Gregor Mendel assisted our understanding of the inheritance of characteristics. What did he determine?
    (A) Crossing over and sex-linkage.
    (B) Inheritance ratios and mutations.
    (C) Blending of characteristics.
    (D) Independent assortment of factors.
11. Gregor Mendel concluded that each pea plant possessed two hereditary factors. What do we now call these 'factors'?
    (A) Chromosomes. (B) Nucleotides.
    (C) Genes. (D) DNA.

12. What is the name for the alternative genes for a particular characteristic?
    (A) Nucleotide.
    (B) Homologous pair.
    (C) Locus genes.
    (D) Alleles.
13. Peas can self-pollinate. Why is this an important factor in the success of Mendel's experiments?
    (A) It was easy for Mendel to produce pure strains.
    (B) It means a large number of offspring must be produced.
    (C) It means a small number of offspring must be produced.
    (D) It decreases the time between generations.

Figure 55.3 Edible pea plant, *Pisum sativum*.

The following information is to be used for the next THREE questions. Mendel investigated the inheritance of seven pairs of traits in edible peas, *Pisum sativum*. He studied first one pair at a time, then two pairs at a time. In 1865 he published his hybridisation experimental results. His results for F_1 crosses are shown in the table.

Feature	Dominant trait	Recessive trait	F_2 generation dominant : recessive	Ratio
Flower colour	Purple	White	705 : 224	3.15 : 1
Flower position	Axial	Terminal	651 : 207	3.14 : 1
Seed colour	Yellow	Green	6022 : 2001	3.01 : 1
Seed shape	Round	Wrinkled	5474 : 1850	2.96 : 1
Stem length	Tall	Dwarf	787 : 277	2.84 : 1
Pod colour	Green	Yellow	428 : 152	2.82 : 1
Pod shape	Inflated	Constricted	882 : 299	2.95 : 1

14. Peas have seven chromosomes and each of the traits Mendel studied is carried on a different chromosome. What would have happened if Mendel had studied eight traits?
    (A) His results would have been more statistically valid.
    (B) He would probably not have worked out the law of independent assortment.
    (C) He would have worked out sex-linkage.
    (D) His work would have been recognised at the time instead of many years later.
15. If a scientist crossed plants that were homozygous for all three traits, with axial flower position and yellow constricted pods, with plants that were homozygous for all three traits, with terminal flower position and green constricted pods, what would you expect to be the appearance of the offspring?
    (A) All with axial flower position with green inflated pods.
    (B) All with axial flower position with green constricted pods.
    (C) All with terminal flower position with yellow constricted pods.
    (D) All with green inflated pods, some with terminal flower position, some with axial flower position.
16. Two purebreeding dwarf pea plants were used in an experiment. When the coleoptiles were 1 cm tall, one was treated with plant growth hormone and it grew to the height of purebreeding tall pea plants. If the F_1 plants were cross-fertilised, what would you expect the F_2 to be?
    (A) All tall plants. (B) 1 tall : 3 dwarf.
    (C) 2 tall : 2 dwarf. (D) All dwarf plants.
17. When 210 round peas were sown and the resulting flowers were self-pollinated, scientists collected 6590 seeds. There were 1646 wrinkled seeds and 4944 round seeds. If the symbol R represents round seeds and the symbol r represents wrinkled seeds, what would be the most likely genotype of the original 210 round seeds?
    (A) Both RR. (B) RR and Rr.
    (C) Both Rr. (D) RR and rr.
18. Mendel concluded from his experiments that the pairs of factors from each parent separated at one stage. We now know that this segregation occurs during:
    (A) Mitosis. (B) Meiosis.
    (C) Fertilisation. (D) Pollination.
19. Mendel began his breeding experiments using honey bees but abandoned this work due to difficulties in handling the bees. Which of the following identifies one way in which garden peas were a more suitable choice of experimental organism?
    (A) Garden peas produced a predetermined number of offspring each generation.
    (B) Garden peas have only two chromosomes which are extremely large.
    (C) Garden peas are grown quickly and easily.
    (D) Garden peas cannot be cross-pollinated.

56 Writing Genotypes

The genotype shows the genetic make-up or set of alleles of an organism that control a characteristic.

Rules for writing genotype

When writing the genotype it is customary to:

- Represent the gene with a letter of the alphabet, e.g. for plant height the letter T can be chosen. It is a good idea to choose a letter of the alphabet that looks different in upper case and lower case.
- The genotype will have two letters representing the two alleles – one from each parent.
- The dominant allele is capitalised, e.g. B = brown eyes and the recessive allele is lower case, e.g. b = blue eyes.
- The dominant allele is written first.

Homozygous and heterozygous

When both alleles are the same in the organism, e.g. TT or tt, the organism is **homozygous** for that trait. If the alleles are different, e.g. Tt, then the organism is **heterozygous** for that trait.

Monohybrid cross

A monohybrid cross is a mating where a single pair of characteristics is studied.

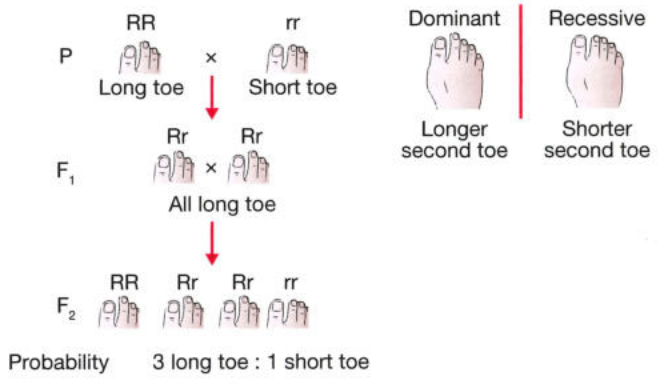


Figure 56.1 Monohybrid cross – long second toe/short second toe.

A **monohybrid test cross** is carried out to find out an unknown genotype. Organisms showing a dominant trait can be either homozygous (TT) or heterozygous (Tt). In a monohybrid test cross the organism showing the dominant trait is crossed with the homozygous recessive. The genotype is determined by the ratio of phenotypes in the offspring. If any offspring show the recessive trait then the unknown genotype is heterozygous. If sufficient numbers of offspring are produced and none show the recessive trait then the genotype is likely to be homozygous.

Remember that these are probability models and the statistical ratios need large samples to be used. The probability of an offspring showing the recessive trait for a heterozygous genotype in a test cross is 50% but just as the chance of a child being 50% boy or 50% girl, some couples have all daughters and no sons while other couples have all sons and no daughters. Probability ratios do not always show in real life situations with small sample size.

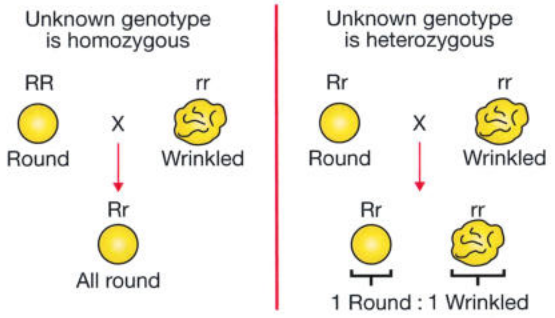


Figure 56.2 Monohybrid test cross.

QUESTIONS

1. Define genotype.
2. Outline the rules for writing genotype.
3. Distinguish between homozygous and heterozygous.
4. Define a monohybrid cross.
5. (a) Outline what is involved in a monohybrid test cross.
   (b) Explain why monohybrid test crosses are carried out.
   (c) How is the unknown genotype determined in the test cross?
6. Discuss why the ratios given in probability models do not always happen in real life situations.
7. In the fruit fly, *Drosophila melanogaster* straight wing is dominant over curly wing.

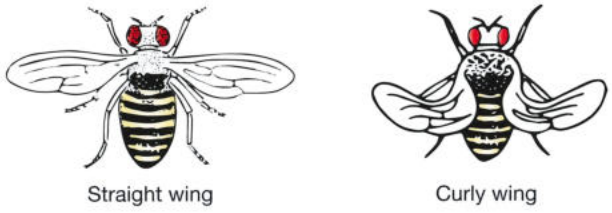


Figure 56.3 *Drosophila* wings.

If you were given a fruit fly with straight wings outline how you could determine the genotype of this fruit fly.

8. In humans a tooth gap (diastema) between the front two teeth is a dominant trait. If a couple, the man with a diastema and the female without a distema have two children one with a diastema and one without a diastema which of the following shows the probability of offsping genotype?
   (A) 2Hh : 2 hh
   (B) 1HH : 2 Hh : 1 hh
   (C) 2HH : 2 Hh
   (D) 2HL : 2hl

57 Probability Models

The work of Gregor Mendel led to mathematical calculations to explain the outcome of a particular cross. Reginald Crundall Punnett (1875-1967) devised the 'Punnett square' to summarise the fusion of gametes in genetic crosses and also contributed to the use of statistics in Mendelian genetics.

Figure 57.1 RC Punnett.

QUESTIONS

1. Use the work of Gregor Mendel to show how his work with pea plants influenced the direction and nature of scientific thinking.
2. Describe how you could establish whether a tall pea plant was homozygous or heterozygous.
3. In mice the long-eared trait is dominant over the short-eared trait. If a homozygous long-eared mouse was crossed with a homozygous short-eared mouse, what are the genotype and phenotype ratios of F_1 and F_2?
4. In humans there is a gene to taste phenylthiourea (PTU). 'Tasters' (TT or Tt) detect an extremely bitter taste, while 'non-tasters' (tt) detect no change or taste. If a person who is heterozygous for phenylthiourea marries a 'non-taster', what is the probability that their offspring will be 'non-tasters'?
5. In humans long eyelashes are dominant over short eyelashes.

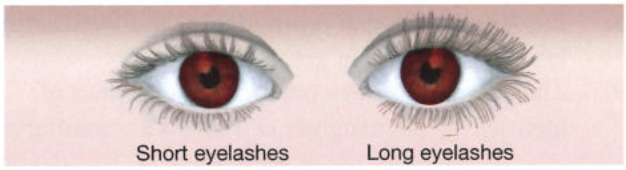


Figure 57.2 Long and short eyelashes.

If two people who both had long eyelashes had a child with short eyelashes, determine the genotype of the parents.

6. (a) In a genetics experiment a normal-coloured rabbit was crossed with an albino rabbit. The experiment was repeated three times using the same parental rabbits and the F_1 totalled 11 normal coloured offspring and 12 albino offspring. Draw a Punnett square to explain this inheritance.
   (b) The experimenters then crossed two normal rabbits several times and the total offspring from this cross was 18 normal rabbits and six albino rabbits. Draw a Punnett square to explain this inheritance.
7. In humans, the gene for thick lips (T) is dominant over the gene for thin lips (t).

Figure 57.3 Thin and thick lips.

   (a) What is the genotype(s) for a person with (i) thin lips or (ii) thick lips?
   (b) Explain why two people with thin lips can only have children with thin lips, while two people with thick lips could have children with either thin lips or thick lips.
8. In humans the blue eye colour (represented by b) is recessive to brown eye colour (represented by B). In one family, the father has brown eyes, the mother has blue eyes and they have one daughter with blue eyes and one son with brown eyes. What would be the genotypes of the father and son respectively?
   (A) BB and BB (C) Bb and Bb
   (B) BB and Bb (D) Bb and BB
9. What is the relationship between a gene and an allele?
   (A) Each gene exists in two or more forms of alleles.
   (B) A gene is one characteristic while an allele is several characteristics.
   (C) Genes are on chromosomes while alleles show different characteristics.
   (D) Alleles determine the characteristic while genes determine the particular form of that characteristic.
10. Albinism, or lack of skin pigmentation, in humans is caused by two recessive alleles. If a phenotypically normal couple have two children, one non-albino and one albino, what is the probability that their next child is albino?
    (A) 25% chance
    (B) 50% chance
    (C) 75% chance
    (D) 100% chance
11. A woman with a genetic disease marries a phenotypically normal man. They have two daughters and one son none of whom show symptoms of the disease. From this data you can conclude:
    (A) The disease has late onset symptoms.
    (B) The disease is caused by a recessive allele.
    (C) The man must be heterozgyous for the disease.
    (D) The man must be homozgyous normal for the disease.

58 Variation in Genotype

There are several causes for variation in genotypes of offspring. Mutation and the processes of meiosis and fertilisation lead to offspring being different to either parent and to other offspring from the same parents.

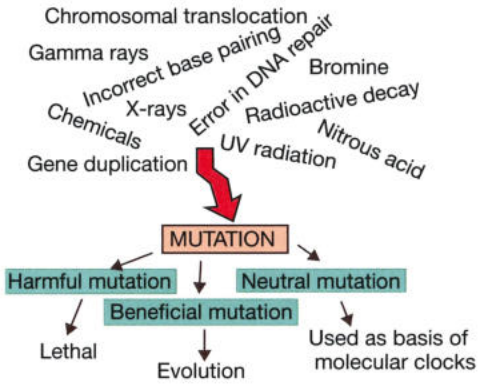


Figure 58.1 Mutation causes and effects.

Mutation

A mutation is a permanent change in the genetic information. Mutations can be small scale, e.g. point mutations with single nucleotide-pair substitution, insertions or deletions, or large scale chromosomal mutations with duplication and alteration of chromosome structure. Mutations may change the phenotype and the appearance of an organism and the change may lead to the formation of a favourable characteristic which aids survival in a changing environment and leads to natural selection and evolution. DNA repair mechanisms help prevent or correct changes to the original genetic information.

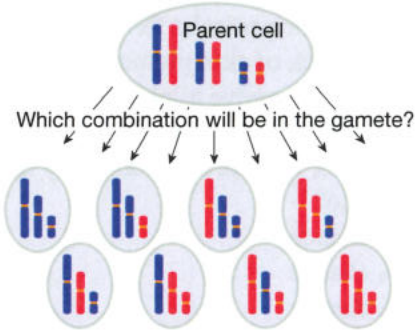


Figure 58.2 Random segregation.

Meiosis

Meiosis is a type of cell division in sexually reproducing organisms that leads to the formation of haploid daughter cells. Crossing over and random segregation of chromosomes are two processes that lead to variation. The **law of segregation** states that the two alleles for each gene separate during meiosis and end up in different gametes.

Fertilisation

Fertilisation is the union of two haploid gametes to form a diploid zygote.

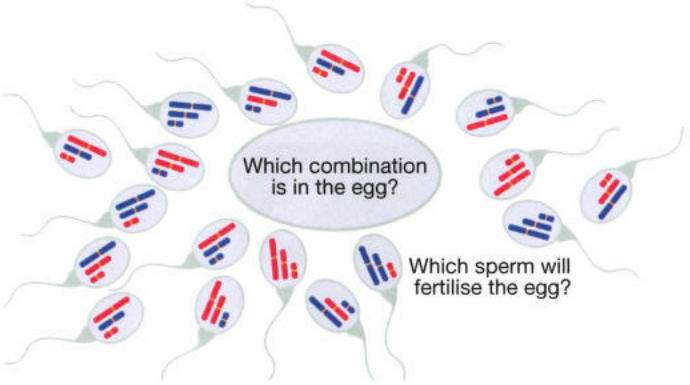


Figure 58.3 Fertilisation.

QUESTIONS

1. List four processes that lead to variation in the genotypes of offspring.
2. Identify some causes of mutation.
3. Name some ways in which DNA can be changed.
4. Explain how mutation can lead to evolution.
5. What is random segregation?
6. If a parent cell has a diploid number of six ($2n = 6$). How many different gametes can form during meiosis due to random segregation?
7. Study the following diagram of a process that occurs in cells and explain how it leads to variation in genotype.

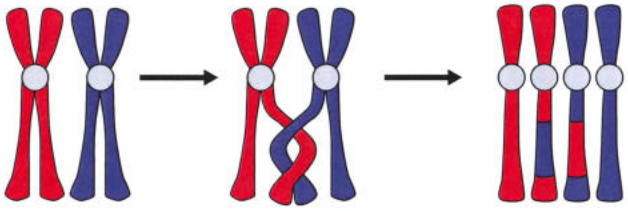
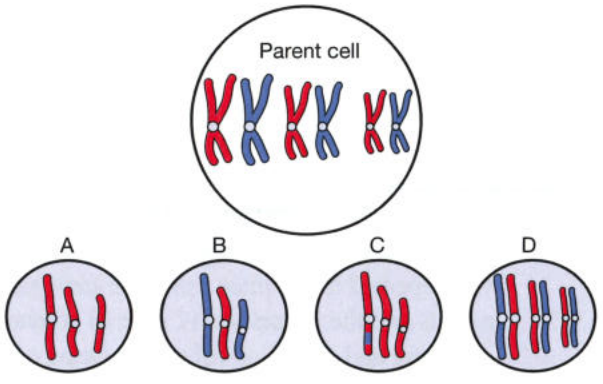


Figure 58.4 Process that occurs in cells.

8. Explain why sexual reproduction and fertilisation leads to variation in genotype.
9. The diagram shows a parent cell at the start of meiosis. Which daughter cell involves crossing over?



59 Karyotyping

A **karyotype** is an arrangement of chromosome pairs according to size and shape. It is a property of a cell giving the number and type of chromosomes present in the nucleus and not a photograph or diagram of them. A karyotype is prepared by collecting cells, culturing them and then halting cell division at metaphase. The analysis is carried out using cells that are dividing as this is when the chromosomes are clearly visible under the microscope. The cells are stained and then the chromosomes sorted according to size and structure. Cells likely to be collected for karyotyping include embryonic cells collected by amniocentesis or chorionic villus sampling to check for abnormalities in the developing child. Blood cells from adults or children are collected for diagnosis of chromosome abnormalities.

A **karyogram** is a photograph or diagram of the chromosomes in a cell. The chromosomes are arranged in homologous pairs of decreasing length in a karyogram. The longest pair are chromosome 1.

Uses of karyotyping

Karyotyping is useful to look for chromosome anomalies, e.g. trisomies where there is an extra copy of a chromosome making three copies instead of two or translocations where part of a chromosome has duplicated and been added to another chromosome. Karyotyping is also carried out if a female has recurrent miscarriages.

Karyotyping determines the number of autosomes which are chromosomes that are not directly involved in determining sex and also identifies the sex chromosomes present which in humans are one pair of chromosomes responsible for determining the sex of an individual.

Case study – Klinefelter syndrome

Klinefelter syndrome was first diagnosed by Harry Klinefelter in 1942. It is a chromosomal disorder that occurs when males have two or more X chromosomes, e.g. a male has XXY. It is a congenital disorder detected by karyotyping. It is not an inherited disorder and occurs in about 1 in 500 men to 1 in 1000 men in Australia with many not being diagnosed. About 10% of Klinefelter syndrome cases are detected by prenatal diagnosis. Some men are mosaics with some cells in the body having an extra X chromosome and other cells having the normal 46 chromosomes.

The mother always passes an X chromosome and the father will pass on a Y chromosome to make the child male. Klinefelter syndrome can be caused by an additional X chromosome from the egg or sperm, e.g. due to non-disjunction during meiosis in gamete formation or added when an error occurs within the fertilised egg or dividing cells as the embryo develops.

The additional X chromosome causes sterility and the male may have weaker muscles, poor coordination, less body hair, smaller genitals and breast growth. There is often problems with speech and reading and children often grow taller than average. The additional X chromosome does not seem to affect sexual orientation and in many cases it is diagnosed at puberty when symptoms are more noticeable. If there are three or more X chromosomes present, e.g. XXXY, the symptoms are more severe.

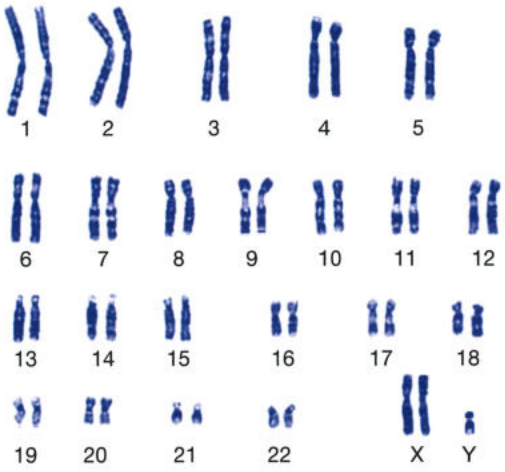


Figure 59.1 Karyotype of Klinefelter syndrome.

There is no cure for Klinefelter syndrome but treatment tries to assist some aspects such as hormone therapy, e.g. testosterone to aid puberty changes and increase physical strength, cosmetic surgery to reduce breast size. Speech therapy and special reading programs can help improve language skills and reading and writing. Treatment usually begins around ages of 11 to 12 years old.

Case study – Turner syndrome

Turner syndrome (TS) was first described by Henry Turner in 1938. It is a chromosomal disorder that occurs when females are missing part or a whole X chromosomes, e.g. a female has XO. The female will have 45 chromosomes instead of 46 chromosomes. The person may show mosaicism with some cells normal and some cells lacking the second X chromosome. Mosaic Turner syndrome occurs in about 20% of cases. Turner syndrome affects about 1 in every 2000 girls born in Australia with about 50% having XO in all their cells.

Mosaicism occurs when there is a mistake in cell division mitosis during embryonic development. The severity of the symptoms is related to the number and distribution of cells that are XO.

It can be diagnosed by karyotyping, e.g. in prenatal diagnosis, at birth if the baby shows certain features, during childhood if growth rate is reduced, during the teenage years if puberty does not occur or during adulthood if testing for reasons for infertility.

The lack of the second X chromosome affects development in females often causing reduced adult height, low-set ears, low hairline, reduced breast development, heart defects, lack of menstrual periods and inability to produce children. It can also cause spatial awareness problems and hearing problems. About 30% of females with Turner syndrome have a webbed neck – extra folds of skin on the neck. At birth the baby girl may show puffiness on the back of the hands and top of the feet and have problems sucking and feeding due to a high-arched palate.

There is no cure for Turner syndrome but treatments, e.g. hormone replacement therapy can aid growth development and breast development and plastic surgery can remove the webbing on the neck.

Case study – Trisomy 18 Edwards syndrome

Edwards syndrome was first described by John Edwards in 1960. It is a chromosomal disorder when there is three copies of chromosome 18. The additional copy of chromosome 18 is due to non-disjunction during meiosis. It can also occur due to translocation of part or the whole copy of chromosome 18 attaching to another chromosome. This leads to three types of Edwards syndrome – full trisomy 18 with every cell in the body having the extra chromosome, partial trisomy 18 with only part of chromosome 18 additional and attached to another chromosome and mosaic trisomy 18 with some cells having the additional chromosome and some cells having the normal number of 46 chromosomes per cell.

Edwards syndrome occurs in about 1 in 6000 live births with about 80% of those affected being female. Most foetuses with the disorder die before birth and the incidence of the disorder increases with increasing age of the mother.

The additional chromosome 18 disrupts normal development causing low birth weight and defects in the kidneys, ureters, heart, lungs and diaphragm. The child is likely to have a small head with cleft lip or cleft palate. Hands and feet can be malformed with missing thumbs, thumb feet and webbing between the fingers and toes. Survival beyond the neonatal period, e.g. within their first month is rare mainly due to the heart and kidney abnormalities. About 5% to 10% of children with Edwards syndrome live past their first year but have severe intellectual disabilities.

QUESTIONS

1. Define a karyotype.
2. Outline how a karyotype is prepared,
3. Explain why karyotyping is carried out using dividing cells.
4. What is a karyogram?
5. Outline some uses of karyotyping.
6. Describe Klinefelter syndrome.
7. Describe Turner syndrome.
8. Outline the treatment and cure for Turner syndrome.
9. What is Edwards syndrome?
10. Outline the three types of Edwards syndrome.
11. Many miscarriages in the first trimester of pregnancy are due to trisomy 2. Babies that are born with Trisomy 2 are mosaics.
    (a) What is meant by trisomy 2?
    (b) What is meant by a child being a 'mosaic'?
12. Suggest why children that are mosaic trisomy 2 survive while those with full trisomy 2 miscarry during pregnancy in a spontaneous abortion.
13. The diagram shows four karyograms. Which karyogram shows Turner syndrome?

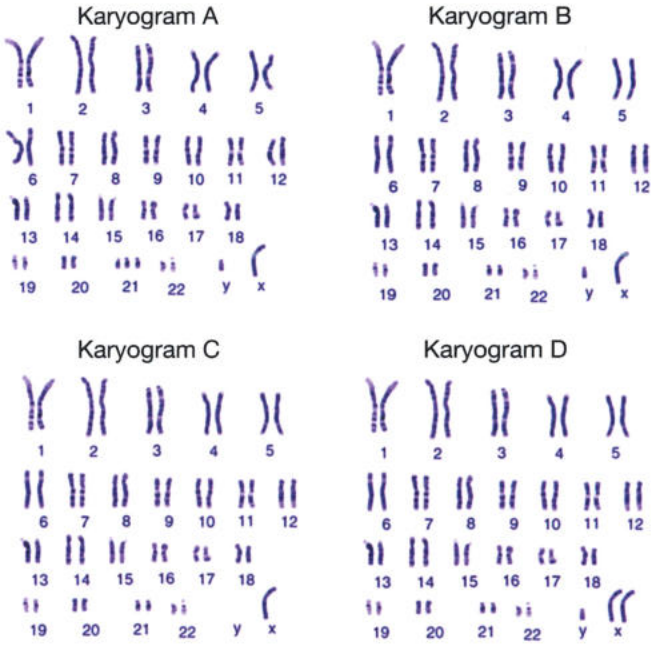


Figure 59.2 Four karyograms.

14. The diagram shows an amniocentesis.

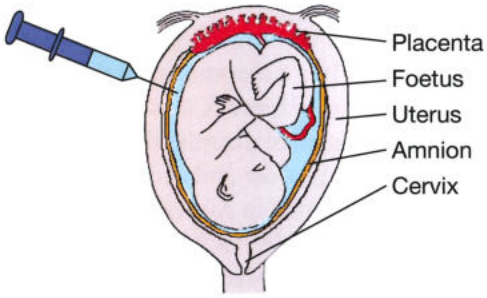


Figure 59.3 Amniocentesis.

What is the function of an amniocentesis?
(A) Inject hormones to improve the uterine conditions.
(B) To check for foetal abnormalities.
(C) Allow gene therapy to change the baby's genome.
(D) To purify the amniotic fluid of infection.

60 Sex Chromosomes

For a species that has male and female sexual forms, it has been found that they have all chromosomes the same, called **autosomes**, and one pair which is different, called the **sex chromosomes**. In many instances the female has two similar chromosomes called X chromosomes and the male has two dissimilar chromosomes, an X and a Y chromosome. Thus females are XX and males are XY.

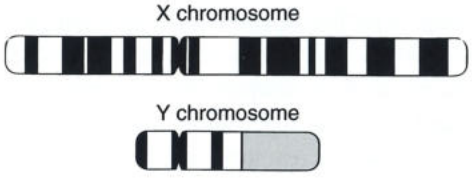


Figure 60.1 Human X and Y chromosomes.

A **hermaphrodite** is an organism that has both male and female sexes. Some people are hermaphrodites, though the word **intersex** is now used for humans as the term hermaphrodite is misleading and some believe discriminating. A **true intersex** person has a karyotype with both XX and XY chromosome pairs. There are records of some people who have both testicular and ovarian tissues able to produce both functional sperm and functional eggs. Intersex children with ambiguous outer genitalia are often surgically or hormonally treated so they will have more socially acceptable sex characteristics.

For various reasons some people believe they have been born into the wrong sex. The term LGBT (lesbian, gay, bisexual and transgender) was coined in the 1990s and there is some activism to recognise intersex people creating an LGBTI community.

SRY gene

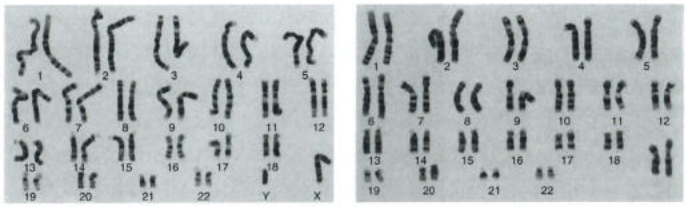
In 1990 the *SRY* gene (sex-determining region Y) on the Y chromosome was identified. If this gene is present the gonads develop into testes. If this gene is not present the gonads develop into ovaries. The protein made by the *SRY* code starts male sex determination and regulates other genes which are also involved in developing fully functioning sex organs. Mutations of this gene lead to several types of sex-related disorders.

Pseudoautosome region of Y chromosome

The pseudoautosome region of the Y chromosome is homologous to a region on the X chromosome. It is needed so that cell division, e.g. meiosis can correctly occur in males with the X and Y chromosomes pairing together. The genes in this region are not considered to be sex-linked as males have two genes for these traits. Failure of correct pairing of the X and Y chromosomes in meiosis in males leads to male sterility.

QUESTIONS

1. Distinguish between autosomes and sex chromosomes.
2. In humans, what are the sex chromosomes?
3. The following diagram shows the karyotype of two individuals – P and Q.



P Q

Figure 60.2 Karyograms of two people.

Identify the sex of P and Q and explain your reasoning.

4. Draw a Punnett grid to show why the probability of having a daughter or a son is 50 : 50.
5. In genetic terms, which parent determines the sex of the child?
6. There are several traits, such as haemophilia, which are carried on the X chromosome as a recessive gene. Explain why males are more likely to show this characteristic than females.
7. What is a hermaphrodite?
8. Outline the function of the *SRY* gene.
9. A family has five sons and the mother is pregnant. What is the probability that this child is a daughter?
   (A) 25% (B) 50%
   (C) 75% (D) 100%
10. In humans, the gene for red-green colour blindness is found on the X chromosome. In the population, how would you expect the number of red-green colourblind women to compare to the number of red-green colourblind men?
    (A) More red-green colourblind men than women.
    (B) More red-green colourblind women than men.
    (C) Numbers of red-green colourblind men and red-green colourblind women would be approximately the same.
    (D) Cannot be determined from this information as the ratio will depend upon other environmental factors such as temperature.
11. What is the main determining factor for a child to be male or female?
    (A) If a child inherits an X egg or a Y egg from its father.
    (B) If the child inherits an X egg or a Y egg from its mother.
    (C) If the child inherits an X sperm or a Y sperm from its father.
    (D) If the child inherits two X chromosomes from its mother or an XY from its father.

61 Polygenic Inheritance

Polygenic inheritance refers to a trait that involves several genes, e.g. human height and human weight are not the result of a single height gene or a single weight gene.

Polygenic inheritance usually produces a large variation in phenotypes showing a **continuous variation** between two extremes. In a graph showing the frequency of phenotype distribution the graph for polygenic distribution usually has the bell-shape curve of a normal distribution. This shows most individuals fall in the middle range and there are fewer numbers at either end.

When Mendel studied pea plants, he found that only a few phenotypes were produced, e.g. tall or dwarf pea plants. This is called **discontinuous variation**. If you look at a graph of the number of cases of this trait you will find two columns, one for the dominant trait and one for the recessive trait.

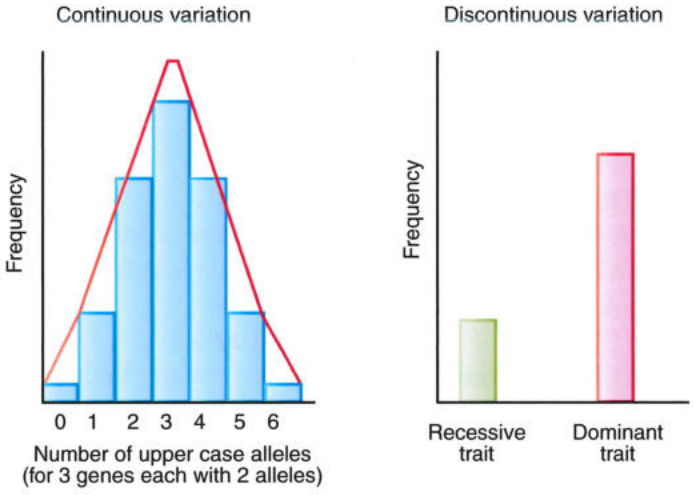


Figure 61.1 Graphs for continuous and discontinuous variation.

Quantitative traits

Quantitative traits are heritable features that vary continuously over a range rather than an either-or manner. The phenotype of the trait depends upon the combination of each involved gene with each gene adding and contributing to the final appearance.

A **polygene** or quantitative gene is non-allelic and each gene affects the trait. Polygenes are often pleiotropic ie. They have multiple effects on the body.

Quantitative trait locus (QTL) analysis is a statistical method that links the phenotypic data with measurements of the variations of the trait with genotypic data and specific molecular markers, e.g. particular nucleotide sequences in DNA. QTL analysis is used to explain the genetic basis of variation in complex traits.

The development of **new technologies** especially in DNA sequencing methods and bioinformatics have increased both the amount of new data about quantitative traits and genomes and increased the rate at which new data is discovered. The new data is fed into databases which analyse the relationships of the amino acid sequences, the nucleotide sequences and phenotype distribution.

Widespread international cooperation and the sharing of information by the scientific community has also increased the rate at which new knowledge and understanding is gathered and analysed. Complex models are made to compare the sequences between individuals within a species and to compare the sequences between different species. These discoveries have influenced the direction of further research and are being used to determine the relationship between allelic differences and the fitness and functionality of an organism.

Figure 61.2 Hair colour.

Human hair colour

Human hair colour is a continuous trait that is determined by the combined effect of several pairs of genes. It is not possible to explain the variation in human hair colour using a simple two-allele model.

For each gene there are a variety of alleles that have a slightly different sequence of nucleotide in the DNA sequence. The DNA codes form enzymes and the differences in the enzymes leads to slightly different metabolic activity which affects pigment colour.

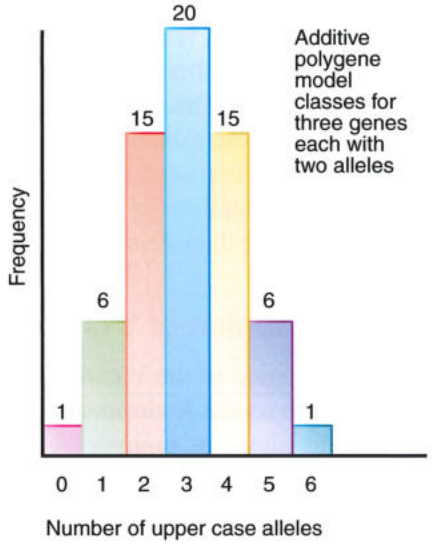


Figure 61.3 Frequency of phenotype distribution.

Human height

Human height is controlled by perhaps ten or more genes. However, appearance, especially body size is determined not only by genotype but also by the environment, e.g. poor nutrition may inhibit a person from reaching their full height potential. If the height of men is surveyed and plotted against the number of men found to have that height, the resultant graph has a bell-shaped curve showing a normal distribution between the extremes of short and tall. Environmental conditions often cause the shape of the graph to become a smoother curve. The greater the number of genes involved, the greater the variation in the possible phenotypes.

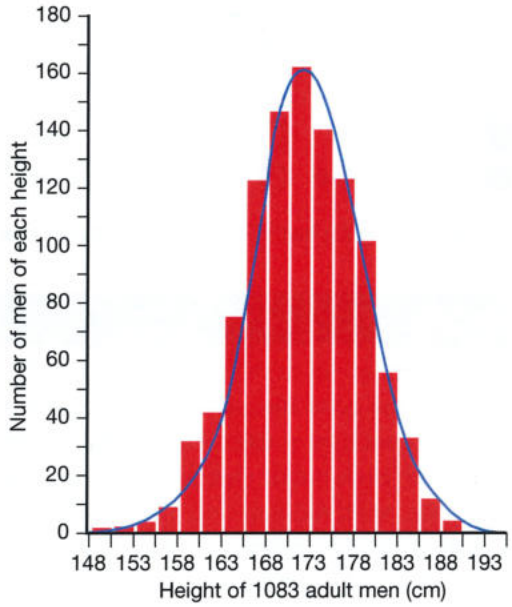


Figure 61.4 Normal distribution of height.

Human skin colour

Skin colour in humans is controlled by a number of genes. These genes code for different types of melanin so that different combinations provide different degrees of pigmentation.

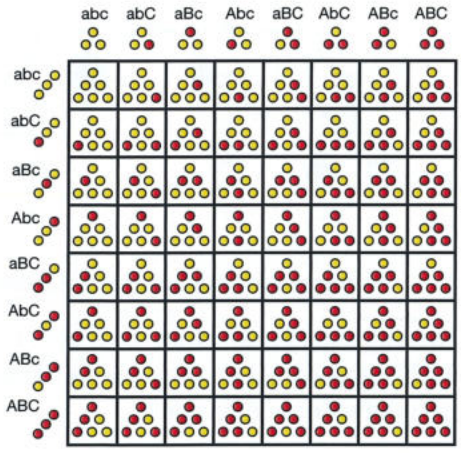


Figure 61.5 Human skin colour.

QUESTIONS

1. Define polygenic inheritance and give one human example.
2. Distinguish between continuous and discontinuous variation in phenotype.
3. When graphing phenotype distribution compare the graphs produced by discontinuous variation and continuous variation.
4. What is a quantitative trait?
5. What is a polygene?
6. Define pleiotropy.
7. What is QTL analysis?
8. Suggest why knowledge about quantitative traits has rapidly increased in the 21st century.
9. A human trait is controlled by two gene pairs so that dark colours are homozygous AABB and light colours are homozygous aabb. Identify three genotypes which you would expect to show the same phenotype.
10. Distinguish between polygenic characteristics and those caused by multiple alleles.
11. Identify a trait in humans that is polygenic in origin and a trait in humans that is determined by multiple alleles.
12. Which of the following graphs would most likely showing the frequency of polygenic inheritance?

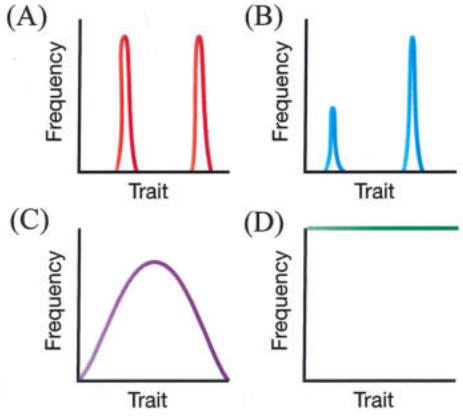


Figure 61.6 Inheritance graphs.

13. What is polygenic inheritance?
    (A) A trait that involves several genes.
    (B) The ability of a single gene to have multiple effects.
    (C) When there is more than one form of a gene for a particular characteristic.
    (D) The presence of two or more forms in the same population.
14. What is an example of polygenic inheritance?
    (A) Roan cattle having white and red hair.
    (B) Skin colour in humans.
    (C) Pea plants are tall or dwarf.
    (D) ABO blood types in humans.

62 Pedigrees

Pedigrees are graphical ways of picturing the ancestry of living things. They assist the study of the inheritance of genes by showing the genetic family history of specific traits.

Pedigrees track hereditary diseases

For families that carry genetic diseases, pedigrees are used by doctors to help advise these families about the chance of future children having a particular disease. A study of pedigrees has given valuable information about specific diseases, e.g. breast cancer, albinism and haemophilia.

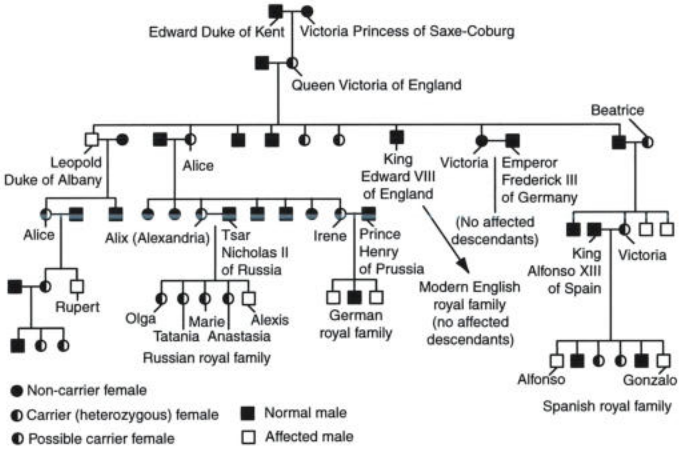


Figure 62.1 Pedigree of Queen Victoria.

Queen Victoria was a carrier for the blood-clotting disease haemophilia and a study shows that it must be sex-linked as it only appears in males.

How to draw pedigrees

In a pedigree chart males are represented by squares and females are represented by circles. Shading indicates the particular phenotype being investigated.

Figure 62.2 Males and females in pedigree charts.

Each generation is set out in one line and the birth sequence in a family is read from left to right. The next generation will be on the line below.

Interpreting pedigree charts

A pedigree chart can be drawn for a recessive trait, e.g. inheritance of albinism or it can be drawn for a dominant trait, e.g. brown eyes.

If you are not told if the trait represented in a pedigree chart is dominant or recessive you can work it out by analysing the inheritance patterns. You need to look for two parents that are the same and have a child that is different.

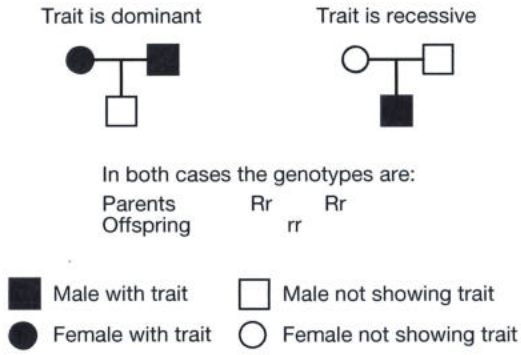


Figure 62.3 Determining if a trait in a chart is dominant or recessive.

QUESTIONS

1. Outline the rules for drawing pedigree charts.
2. In humans, the gene for red hair is recessive to the gene for brown hair. A man with red hair marries a woman with brown hair and they have two children – a daughter with red hair and a son with brown hair. His mother and his mother's father have red hair. His wife's father and sister have red hair.
   (a) Draw a family tree for red hair in this family.
   (b) Draw a Punnett grid and predict the probability of future children of the couple having red hair.
3. Explain how you would draw twin sons in a family tree. Draw a diagram to show this on a pedigree.
4. The diagram shows a family tree for a particular characteristic.

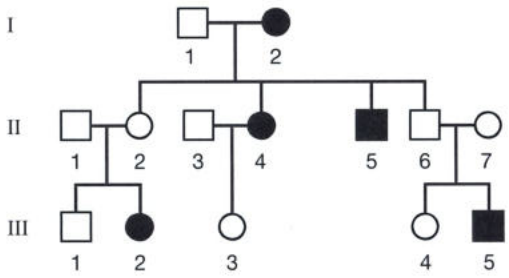


Figure 62.4 Pedigree for a particular characteristic.

(a) Identify each person with the trait.
(b) Is this trait dominant or recessive? How did you draw this conclusion?

5. Explain why pedigree charts are drawn.
6. Outline why doctors and counsellors may draw pedigree charts for couples who are hoping to start a family.
7. Explain why a family tree of the British royal family is often used to show how pedigrees provide important information.
8. In humans, normal colour is dominant to albino. Albino people have white hair and pink eyes. Which pedigree shows conclusively that normal colour is a dominant trait?

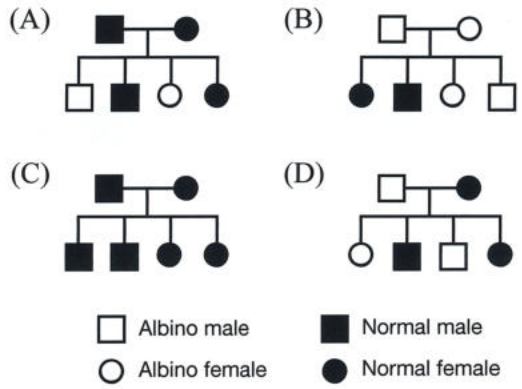


Figure 62.5 Pedigree of normal colour in people.

9. The family tree for the Smith family shows Mrs Smith who has red hair and Mr Smith who has black hair. They have two children and one of their children is married with two children.

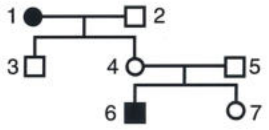


Figure 62.6 Family tree of Smith family.

What can you conclude for hair colour?
(A) Individuals 3 and 7 must be heterozygous.
(B) Individual 2 could be either homozygous or heterozygous.
(C) Individual 4 must be heterozygous.
(D) Individual 5 is homozygous.

10. Which of the following pedigrees shows the inheritance of a trait that must be recessive?

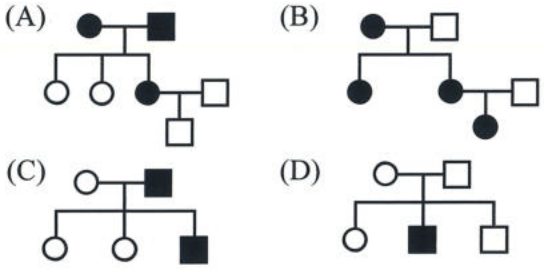


Figure 62.7 Patterns of Inheritance.

11. In a pedigree chart who is represented by a shaded circle?
(A) Male with the trait.
(B) Male not showing the trait.
(C) Female with the trait.
(D) Female not showing the trait.

12. When drawing pedigrees, what is the respective meaning of a shaded square and a shaded circle and a horizontal line connecting a square and a circle?
(A) Female with trait, male with trait and a mating line.
(B) Male with trait, female with trait and a mating line.
(C) Female without trait, male without trait and offspring.
(D) Male with trait, female with trait and offspring.

13. The pedigree shows the inheritance of a trait within a family.

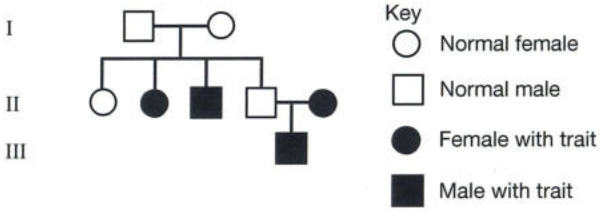
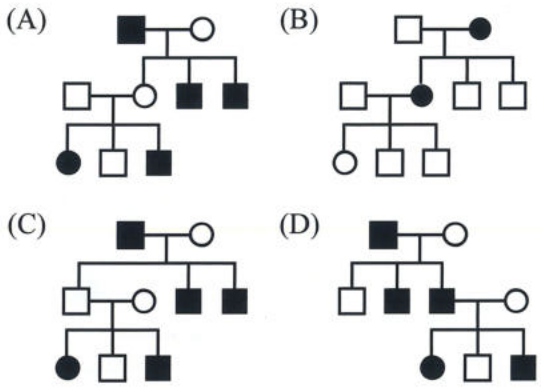


Figure 62.8 Family tree.

Is this trait dominant or recessive?
(A) Dominant.
(B) Recessive.
(C) Dominant in males and recessive in females.
(D) Insufficient information to determine.

14. Some senior Human Biology students were asked to draw a family tree for a particular trait using the following information.
'A couple have three children – a girl and two sons. The daughter marries a man and has three children a girl and two sons. The father shows the trait and his two sons show the trait. For the daughter's children her daughter and youngest son show the trait.
Which student drew the correct family tree?



(A) Student A.
(B) Student B.
(C) Student C.
(D) Student D.

63 Codominance

Mendel's work with pea plants showed that one form of a trait was always dominant over the other form of the trait, e.g. tall is dominant over dwarf. However, not all relationships between phenotype and genotype follow this simple model.

In some instances the heterozygous form has a different phenotype to both homozygous forms. This is called **incomplete dominance**. When both alleles are expressed in the phenotype it is called **codominance**. Codominance is a type of incomplete dominance.

Tay-Sachs disease

In humans Tay-Sachs disease is an inherited metabolic disorder that shows codominance. In this disease certain lipids accumulate in the brain and spinal cord and destroy neurons leading to muscle weakness, loss of muscle coordination and mental illness. A person who is heterozygous for this disease will make both the needed enzyme and the abnormal form of enzyme that breaks down the toxic fatty substances (e.g. GMS ganglioside) that accumulate in the lysosomes of neurons.

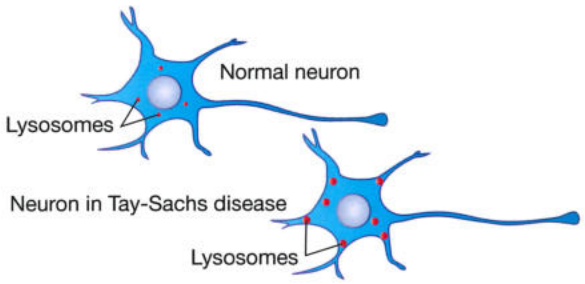


Figure 63.1 Large lysosomes in neurons in Tay Sachs disease.

Human blood groups

In human ABO blood groups a person with AB blood type has both alleles expressed due to codominance. A person will AB blood will make both the antigen A and antigen B which are both expressed on the surface of red blood cells.

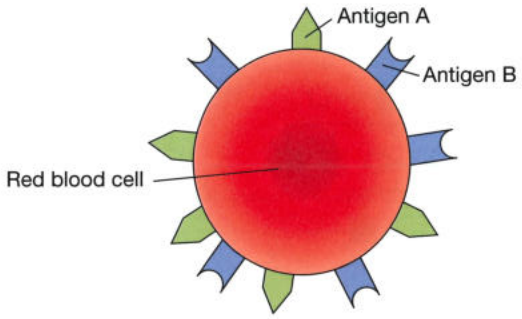


Figure 63.2 Red blood cell of a person with AB blood group.

Sickle cell anaemia

Sickle cell anaemia is a genetic disease that affects the shape and functioning of red blood cells. It is due to a mutation of the Hb gene that causes haemoglobin S being present in red blood cells instead of the normal haemoglobin A. The different protein causes the red blood cell to become sickle shaped and less able to carry oxygen than normal donut shaped red blood cells. However, the sickle cell shape gives some resistance to malaria giving an advantage to people with a heterozygous genotype as they have both types of red cells meaning they do not have the disease sickle cell anaemia but also have some reistance to malaria.

QUESTIONS

1. Define incomplete dominance.
2. (a) What is codominance?
   (b) Explain why codominance is a type of incomplete dominance.
3. List some examples of codominance in humans.
4. A man has blood group AB and has been called to the law courts in a paternity suit. The mother has blood group A and the child has blood group O. Will he be charged as the father of the child? Show how you came to this conclusion.
5. In humans the gene for straight hair is incompletely dominant over the gene for curly hair. A person with the heterozygous genotype has wavy hair. If a man with wavy hair and a woman with straight hair marry, what would be the probability ratio of hair type of their children?
   (A) 3 straight : 1 curly
   (B) 2 curly : 2 wavy
   (C) 2 curly : 2 straight
   (D) 1 straight : 2 wavy : 1 curly
6. A parent with a large lip protrusion marries a woman with a small lip protrusion and all their children have an average lip protusion. What type of inheritance is this?
   (A) Incomplete dominance.
   (B) Transitional inheritance.
   (C) Mendelian inheritance.
   (D) Pedigree dominance.
7. The diagram shows a blood sample from person X. What genotype would you expect person X to be?
   (A) Homozygous sickle cell anaemia.
   (B) Homozygous normal red blood cell.
   (C) Heterozygous carrier sickle cell anaemia.
   (D) Not enough information is provided to determine.

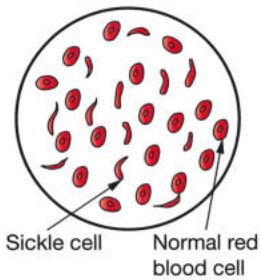


Figure 63.3 Blood sample person X.

64 Multiple Alleles

An **allele** is a form of a gene for a particular characteristic having the code for a particular phenotype. They occupy the same locus on a chromosome.

Single genes may have only two possible alleles. In humans about 30% of the genes have two forms (they are diallelic) and about 70% exist in one form showing no variation (they are monoallelic).

However, some characteristics may have many possible alleles, e.g. in humans there are three alleles for blood type – I^A, I^B and i. The ABO blood system in humans is controlled by a triallelic gene.

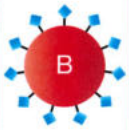
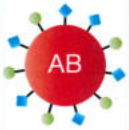
	Group A	Group B	Group AB	Group O
Red blood cell type				
Genotype	I^AI^A or I^Ai	I^BI^B or I^Bi	I^AI^B	ii

Figure 64.1 Multiple alleles for ABO blood system.

Very few genes are polyallelic and have more than two forms. Many of the genes which have multiple alleles are concerned with tissue types and thus can be used for genetic fingerprinting. Those concerned with the immune system help distinguish between invading pathogens and our own cells.

Alleles in an individual

Even though the gene pool for a particular species may contain multiple alleles for a particular gene, an individual can only have *two* of those alleles. This is because each person inherits one allele from one parent and the other allele from the other parent, so there are two alleles for every trait in each individual.

Dominance with multiple alleles

When there is a series of possible alleles for a certain trait, one allele can be dominant or recessive depending on which other allele is present. To show this situation the dominance is often written as a series. For example, hair colour is controlled by more than one gene and the genes have multiple alleles that give the continuous range of hair colour found in the human population. When looking at multiple alleles there can be an order or dominance, e.g. if black is dominant to brown and brown is dominant to albino then this can be written: Black > brown > albino.

Increasing the number of alleles increases the number of genotypes and phenotypes and for convenience one letter is often chosen to represent each allele and numbers are used in superscript to show the different alleles. For example, if there are four alleles for a particular trait, E^1 is appearance 1, E^2 is appearance 2, E^3 is appearance 3 and E^4 is appearance 4. The dominance pattern can then be given as $E^1 > E^2 > E^3 > E^4$. This means there is an increased number of possible phenotypes. For example, if there are four alleles for brown hair colour the variants could be dark brown, mahogany brown, honey brown and light brown.

QUESTIONS

1. Define allele.
2. Give an example of multiple alleles.
3. Explain why individuals can only have two alleles for a particular trait.
4. How common are multiple alleles for a particular trait in humans?
5. Outline how multiple alleles can be applied to society.
6. Assume there are four alleles for hair colour so that: Dark brown > mahogany brown > honey brown > light brown.
   Construct a table to relate phenotype to genotype for hair colour using the allele codes – dark brown = C, mahogany brown = c^m, honey brown = c^h and light brown = c.
7. (a) What is meant by a monoallelic gene?
   (b) What is the amount of monoallelic genes in humans?
8. What are the possible blood types of a child of a man with blood type A and a woman, whose parents were both AB, of blood type B?
   (A) B only
   (B) A or B only
   (C) B or O only
   (D) B or AB only
9. If the first child in a family has blood group AB and the second child in the family has blood group O, what are the genotypes of the parents?
   (A) I^Bi and I^Ai
   (B) I^AI^B and ii
   (C) I^BI^B and I^Ai
   (D) I^AI^A and I^BI^B
10. In humans the ABO blood system has three possible alleles. How many alleles in one individual determine their blood group?
    (A) One
    (B) Two
    (C) Three
    (D) Can be two or three

65 Morgan and Sex-Linkage

Thomas Hunt Morgan studied the fruit fly, *Drosophila melanogaster*, in genetics experiments. It was an ideal species for breeding experiments as it has easily observable characteristics that are carried on only four pairs of chromosomes (three pairs of autosomes and one pair of sex chromosomes). They produce hundreds of offspring in a single mating, and can be easily bred in a laboratory, producing a new generation in 10 to 12 days.

Figure 65.1 Thomas Hunt Morgan.

After a year of breeding flies, looking for new characteristics, Morgan found a male fly with **white eyes** in his breeding stock. The normal or 'wild' eye colour is red. The white-eyed individual was probably a mutant. He mated this white-eyed male with a red-eyed female and all F_1 had red eyes, showing that red eyes are dominant over white.

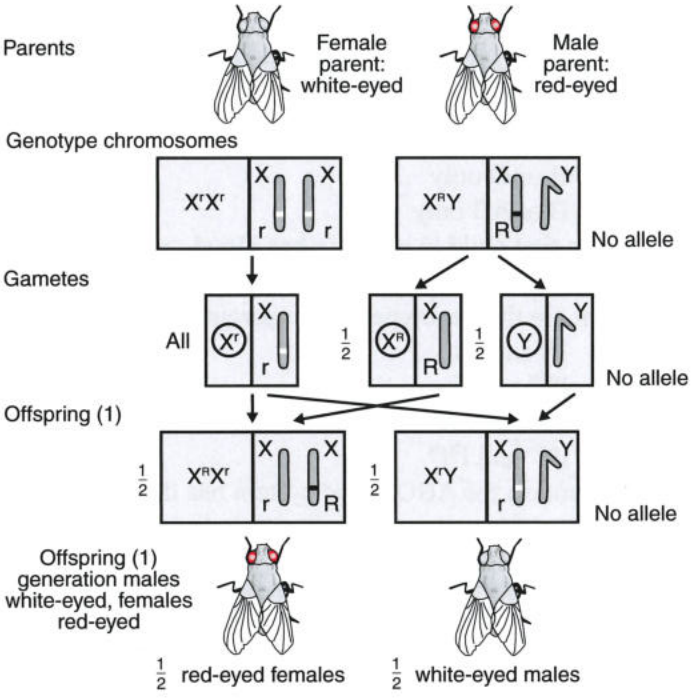


Figure 65.2 Thomas Hunt Morgan's experiment with white-eyed and red-eyed fruit flies.

When Morgan crossed two of the F_1 generation the offspring had a phenotypic ratio of 3 red-eyed offspring : 1 white-eyed offspring. However, the white-eyed trait was only present in the males, with half the males having white eyes and half having red eyes. Using this and other evidence, Morgan worked out that the gene for eye colour is located on the X chromosome and there is no corresponding locus on the Y chromosome. Female fruit flies have XX and males have XY. Morgan concluded that the gene for eye colour was sex-linked, and that it was located on the X chromosome.

Sex-linked genes are located on the chromosome, being either X-linked or Y-linked. In humans X-linked genes include red-green colour blindness and haemophilia, while Y-linked genes include those for male characteristics and a hairy pinna.

Probability models

When solving genetics problems dealing with sex linkage it is usual to use superscripts for the alleles next to the X chromosome, if it is on the X chromosome. For example, for haemophilia, which is carried on the X chromosome, a female homozygous for the normal condition would be X^NX^N, a carrier female would be X^NX^n, a normal male would be X^NY, and a haemophiliac male would be X^nY.

QUESTIONS

1. Throughout the history of science, many important discoveries have occurred when scientists have chosen a very suitable experimental organism, either by luck or by insight. Mendel chose edible garden peas. Morgan chose the fruit fly *Drosophila melanogaster*. List three reasons why the fruit fly was a suitable experimental choice.
2. Assess how the work of TH Morgan contested the existing model of inheritance and led to a new model that had greater explanatory power.
3. The normal eye colour in the fruit fly is red, often called the wild type, and the alternative trait, white eyes, is often called the mutant phenotype. Draw a Punnett grid to show the genotypes and phenotypes of the F_1 and F_2 if a pure-breeding red-eyed fruit fly is mated with a pure-breeding white-eyed fruit fly. Show all working.
4. The first mutant phenotype was a white-eyed male fruit fly discovered after Morgan had been investigating fruit flies for about one year. When he crossed this white-eyed male with a red-eyed female, all the F_1 offspring had red eyes. When he crossed two of the F_1 generation the offspring were 3 red-eyed : 1 white-eyed; however, all the white-eyed offspring were males.
   (a) What conclusion could Morgan draw from this result?

(b) Using this information, redraw the Punnett grid from Question 2, showing the F_2 generation with sex-linkage to show Morgan's results.
(c) Is it possible for a female fruit fly to have white eyes? Show all workings to explain your answer.

5. Fruit fly, *Drosophila melanogaster*, has four pairs of chromosomes – three pairs of autosomes and one pair of sex chromosomes whereby females are XX and males are XY.
(a) How many different types of gametes could one parent form, excluding mutation and crossing over?
(b) Describe how fertilisation increases variability.
(c) Name one other source of variability in organisms.

6. The pedigree shows the inheritance of curly wing in fruit fly.
(a) Is curly wing dominant or recessive? Use examples from the pedigree to explain your reasoning.
(b) Choose a letter and identify the genotypes of individuals 5, 6 and 20.

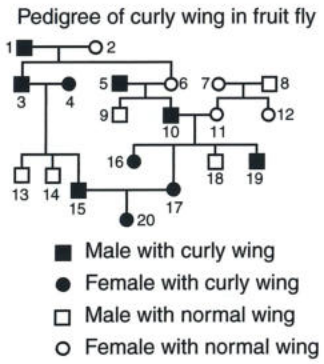


Figure 65.3 Pedigree of curly wing in fruit fly.

(c) The curvature depends on the difference in the contraction of the upper and lower epithelia of the wings when the adult emerges from the pupa. It has been found that some fruit flies which are heterozygous for the trait have normal wings, especially if the temperature is lower, e.g. most have curly wings at 25°C but few have curly wings at 19°C. Draw a Punnett grid to identify which of the offspring would develop the 'wrong' wing curvature if the environmental temperature was kept at 19°C for the pupa and explain how this affects Mendel's ratios.

7. Fruit fly, *Drosophila melanogaster,* has three pairs of autosomes and one pair of sex chromosomes and are like humans in that females have an homologous pair of X chromosomes and males have one X chromosome and one Y chromosome. After one year of investigating inheritance in fruit flies, Thomas Hunt Morgan discovered a single male fly with white eyes instead of the usual red. He mated it with a red-eyed female. All the offspring had red eyes. When he bred the F_1 flies, the offspring showed a ratio of 3 red eyes : 1 white eyes with only males having white eyes. From this evidence what could Morgan deduce about eye colour?
(A) It is on the X chromosome with no corresponding locus on the Y chromosome.
(B) It is sex-linked, found only on the Y chromosome.
(C) It is a codominance trait.
(D) It is sex-linked.

8. In the fruit fly, *Drosophila melanogaster*, long wings are dominant to vestigial (short) wings. When two long-winged fruit flies were mated they produced 104 long-winged and 30 vestigial-winged offspring. What does this show?
(A) Both parents were homozygous for wing length.
(B) Both parents were heterozygous for wing length.
(C) One parent was heterozygous and the other was homozygous for wing length.
(D) Mutation occurred during fertilisation.

9. Red-green colour blindness is an X sex-linked recessive trait in humans. A carrier female marries a man with normal vision. What is the probability of their children showing this trait?
(A) Half their sons will be colourblind; all daughters with normal vision.
(B) All their sons will be colourblind; all daughters with normal vision.
(C) Half their daughters and half their sons will be colourblind.
(D) All their daughters and half their sons will be colourblind.

10. ABO blood groups in humans are determined by three alleles, I^A, I^B and i. The alleles I^A and I^B are co-dominant and i is recessive. If a man with blood group AB marries a woman with blood group O, what are the probabilities for their children's blood groups?
(A) All girls will have A blood group; all the boys will have B blood group.
(B) All the offspring will be AB.
(C) 25% will be A; 25% will be B; 50% will be O.
(D) 50% will be A; 50% will be B.

11. Baldness in men is inherited on the X chromosome. The gene for baldness is recessive to the gene for normal hair. If a bald man marries a woman carrier, what is the probability of their children being bald?
(A) Half their daughters and half their sons will be bald.
(B) All their sons and none of their daughters will be bald.
(C) Half their sons and none of their daughters will be bald.
(D) All their daughters and none of their sons will be bald.

12. Thomas Hunt Morgan investigated the inheritance of characteristics in fruit flies. The discovery of which mutant fly led to his identification of sex-linkage?
(A) Red-eyed female. (B) Red-eyed male.
(C) White-eyed male. (D) White-eyed female.

66 Human Blood Groups and Rhesus Factor

There are three alleles for the gene that determines human blood. The alleles are I^A, I^B and i. The allele 'i' is recessive to both I^A and I^B. I^A and I^B are codominant. There are four possible phenotypes – A, B, AB and O which are inherited through genes on chromosome 9. Two antigens and two antibodies are mostly responsible for the ABO types. The antigens are on the surface of the red blood cells and the antibodies are in the blood plasma and are also found in saliva and other body fluids except spinal fluid for some people.

Table 66.1 Human blood types.

Blood type	Possible genotype	Antigens on RBC	Antibodies in blood
A	I^AI^A I^Ai	A	Anti-B
B	I^BI^B I^Bi	B	Anti-A
AB	I^AI^B	A and B	Neither
O	ii	Neither	Anti-A and anti-B

If red blood cells carrying one or both antigens contact their corresponding antibodies, e.g. antigen A contacts Anti-A, the red blood cells agglutinate (clump together) which can cause death. This means that blood types must be carefully matched when carrying out blood transfusions. It is theoretically possible to use O blood in a transfusion to any other blood group, however unless there is careful matching, antibodies in the donated plasma can damage the red blood cells of the recipient. Blood type AB can donate to the fewest groups but can receive blood from all other groups.

Table 66.2 Frequency of blood groups for some populations.

Population	Blood group (% population surveyed)			
	AB	A	B	O
American Indian	0	4%	2%	94%
Hong Kong Chinese	6%	24%	27%	44%
Japanese	9%	38%	22%	31%
Eskimos	4%	45%	6%	47%

To determine a person's blood group, you need a few drops of blood and to one sample add a serum containing anti-A antibodies and to another sample add a serum containing anti-B antibodies. Agglutination shows blood type, e.g. agglutination for anti-B antibody but not anti-A antibody means the person is blood type B and has antigen B present. In paternity cases blood grouping can indicate who *cannot* be the father, however conventional blood typing does not give enough specific evidence to identify the actual father. DNA sequencing is now used in legal cases to accurately determine paternity.

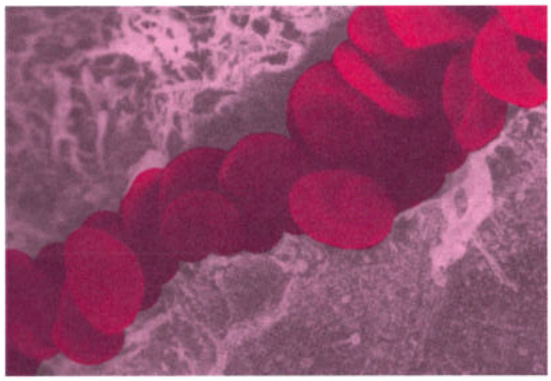


Figure 66.1 Red blood cells.

Rhesus factor

The Rhesus factor was first found in rhesus monkeys and hence its name. The Rhesus factor refers to a group of transmembrane antigens found on the surface of red blood cells. If you possess the main Rhesus factor you are Rh+ (Rh positive), and if you do not have this factor you are Rh− (Rh negative) and produce antibodies if your blood comes into contact with this factor. The Rh antigens seem to be needed for the transport of carbon dioxide and/or ammonia across the plasma membrane. If you are Rh+ you may have alleles for both positive and negative. If you are Rh−, you have only Rh− alleles. About 15% of the human population are Rh− and have no Rh antigens.

The presence of Rh antigens is very important during pregnancy if there is incompatibility between the mother and foetus. If the baby is Rh+ and the mother is Rh− and there is leakage of the baby's blood into the mother's circulation during childbirth, the mother will develop antibodies again the Rh antigen. This means that during any *future* pregnancy for that mother, if the new foetus is Rh+, the mother's antibodies will cross the placenta and attack the foetus' red blood cells. The antibodies will destroy the red blood cells causing anaemia and jaundice and the effects can be so severe the foetus or newborn can die. This is called haemolytic disease. To solve this disease it is usual practice today to give the Rh− mother an injection of anti-Rh antibodies after the birth of an Rh+ baby. Once in the mother's bloodstream these acquired antibodies destroy any foetal cells that entered during childbirth and hence stop the mother's immune response becoming active. Thus the mother will not produce antibodies during her next pregnancy if the next foetus is Rh+. There is not a problem for mother and foetus for ABO compatibility.

In 1930 Karl Landsteiner won the Nobel Prize in Physiology or Medicine for the discovery of human blood groups. In 1909 he had developed the modern system of classification of blood groups after conducting many experiments on the clumping of blood when blood serum from different people was mixed. In 1940, Karl Landsteiner with Alexander S Weiner discovered the Rhesus factor.

QUESTIONS

1. Construct a table to show blood type and the possible allele combination for each of these blood types.
2. Identify the location of the antigens and antibodies that cause the differences in human blood.
3. What is meant by the phrase 'red blood cells agglutinate'?
4. Explain why blood type O is often called the universal donor and why blood type AB is often called the universal recipient.
5. Describe how you can determine a person's blood group.
6. How was the Rhesus factor named?
7. What is meant by Rh+ and Rh−?
8. Outline the function believed to be carried out by the Rh antigens.
9. What proportion of the population is Rh+?
10. Explain why it is important to know the Rh compatibility of an expectant mother and her unborn baby.
11. Discuss how current medical practices solve the problem of Rh haemolytic disease.
12. Construct a table to show the blood type of parents, possible genotypes of parents and possible blood types of children from those parents, for each of the following parent combinations – A and A, A and B, A and O, A and AB.
13. If the father is Rh+ and the mother is Rh−, what is the possible Rh status of their children?
14. The diagrams in Figure 66.2 show human red blood cells before and after adding serum contain anti-A antibodies.

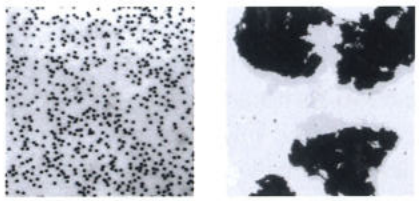


Figure 66.2 Blood cells before and after adding anti-A antibody serum.

Identify which diagram shows the 'before' serum appearance of red blood cells and which diagram shows the 'after' serum appearance. Explain what has happened in the 'after' diagram.

15. A person's blood group relies solely on the genes inherited from their parents. However, some scientists believe the environment has played a part in the inheritance of blood groups due to the linking of specific ABO blood types to particular diseases. For example, it has been suggested the type A blood people have a higher risk of developing cancer of the pancreas, oesophagus and stomach than do people with type O blood. How could you determine if these claims have validity?
16. An analysis of the frequency of different blood groups in different countries has shown that about 49% of individuals in some European countries have Type O blood, while in some populations of North American Indians there is 97% type O blood. Suggest a reason that would account for these differences between human populations.
17. Terumi has blood group B. She has a child who is blood group O. She believes that Marcus is the father of her child but he says he cannot be the father because he is blood group A. Use genetic diagrams to explain if Marcus could/could not be the father of Terumi's child.
18. Explain how it is possible for two parents who both have AB blood type can have three children who each have a different blood group to each other.
19. In the Chowdhury family there are three children. Tom has blood group O, Sam has blood group A and Chris has blood group B. Suggest possible genotypes and blood groups for Mr and Mrs Chowdhury.
20. There is an interesting condition called blood chimera. These people have two different blood types. Most of these people shared a blood supply with a non-identical twin before birth. It is believed that about 8% of non-identical twins have chimeric blood. Explain how a person could have chimeric blood but was not born with a living twin.
21. If both parents have blood type O, why can you predict the blood type of all their offspring with absolute certainty?
22. Explain why early attempts to transfuse blood met with variable success.
23. Describe two blood group systems that must be considered when carrying out transfusions.
24. Discuss how the work of Karl Landsteiner led to a major advance in scientific understanding and influenced the direction of biological research.
25. The table shows donor-recipient compatibility.

Table 66.3 Donor-recipient compatibility.

		Recipient			
	Blood Type	A	B	AB	O
Donor	A	Safe	Not safe	Safe	Not safe
	B	Not safe	Safe	Safe	Not safe
	AB	Not safe	Not safe	Safe	Not safe
	O	Safe	Safe	Safe	Safe

From this table the universal donor is:

(A) Blood type A.
(B) Blood type B.
(C) Blood type AB.
(D) Blood type O.

67 Autosomal Genetic Diseases

There are several human genetic diseases due to abnormal alleles on autosomes and the frequencies of genotypes and phenotypes of offspring can be predicted using probability models.

Tay-Sachs disease

Tay-Sachs disease is a rare genetic disease caused by a range of mutations of the *HEXA* gene on chromosome 15. This gene produces the lipid-metabolising enzyme hexosaminidase A which is found in lysosomes. A baby with two Tay-Sachs alleles will have the disease and will be unable to metabolise certain lipids. As the lipids accumulate in brain cells the child will have blindness, seizures, deteriorating motor and mental functions and shortened life expectancy.

Since two alleles are needed to show the disease and a person with one Tay-Sachs allele and one normal allele (heterozygote) does not have the disease, the disease is considered to be **autosomal recessive on an organisational level**. However, an analysis of the enzymes produced by a person heterozygous for Tay-Sachs disease shows that they produce equal quantities of the normal enzyme and the dysfunctional enzymes. This means that **on a biochemical level the Tay-Sachs allele is codominant** with the normal allele. The amount of normal allele produced by the heterozygote is sufficient for the person not to develop symptoms of the disease.

Tay-Sachs disease has high frequencies in certain populations, e.g. Ashkenazi Jews, Cajuns of southern Louisiana and French Canadians of south-eastern Quebec. The construction of pedigrees and the analysis of family trees for Tay-Sachs disease has shown how mutations can spread over several generations from a small founder population into a larger community and population.

Huntington's disease

Huntington's disease is a degenerative disease of the nervous system caused by a **lethal dominant allele** on the short arm of chromosome 4. The Huntington gene (*HTT*) has a trinucleotide repeat and the defective gene has an increased count of the trinucleotide repeat which increases the length of the gene. The number of trinucleotide repeats affects the disease status and the ability of the person to make the functional protein. The mutant protein increases the decay of certain types of neurons.

Symptoms of Huntington's disease usually start to show between ages 35 and 45 years old. Since it is a dominant allele, a person with only one mutant allele will have the disease and since many people have offspring before the onset of symptoms, the frequency of the dominant mutant allele can stay high in a population.

The disease causes the death and decay of certain types of neurons in the brain in certain areas of the brain leading to gradual loss of muscle coordination with twitching that progresses to jerky movements and loss of thinking and emotional abilities. The increased mental deterioration leads to dementia and decreased control of body movements make normal movement extremely difficult. Life expectancy is reduced from the onset of symptoms and there is no cure at the moment.

New technologies that can analyse DNA samples means that people with a family history of Huntington's disease can be tested for the mutant gene. This diagnostic testing has then led to ethical, legal and social concerns about disclosure and confidentiality of the test results.

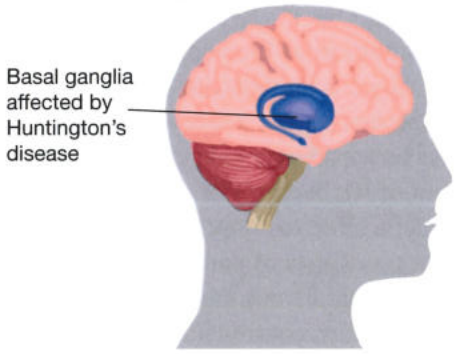


Figure 67.1 Huntington's disease.

Phenylketonuria

Phenylketonuria (PKU) is an autosomal genetic disorder caused by two **recessive** alleles for the gene that produces the enzyme phenylalanine hydroxylase (PAH). The PAH gene is on chromosome 12 and there are many types of mutations that lead to different forms of this disease. The enzyme PAH is needed to convert the amino acid phenylalanine into the amino acid tyrosine. Without treatment liver cells will not be able to properly break down phenylalanine so it and its by-product phenylpyruvate will accumulate in the blood reaching toxic levels.

Symptoms of PKU include stunted and delayed growth, hypersensitivity, poor bone strength, limb and hand tremors that lead to mental retardation and seizures. The damage is mostly irreversible.

Newborn children in Australia are screened for PKU usually between 2 to 7 days after birth. There are several ways to test for PKU, e.g. the **Guthrie test** uses a bacterial inhibition assay. The effects of PKU can be controlled by controlling the diet of the person with the disease. The 'diet for life' restricts the intake of foods high in phenylalanine, e.g. red meat, chicken, fish, eggs, nuts, cheese, milk and other dairy products. Medication and dietary supplements can also assist. The strict diet prevents the mental retardation and other issues. PKU levels can be monitored through home blood tests which are sent to screening laboratories for analysis.

Cystic fibrosis

Cystic fibrosis (CF) is a hereditary disease caused by a mutation of the CF gene on chromosome 7 and is one of the most common life threatening inherited diseases. It is caused by a recessive gene that lacks the code for one amino acid – phenylalanine and occurs equally in males and females. There are more than 500 mutations which lead to this disease. The cystic fibrosis transmembrane conductance regulator (CFTR) gene is involved in forming chloride ion channels in cell membranes. Dysfunctional channels cause mucous glands to secrete very thick sticky mucus which accumulates leading to problems with the respiratory, digestive and reproductive systems, e.g. restriction of air passages in the lungs and problems with the release of enzymes from the pancreas affecting food digestion.

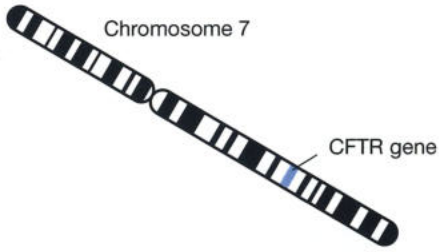


Figure 67.2 Chromosome 7 with CFTR gene that causes cystic fibrosis.

Symptoms include persistent cough with difficulty breathing, tiredness and poor appetite. There is no cure for cystic fibrosis but treatment can slow the development of the disease.

Genetic testing can use probes that are short sequences complementary to the mutated sequence. The probe binds to the sequence and flags the mutation. It is usually diagnosed at birth.

Detection of genetic diseases

DNA samples can be taken to determine if a particular person is a carrier of a particular disease. Tissue or blood samples can be analysed to produce a DNA profile which shows the presence of particular DNA sequences that are indicative of particular hereditary diseases.

If a person is concerned about a hereditary condition, they can contact the Genetic Services of Western Australia which is part of the Department of Health, Genomics Directorate. The Genetic Services provides clinical and laboratory genetic service. They can deal in the prevention and treatment of genetic diseases.

For people with family histories of particular genetic diseases who undergo genetic testing and find they are carriers of a particular disease there are genetic counsellors who can help to explain the disorder, provide information about probable frequencies of genotype and phenotype of offspring and provide information about future options.

QUESTIONS

1. Identify a human genetic disease for each of the following.
   (a) Autosomal dominant.
   (b) Autosomal recessive.
   (c) Autosomal codominant at molecular level.
2. Explain why Tay-Sachs disease can be considered to be autosomal recessive on an organisational level but codominant on a biochemical level.
3. (a) What is Huntington's disease?
   (b) What causes Huntington's disease?
   (c) If a person has the Huntingdon disease allele, what is the chance that their child will have the disease?
4. In Australia it is common for newborn children to be screened for phenylketonuria. Assess the importance of newborn screening for phenylketonuria.
5. List some of the symptoms that occur if PKU is untreated.
6. (a) What causes cystic fibrosis?
   (b) Why is cystic fibrosis considered to be an autosomal recessive genetic disorder?
   (c) How can genetic testing for cystic fibrosis before birth aid children?
7. Construct a table to summarise the symptoms of an autosomal dominant disease, autosomal recessive disease and autosomal codominant disease.
8. The diagram shows a pedigree for the inheritance of a genetically inherited blood disorder.

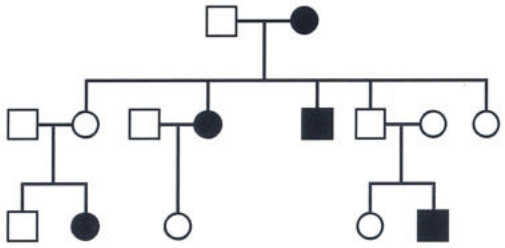


Figure 67.3 Family tree for genetic disease.

What is the most likely mode of inheritance for this disorder?
(A) X-linked recessive.
(B) Y-linked recessive.
(C) Autosomal dominant.
(D) Autosomal recessive.

9. A couple who do not have cystic fibrosis have a son who does not have cystic fibrosis and then a daughter who does have cystic fibrosis. If you were the genetic counsellor for this couple, what would you tell them is the probability that their next child will have cystic fibrosis if they did not use assisted reproductive technologies?
(A) 25% (B) 50% (C) 75% (D) 100%

68 Sex-Linked Genetic Diseases

Abnormal alleles on the sex chromosomes can cause genetic diseases. Traits controlled by genes on the X chromosome are called X-linked and for these traits females have two alleles for the trait (XX) while males only have one allele for the trait (XY).

Red-green colour blindness

The gene for red-green colour blindness is on the X chromosome. It is a recessive allele and this means that red-green colour blindness is more common in males as they only have one allele and the recessive trait will always show in the phenotype. Females are XX and have two alleles for red-green colour blindness which means they can be carriers and need to be homozygous for the trait to show red-green colour blindness phenotype. Statistically about 8% of all males and 0.5% of all females are colourblind.

Colour blindness can result from the absence or impairment of one or more of the type of cones in the retina of the eye. Normal colour vision uses the three types of cone – red, green and blue with normal vision called trichromat. The gene for blue-yellow colour blindness is on an autosome and affects males and females equally.

There are several tests that can be carried out to determine colour blindness (e.g. Ishihara's test for colour blindness which has a series of dots in different colours). Colour blindness is not always self-evident. Many people only realise they have some form of colour blindness when they have done a colour blindness test. Certain drugs (e.g. barbiturates, cocaine, caffeine and tobacco) and certain diseases of the retina or optic nerve can harm normal colour vision.

There is no cure for hereditary colour blindness although some strategies can help, such as specially tinted glasses to help distinguish confusing colours or the person developing their own system using location, texture, brightness etc. Many professions (e.g. pilot, electrician, dye manufacturing) require good colour vision.

Haemophilia

Haemophilia is a genetic disorder caused by an X-linked recessive allele that causes insufficient amounts of clotting factor to be present causing the blood not to clot properly. Most bleeding is internal with episodes affecting the joints or muscles. These incidents can occur spontaneously without being triggered by an injury or trauma.

Haemophilia A (or classical haemophilia) is the most common form of haemophilia and is a deficiency in Factor VIII. This can caused by the factor being missing or defective with about 1 in 3 cases due to a spontaneous mutation changing the allele for the gene. It leads to increased bleeding and longer bleeding without clotting in the joints, muscles, digestive tract and brain.

Haemophilia B (or Christmas disease) is a deficiency in factor IX leading to increased bleeding without clotting in joints and muscles.

As the gene is on the X chromosome males only need one defective allele to have the disease while females need two defective alleles. Sever haemophilia is very rare in females though a few females have low levels of clotting factors and are considered to have haemophilia.

The main treatment is recombinant factor products which can be given intravenously through a vein in the arm or a port in the chest. The Haemophilia Foundation Western Australia assists people and their families living with the disease.

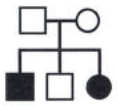
When solving genetic problems with sex-linkage it is first necessary to write out possible genotypes and phenotype.

Below is an example of script to use for X chromosome sex-linkage genetic problems, e.g. haemophilia.

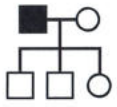
X^NX^N = female with normal blood clotting
X^NX^n = carrier female with normal blood clotting
X^nX^n = female with haemophilia
X^NY = male with normal blood clotting
X^nY = male with haemophilia

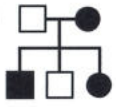
QUESTIONS

1. What is meant by a sex-linked genetic disease?
2. Name two sex-linked genetic diseases.
3. Explain why more males are red-green colourblind than females but there are equal numbers of blue-yellow colourblind men and women.
4. Why can people not know they are colourblind?
5. What is the difference between congenital colour blindness and acquired deficiency?
6. Explain why an electrician needs good colour vision.
7. Which of the following indicates an X-linked dominant trait?

(A) 

(B)

(C) 

(D) 

8. A couple who do not have haemophilia have a son with haemophilia. If you were the genetic counsellor, what would you tell this couple is the probability that their next son will have haemophilia if they did not use assisted reproductive technologies?
   (A) 50% of sons have haemophilia.
   (B) 50% of children have haemophilia.
   (C) 75% of children have haemophila.
   (D) 100% of sons have haemophilia.

69 DNA Profiling

Every person has a unique DNA profile – except for identical twins. For each individual the DNA profile (or DNA fingerprint) is the same for the cells, tissues and organs of their body. This means each individual can be identified by their DNA fingerprint.

When the courts decided that DNA profiling was a reliable method to identify criminals, the use of DNA profiling rapidly expanded to include solving paternity cases, identifying unknown bodies, diagnosing inherited disorders and solving old crimes.

Making a DNA fingerprint

To make a DNA fingerprint, you need to carry out the following.

- Take DNA from some cells or tissues, e.g. hair, skin or blood.
- **Restriction enzymes** are then used to cut the DNA at specific points.
- The segments are sorted by size by a sieving process called **electrophoresis**.
- The DNA segments are next transferred to a nylon sheet and radioactive dyes or coloured probes are added to the nylon sheet to produce a pattern called the DNA profile.
- Several probes (e.g. 5 to 10 or more) are used to build the DNA fingerprint.
- When the pattern is exposed to X-ray film the result is a series of bars.

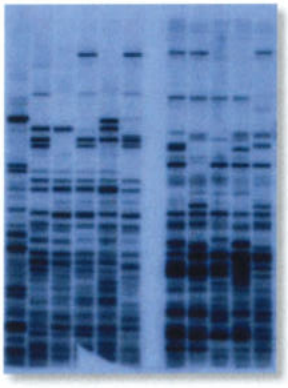


Figure 69.1 DNA fingerprint.

Since 99% of human DNA is identical for all individuals, DNA fingerprinting needs to use the sites of polymorphism to identify particular people. Restriction enzymes are used to cut DNA at specific sections. The length of each of these fragments depends on the individual. They are commonly called 'reflips' (RFLPs – **restriction fragment length polymorphism**).

A more recent development in DNA fingerprinting is the use of **short tandem repeats** (STRs). There are certain base sequences within the genome that are highly variable in length from person to person and are found at specific loci on the DNA molecule. These polymorphic loci are used as markers and the number of repeats at each locus is recorded. The greater the number of markers studied, the greater the ability to identify a particular person.

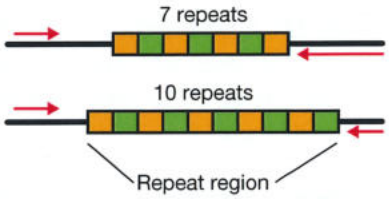


Figure 69.2 Short tandem repeats show a variable repeat region between samples while surrounding areas are constant. A homozygote has both alleles with the same length and a heterozygote has alleles with different lengths.

When using DNA profiling data it is preferable to use multiple stretches of DNA to make sure the results are statistically significant, especially if linking the data to a person in a case before the courts.

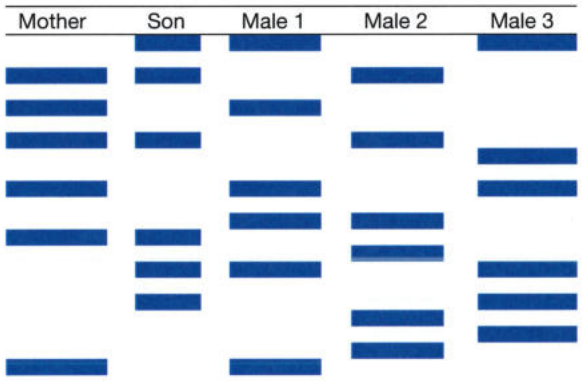
Uses of DNA profiling

DNA profiling is a way of establishing identity and has widespread uses.

- Identification of a particular hereditary disease, e.g. to determine if a particular person is a carrier of a particular disease, e.g. cystic fibrosis or sickle cell anaemia or of presymptomatic testing for predicting adult onset disorders, e.g. Huntington's disease.
- Use in paternity disputes.
- Identification of siblings, e.g. for adopted people to help find biological relations.
- Following ancestry and family trees.
- Solving crimes, e.g. forensic science data is allowed to be used in the courts and the justice system.
- Establishing evolutionary links and constructing evolutionary trees, e.g. evolution of humans and the relationships of humans to other primates.
- By animal breeders to accurately identify parentage of offspring, e.g. CSIRO Livestock Industries developed the technology for DNA profiling that has been applied to Merino sheep. This means that accurate records for pedigrees can be kept and it is predicted that there could be a 25% increase in the rate of genetic gain. In some countries dog breeders use DNA fingerprinting to guarantee bloodlines for a particular breed, e.g. in the USA the American Kennel Club requires DNA certification for all studs used over five times a year.

QUESTIONS

1. Define DNA profiling.
2. Describe how you would make a DNA profile.
3. Explain why DNA polymorphisms are important in DNA profiling.
4. Describe how restriction enzymes are used in DNA profiling.
5. What is a 'reflip'?
6. The following DNA profile shows a mother and her son. There are three men who could be the father of this child. Which man is the father and explain why you made this decision.



7. Discuss why DNA profiling is important in forensic science.
8. Explain why DNA profiling provides stronger evidence in a court of law than same blood or tissue type.
9. Explain why it is important to gather information about the frequency of STRs and markers in different ethnic groups.
10. Discuss how DNA profiling can be used for determining the pedigree of animals.
11. The diagram shows a technology used to investigate DNA profiles.

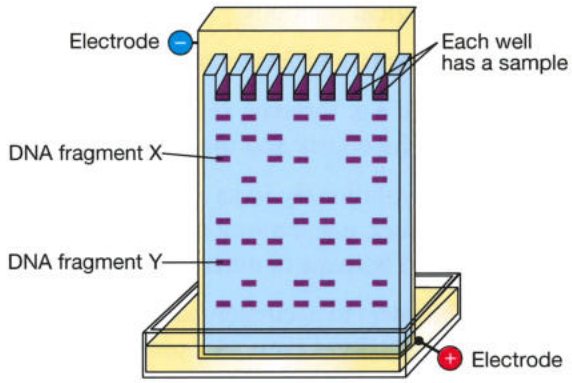


Figure 69.3 DNA technology.

(a) Name this technology.
(b) How is this technology used in DNA profiling?
(c) What is the difference between DNA fragment X and DNA fragment Y?

12. The diagram shows the process of DNA profiling.

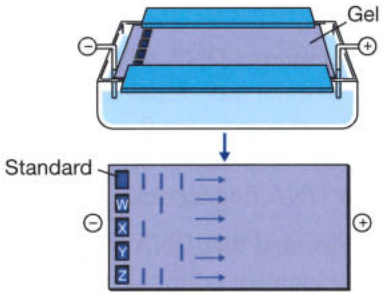


Figure 69.4 Process in DNA profiling.

(a) In this process how many DNA samples are being tested?
(b) Why is a standard placed in one of the wells?
(c) The diagram shows the movement of the DNA fragments when an electric field is switched on. Which way is the movement?
(d) From the diagram which sample has the shortest fragment?
(e) From the diagram which sample has the highest number of long DNA fragments?

13. The following notes show the procedure of making a DNA profile.
(i) Restriction enzymes cut DNA into fragments.
(ii) Radioactive probes added and autoradiography used.
(iii) PCR used for amplification.
(iv) Computer analyses data.
(v) Blood sample taken.
(vi) DNA extracted from blood.
(vii) Gel electrophoresis.
What is the correct order for making a DNA profile?
(A) v, iv, i, vii, ii, iii, vi (B) vi, i, v, vii, iii, ii, iv
(C) i, ii, ii, v, vi, vii, iv (D) v, vi, i, vii, iii, ii, iv

14. Why are radioactive probes or fluorescent dyes used in DNA profiling?
(A) To cut the DNA into fragments.
(B) To amplify the number of fragments.
(C) To make the DNA fragments detectable.
(D) To make the DNA fragments able to move in the electric field.

15. The following set of DNA profiles was made to determine which of the men fathered the child. Which man is the most likely father?
(A) Man A
(B) Man B
(C) Man C
(D) Man D

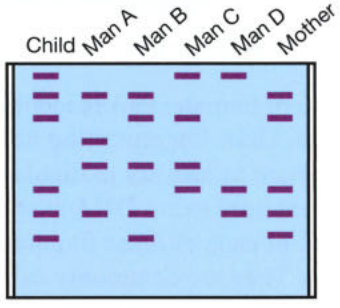


Figure 69.5 DNA fingerprint.

70 Technology for DNA Profiling

DNA profiling also called DNA fingerprinting or genetic fingerprinting identifies particular DNA sequences and the resulting pattern, which looks like a series of bar codes can be used to identify individuals. It refers to the unique set of genetic markers in the genome of an individual.

The process relies on DNA sequencing technology, electrophoresis, nucleic acid probes and PCR. The techniques use repetitive sequences of nucleotides that are variable from person to person.

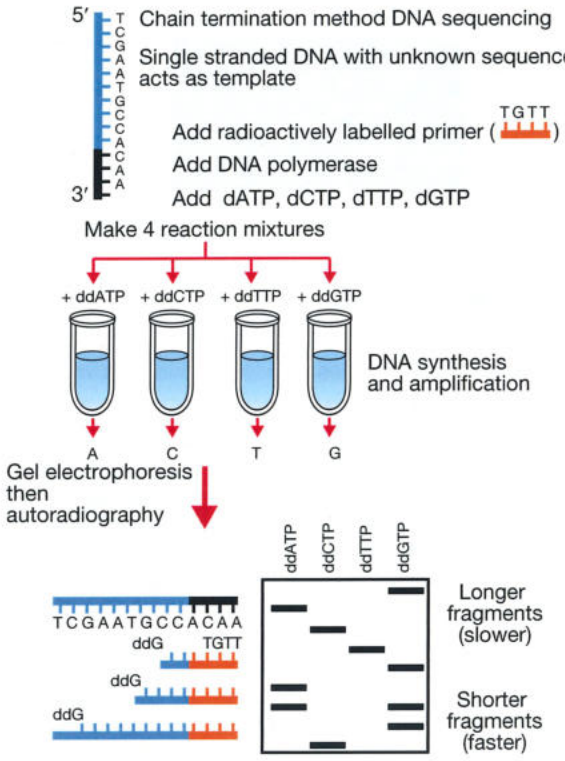


Figure 70.1 Sanger method.

Sanger method

The **Sanger method** of DNA sequencing or chain termination method is a popular method developed by Frederick Sanger in 1977 and is used to determine the sequence of nitrogenous bases. The process involves four basic steps.

1. **Denaturation** – the DNA is denatured into single strands using heat.
2. **Primer attachment and extension of bases** – the gene of interest is cloned using recombinant DNA techniques and polymerase chain reaction (PCR). The DNA fragments are mixed with four nucleotides, DNA polymerase and a radioactively labelled primer that attaches to a known sequence at the beginning of the fragment. The mixture is divided between four test tubes and a different nucleotide is added to each.
3. **Termination** – replication in the test tube stops when a replication terminating nucleotide, e.g. dideoxynucleotide is added and is inserted into the growing polynucleotide chain. This results in a mixture of DNA copies of different lengths.
4. **Gel electrophoresis** and **Southern blotting** are then used to separate and show the resulting sequences. Analysis of the sequence allows you to determine the original template.

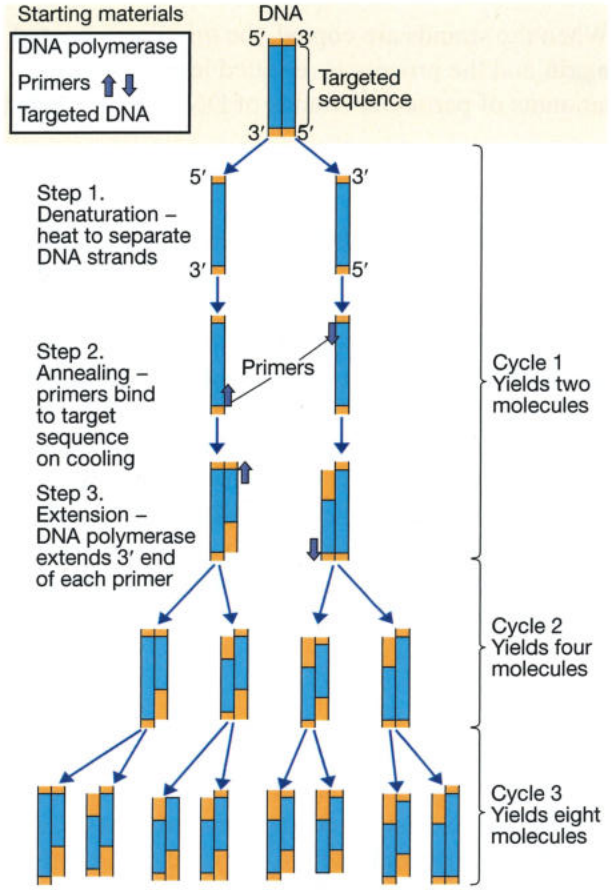


Figure 70.2 Polymerase chain reaction.

Polymerase chain reaction

Polymerase chain reaction (PCR) is a method used to amplify a specific DNA sequence by repeated cycles of synthesis using the enzyme DNA polymerase and specific primers. **DNA polymerase** is an enzyme that catalyses the elongation of new DNA at the replication fork by adding DNA nucleotides to an existing chain, e.g. in DNA replication.

Thermophilic bacteria that live in hot springs provide the DNA polymerase for the polymerase chain reaction. As high temperatures cause DNA to unwind into single strands the discovery of these bacteria was extremely important as the enzymes in these bacteria do not denature at high temperatures. The bacterial enzyme is used in PCR to catalyse the formation of more DNA from the nucleotides present and the DNA fragment that needs to be copied.

The basic steps in PCR are as follows.

1. **DNA is heated** to separate the double helix into single strands.
2. The mixture is cooled to allow the primers (short pieces of synthetic single stranded DNA) to bind to the ends of the genes.
3. **DNA polymerase** and free nucleotides are added and the DNA polymerase copies the DNA strands, starting with the primers.
4. When the strands are copied, the mixture is **heated again** and the process is repeated leading to large amounts of particular strands of DNA being created.

Gel electrophoresis

Gel electrophoresis uses short fragments of DNA that have been cut using restriction enzymes. The fragments are placed at one end of a gel layer coated onto a glass plate. There are electrodes at each end of the gel and when the electrical current is switched on the fragments move through the gel with distance travelled dependent of fragment size – short fragments move farther and fastest. When the voltage is turned off the fragments stop moving. A standard mixture is provided in its own lane to allow comparison. A fluorescent dye or radioactive probes are added so the result can be studied under UV light or by X-ray respectively.

Gene probes

Gene probes are usually short sequences of DNA which are complementary to a mutated sequence that causes a particular disease. In a genetic test, the probe will bind to a mutated sequence and flag the presence of the disease-causing sequence. **Oligonucleotides** are short segments of RNA or DNA that are very useful as genetic probes as they readily bind to their complement. They are useful as probes as they can be chemically synthesised to match a region where a known mutation occurs. They are also used as primers in PCR.

There are several limitations and problems associated with the use of genetic probes. For example, if a young person, who has a strong family medical history of an adult onset disorder (e.g. Huntington's disease, Alzheimer's disease) or some cancers (e.g. breast cancer gene *BRCA1*) and the result is positive, it does not necessarily mean the young person will definitely develop the disorder in later life. The test gives a probability or likelihood for the development of the disorder. Receiving the genetic information at a young age may require psychological adjustment and counselling especially if there are no therapies or diagnostics for treatment of that particular disorder.

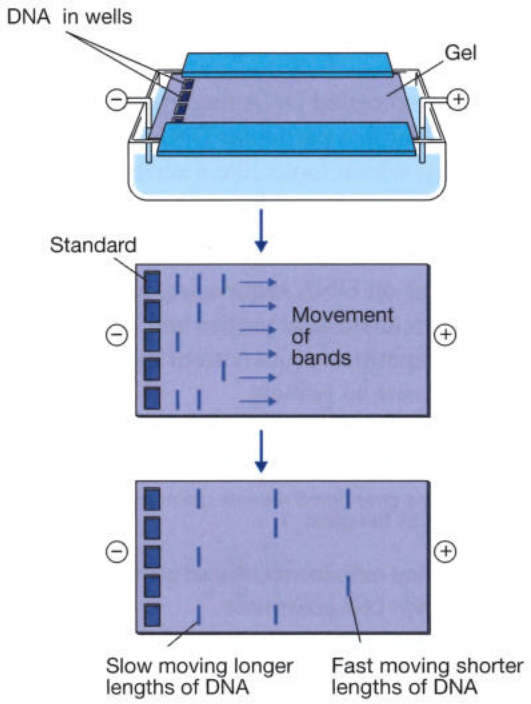


Figure 70.3 Gel electrophoresis.

Table 70.1 Different technologies.

Technology	Features
Restriction fragment length polymorphism (RFLP) analysis	Uses restriction enzymes to cut restriction fragments which are separated by gel electrophoresis. The Southern blot test is used and when the result is exposed to X-ray film the dark bands show the presence of different repeat sequences of nucleotides.
Short tandem repeat (STR) analysis	Based on PCR, STR analysis is useful when comparing unrelated individuals. Gel electrophoresis and capillary electrophoresis are used to separate the DNA fragments.
Amplified fragment length polymorphism (AmpFLP) analysis	AmpFLP analysis uses PCR to amplify DNA samples and variable number tandem repeat (VNTR) polymorphisms to identify different alleles using an allelic ladder on a polyacrylamide gel.

QUESTIONS

1. Define DNA profiling.
2. (a) What is the Sanger method?
   (b) Identify the four main steps in the Sanger method.
3. What is the polymerase chain reaction?
4. Why was the discovery of thermophilic bacteria important?
5. What is DNA polymerase?
6. Outline the basic steps in PCR.
7. Describe gel electrophoresis.
8. What are gene probes and oligonucleotides?
9. Identify one problem with using gene probes.
10. Identify some different types of fragment analysis techniques used in profiling.

71 Implications of Genetic Testing

Genetic testing and genetic screening involve the use of the same testing procedures, e.g. using recombinant DNA technologies, DNA sequencing, gel electrophoresis and chromosome microarray testing.

Genetic screening is done for a particular condition in individuals, groups or populations without family history of the condition while **genetic testing** is done for a particular condition where an individual is suspected of being at increased risk due to their family history or the result of a genetic screening test.

Ethical issues

There are both advantages and disadvantages to genetic testing. The decision making process is not always a straightforward process. Genetic tests, e.g. prenatal diagnostic testing can identify if a baby has a particular genetic condition. Parents can then receive genetic counselling and have an opportunity to discuss the result and receive assistance in their decision making. This can lead to ethical issues for the parents and their families as they have choices in how they wish to proceed.

Issues also arise as some genetic tests may not provide all the needed information, giving a limited perspective of the situation, e.g. cystic fibrosis (CF) carrier testing can identify couples who are both carriers but the presence of two CF mutations does not exactly predict the severity of CF in the baby. This means couples need to make decisions based on partial knowledge of the situation and they can have moral and ethical concerns about the situation.

There are several limitations and problems associated with the use of genetic probes. For example, there are commercialised tests for adult-onset disorders such as some cancers (e.g. breast cancer gene *BRCA1*) and Alzheimer's disease. If a young person, who has a strong family medical history of one of these diseases, takes the test and the result is positive, it does not necessarily mean the young person will definitely develop the disorder in later life. The test gives a probability or likelihood for the development of the disorder. If the result is positive there can be problems associated with psychological adjustment and there can be problems if the gene is identified and there are no therapies or diagnostics for treatment.

Genetic testing brings about several privacy and confidentiality issues. Confidentiality is needed for the health care issues and to prevent genetic discrimination, e.g. in gaining insurance coverage, employment and activity participation. This means that policy makers, clinicians, patients and the general public need to be aware of the accuracy, limitations, social issues and ethical issues in genetic testing.

Sex selection issues

The use of sex selection raises several social concerns. In Australia it is illegal for sex selection unless there is a medical reason, e.g. if there is a risk of the child having a genetic disorder and the NHMRC has ethical guidelines on the use of assisted reproductive technology. Sex selection creates moral, social and economic problems. Some cultures have a preference for sons which leads to discrimination against females and can lead to a distortion of the males to females ratio in the population. This can affect employment, crime rates, social attitudes and demographics.

Human evolution

Genetic testing could have an impact on human evolution as it can affect allele frequencies in the gene pool. In many Western societies infertile couples can use a variety of reproductive technologies which means their genes will stay in the gene pool and be passed on to the next generation.

Modern medicine has reduced infant mortality and increased life expectancy with a reliance on effective drugs such as antibiotics to fight infections and medical procedures, e.g. renal dialysis for kidney failure and supplementary hormones, e.g. insulin for diabetes. Harmful genes are no longer removed from the gene pool and can even increase in frequency in some populations depending on the number of children born to the people carrying these genes. This can create moral, social and ethical issues for the families and communities involved in the situation.

There are some localised studies that link human evolution with fertility-related natural selection. Some research projects have found a link between a woman's fertility and particular features, e.g. a woman's height, blood pressure and amount of body fat and cholesterol is linked to the number of children she has. Using these results they have predicted how allele frequency will change very slowly over time for these traits.

QUESTIONS

1. Distinguish between genetic screening and genetic testing.
2. Prenatal screening tests may identify a baby at being at an increased risk of having a particular problem. Explain why ethical issues can arise with this type of testing.
3. Use an example to discuss how finding out that a person has a variation in a gene involved in a particular condition does not always relate to how the person is, or will be affected by that condition.
4. Explain why genetic testing can lead to sex-selection issues.

72 The Human Genome Project

The Human Genome Project (HGP) was a program which wanted to map all the genes in the human genome. The human genome contains about three billion base pairs in a haploid nucleus (23 chromosomes).

The idea of working out the complete array of DNA sequences for all human chromosomes was conceived in the 1980s after Frederick Sanger developed DNA sequencing techniques in the mid 1970s. The DNA sequencing process was automated in the 1980s which meant working out the human genome was now a feasible goal. The HGP could not be achieved by studying linkage maps as these maps only give relative locations of genes rather than absolute position. Another reason why linkage maps could not be used to determine the human genome is that linkage maps are created using the three-factor cross principle and you cannot use humans in such breeding experiments.

An early genome project was set up in 1987 by the United States Department of Energy. Many new techniques were devised during the course of the HGP, e.g. the use of the polymerase chain reaction, restriction fragment length polymorphisms, and pulsed field gel electrophoresis. Many methodologies also needed to be automated. These technologies enabled the DNA sequencing process to be cheaper and faster.

During the project genetic maps were constructed to trace the inheritance of diseases, then physical maps of chromosome DNA were made to show the loci of specific genes and then nucleotide-by-nucleotide sequences of the whole genome were compiled.

The complete human genome sequence was finished in 2003.

The DNA sequences are stored in databases on the Internet and computer programs are being written to analyse the data. The information is freely available.

Benefits of HGP

Many of the benefits of the HGP will occur when researchers who are using the data from the HGP enable physicians to be able to diagnose, treat and prevent disease. Knowing the DNA sequence is the first step. The data has already been used to create genetic tests that indicate a predisposition for particular diseases, e.g. breast cancer, cystic fibrosis and some haemostasis disorders. Understanding gene expression and how phenotype is influenced by the DNA sequencing will provide the genetic basis of health. Preventive medicine will benefit from knowledge of an individual's genome, e.g. diet and lifestyle can be structured to maintain health for a particular individual. Identifying the genes related to particular diseases, e.g. asthma, diabetes, specific cancers will benefit researchers trying to find effective drugs, vaccines and other treatments for these diseases.

One benefit of the HGP that has aided areas besides medical research is computing. To hold and analyse the large volume of data encoded in the human genome computing technology rapidly advanced.

Limitations of HGP

The HGP aimed to map the human genome and this has been achieved. However, it is the next stage of research that will provide the main benefits. Understanding gene expression and how this influences the health of an individual is needed to successfully treat many medical conditions. The HGP has raised many legal, ethical and social questions. For example, DNA information is now used in law courts in forensics to identify criminals, but can the DNA information of an individual be used for insurance purposes or by employers to discriminate against particular people? Some people are concerned that the data collected by the Human Genome Project could be used in eugenics programs to selectively breed people with 'better genes'. Other people are concerned that the development of technology in this area could lead to genetic engineering of humans to 'improve' the gene pool of the population. Adolf Hitler had a eugenics program to produce a 'master race'.

QUESTIONS

1. What is the Human Genome Project?
2. How many base pairs are there approximately in a haploid nucleus?
3. Differentiate between haploid and diploid.
4. Use an example to show how technological advances influenced the Human Genome Project.
5. When was the complete human genome finished?
6. Outline why linkage maps could not be used to determine the human genome.
7. Describe one limitation of the Human Genome Project.
8. Describe one benefit of the Human Genome Project.
9. The collection of the data involved in the Human Genome Project raises questions about intellectual property rights. Identify possible groups/individuals who could claim intellectual property rights of this information.
10. Identify some steps that can be taken to protect individuals from the misuse of genetic information gathered in the Human Genome Project.
11. Outline why some people are concerned that data collected in the Human Genome Project could be used in eugenics programs.
12. Assess how advances in bioinformatics and widespread international cooperation led to the Human Genome Project being completed ahead of schedule.

Topic Test

Section A – Multiple Choice (35 marks)

1. The diagram shows two stages in mitosis.

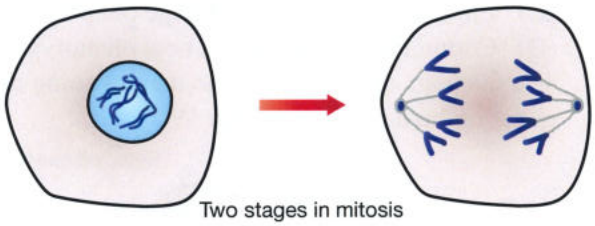


Figure TT.1 Two stages of mitosis.

What has happened *between* these two stages?
(A) Cytokinesis has occurred.
(B) The nucleoli has disappeared.
(C) Chromatids lined up at the equator.
(D) The spindle fibres have lengthened.

2. The diagram shows a model of mitosis made by a student.

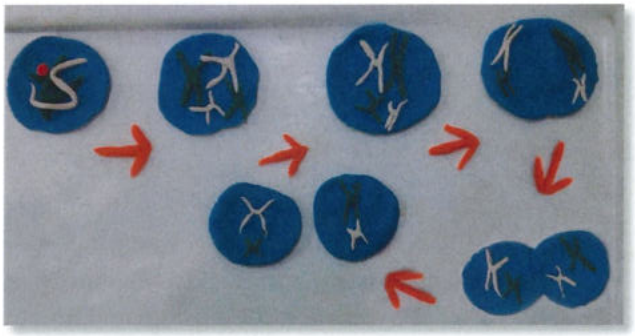


Figure TT.2 Student model of mitosis.

Which of the following statements would the teacher have probably made about this model?
(A) It is a poor model as it does not show chromosomes.
(B) It is a poor model as the chromatids have not separated.
(C) It is a good model as it shows the production of two identical daughter cells.
(D) It is a good model as it shows metaphase with chromatids lined up at the equator.

3. What is the longest phase of the cell cycle?
(A) Mitosis (B) G_1 phase
(C) Interphase (D) Cytokinesis

4. In which organelle(s) can you find DNA?
(A) Nucleus only.
(B) Nucleus and mitochondria.
(C) Nucleus and Golgi bodies.
(D) Nucleus and endoplasmic reticulum.

5. What are histones?
(A) Small proteins that bind to DNA.
(B) A chemical that causes dilation of blood vessels.
(C) An iron-containing pigment that carries oxygen.
(D) Proteins that coat fats for lipid transport.

6. How does DNA appear in humans as compared to bacteria?

	Human DNA	Bacterial DNA
(A)	Unbound, circular DNA	Bound to proteins, circular DNA
(B)	Double stranded DNA bound to protein	Unbound circular DNA
(C)	Unbound, double stranded DNA	Double stranded, circular DNA
(D)	Double stranded circular DNA	Unbound linear DNA

7. The diagram shows data collected by a relative new technology.

Figure TT.3 Data collected.

What is this technology and what is shown?
(A) Ultrasound checking foetal development.
(B) Ultrasound checking knee ligament tearing.
(C) X-ray checking very early foetal development.
(D) X-ray checking knee ligament and tendon tearing.

8. What sequence contains the genetic information for different amino acids in the DNA molecule?
(A) Three bases.
(B) Bonds joining six bases together.
(C) Three sugars and three phosphates.
(D) A short section of DNA, which can be of different length depending on the characteristic.

9. Before undergoing meiosis a cell contained the following section of DNA.

```
C A T G G A
| | | | | |
G T A C C T
```

Assuming there was no crossing over, which of the following is most likely to appear in one of the gametes?

(A) C A T G G A (B) C T T C G T

(C)
```
C A T G G A
| | | | | |
G T A C C T
```
(D)
```
C A T G G A
| | | | | |
C A T G G A
```

Use the following diagram for the next TWO questions.

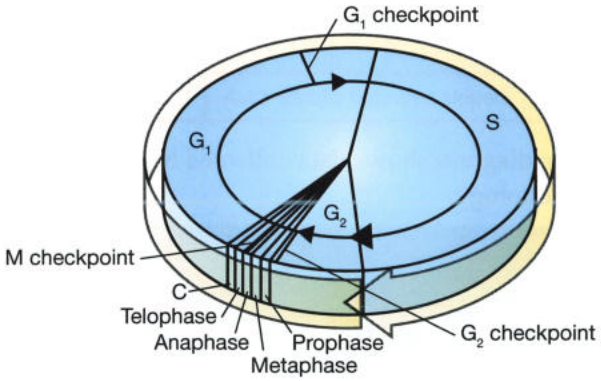


Figure TT.4 Cell cycle.

10. At which checkpoint is DNA damage checked to see if sister chromatids are correctly attached?
(A) S checkpoint (B) M checkpoint
(C) G_1 checkpoint (D) G_2 checkpoint

11. In which phase is DNA replicated and the amount of DNA in the nucleus doubles?
(A) G_1 phase (B) G_2 phase
(C) Mitosis (D) S phase

12. The diagram shows a foetus just before birth.

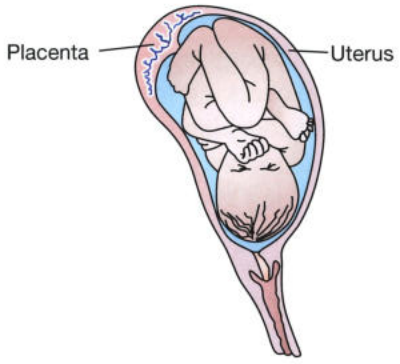


Figure TT.5 Foetus just before birth.

What is the first stage of labour?
(A) Delivery of child.
(B) Expulsion of placenta.
(C) Release of amniotic fluid.
(D) Dilation of cervix forming birth canal.

13. Which of the following cannot cross the placenta?
(A) Glucose. (B) Oxygen.
(C) Red blood cells. (D) Rubella virus.

14. When does Mendel's second law of independent assortment not apply? When:
(A) There is incomplete dominance.
(B) Both genes are on the same chromosome.
(C) Environmental factors change phenotype.
(D) Codominance produces a new phenotype.

15. The diagram shows the genetic map of some of the genes on human chromosome 19.

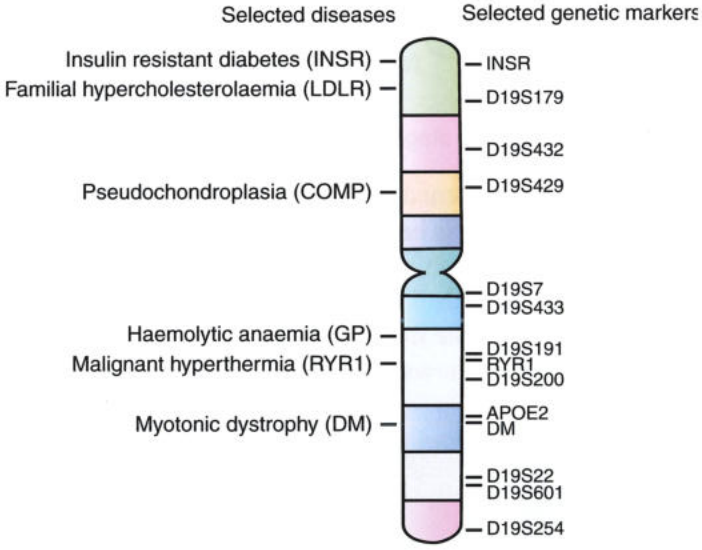


Figure TT.6 Chromosome 19.

If crossing over occurs, which of the following genes are most likely to remain together in most instances?
(A) Insulin resistant diabetes and myotonic dystrophy.
(B) Familial hypercholesterolaemia and myotonic dystrophy.
(C) Pseudochondroplasia and myotonic dystrophy.
(D) Malignant hyperthermia and myotonic dystrophy.

16. The diagram shows a condition known as syndactyly.

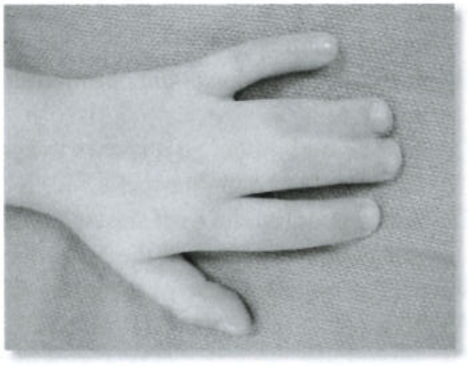


Figure TT.7 Syndactyly.

What is the most likely cause of this condition?
(A) Genetic mutation on the X chromosome.
(B) Genetic mutation on the Y chromosome.
(C) Additional localised tissue differentiation.
(D) Lack of apoptosis during limb development.

17. What is the name of the process that causes cells with the same genotype to become specialised with distinct structures to carry out particular functions?
(A) Transcription and RNA processing.
(B) Directional selection.
(C) Differential gene expression.
(D) Tissuisation.

18. What can lead to variations in genotype?
(A) Meiosis.
(B) Fertilisation.
(C) Mutation.
(D) All of the above.

19. Trisomy 18 or Edwards syndrome is a chromosome disorder usually caused by non-disjunction. Which of the following karyotypes shows a person who has Edwards syndrome?

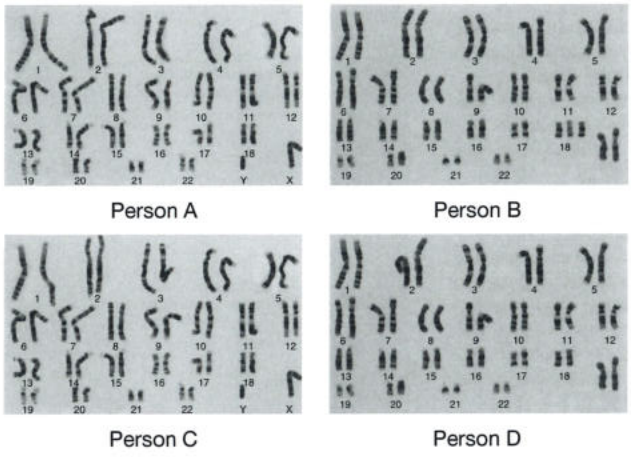


Figure TT.8 Four karyotypes.

20. In humans thick lips are dominant over thin lips.

Figure TT.9 Thick lips and thin lips.

If a man with thick lips has a child with thin lips, what conclusion can be drawn about the mother?
(A) She must have thin lips.
(B) She must have thick lips.
(C) She could have thin lips or thick lips.
(D) She must be homozygous for this trait.

21. At which stage of the ovarian cycle is the corpus luteum active?
(A) During the menstrual period days 1 to 5.
(B) During follicular phase days 5 to 12.
(C) During ovulation days 12 to 16.
(D) Post ovulation days 16 to 21.

22. If a person is heterozygous for a trait this means that:
(A) They will show the recessive phenotype.
(B) They will show the dominant phenotype.
(C) None of their children will inherit the recessive allele.
(D) They will always show a third phenotype different to either parent.

23. Which chemical is responsible for adding nucleotides to the end of new DNA strands during DNA replication?
(A) Primer. (B) Primase.
(C) DNA helicase. (D) DNA polymerase.

24. In 1909 Archibald Garrod suggested that inherited diseases relate to a person's inability to make a particular enzyme, giving an example of alkaptonuria – where urine containing alkapton turns black when exposed to air. Normal individuals have an enzyme which breaks down alkapton. Garrod's work was ahead of its time. Which of the following is the best explanation of why an inherited disease prevents a person making a particular enzyme?
(A) Their DNA does not contain the code for a particular amino acid.
(B) Individuals with this disease were exposed to exogenous mutagens.
(C) Sections of DNA translocate from one chromosome to another.
(D) Individuals with this disease had a spontaneous mutation during childhood.

25. The diagram shows a family tree.

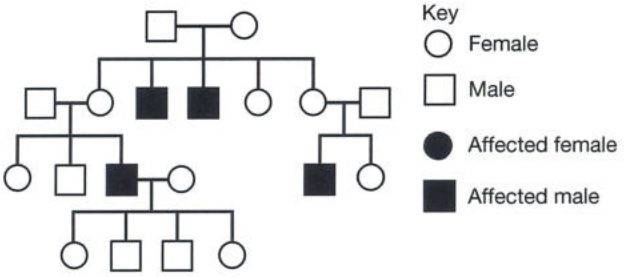


Figure TT.10 Family tree.

What type of inheritance is best represented by this family tree?
(A) Sex-linked dominant. (B) Sex-linked recessive.
(C) Autosomal dominant. (D) Autosomal recessive.

26. When conducting an investigation, which of the following is true for experimental error?
(A) Multiple trials will reduce the impact of experimental errors on the final result.
(B) Making sure all instruments are correctly zeroed prevents experimental errors from happening.
(C) Ensuring there is an independent variable and a dependent variable reduces experimental error.
(D) Having a very large number of controlled variables eliminates experimental error.

27. What is the function of the umbilical artery?
(A) Connect maternal blood to foetal blood supply.
(B) Carry deoxygenated, nutrient depleted blood from the foetus to the placenta.
(C) Carry oxygenated, nutrient rich blood from the placenta to the foetus.
(D) Carry oxygenated, nutrient rich blood from the foetal heart to other body areas.

28. The diagram shows two techniques used in reproductive technologies.

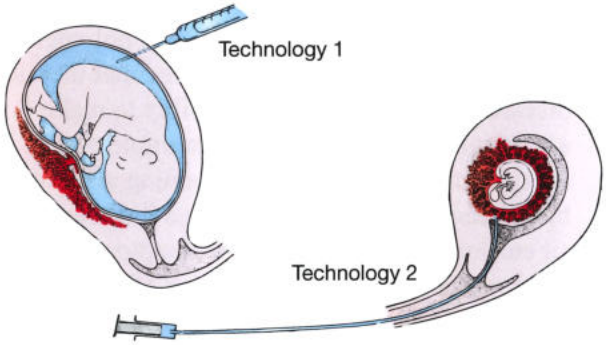


Figure TT.11 Two reproductive techniques.

Identify each of these technologies.

	Technology 1	Technology 2
(A)	Artificial insemination	Ultrasound
(B)	Amniocentesis	Chorionic villus sampling
(C)	Ultrasound	Amniocentesis
(D)	Chorionic villus sampling	Artificial insemination

29. The diagram shows the karyogram for a normal male.

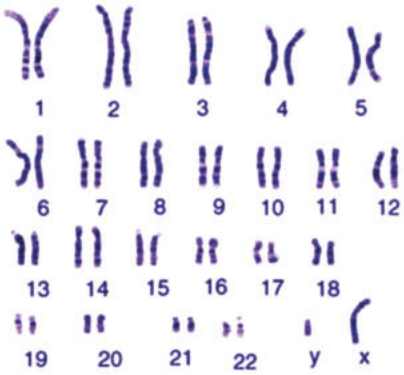


Figure TT.12 Karyogram of normal male.

How would the karyotype differ if the male had Klinefelter syndrome?
(A) There would be an extra X chromosome.
(B) There would not be an X chromosome.
(C) There would be an extra chromosome 21.
(D) There would be an extra Y chromosome.

30. What is a result of the Human Genome Project?
(A) Faulty genes can be repaired.
(B) Genetic diseases can now be cured.
(C) The genes in all 46 chromosomes have been mapped.
(D) Genetic codes can be identified for all proteins and lipids.

31. Why is it important for the chromosome number to be halved in meiosis?
(A) To ensure new cells for growth and repair can produce more daughter cells.
(B) To maintain the correct chromosome number in offspring.
(C) To allow crossing over to adjust gene combinations.
(D) To allow doubling of the DNA in the S phase.

32. The diagram shows the relationship between Down syndrome and the mother's age

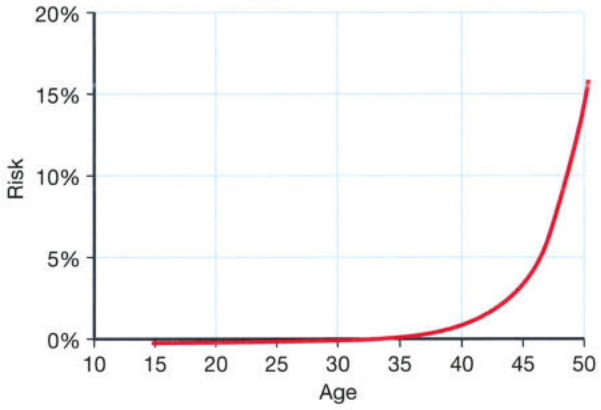


Figure TT.13 Mother's age and risk of Down syndrome.

At what age of the mother is there a 10% risk that the child will have Down syndrome?
(A) At 47 years old (B) At 40 years old
(C) At 50 years old (D) At 20 years old

33. If a person is heterozygous for a trait:
(A) Their offspring cannot receive a recessive allele.
(B) Their offspring will also be heterozygous.
(C) Their phenotype will not show the dominant trait.
(D) They will be a carrier for a recessive allele.

34. The diagram shows a type of graph.
Name this type of graph.
(A) Column graph.
(B) Bar graph.
(C) Histogram.
(D) Block graph

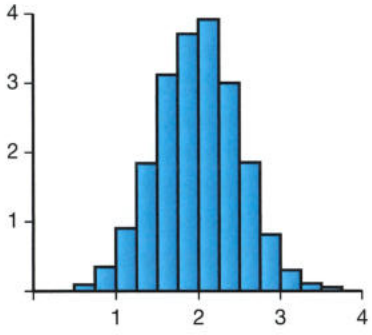


Figure TT.14 Type of graph.

35. The diagram shows a device used by some people.

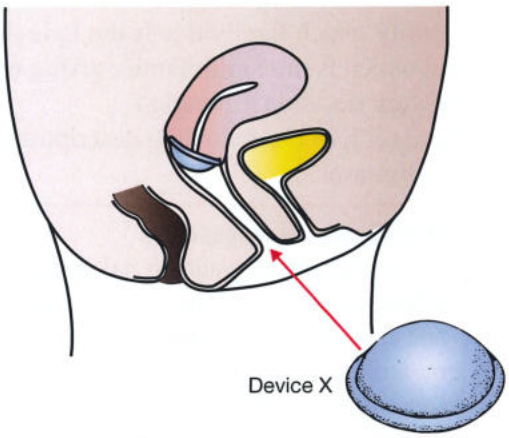


Figure TT.15 Device X and its use.

What is device X and its use?

(A) An IUD to function after coitus.
(B) A female condom to function after coitus.
(C) A chemical spermicide to immobilise sperm.
(D) A diaphragm forming a physical barrier between gametes.

Section B – Written Response (65 marks)

36. The diagram shows the cell cycle.

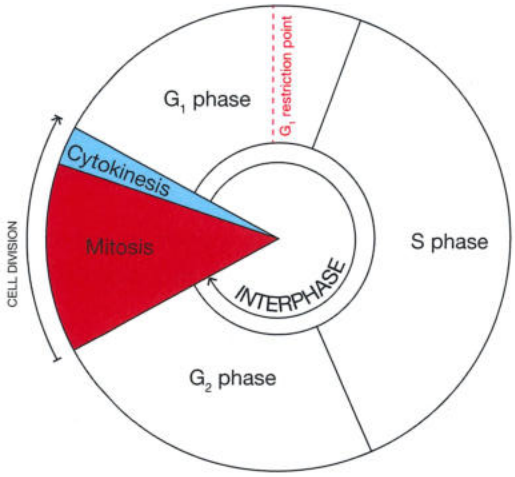


Figure TT.16 Cell cycle.

(a) What is the cell cycle and what does it consist of? (2 marks)
(b) Outline the significance of the G_1 checkpoint. (2 marks)

37. Phenylketonuria (PKU) is a genetic disorder that prevents the normal breakdown of the protein phenylalanine that is found in some foods.
(a) PKU is an autosomal recessive condition. What does this mean? (2 marks)
(b) Explain why a newborn child that is screened and found to have PKU is put on a special restricted diet. (3 marks)

38. (a) Define homologous chromosomes. (1 mark)
(b) When do homologous chromosomes pair in meiosis? (1 mark)
(c) Draw a labelled flow diagram to show crossing over in meiosis. (3 marks)

39. The human ABO blood group system is an example of codominance and multiple alleles. The table shows inheritance of different ABO blood group alleles.

Table TT.1 Table for human ABO blood groups.

Blood group	Allele	Dominant/recessive
A	Allele I^A	Codominant with I^B
B	Allele I^B	Codominant with I^A
O	Allele i	Recessive to both I^A and I^B

(a) Explain what is meant by codominance and multiple alleles. (2 marks)
(b) Use a Punnett grid to explain how it is possible for two parents, one with blood type A and one with blood type B to have four children each with a different blood type. (2 marks)

40. Copy the diagram which shows the process of oogenesis. (6 marks)

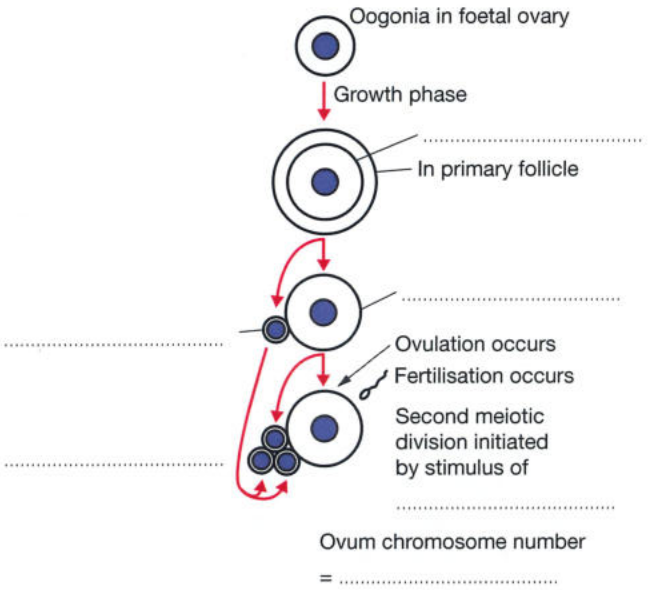


Figure TT.17 Oogenesis.

Complete the diagram by filling in the missing labels.

41. Distinguish between genetic screening and genetic testing. (2 marks)
42. Construct a table to compare the causative agent, effects and treatment and control of two common STIs. (8 marks)

43. The diagram shows how metastasis can occur from the occurrence of an abnormal cell in breast tissue.

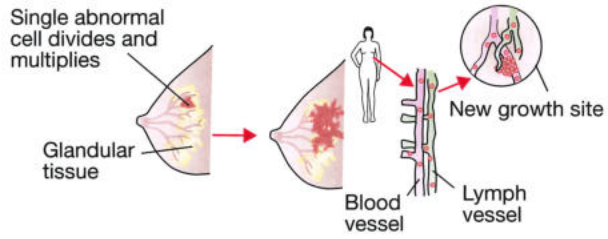


Figure TT.18 Breast cancer.

Use the diagram to explain how metastasis has occurred and how the problem could have been detected earlier. (4 marks)

44. The diagram shows foetal circulation.

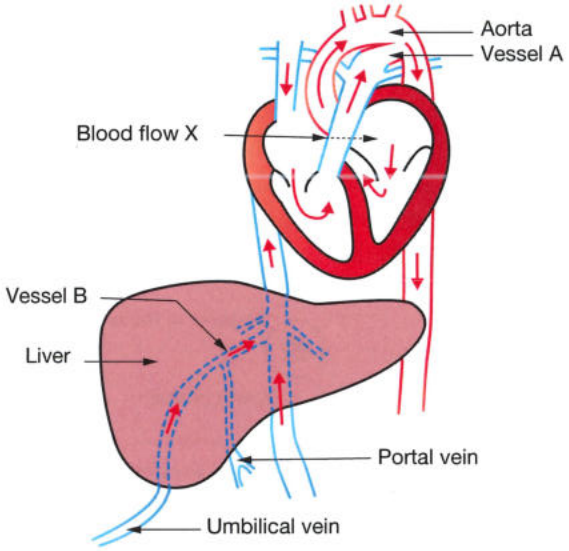


Figure TT.19 Foetal circulation.

(a) Identify vessel A and vessel B. (2 marks)
(b) Explain what is happening with blood flow X. (4 marks)

45. A person constructing family trees for two families forgot to label the trees. The Smith family had a history of Huntington's disease and the Lyle family had a history of phenylketonuria (PKU).
The diagram shows the two trees.

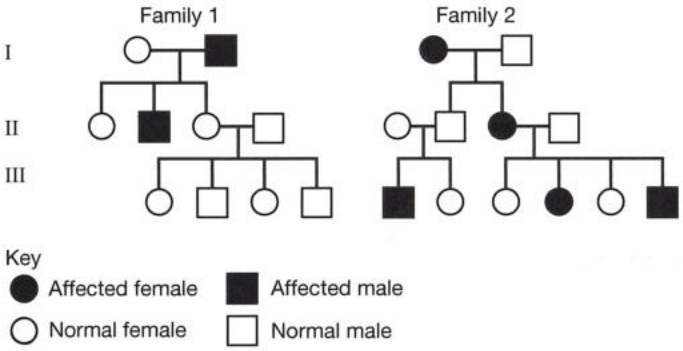


Figure TT.20 Two family trees.

(a) If Huntington's disease is caused by a dominant allele and PKU is caused by a recessive allele, identify which family tree is the Lyle family and which is the Smith family giving reasons for your decision. (2 marks)
(b) Read each of the following descriptions of the two diseases.

> **Huntington's disease** is a neurodegenerative autosomal disease that affects muscle coordination causing jerky and random movements and also causes mental decline with reduced cognitive functions, memory deficiency leading to dementia. Symptoms usually begin between ages 35 to 44 years old.

> **Phenylketonuria** is an autosomal genetic disorder that can lead to mental retardation and seizures if left untreated. Screening newborns for PKU, e.g. with the Guthrie test can identify babies with the disease. Once identified the child can go on the 'diet for life' restricting foods high in phenylalanine, e.g. meat, eggs, nuts and dairy products.

(i) The female in generation III who has the genetic disease has requested genetic counselling. If her partner has no family history of this disease, what is the chance of their children showing the disease? (3 marks)
(ii) Explain why a disease such a Huntington's disease can stay in a population? (3 marks)

46. Evaluate the use of the Human Genome Project and new technologies to detect harmful genetic diseases. (5 marks)

47. Construct a table to show why and when ultrasound, amniocentesis and chorionic villus sampling are used to monitor the foetus within the womb. (8 marks)

Answers

1 Assumed Knowledge

1. (a) Mitosis is a process during cell division in which the cell nucleus divides into two.
   (b) Meiosis is cell division to produce haploid daughter cells.
   (c) Meiosis produces four daughter cells.
2. Information is transferred as DNA on chromosomes when cells reproduce themselves.
3. DNA stands for deoxyribose nucleic acid.
4. The basic unit of DNA is the nucleotide.
5. Most DNA is located in the nucleus. DNA is also found in mitochondria.
6. The structure of the DNA molecule is a double helix.
7. A gene is a certain length of DNA that has the code for one characteristic.
8. DNA needs to be able to replicate itself exactly so that cell division can form identical new cells for growth, repair and maintenance of the body of a multicellular organism. Exact replication is also needed to maintain the genetic code for a species and hence keep its integrity as a distinct unit in nature.
9. A mutation is a change in the chemical structure of the DNA.
10. A pedigree is a graphical way of picturing the ancestry of living things. It shows genetic history.
11. System A is the female reproductive system and system B is the male reproductive system.
12. Biotechnology is the use of biological processes by industry or agriculture to change organisms in order to produce useful products or provide services.
13. A somatic cell is a body cell.
14. A stem cell is a relatively unspecialised cell that can divide by cell division to produce identical daughter cells and to differentiate to form different specialised cells.
15. A germ layer is a group of cells in an embryo that will give rise to tissues and organs of the body.
16. Human gestation is on average 266 days (approximately 38 weeks). Since timing is related to the menstrual cycle it is 40 weeks since the start of the last menstrual cycle.
17. Parturition means giving birth to a baby.
18. It is a sperm and it is the male reproductive sex cell (gamete).
19. An embryo is the early stage of development and in humans after 6 to 8 weeks when all the organs have formed the embryo is called a foetus.
20. A mutagen is a chemical or physical agent that causes a mutation changing DNA and the genetic information.
21. Gregor Mendel experimented with pea plants and worked out the basic laws of inheritance. His work led to the study of genetics and hence he is often referred to as the 'father of genetics'.
22. Both genes and environmental factors determine the features of an organism.
23. Many different examples can be used to show how the environment influences the appearance of a person. If a person changes their diet, the amount and type of exercise they do every day and their exposure to UV sunlight their appearance can change with body shape, weight, muscle tone and skin colour with sunburn and tanning.
24. Studying eye colour with two alleles for a particular gene – brown eyes (B) and blue eyes (b). Given that all other environmental factors are the same, a person with the genetic code BB or Bb will have brown eyes, while a person with the code bb will have blue eyes.
25. Watson and Crick discovered that DNA had a double helix structure.
26. The genome is the complete genetic information of an organism.
27. A chromosome is a cellular structure that holds genetic information in the coding of the DNA molecule.
28. Genotype is the genetic make-up of an organism, or a set of alleles of an organism.
29. Fertilisation is the union of two gametes.
30. Gametes fuse to form a zygote. It is essential that gametes contain only half the number of chromosomes to maintain the chromosome number of the species. Otherwise the number of chromosomes would double every generation.
31. Part A is the ovary which produces eggs (female gamete) and part B is the uterus which is where the embryo/foetus will grow and develop until birth.

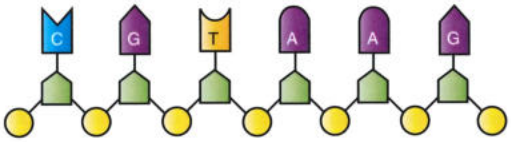
2 Discovering DNA

1. Nucleic acids were first identified in 1869 and were called 'nuclein' as they came from the nucleus.
2. When Frederick Griffith mixed the remains of dead pathogenic (disease-causing) bacteria with living non-pathogenic bacteria he found that some of the living bacteria transformed into pathogenic bacteria and that their offspring were pathogenic. He called the phenomena bacterial transformation.
3. (a) A nucleotide is the basic unit of DNA and consists of a phosphate, sugar and base.
   (b) Phoebus Levene named nucleotides.
   (c) When nucleotides were first named it was commonly believed that the carrier of genetic information was proteins.
4. William Astbury used X-ray diffraction to study DNA molecules and found that the structure repeated every 2.7 nanometres and that the bases were arranged in flat stacks that were 0.34 nanometres apart. Data like this is crucial in determining the structure of a molecule as it provides the dimensions for any proposed model.
5. Beadle and Tatum carried out experiments with the bread mould *Neurospora* and identified the link between genes and enzymes proposing the one gene – one enzyme hypothesis.
6. Chargaff's rules for the composition of DNA are: 1. In natural DNA the number of guanine units is the same as the number of cytosine units and the number of adenine units is the same as the number of thymine units. 2. The number of units of adenine, thymine, cytosine and guanine units is different for different species.
7. The Hershey-Chase experiment used two sets of bacteriophages, one with proteins labelled with radioactive sulfur (using ^{35}S) and one with DNA radioactively labelled with phosphorus (using ^{32}P) to infect *Escherichia coli* which is a bacteria that lives in the intestines of mammals. They found that the radioactively labelled DNA entered the bacteria cells while the radioactively labelled proteins did not enter the bacteria cells. This confirmed that DNA and not protein was the genetic material.
8. Rosalind Franklin took X-ray diffraction images of DNA that clearly showed the helical structure of DNA. The evidence in her photograph was used by Watson and Crick to finally determine the double helix structure.
9. (a) In 1953, Watson and Crick suggested the double helix structure of DNA with two phosphate-sugar strands winding around the outside of nitrogenous bases with the pairs A-T and C-G. Rosalind Franklin took X-ray diffraction photographs of DNA. Maurice Wilkins was Franklin's colleague and showed her X-ray photographs to Watson. The work of all these scientists needed to be put together as each piece of information and evidence contributed to the final determination of the structure of DNA.
   (b) Watson and Crick collaborated to produce the final double helix structure and Wilkins collaborated when he showed the X-ray diffraction photographs to Watson which led to an understanding of the physical shape of the molecule. For this joint effort the three men were awarded the 1962 Nobel Prize for Physiology. Franklin took the photographs but there was friction between her and her colleague Wilkins. She died of cancer in 1958 and could not be given credit posthumously for the Nobel Prize.

10. In the early 1950s the three groups – Wilkins and Franklin at the University of London, Pauling at Caltech and Watson and Crick at Cambridge University – were all trying to determine the structure of DNA. They were competing to be the first to unravel the mystery about how genetic information is coded and able to be passed from one generation to the next. It was a scientific race. A number of scientists had collected experimental evidence and information about the structure of DNA, e.g. Astbury had determined the dimension of the repeating structure and Chargaff's rules showed there were equal amounts of adenine and thymine and equal amounts of cytosine and guanine. The three groups were trying to collect the final piece of evidence that would show how all the pieces of information fitted together. There was competition but there was also collaboration, e.g. Wilkins showed Watson and Crick Franklin's photograph and the structure of DNA was able to be finally determined.
11. (a) A bacteriophage is a virus that can infect bacteria.
    (b) Bacteriophages were used in the Hershey-Chase experiment to determine that DNA was the carrier of genetic information and not protein. Bacteriophages labelled with radioactive DNA ie with radioactive phosphorus (^{32}P) and other bacteriophages labelled with radioactive protein ie with radioactive sulfur (^{35}S) were allowed to infect bacteria. Since bacteriophages can inject their DNA into bacteria and their protein coat stays outside the bacteria, the experiment with bacteriophages proved that it was the DNA that was injected into the bacteria to cause bacterial transformation and not the protein.
12. (a) Molecule is DNA – deoxyribose nucleic acid.
    (b) L = phosphate, M = sugar, N = nitrogenous base.
    (c) Watson and Crick.
    (d) DNA encodes a large amount of hereditary information in the sequences of the nitrogenous bases. It is able to replicate itself with the complementary pairs AT and CG. Mutations or errors can occur during replication, e.g. base-pair substitution, which is a source of genetic variation.
13. C
14. A
15. B
16. B

3 The Structure of DNA

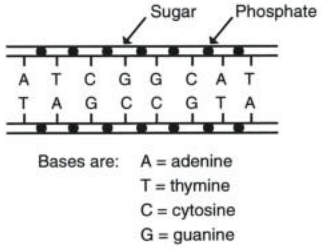
1. A gene is a certain section of DNA in a chromosome that represents a particular characteristic.
2. DNA stands for deoxyribose nucleic acid.
3. A major advance in scientific understanding occurred when in 1944 Oswald Avery demonstrated that genes were made of DNA. This directed future biological research into the structure of DNA, leading to Rosalind Franklin, with Maurice Wilkins, using X-ray diffraction crystallography in 1952 to show that the DNA molecule was helical. This major advance then led to James Watson and Francis Crick proposing in 1953 that DNA molecules had a double helix structure. The work of each scientist built upon the work of another scientist to unravel the mystery of the structure of DNA.
4. DNA is a double helix or twisted ladder shape.
5. A nucleotide is the basic unit of DNA. It consists of a sugar, a phosphate and a nitrogenous base.
6. The phosphate group gives the nucleotide its acidic properties.
7. Adenine pairs with thymine and cytosine pairs with guanine.
8. 1. DNA is able to encode a large amount of information. 2. It is chemically stable. 3. It is able to make an accurate replication of itself. 4. It controls and directs protein synthesis. 5. Occasional errors (mutations) occur.
9. Covalent bonds called phosphodiester linkages hold the sugar of one nucleotide to the phosphate group of the next nucleotide.
10. Hydrogen bonds hold the nitrogenous bases on the rungs of the ladder together.
11. The two stands of DNA are called antiparallel a one sugar-phosphate backbone runs 5′ → 3′ direction while the other strand runs 3′ → 5′ direction.
12. The hydrogen bonds holding the nitrogenous bases together on the rungs of the ladder break during DNA replication.
13. Each gene is a certain section of a chromosome which means it is a certain length of the DNA molecule. The coding of the DNA molecule passes hereditary information from one generation to the next.
14. The significance of DNA replication is that large amounts of coded information can be copied and passed onto the next generation, providing continuity of a species. The method also allows for some changes (mutations).
15. (a) Part X is phosphate, Part Y is sugar.
    (b)



    (c) The segment of DNA has the base code GCATTC. The messenger RNA uses the DNA as a template, with uracil replacing thymine, forming the code CGUAAG. mRNA moves from the nucleus to the cytoplasm and attaches to a ribosome. The sequence of codons along mRNA is translated into amino acids by tRNA, e.g. CGU forms one amino acid and AAG forms the next amino acid.
16. B
17. A
18. B
19. B
20. D
21. C
22. A

4 Activity – Making a Model of DNA

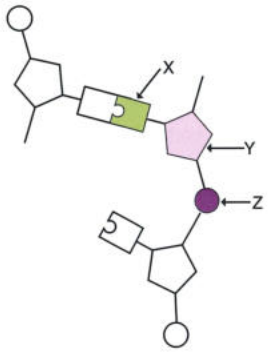
1. In eukaryotic cells each chromosome is a long linear DNA molecule. In prokaryotic cells there is a single circular DNA molecule.
2. There are many ways you can construct a model to show the structure of DNA. Models can be made of paper, cardboard, pipecleaners, plastic sticks, coloured balls or created using multimedia graphics.
3. A model of DNA must show: 1. One nucleotide of DNA must contain one phosphate, one deoxyribose sugar and one nitrogenous base, e.g. adenine (A), thymine (T), cytosine (C) or guanine (G). 2. The nitrogenous bases must also be in complementary pairs, e.g. A-T and C-G. 3. The nucleotides need to link together to form a ladder structure with the sugar-phosphates forming the sides of the ladder and the nitrogenous bases forming the rungs of the ladder.
4. The triplet code is three bases next to each other in a molecule of DNA and specifies a particular amino acid or instruction.
5. Not all triplet codes specify an amino acid. There are three codes that are 'stop' signals – DNA triplet code ATT, ATC and ACT. The DNA code TAC is the code for methionine but also a 'start' signal.
6. (a) The mRNA code is used in definitions and to identify amino acids.
    (b) The mRNA code is complementary to the DNA code.
7. A fully labelled diagram of DNA with eight base pairs:

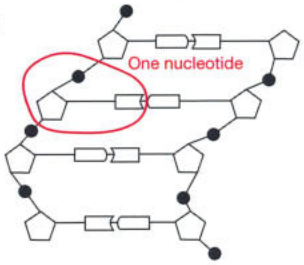


8. The data shows the pairing of the nitrogenous bases in the DNA molecule. The percentage of the bases shows that adenine only pairs with thymine and cytosine only pairs with guanine.
9. If the DNA from a turtle cell contains 30% adenine, then it must contain 30% thymine since A-T are complementary pairs. This totals 60% DNA. The rest of the DNA (40%) must be C-G. Thus guanine would be 20% DNA of a turtle cell.
10. An advantage of making a 3-D model of DNA is that you can see the double helix structure as the two strands twist around each other. A disadvantage of making a 3-D model of DNA is that it is not particularly easy to use to show DNA replication and proteins synthesis as when you try to separate the complementary nitrogenous bases to separate the two strands the structure becomes unstable and tends to fall apart.
11. Scientists construct models, e.g. Watson and Crick constructed a 3-D model of DNA as a visual representation of a molecule that could not be directly seen at that time. The model made it easier for Watson and Crick to explain their ideas to other people and to show how DNA could hold a vast amount of genetic information and could also replicate to pass information to the next generation. A 3-D model of DNA shows how each component of a nucleotide links to each other component and the model could be used to explain other processes, e.g. DNA replication and protein synthesis.
12. There are four basic nucleotides in DNA.

Nucleotide	Description
1	Phosphate – sugar – adenine
2	Phosphate – sugar – thymine
3	Phosphate – sugar – cytosine
4	Phosphate – sugar – guanine

13. (a) X = nitrogenous base, Y = sugar, Z = phosphate
    (b) Diagram with two other nucleotides attached:



14. To show the structure of DNA, you can use jigsaw cardboard cut-outs with different shapes and colours for the nitrogenous bases, adenine, thymine, cytosine and guanine and a different shape for the sugar and for the phosphate. This model is useful as it shows the pairing of the DNA bases (A-T and C-G) to form the rungs of the ladder of the double helix and it also shows how the arrangement of the sugars and phosphates forms the sides of the ladder. The model has limitations as it is only a two-dimensional model of a molecule that is three-dimensional and it does not show how the structure is twisted to form the double helix. Making a model is important as it means a complex structure, too small to be seen by the naked eye, can be simplified and studied more easily. However, relative size can be misrepresented and there is no information about the bonding between nucleotides and within the nucleotides.
15. A codon, or triplet, is the basic instruction of the genetic code consisting of three bases next to each other in a molecule of DNA or mRNA and specifying a particular amino acid.
16. The sequence of base pair coding in a section of DNA is important because the sequence of codons gives instructions for the sequence of amino acids which will be joined together to form a polypeptide. Since proteins determine the different appearance and type of cells found in the body and also determine the reactions that occur in cells a section of DNA is very important as it determines your appearance and phenotype.
17. 

18. B
19. D

5 DNA in the Nucleus and Organelles

1. Extranuclear genes are genes found in organelles in the cytoplasm of eukaryotes.
2. The universality of the genetic code means that DNA is similar in its coding of information with the nucleotide complementary pairing – adenine pairing with thymine and cytosine pairing with guanine. The triplet codes in DNA for amino acids are also universal
3. A person inherits nuclear DNA with one set of chromosomes from their father and one set of chromosomes from their mother.
4. Chromatin is the name given to the complex of DNA and protein that makes up chromosomes in the nucleus of eukaryotic cells.
5. Nucleosomes are bead-like structures with a protein core of histones and DNA wound twice around the group of histones.
6. Histones are important in the structure and packaging of chromatin in the nucleus. Without histones the DNA would be very long compared to the size of the cell. Histones also aid the ability for certain genes to be expressed in the cell while other genes stay dormant as the histone tails are accessible to the enzymes that begin gene expression and transcription to start protein synthesis.
7. Mitochondria are cylinder shaped organelles that are the site of aerobic respiration. They have a double membrane with the inner membrane infolded to form cristae.
8. mtDNA is a ring of DNA similar to DNA found in prokaryotes.
9. The endosymbiotic theory suggests that mitochondria arose as a symbiotic relationship between a free living heterotrophic bacterium with aerobic respiration that entered a larger cell – either a larger prokaryote or an eukaryote and both then existed in a mutualistic partnership.
10. Human mtDNA is inherited through the female line as the mitochondria are in the egg. This means that scientists have been able to compare the nucleotide sequences of different populations of humans to determine the possible timelines and paths of human migration across the Earth.
11.

Nuclear DNA	Mitochondrial DNA
DNA in the nucleus (nDNA) of human cells (eukaryotic cells) is bound to proteins to form bead-like structures called nucleosomes and the nucleosomal chromatin is organised to form chromosomes.	Mitochondrial DNA (mtDNA) is not bound to proteins – it lacks histones.
nDNA has two sets of each chromosome – one paternal and one maternal.	In humans mtDNA is maternal.
nDNA has many more genes in its structure than mtDNA.	mtDNA is relatively small (16 500 bases).
Has non-coding exons.	mtDNA is circular and does not have the non-coding regions (exons) found in nDNA.

12. (a) The diagram suggests that the sequence of events that led to the origin of mitochondria and chloroplasts were: 1. There was a cell with linear DNA and histone protein. 2. There was infolding of the cell membrane which led to the formation of endoplasmic reticulum and the nuclear membrane. 3. Aerobic bacteria were engulfed and retained within the cytoplasm living in a mutualistic relationship with the host cell. 4. Some of these cells went on to become eukaryotic animal cells. 5. Other cells engulfed photosynthetic bacteria which were retained in the cytoplasm living in a mutualistic relationship with the host and the host became eukaryotic plant cells.
    (b) Mitochondrial DNA and chloroplast DNA are both circular rings of DNA that are similar to prokaryotic DNA.
13. (a) The structure is a nucleosome.
    (b) Part X shows a histone tail. Part Y shows the histone protein and part Z is the double helix of the DNA molecule.
14. C
15. A
16. B
17. D
18. A
19. B
20. C

6 Genome, DNA and Bioinformatics

1. The genome is the sum total of an organism's DNA measured in the number of base pairs contained in a haploid set of chromosomes.
2. DNA stands for deoxyribose nucleic acid.
3. DNA is a double helix or twisted ladder shape.
4. A nucleotide is the basic unit of DNA. It consists of a sugar, a phosphate and a nitrogenous base.
5. Part X is phosphate, part Y is sugar.
6. The base pairs of DNA are A-T (adenine always pairs with thymine) and C-G (cytosine always pairs with guanine).
7. A genome map is less detailed than a genome sequence identifying short DNA sequences as markers in a genome. A genome sequence is the complete order of the As, Ts, Cs and Gs in the DNA code.
8. Bioinformatics uses computers, software and mathematical models to process and integrate biological information from large data sets.
9. Bioinformatics is important in genome sequencing as the computers, software and mathematical models can manipulate and analyse the large data sets of the genome sequences to help determine the genetics of disease, to understand gene expression and protein synthesis, to determine the effects of mutations, to understand the processes involved in the formation of tumours and cancers and to trace evolutionary changes in DNA and determine evolutionary relationships between different species.
10. D
11. C
12. B

7 DNA Replication

1. Watson and Crick proposed the semiconservative model of DNA replication.
2.

Model of DNA replication	Process
Semiconservative model	Each of the two daughter molecules have one old strand from the parental molecule and one newly made strand.
Daughter cells are haploid	The parental strands reassemble after forming a template that forms a new double helix.
Daughter cells are different to parent cells	Both daughter molecules contain a mixture of old and newly synthesised DNA

3. The Meselson-Stahl experiment showed that DNA replication followed the semiconservative model. In this experiment *Escherichia coli* labelled with radioactive nitrogen (^{15}N) was used to investigate the three models of DNA replication. Matthew Meselson and Franklin Stahl found that the amount of ^{15}N in the each generation after cell division and replication over several generations was consistent with the semiconservative model and not with the other two models – conservative model and dispersive model.
4. The DNA molecule unwinds and the double helix separates into two single strands. Each single strand is a template for the synthesis of a complementary strand so that adenine is opposite thymine and guanine is opposite cytosine. Two new daughter DNA molecules are produced.
5. An origin of replication site is section of DNA with a specific code where the process of replication begins. There are many origin of replication sites along a chromosome.
6. A replication 'bubble' is a section of DNA where replication has begun and the two strands have separated to form a 'bubble'.
7. Helicase enzymes cause the bonds between the two strands of DNA to break and the double helix to untwist.
8. (a) A primer is a short RNA chain with a free 3′ end that is bound by complementary base pairing to the template strand to which nucleotides are added in DNA replication.
    (b) RNA primase is an enzyme that joins RNA nucleotides to make a primer during DNA replication using the parental DNA strand as a template.
9. DNA polymerase enzymes catalyse the addition of nucleotides to the 3′ end of an existing DNA chain. They proofread each nucleotide pair and remove and replace any incorrect nucleotide.
10. The leading strand is the new DNA strand that is continuously synthesised and goes towards the replication fork. The lagging strand is discontinuously synthesised and grows away from the replication fork.
11. Okazaki fragments are short sections of DNA that are synthesised on the lagging strand and become linked together by ligase enzymes to complete a new single DNA strand.
12. DNA ligase enzymes catalyse the joining of DNA fragments joining the sugar-phosphate backbone of DNA molecule with covalent bonds.
13. The significance of DNA replication is that large amounts of coded information can be copied and passed onto the next generation, providing continuity of a species. The method also allows for some changes (mutations).
14. D
15. D

8 DNA Repairs Itself

1. DNA exposed to mutagens, radiation etc becomes damaged and needs to be repaired.
2. Two causes of DNA damage include: 1. Certain wavelengths of radiation, e.g. X-rays and gamma rays. 2. Chemical mutagens, e.g. some hydrocarbons found in cigarette smoke and nitrogen mustard (the poisonous part of mustard gas).
3. DNA can be damaged in several ways: 1. One of the nitrogenous bases in DNA is modified. 2. Mismatching of normal bases in DNA. 3. Crosslinks between DNA bases either on the same DNA strand or between different DNA strands. 4. Breaks in the backbone.
4. UV radiation can be absorbed by the bases in DNA. The changes in DNA caused by UV can lead to mutations, lesions, and can be lethal if not repaired. People with the hereditary disease xeroderma pigmentosum (XP) have XP cells with defective DNA repair. This means the sections of DNA damaged by UV are not able to be removed and replaced. These people will then have mutations, lesions, and could die if overly exposed to UV light.
5. (a) The thymine dimer distorts the DNA molecule and will block transcription and thus stop DNA replication.
    (b) Enzymes involved in the DNA repair process include a glycosylase to cut the damaged DNA, a DNA polymerase to replace the section and a DNA ligase to seal the section.

6. DNA has considerable ability to repair itself. Damage can be caused by chemical mutagens or radiation. Damage includes modification of a nitrogenous base, mismatching of normal bases, crosslinks forming between DNA bases or breaks in the DNA backbone. The damage interferes with DNA replication and can lead to mutation and diseases. To repair the damage, mechanisms include the direct reversal of the damage, excision repair where the damaged section is cut out and replaced with a correct sequence or a method is found to tolerate the damage for a period of time. Different enzymes are involved in the processes, e.g. glycosylase, DNA polymerases and DNA ligases.

9 The Beadle and Tatum Experiment

1. *Neurospora crassa* is a bread mould, a fungus with a haploid genome.
2. Beadle and Tatum produced mutant forms of *Neurospora* by exposing the wild type to radiation, e.g. X-rays.
3. The control is the normal type which can be compared to the other forms to show differences.
4. The results show that each mutant had a metabolic block at a different step in the pathway – mutant 1 did not have enzyme 1, mutant 2 did not have enzyme 2 and mutant 3 did not have enzyme 3. If each was given the chemical of the next stage, then the pathway would resume.
5. Mutations are abnormal variations of different genes and each gene caused the production of one enzyme.
6. The hypothesis was modified to the one gene-one polypeptide hypothesis.
7. Since most enzymes are proteins, the hypothesis was first modified to one gene-one protein, although not all proteins are enzymes. Since proteins consist of several polypeptide chains and each subunit has its own gene, the statement became one gene-one polypeptide.
8. In 1958 Beadle and Tatum received the Nobel Prize in Physiology or Medicine for their discovery that genes act by regulating definite chemical events.
9. Such substances as salts and a carbon source would be included in a minimal medium as the bread mould is a heterotroph, requiring a source of organic nutrients to survive. Vitamins would not be included as that was part of the control of the experiment – some of the mutants could not live without a particular vitamin or amino acid in their food source, e.g. normal mould requires only the vitamin biotin, while some mutants also required thiamine or choline.
10. C
11. B

10 Introns and Exons

1. Introns are non-coding segments of nucleic acid found in eukaryotic DNA. While exons are coding segments of nucleic acid found in eukaryotic DNA that are involved in gene expression.
2. Gene expression the process where DNA directs protein synthesis.
3. A gene is a section of DNA with the code for a particular polypeptide.
4. (a) Gene expression and protein synthesis involves the two steps of transcription and translation.
   (b) In eukaryotic cells transcription occurs in the nucleus and translation occurs in the cytoplasm.
5. The triplet code is a sequence of three nucleotides on a strand of DNA that are read as a single unit in the flow of information from gene to polypeptide.
6. The base pairing rules in RNA are cytosine pairs with guanine (same as DNA) but adenine pairs with uracil. There is no thymine in RNA.
7. RNA processing involves the modification of the RNA transcript taken from the DNA gene before it leaves the nucleus as mRNA.
8. Pre-mRNA has the primary transcript taken from the DNA gene and has both exons and introns. It is in the nucleus. mRNA has exons and a functional code for the production of a polypeptide and takes the code from the nucleus to the cytoplasm.
9. In RNA processing introns are removed and the segments of exons are spliced together to form a functional code for the production of a polypeptide.
10. B
11. A

11 Gene Expression

1. Gene expression is the process where DNA directs protein synthesis. It refers to a gene being 'turned on' to produce a polypeptide, i.e. it changes the DNA information into the functions and structures of a cell.
2. Differential gene expression occurs when different genes are expressed in different sets of cells with the same genome.
3. Most control occurs in the transcription of DNA into RNA.
4. Areas of DNA that are highly condensed and packed are genes that are switched off, sometimes permanently.
5. A promoter is a regulatory area at the start of a gene that acts as the binding site for RNA polymerase.
6. The presence of some chemical groups can block gene expression, e.g. methyl groups ($-CH_3$) while the presence of other groups can allow gene expression to occur more easily, e.g. acetyl groups (–COOH).
7. A cancerous cell will have different gene expression to a normal cell. We can use cDNA hybridisation techniques to compare the gene expression in these cells. This information shows where the transcription change(s) occurred to change the cell from normal to cancerous. This information can then be used to work out a way to stop the cancer.
8. Gene expression can be indirectly measured using DNA microassay technology to give an approximate cellular concentration of different mRNAs present.

12 Polypeptide Synthesis – Transcription

1. A codon is a triplet of nitrogenous bases that gives the basic instruction of the genetic code specifying a particular amino acid.
2. Both transcription and translation are involved in protein synthesis. Transcription precedes translation and changes the code from DNA to mRNA. Translation directs the code on the mRNA to sequence the amino acids in the formation of a polypeptide.
3. The template strand is the side of DNA that is used in transcription to make mRNA.
4. Scientific knowledge about the process of gene expression and polypeptide synthesis has rapidly increased as new research methods and developments in DNA technology have enabled biologists to investigate the structures and functions of molecules in cells.
5.

Feature	DNA	RNA
Number of strands	Two strands	Single strand
Sugar present	Ribose	Deoxyribose
Nitrogenous bases present	Adenine, guanine, cytosine, thymine	Adenine, guanine, cytosine, uracil

6. In eukaryotes transcription needs to be completed and the mRNA takes the message from the nucleus to the cytoplasm for translation. Since prokaryotes do not have nuclei translation starts before transcription is completed.
7. Transcription occurs in the 5′ → 3′ direction from the DNA template strand.
8. (a) The promoter is a region of DNA that has a specific nucleotide sequence that signals the beginning of a gene and where transcription begins.
   (b) The sequence of nucleotides in the promoter is slightly different for different genes. This is significant as it means that a cell can direct several different genes to be transcribed at the same time.
9. RNA polymerase is the enzyme which causes the two strands of DNA to separate and then links the RNA nucleotides together to form mRNA.
10. Base pairs form the 'rungs' of the ladder in the DNA molecule and the number of base pairs gives an indication of the length of the gene on the chromosome. The average human gene is about 3000 bases long, with the largest gene being over 2 million base pairs in length and the smallest gene being a few hundred base pairs long.

11. Less than 10% of the human genome is involved in actual coding for protein synthesis even though it is a critical function of the coding.
12. B
13. B
14. D
15. A
16. D
17. B
18. C

13 Polypeptide Synthesis – Translation

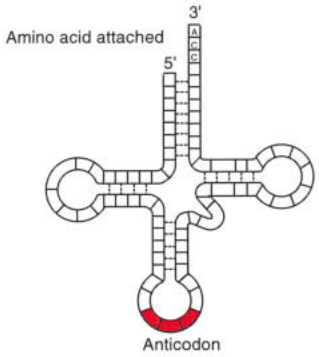
1. (a) Translation is a cellular process that uses the genetic code on mRNA to synthesise a polypeptide.
   (b) Translation occurs in the cytoplasm on ribosomes.
   (c) The translator is tRNA (transfer ribonucleic acid).
2. (a) The three main stages of translation are initiation, elongation and termination.
   (b)

Stage of translation	What happens
Initiation	In initiation the mRNA, tRNA with the first amino acid of the polypeptide that is to be formed and the two subunits of a ribosome are brought together to form the initiation complex.
Elongation	In elongation the amino acids are brought to the mRNA and added to the growing polypeptide chain.
Termination	In termination a stop codon (e.g. UAG, UAA, UGA) in the mRNA is recognised by a protein release factor causing the polypeptide to be completed and released and the translation assembly comes apart.

3.

DNA code	mRNA code	Amino acid
GAC	CUG	Leucine
TCT	AGA	Arginine
GTA	CAU	Histidine
CCA	GGU	Glycine
TTA	AAU	Asparagine
GTC	CAG	Glutamine

4. (a) tRNA has a distinctive shape which looks a 3-leaf cloverleaf when flattened to draw in two dimensions. One end of the tRNA has a nucleotide triplet called the anticodon which is complementary to the mRNA code. The other end of the tRNA carries the amino acid and takes it to the ribosome for translation.
   (b)



5. In eukaryotes, tRNA is formed in the nucleus.
6. There are 20 different synthetase enzymes that relate to the 20 different amino acids. In translation the synthetase enzymes link the correct amino acid to the tRNA with the correct anticodon. Once a tRNA has delivered its amino acid at the ribosome, the synthetase enzyme can link another amino acid to the t RNA thus allowing a tRNA to be used many times.
7.

(a) mRNA	(b) tRNA	(c) Ribosomes
mRNA carries the code from the DNA that will be translated into an amino acid sequence.	tRNA has an anticodon at one end to connect to the mRNA and an amino acid on the other end which will become part of the sequence in the protein.	Ribosomes have grooves where the mRNA strand can fit. They provide the environment for attachment. The ribosome moves along the mRNA strand, binding the correct tRNA to the mRNA to form the correct amino acid sequence for the protein.

8. Ribosomes are made up of two subunits which consist of proteins and ribosomal RNA (rRNA) with about two-thirds of the mass of a ribosome consisting of rRNA.
9. Ribosomes consist of two subunits. The large subunit binds with tRNA and the smaller subunit binds with mRNA.
10. A ribosome has three binding sites for tRNA: 1. 'P site' holds the tRNA with the growing chain. 2. 'A site' holds the tRNA carrying the amino acid about to be added to the chain. 3. 'E site' has the exiting tRNA that has delivered its amino acid and is about to leave the ribosome. There is also the binding site for mRNA.
11. Translation occurs in the 5′ → 3′ direction.
12. Codes for stop codons include UAG, UAA and UGA.
13. (a) A polysome is a cluster of ribosomes bound to one mRNA.
    (b) The formation of a polysome means that multiple copies of a polypeptide can be quickly synthesised. Thus increasing the efficiency of polypeptide production.
14. DNA molecule unwinds → mRNA forms using DNA template → mRNA moves from nucleus to cytoplasm → mRNA binds to ribosome → tRNA with amino acid is drawn to ribosome → tRNA adds amino acid to chain, then leaves → line-up of amino acids leads to the synthesis of a polypeptide.
15. C
16. A
17. B

14 Activity – Making a Model of Protein Synthesis

1. (a) Transcription is a process that copies the information from DNA onto mRNA.
   (b) During transcription the DNA unzips and the enzyme RNA polymerase binds to one of the DNA strands. The enzyme makes a long chain of RNA nucleotides that are complementary to the DNA nucleotides. This forms the messenger RNA (mRNA) which is released from the enzyme.
2. During translation the coding on the mRNA is used to make proteins. Translation occurs on ribosomes in the cytoplasm. The ribosome provides the structural site for the mRNA and a tRNA brings an amino acid to this site. The tRNA has a triplet code which is complementary to the code on the mRNA. The ribosome moves to the next triplet code and the amino acids are added to the growing peptide chain.
3. In eukaryotes transcription occurs in the nucleus while translation occurs in the cytoplasm on a ribosome.
4. Protein synthesis is a complex biological process in eukaryotes that involves transcribing the DNA code in the nucleus onto mRNA and the passage of the mRNA from the nucleus into the cytoplasm. The mRNA binds to a ribosome in the cytoplasm and tRNA delivers amino acids in an order that matches up the complementary base-pairing between the mRNA and tRNA. A 3-D model of this complex sequence of events would be very hard to build, keep pieces together and show all details of base-pairing. Other details, e.g. action of enzymes would be hard to attach which means some details would not be present. Thus all groups reached the same conclusion about not building a 3-D model.
5. Protein synthesis is a biological process that involves a complex sequence of steps that each need to be understood to understand how a DNA is taken from the nucleus of a eukaryote and used to produce a protein that will be used by the cell. The construction of a model means each step can be detailed and understood before moving to the next step. By working in a group there can be discussion about what is happening, why it is happening and how each step fits into the sequence. This aids understanding, learning and remembering. Thus group work to construct a group model will assist student understanding of this biological process.

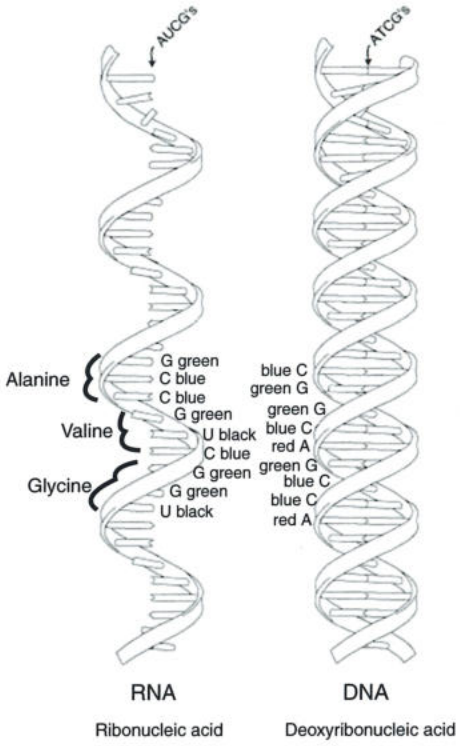
6. An advantage of constructing a model as group work is that the model can be more efficiently constructed with each person in the group being given individual tasks. This can save time and effort with students pooling their resources to complete the project. A disadvantage that arises from each person carrying out only one section of the task and then everyone pooling resources is that some students may not take enough 'interest' in the work done by others in their group and thus may not understand and learn all steps in protein synthesis. This will give these students a patchy and incomplete understanding of the biological process.
7. (a) The most favoured form of representing the process of protein synthesis was a PowerPoint slideshow presentation.
   (b) PowerPoint slideshow presentations allow each step in the process of protein synthesis to be on individual slides. Highly detailed and coloured diagrams can be inserted to show what is happening and animations can be used to label different features and indicate specific details. Thus PowerPoint slideshow presentations were suitable for this task and were preferred by 2/5 groups.
8. Two types of RNA are mRNA and tRNA.
9. (a) Glycine, serine, valine.
   (b) Codons for isoleucine include AUC, AUU and AUA.
   (c) The new code on the mRNA would read GGUUCAAUG and this changes the code to read glycine, serine and methionine.
10. A codon is a triplet code of three bases that gives the genetic information in either a molecule of DNA or mRNA to specify a particular amino acid. An anticodon is a group of three bases on a tRNA molecule that pairs with the complementary codon on the mRNA.
11. In DNA there are four kinds of bases – adenine, thymine, guanine and cytosine. These bases form the 'rungs' of the 'ladder' of the double helix, always pairing adenine-thymine and cytosine-guanine. RNA is only a single strand of nucleic acid and has the bases adenine, cytosine and guanine, however, uracil replaces the thymine.
12. Both mRNA and tRNA are single strands of nucleic acid with the bases adenine, uracil, cytosine and guanine. They differ in that mRNA carries the codon sequence from the nucleus to the ribosomes in the cytoplasm, while tRNA gathers and assembles amino acids in their correct order along the mRNA.
13.

Single strand DNA	Transcribed mRNA	Anticodon of tRNA	Amino acid
AAA	UUU	AAA	Phenylalanine
TAG	AUC	UAG	Isoleucine
TGG	ACC	UGG	Threonine
CCA	GGU	CCA	Glycine
AGC	UCG	AGC	Serine

14. Transcription begins when an enzyme, RNA polymerase binds to the DNA at the beginning of a gene and moves along to the end of the gene causing the formation of a RNA transcript which is released at the end of the gene.
15. There are several possible answers to this question. The following table shows two possibilities.

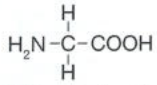
Amino acid	Arginine	Lysine	Glutamine	Cysteine	Trypsin
DNA code 1	GCA	TTT	GTT	ACA	ACC
DNA code 2	GCT	TTC	GTC	ACG	ACC

16. To code 51 amino acids you would need 51 codons which is 153 bases, but you would also need a stop and start code, so the minimum number of bases is 159 bases. This number of bases relates to exons needed to code for the protein. Since genes have many areas of introns, the gene probably has a larger number of bases present.
17. The diagram shows translation. The ribosome is moving along the length of a mRNA strand 'reading' the codons. Two tRNAs are held at any one time on the ribosome and the anticodon matches up with the codon on the mRNA. The polypeptide chain is being assembled.
18. The students were required to make a model of protein synthesis. The group that added the additional information about free and bound ribosomes had researched further information about protein synthesis and found out that proteins synthesised on free ribosomes were mainly used within the cell while proteins synthesised on bound ribosomes were mainly secreted from the cell. This describes what happens at the end of protein synthesis and thus this group felt it was necessary to add the information to complete the story of protein synthesis.
19. There are several possible answers for this as there are several codes for each of those amino acids. The following answer is just one possibility using the mRNA codes alanine (GCC), valine (GUC) and glycine (GGU).



20. D

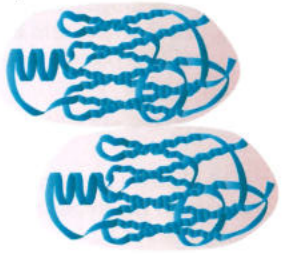
15 Protein Structure

1. A protein is a macromolecule that always contains the elements carbon, hydrogen, oxygen and nitrogen.
2. (a) Each amino acid has an amino group (NH_2), a carboxyl group (COOH) and a side chain (R).
   (b) 
3. The different chemical properties of different amino acids, e.g. polar (hydrophile) or non-polar (hyrodphobe) depend on the properties of R.
4. Glycine is the simplest amino acid with R = H.
5. A dipeptide is 2 amino acids joined together by a peptide bond. A tripeptide is 3 amino acids joined together by peptide bonds. A polypeptide is many amino acids joined together by peptide bonds.
6. (a) The distinguishing test for protein is the Biuret test. Biuret solution (equal amounts of 1% copper sulfate solution and 5% potassium hydroxide solution) will turn from pale blue to purple in a positive test for protein.
   (b) The Biuret test is actually detecting peptide linkage – which is the bond that holds amino acids together in a protein.

7. (a) The primary structure of a protein is the unique linear sequence of amino acids in its molecule and is determined by genetic information.
   (b) The secondary structure of a protein refers to the segments of the polypeptide chain that has coils or folds caused by hydrogen bonds between repeating CO and NH groups.
   (c) The tertiary structure of a protein is formed by the folding of the secondary structure into a complex and compacted shape.
   (d) The quaternary structure of a protein involves the interaction between several polypeptide chains that aggregate into one functional macromolecule.
8. The coils/folds of secondary structures of proteins are the helix and the beta pleated sheet.
9.

Feature	Fibrous protein	Globular protein
Shape	Long thread	Globe-like
Polypeptide chain	Extended chain and threadlike	Twisted, looped and folded back on itself
Type of structure	Secondary structure	Tertiary structure
Example	Collagen	Haemoglobin

10. The three types of bonds that hold the tertiary configuration in shape are: 1. Hydrogen bonds. 2. Ionic bonds. 3. Disulfide bonds.
11. Disulfide bonds are strong covalent links between sulfur containing amino acids, e.g. between the –SH groups of amino acids that are distant in the chain but lie next to each other due to the folding and twisting of the chain. These bonds are important as they stabilise the structure of the protein and provide structural strength.
12. (a) Denaturation occurs when a protein loses its shape and the chains unravel. This occurs when the weak hydrogen bonds in the structure are broken.
    (b) Denaturation can occur due to physical or chemical changes in the environment, e.g. excessive heat, salt concentration, temperature, pH.
13. (a) The diagram shows the primary structure of the protein as it shows the long chain of the polypeptide. However, it does not show the actual sequence of amino acids in the polypeptide which is what makes the precise primary structure of the protein. The diagram shows the secondary structure of the protein as it shows the segments of the polypeptide that are coiled or folded. The diagram shows the tertiary structure as it gives the overall globular shape of the polypeptide.
    (b) Structure X is a pleated sheet and Structure Y is a helix coil.
    (c) It is a globular protein as the secondary structure is folded into a complex and compacted shape.
    (d) One possible arrangement of two of these polypeptides which join together to form a quaternary structure is shown in the diagram.



14. B
15. D
16. A

16 Polar and Non-Polar Amino Acids

1. Amino acids consist of an amino group ($-NH_2$), a carboxyl group (–COOH) and a side chain (–R).
2. (a) The amino group is basic.
   (b) The carboxyl group is acidic.
3. An amphoteric substance is both acidic and basic.
4. The functional groups of amino acids are the amino group and the carboxyl group.
5. In an amino acid the α-carbon atom is the carbon that has the two functional groups attached to it – the amino group and the carboxyl group are attached to this atom.
6. If the amino acid has one amino group and one carboxyl group it will be neutral. If the amino acid has additional amino groups it will be a basic amino acid while additional carboxyl groups will make it an acidic amino acid.
7. Buffers are substances that stabilise the pH of a solution. In an acid solution an amino acid will pick up H^+ ions and become positively charged. In an alkaline solution an amino acid donates H^+ ions to the surroundings and becomes negatively charged. Thus proteins which consist of many amino acids will act as buffers in cells and in body fluids.
8. Amino acids with polar side chains are likely to be hydrophilic while amino acids with non-polar side chains are likely to be hydrophobic.
9. A

17 The Role of Proteins in Human Biology

1. An enzyme is an organic catalyst that changes the rate of a reaction without being consumed by the reaction.
2. Enzymes control the rate of specific reactions. An enzyme can control the rate of only one particular reaction, e.g. the enzyme sucrase only splits sucrose, or an enzyme can control a type of reaction, e.g. peroxidase catalyses several different peroxides, or an enzyme can control a reaction that involves a certain type of bond, e.g. pancreatic lipase splits the ester bonds connecting the glycerol and fatty acids of a large range of fats.
3. Most enzymes are globular proteins.
4.

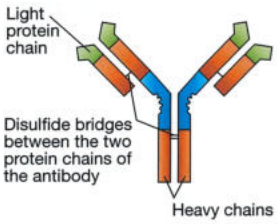
Structural protein	Function
Collagen	Collagen is a fibrous support protein found in connective tissue, e.g. cartilage.
Keratin	Keratin is a fibrous support protein that gives tough support in claws, horns, beaks, nails, hooves and feathers.
Silk	Silk is a fibrous support protein that is used to make cocoons and webs by arthropods.

5. Membrane proteins are a range of different proteins embedded in the lipid bilayer of the cell membrane. Different types of cells have different sets of membrane proteins with some membrane proteins held in place by connections to the internal cytoskeleton and some membrane proteins attached to the extracellular matrix. In animal cells these attachments to the membrane proteins provide support and give the cell a stronger framework.
6. Haemoglobin is an iron-protein that can pick up oxygen when there is a high concentration of oxygen, e.g. lungs or gills. The haemoglobin carrying the oxygen travels in the bloodstream and will release the oxygen in an area that has low oxygen concentration. Thus haemoglobin can be classified as a transport protein.
7. Membrane proteins include channel proteins and carrier proteins which aid the movement of specific substances across the cell membrane. Channel proteins form a hydrophilic channel that allows certain molecules or ions to pass, e.g. aquaporins allow water to cross the membrane. Carrier proteins shuttle molecules across the membrane by changing shape, e.g. glucose carrier proteins. Thus these two types of membrane proteins can be classified as transport proteins.
8. (a) A hormone is formed in specialised cells in multicellular organisms and is a chemical signal that travels in body fluids to act on specific target cells to change the functioning of the target cells
   (b) You cannot give an exact description of the chemical structure of hormones as a hormone is described by function and there is a range of chemical structures for the group.
   (c) Hormones that are proteins are the peptide hormones, e.g. insulin, growth hormone and ADH. The glycoprotein hormones are amino acid chains with side chains of carbohydrate and include luteinising hormone and thyroid stimulating hormone.
9. Insulin is a hormone secreted by the pancreas and regulates the concentration of sugar in the blood of vertebrates by causing cells in fat tissue and skeletal muscle to absorb glucose from the blood.
10. Storage proteins include ovalbumin (in egg white), casein (in mammalian milk), ferritin (iron storage protein)
11. The prolamines are a group of storage proteins that contain a high amount of the amino acid proline, e.g. seeds of cereal crops contain prolamines.

12. (a) Receptor proteins are found on the surface of a cell or inside the cell, e.g. some membrane proteins act as receptors.
    (b) Receptors in the dendrites of neurons detect the neurotransmitter released by other neurons and respond by continuing the nervous impulse.
13. Proteins involved in movement include actin and myosin.
14.

Type of defence protein	Function
β-defensins	Secreted by skin and respiratory tract to combat bacteria and prevent infection.
Complement system	Part of the innate immune system to help remove pathogens from the body.
Interferons	Made by host cell in response to a pathogen and communicates to other cells to resist infection.
Antibodies	Produced by plasma cells in response.

15. Labelled diagram of an antibody.



16.

Type of protein	Function
Enzymes	Catalyse specific reactions
Structural proteins	Provide support
Transport proteins	Aid movement of other substances
Storage proteins	Provide a supply of amino acids
Hormones	Involved in the coordination of different activities in the organism
Receptor proteins	Involved in the response of a cell to chemical stimuli
Contractile and motor proteins	Involved in movement
Defence proteins	Aid protection and defence against disease

17. The diagram shows that in X-ray crystallography an X-ray beam is passed through a crystal of the protein under investigation and the diffracted rays are caught on photographic film. The X-ray diffraction pattern is then analysed to determine the 3-D structure of the protein. A 3-D model of the protein can then be constructed.

18 Phenotype and Epigenetics

1. Epigenetic inheritance refers to the inheritance of traits in ways that do not directly involve the nucleotide sequence.
2. Phenotype refers to the actual appearance of an organism.
3. The genotype refers to the genetic constitution of an organism as well as being used to identify a particular set of alleles present in an organism. The genome always refers to the complete complement of an organism's genetic information.
4. The phenotype of an individual does not always show the genotype, e.g. a person may have the genotype TT and should grow to a tall height but if they have a poor diet through childhood and growth they may not reach their potential height (phenotype).
5. (a) The norm of reaction for a genotype is the range of phenotypic appearances that is found in the population.
   (b) A norm of reaction for a trait with no breadth is ABO blood group. A norm of reaction for a trait with a wide breadth is human skin colour.
   (c) Polygenic traits are features which are determined by two or more genes. Polygenic traits are more likely to have a broad norm of reaction as there is a wider combination of alleles determining the feature.
6. (a) Pleiotropy occurs when a single gene has multiple effects.
   (b) Many genetic diseases caused by a single pair of alleles have multiple symptoms. For example, people with cystic fibrosis, which is a genetic disease have salty tasting skin, frequent lung infections, shortness of breath and coughing, sinus infections, poor growth, poor weight gain and infertility.
7. A multifactorial trait is a phenotypic feature that occurs due to several factors.
8. The fundamental cause of many cancers is DNA damage that accumulates over a person's lifetime. Contributing factors include age (the older you are the more likely you are to develop a cancer), lifestyle (e.g. diet, exercise, alcohol intake, smoking cigarettes), DNA damage (e.g. by carcinogens in the environment), inheritance of a fault in one of the genes, infection by some types of virus (e.g. infection by human papilloma virus – HPV – is linked to cervical cancer) and problems with the immune system.
9. A gene is a section of DNA that has the code for a particular polypeptide.
10. When a gene is expressed the DNA molecule is 'switched on' and the polypeptide is produced by the cell.
11. The two steps in gene expression are transcription and translation.
12.

Feature	Transcription	Translation
Location	In nucleus of eukaryotic cell.	In cytoplasm of eukaryotic cell.
Process	DNA code for a gene is used as a template to form mRNA.	Polypeptide is synthesised under the direction of the mRNA at a ribosome.
Controls	Chromatin-modifying enzymes have initial control. Transcription factors control the start of transcription. RNA processing controls the final form of the mRNA that leaves the nucleus.	Translation initiation factors control the start of translation.

13. Gene expression can be measured by the amount of functional protein produced by the cell.
14. The steps in protein synthesis and degradation in a human cell are: Chromatin in nucleus → changes revealing DNA of a gene → transcription forms the primary transcript → RNA processing forms mRNA → mRNA moves from the nucleus into the cytoplasm → translation leads to formation of polypeptide → protein processing forms the active protein → at the end of its use the protein is degraded by proteases.
    Regulation and control of phenotypic expression of a gene occurs at each step in the process of protein synthesis and degradation.
15. The appearance of a particular trait can depend on the products of several genes as many metabolic pathways involve a series of reactions with each step being catalysed by a different enzyme. Each enzyme is coded by a particular gene and this means that several genes need to be expressed for the complete metabolic reaction to occur.
16. Coexpressed genes share similar control elements and are expressed together in gene expression.
17. Epistasis is a gene interaction with the presence of a particular allele of one gene pair determining whether certain alleles of another gene are expressed.
18. The environment is all the living and non-living surroundings.
19. There are many examples that can be used to show the effect of environment on phenotype. Poor nutrition with insufficient amounts of food and a lack of vitamins and essential amino acids will lead to stunted growth changing the appearance of the person from its genetic potential.
20. Epigenetic inheritance refers to the inheritance of traits in ways that do not directly involve the nucleotide sequence. An allergic response, e.g. to poison ivy toxin was not inherited with the nucleotide sequence in the DNA. The immune system has responded to the environmental stimulus (poison ivy toxin) which acts as an antigen and causes sensitising of B and T cells to this antigen. Gene expression in response to activated helper and cytotoxic T cells causes the release of many substances that lead to the allergic response.
21. C

19 Genetic Profiling and Screening

1. Prenatal tests involve tests, e.g. using placental tissue and/or amniotic fluid to provide information about the baby before birth.
2. A prenatal test is an ultrasound that is carried out at 8 to 10 weeks. This does not harm the baby and it is important as it will confirm the stage of pregnancy, identify the number of embryos present in the uterus and show how the embryo is developing.
3. (a) A CMA test is a chromosome microarray test that is used to check for extra or missing DNA segments.
   (b) CNVs are copy number variations which are changes in the genome due to a gene or segments of a gene being duplicated or deleted.
   (c) If the diagnostic test finds a copy number variant further testing is carried out to determine which particular genes are involved. Knowledge of the particular genes will then enable the medical practitioners to determine the health risk, if symptoms will develop over time and if offspring will also be affected. Some copy number variants are benign and do not decrease the health of the person or affect development.
4. (a) STRs are short tandem repeats which are short microsatellites that consist of a unit of two to thirteen nucleotides repeated hundreds of times in a row on the chromosome and the pattern of the STRs at particular sites on a chromosome is unique to each person.
   (b) STRs are used to construct DNA profiles of individuals and this is used as evidence in forensics to solve crimes identifying perpetrators and can also be used in paternity cases to identify the father of a child.
5. D

20 The Cell Cycle

1. (a) The cell cycle is a series of events that occur in the life of eukaryotic cells.
   (b) The cycle consists of mitosis, cytokinesis and the stages of interphase which are G_1 stage (growth stage 1), S stage (synthesis stage) and G_2 stage (growth stage 2).
   (c) Mitosis is the shortest phase of the cell cycle.
2. Interphase usually lasts for around 90% of the cell cycle.
3. Mitosis is a process during cell division in which the cell nucleus divides into two.
4. Cytokinesis is the division of the cell's cytoplasm following the division of the nucleus.
5.

Phase	What happens
G_1 phase	Normal cellular activity resumes after mitosis and cell division. In this phase cells grow in size by producing proteins and cytoplasmic organelles.
S phase	Is the synthesis phase when DNA is replicated and the amount of DNA in the nucleus doubles. Chromosomes are not visible and the DNA with its associated proteins exists as chromatin.
G_2 phase	Cell grows in size as molecules are produced in preparation for mitosis.

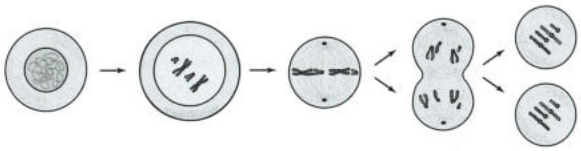
6. A checkpoint is a control point in the cell cycle where stop and go-ahead signals regulate the cycle.
7. The G_1 checkpoint is the restriction point and if it receives the go-ahead the cell will usually complete the cell cycle and divide.
8. The G_2 checkpoint triggers mitosis and makes sure chromosomes have been replicated and DNA is not damaged before mitosis.
9. The M checkpoint is the metaphase checkpoint in mitosis and is the DNA damage point to check if sister chromatids are correctly attached before anaphase. The mitotic spindle needs to be aligned correctly at this checkpoint for the rest of mitosis to occur.
10. Since thymine is in DNA and not RNA, radioactive thymine can be used to track the formation of new DNA. Samples taken over time will show how much radioactive thymine is incorporated into the new DNA that is being synthesised and then during mitosis the thymine can be tracked into the two new daughter cells. G_1 and G_2 will be gap stages as RNA is the messenger that takes the genetic code from the nucleus to the cytoplasm for protein synthesis and RNA does not contain thymine.
11. The cell cycle control system has checkpoint proteins that keep the cycle in order and operating normally.
12. The regulatory proteins involved in the cell cycle control system are kinases and cyclins.
13. The G_0 phase is a non-dividing state that a cell enters when the G_1 restriction point does not give the go-ahead for the rest of the cycle to continue.
14. Chromatin is a three-dimensional network of DNA and associated proteins that makes up a eukaryotic chromosome. In a non-dividing cell the chromatin exists as very long thin fibres that are not visible with the light microscope.
15. (a) This cell cycle is 20 hours long.
    (b) Interphase is the longest phase and during interphase the G_1 phase is the longest section (8 hours).
    (c) At point X in the cycle is the replication of DNA in the nucleus.
16. The cell cycle can be disrupted due to genetic predisposition or the action of mutagens.
17. Genetic predisposition means that the genome of the person contains specific genetic variations, often inherited from a parent that gives the person an increased likelihood of developing a particular disease or disorder.
18. A mutagen is a chemical or physical agent that causes a mutation changing DNA and the genetic information.
19. Proto-oncogenes are a group of normal cellular genes that have the potential to become oncogenes.
20. Proto-oncogenes code for proteins that stimulate cell growth and division, inhibit cell differentiation and stop cell death.
21. An oncogene is a gene that is part of the normal genome or a gene found in viruses that triggers cancerous characteristics.
22. Mutations can cause a proto-oncogene to become an oncogene.
23. When a proto-oncogene is mutated to become an oncogene there will be increased cell division, decreased cell differentiation and inhibited cell death. This means a mass of cells will be produced in a tissue or organ forming a tumour.
24. The uncontrolled cell division will form a mass of cells called a tumour (neoplasm).
25. A benign tumour is a mass of cells that will stay at the original site and have a slower growth rate than a malignant tumour. Whereas the cells of a malignant tumour detach and spread to other organs and tissues and have a faster growth rate with less differentiation than the cells in a benign tumour.
26. B
27. D

21 Mitosis and Cytokinesis

1. (a) Mitosis is a process during cell division in which the cell nucleus divides into two.
   (b) Since all organisms begin life as one cell, a fertilised egg, mitosis is essential to become a multicellular organism. Genetic information is transferred to new cells as DNA and mitosis increases the number of cells. Even when the organism has ceased 'growing', mitosis is still needed. Older cells need replacement as they are damaged or worn out and some parts, even in adults, are continually growing, e.g. hair and fingernails in humans, and root tips and shoot tips in plants.
2. (a) Cytokinesis is the division of the cell's cytoplasm following the division of the nucleus.
   (b) Cytokinesis is important because it stabilises the internal concentration of materials in the two new cells, making sure each new cell has sufficient organelles for growth and carrying out the processes of living.
3. Mitosis is needed to create new cells for growth, repair and reproduction.
4. DNA is present in nuclei, mitochondria and chloroplasts.
5. No. Interphase is called the 'resting' stage and is the time between cell divisions. During this time the cell is active carrying out the normal processes of living and the DNA is replicating in preparation for mitosis.
6. In cytokinesis cleavage occurs when the cells are pinched into two segments by a ring of contractile filaments which appear near the cell surface and contract.

7. The cell organelles are evenly divided and distributed between the two daughter cells.
8. At the beginning of mitosis the nucleus is surrounded by the nuclear membrane. The genetic material has already duplicated, during the S phase, but cannot be seen as the chromosomes have not condensed. A cell at interphase has the 'usual' appearance of a cell with chromatin and nucleoli present in the nucleus.
9. During mitosis the chromosomes become visible as two sister chromatids, the chromatids line up at the equator of the cell, the two sister chromatids separate to become chromosomes. During anaphase the microtubules shorten and the chromosome attached to the centromere moves to the pole. Daughter nuclei form at the poles.
10. The spindle fibres are microtubules with associated proteins which attach to the centromere of the chromatids. The function of the spindle fibres is highly important in mitosis. When the microtubules shorten during anaphase the attached chromosome moves to the pole. Without the spindle fibres the two sets of chromosomes would not be able to move apart to form two daughter cells at the poles.
11. B
12. B
13. C
14. D

22 Experiment – Investigating Mitosis

1. The students using a light microscope to investigate mitosis chose a prepared slide of epithelial cells as these cells are continuously dividing to replace worn out cells.
2. When studying prepared slides with a microscope students would need to handle the microscope with care – using two hands to carry the microscope with one hand holding the arm and the other hand supporting the base. Microscopes are heavy and if dropped could damage legs or feet. Students also need to take care when focusing on slides to make sure not to break the slide. Broken glass needs to be put in the broken glass bin and care taken not to cut yourself which could lead to infection.
3. 

Flow chart of mitosis

4. Most body cells are at interphase of the cell cycle. In interphase most cells are in G_0 phase which is a non-dividing state.
5. A

23 The Need for Mitosis

1. Mitosis is the division of the nucleus in cell division whereas cytokinesis is the division of the cytoplasm in cell division and follows mitosis.
2. Mitosis is very important as it produces two identical daughter cells. The process is needed for growth, maintenance and repair of body cells and tissues.
3. Cells that remain unspecialised are likely to retain the ability to divide while cells that become specialised are likely to lose the ability to divide.
4. Cells that lose the ability to divide include guard cells of plant stomates and some types of cells in the mammalian nervous system.
5. (a) A stem cell is a relatively unspecialised cell that can divide by cell division to produce identical daughter cells and to differentiate to form different specialised cells.
   (b) Mitosis occurs at sites where there are stem cells.
6. The cell cycle is controlled by a group of molecules that act in a coordinated manner forming a cell cycle control system. There are checkpoints in the cycle that either let the cycle continue or halt the cycle. If the cycle is not halted the cell will grow in size and synthesise new components and the DNA will replicate until mitosis and cell division.

24 Stem Cells

1. A stem cell is a relatively unspecialised cell that can divide by cell division to produce identical daughter cells and to differentiate to form different specialised cells.
2. Totipotent stem cells are cells that can give rise to any type of differentiated cell. In mammals totipotent stem cells are those cells formed after the formation of the zygote before the blastocyst has formed.
3. Pluripotent stem cells are able to give rise to any of the embryonic germ layers that form all cell types of the body but not the placenta or umbilical cord.
4. In mammals multipotent stem cells are adult stem cells that are found in various tissues, e.g. bone marrow stem cells give rise to the different types of blood cells and brain stem cells give rise to nerve cells.
5. The blastocyst is an embryonic stage in mammals that is a hollow ball of cells produced about one week after fertilisation in humans.
6. In humans the blastocyst implants in the endometrium around day 7 and implantation is usually completed by day 14.
7. The trophoblast is the outer layer of cells of the blastocyst.
8. Some of the trophoblast develops into the chorion which is an extraembryonic membrane that encloses the entire embryo and forms the placenta.
9. The inner cell mass of the blastocyst become the embryo.
10. In humans the embryo becomes a foetus in week eight.
11. Organogenesis is the development of body organs from the three germ layers.
12. Humans and other vertebrates are triploblastic with 3 germ layers while the jellyfish and corals are diploblastic with only 2 germ layers. The third layer in humans and other vertebrates is the mesoderm which is between the endoderm and ectoderm and gives rise to muscles and most other organs between the digestive tract and the outer covering of the animal.
13. Cell differentiation means the cell becomes specialised with specific structures and functions, e.g. nerve cells have a particular shape and are used to carry nervous impulses.
14. Adult stem cells or tissue specific stem cells are unspecialised cells and found in particular locations in the body.
15. Adult stem cells are needed for growth, repair and maintenance of different tissues and organs of the body.
16. (a) The ectoderm develops into the skin epidermis, sensory receptors in the skin, epithelial lining of the mouth and rectum and the nervous system.
    (b) The mesoderm develops into the notochord, skeletal and muscular system, excretory system, circulatory and lymphatic systems and dermis of the skin.
    (c) The endoderm develops into the epithelial lining of the digestive tract and respiratory system, liver and pancreas, thyroid and parathyroid glands, lining of the urethra, urinary bladder and reproductive systems.
17.

Part of body	Cell types produced at that site
Brain	Oligodendrocyte Astrocyte Neuron
Skin	Epidermis Hair follicle
Gut	Secretory cells
Bone marrow Blood stem cell	Red blood cells White blood cells Platelets
Bone marrow Blood vessel stem cell	Arteries Veins Capillaries
Bone marrow Mesenchymal stem cell	Muscle cell Bone cell Cartilage cell Fat cell

18. Induced pluripotent stem (iPS) cells are cells that have been treated, e.g. with a retrovirus so that they behave like embryonic stem cells.
19. D
20. A

25 Stem Cell Research

1. Unlike typical somatic cells of the body which become specialised to perform specific functions in particular parts of the body, stem cells are unspecialised and can keep dividing, renewing themselves even if they have been inactive for a relatively long period of time. Stem cells also produce daughter cells that can be induced under specific conditions to differentiate into many different cell types while body cells, if they divide by cell division, will divide to produce the same type of cell.
2. Scientists are interested in stem cells for: 1. Investigating the processes involved in normal cell development. 2. Investigating the signals that control the genes that lead to the production of the chemicals that control differentiation. 3. Investigating the effect of new drugs to aid the treatment of specific diseases. 4. For testing new cell based therapies for the treatment of disease and injury.
3. Embryonic stem cells are pluripotent and have the potential to be grown into any type of specialised cells of the body. Research scientists are interested in embryonic stem cells as these cells could be used to investigate normal cell development and to use this knowledge to determine new ways to repair or replace ailing organs or tissues and treat diseases with cell based therapies.
4. IVF is in vitro fertilisation. It is a process where the egg is fertilised by a sperm outside the human body.
5. Research into mesenchymal stem cells (MSC) could find new treatments for many people with diseases associated with bone and cartilage injuries, diseases or degenerative conditions. Transplantation of MSC could hasten recovery or slow down the deterioration of bone and cartilage in the patient. Cell cultures of MSC could be used to produce new body parts that could then be implanted into a person to replace old or damaged parts. Thus research is needed to find new ways to treat and help people with diseases and impairments.
6. There is a debate about using embryonic stem cells as the embryonic stem cells have the ability to produce specialised organs or tissues which could save and extend the lives of people or improve the quality of life for people with degenerative conditions, spinal cord injuries or genetic diseases when used in conjunction with gene therapy. However, there is an ethical issue about the status of embryos, e.g. leftover embryos that were produced for couples on IVF programs and what to do with them and if they should be used as a source for embryonic stem cells. The use of embryonic stem cells has raised concerns on a political level and in religious circles. Many governments have introduced laws to restrict or prohibit certain types of research and use of embryonic stem cells.
7. International groups such as the International Society for Stem Cell Research (ISSCR) play an important role in developing guidelines for the use of stem cells and stem cell research. ISSCR has experts in science, ethics and law with representatives from the academic community, industry and different government bodies. This means that information about the latest developments can be discussed, analysed and evaluated. Decisions and recommendations can then be made about the worthiness of following a particular line of investigation and if there are sound ethical reasons for allowing particular research methods. Decisions can also endorse the continuation or development of methods that have a potential industrial application which would help the sick and ailing. Decisions can also determine that some lines of investigation should not be continued. Thus the international groups are very important in promoting a transparent and uniform practice worldwide and in evaluating the moral and ethical issues of stem cell use and stem cell research.
8. Multiple sclerosis is called a demyelinating disease as the immune system attacks the specialised cells of the central nervous system that produce the myelin which forms a protective sheath around neurons. This reduces the myelin around the axons and affects the conduction of nervous messages and communication between cells.
9. Cardiovascular disease refers to a range of diseases of the heart and blood vessels, e.g. coronary heart disease, heart failure, stroke, hypertension and congestive heart failure.
10. Cell types being investigated in stem cell research into cardiovascular disease include the cardiac muscle cells of the heart, the cardiac pacemaker cells and the endothelial cells that line blood vessels.
11. A difference between these two research investigations is the source of the stem cells. Method I uses induced pluripotent stem (iPS) cells that can be adult stem cells, e.g. could be stem cells that produce skin cells that have been reprogrammed to behave like embryonic stem cells while method II uses RPE-specific stem cells from donated eyes from eye banks. Both these sources of RPE cells can be used to generate many RPE cells that can be used to investigate new treatments for macular degeneration, e.g. how to deliver the cells into an eye and at what stage of differentiation is the most successful for transplantation to replace or induce repair of damaged RPE for a person to recover vision capabilities.
12. D

26 Tumours and Cancers

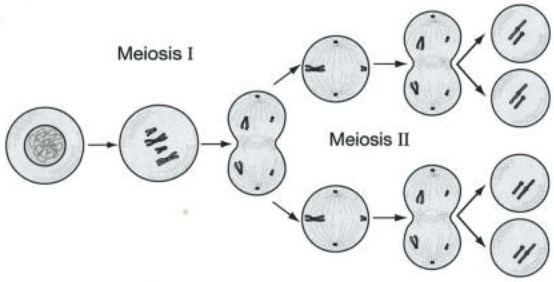
1. A tumour is an abnormal growth of tissue of a part of the body.
2. The tumour cells stop responding to normal control mechanisms and ignore the normal checkpoints in the cell cycle continuing to divide and divide.
3.

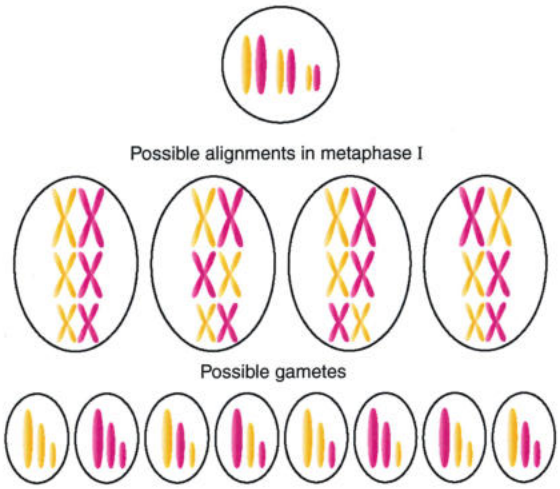
Benign tumour	Malignant tumour
Stays at the original site.	Can spread to new tissues.
Usually does not cause serious problems.	Affects the functions of one or more organs.
The cells of a benign tumour usually appear like relatively normal cells with features of the specialised cells in the tissue where the tumour is located.	The genetic and cellular changes of the abnormal cells of a malignant tumour allow the cells to survive in different tissues and organs with the cells of a malignant tumour tending to be less differentiated and have less specialised features.
Most benign tumours can be removed by localised surgery and in many cases a new growth does not reappear.	A malignant tumour becomes a cancer with a primary growth area and spreads to secondary growths. These areas can be removed by surgery which is usually followed with chemotherapy and radiotherapy.

4. They can be caused by – 1. Genetic mutations – the changes in the DNA can be naturally occurring or due to outside agents, e.g. tobacco smoke, ionising radiation, and environmental pollutants. Translocations, deletions and other chromosome abnormalities can be seen in cancer cells. 2. Inherited genetics – many cancers require two or more mutations before the cancer develops and some families have a hereditary tendency to cancer with one or more mutated genes already in the inherited genome. 3. Lifestyle and behavioural factors, e.g. overweight, lack of exercise and stress factors. Some physical irritants, e.g. some types of food scar the lining of the intestinal tract causing rapid mitosis to replace damaged cells.
5. Cancers begin when one body cell starts to reproduce uncontrollably and can develop from most types of cells. Each cancer has its own set of causes, is associated with different parts of the body and will spread in different ways and at a different rate. Different cancers include non-melanoma skin cancer, breast cancer, kidney cancer, prostate cancer, pancreatic cancer, thyroid cancer, lung cancer.
6. These are both types of non-melanoma skin cancer. Basal cell carcinoma begins in the lowest layer of the epidermis and is the most common form. Squamous cell carcinoma begins in cells called keratinocytes in the upper level of the epidermis. It can develop in scars or skin ulcers. It is more aggressive and can invade fatty tissues beneath the skin spreading to lymph nodes.

27 Meiosis

1. Meiosis is cell division to produce gametes – eggs and sperm.
2. Four daughter cells are produced by meiosis.
3. Meiosis occurs in the gonads, e.g. testes and ovaries in animals, ovaries and anther in flowering plants.
4. The daughter cells formed by meiosis are haploid and not identical to the parent cell.
5. Homologous chromosomes are pairs of the same length and centromere position with one being inherited from the father and one from the mother.

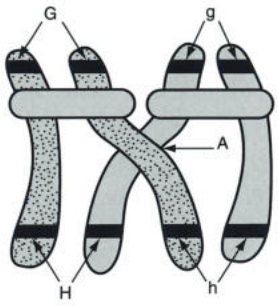
6. The law of independent assortment states that alleles of genes on non-homologous chromosomes sort independently during gamete formation.
7. Linked genes are on the same chromosome.
8. Linked genes can be separated and inherited independently of each other if crossing over occurs during prophase I.
9. The law of segregation states that the two alleles for each gene separate during meiosis and gamete formation.
10. 

11. A testis would have been chosen to study meiosis as this is where meiosis occurs in males in the formation of sperm.
12. 

13. (a) Spermatogenesis is the continuous production of large numbers of sperm in the testis.
    (b) A primary spermatocyte is diploid and is in the prophase of meiosis I whereas a secondary spermatocyte is haploid and is formed after the first meiotic division.
14. B
15. B
16. A

28 Processes in Meiosis

1. Meiosis is cell division to produce haploid daughter cells.
2. Genetics is the study of heredity and hereditary variation.
3. Meiosis is very important as a way variation can occur from one generation to the next. Thus meiosis is studied as part of a genetics study.
4. The diploid number is the normal number of chromosomes for a species, while the haploid number is half the diploid number and is the number of chromosomes found in a gamete of that species.
5. For humans the diploid number is 46 and the haploid number is 23.
6. Homologous chromosomes are the corresponding pairs which look the same, except for the sex chromosomes X and Y.
7. Crossing over is the exchange of certain sections of chromosomes producing new linkage groups.
8. The chiasma is the site of crossing over when two homologous chromosomes exchange genetic sections during meiosis.
9. Crossing over produces gametes which contain a new combination of characteristics containing features of both parents. This provides new genetic variation which is important for natural selection in a changing environment.
10. Crossing over is likely to occur when the homologous pairs of chromosomes come together in prophase I.
11. Meiosis is called the 'reduction division' as the daughter cells have half the number of chromosomes of the parent cell.
12. (a) 23 chromosomes in the cells formed after meiosis.
    (b) 46 chromosomes in the cells formed after mitosis.
13. DNA is replicated during interphase for meiosis.
14. Factors that can lead to variation are: 1. Crossing over during meiosis. 2. Random fertilisation. 3. Random segregation.
15. (a) Diagram of crossing over in cell division.
    (b) Cell division is meiosis.
    (c) Daughter cells could be GH, Gh, gH, gh.
    (d) Crossing over is significant as it is one of the means of providing variation in a population so that in a changing environment there is a greater likelihood of some individuals having a favourable feature for natural selection.
    (e) Meiosis occurs in gonads – ovary or testis.



16. B
17. D
18. A
19. D
20. B
21. B

29 Non-Disjunction

1. Non-disjunction is an error in either mitosis or meiosis where both members of an homologous pair or both sister chromatids fail to properly separate.
2. Aneuploidy is a genetic condition where there is an abnormal chromosome number, e.g. there are one or more additional copies of chromosomes or there is a deficiency in one or more chromosomes.
3. If an homologous pair fail to separate properly during meiosis one of the daughter cells which will become a gamete will not have a particular chromosome while another daughter cell, another gamete, will have two copies of a particular chromosome.
4. The diagram shows non-disjunction in the second division of meiosis. After the first division the two short sister chromatids fail to separate causing cell W to have two copies of the short chromosome and cell X to have no copies of the short chromosomes. After the first division the bottom cell divides normally so that cell Y and cell Z are haploid with correct chromosomes.
5. B
6. A

30 Case Study – Down Syndrome

1. Trisomy is the presence of three copies of one particular chromosome instead of the normal two copies.
2. Jerome Lejeune in 1959 was the first person to identify the cause of Down syndrome as an extra copy of chromosome 21.
3. A congenital condition is one that is present at birth.
4. The karyotype is most likely caused by non-disjunction. It shows three distinct copies of chromosome 21. If it had been caused by translocation, the additional copy of chromosome 21 would be fused to another chromosome.
5. Symptoms of Down syndrome include wide-set eyes; short, flat-bridged nose; spinal weakness; heart defects and intestinal defects.
6. For trisomy 21, the effects of a germline mutation and a somatic mutation influence the degree of effect of the symptoms. A germ line mutation means all cells in the body will be affected while a somatic mutation means only some cells will be affected and the person is a mosaic. A mosaic will show reduced symptoms.
7. Mosaicism in Down syndrome can occur when there is non-disjunction after fertilisation during mitosis in the embryo so that some cells have 47 chromosomes and some cells have 46 chromosomes. If a baby is found to have mosaicism, 20 different cells, e.g. blood, skin, muscle, bone marrow are analysed and the mosaicism is expressed as a percentage of cells, e.g. if five cells have 46 chromosomes and 15 cells have 47 chromosomes, it is called a mosaicism level of 75%.

8. Down syndrome causes an overexpression of genes found on chromosome 21, e.g. increased activity in the superoxide dismutase (SOD) gene. SOD converts oxygen radicals to hydrogen peroxide and water. If these oxygen radicals are not converted they can damage cellular structures but an increase in SOD activity causes a build-up of hydrogen peroxide which also damages cells. Oxidative damage to neurons can cause rapid brain ageing and mental retardation. Chromosome 21 also has the gene for the precursor of amyloid protein. Amyloid plaques damage nerve cells causing a decrease in the number of nerve cells in the cortex and thus brain damage.
9. People with mosaic Down syndrome show a wide range and variety of symptoms as it depends on the number of different types of cells influenced by trisomy 21, e.g. heart, brain, blood, skin and on the parental DNA information on chromosome 21.
10. It is important to give assistance as early as possible for children that are born with Down syndrome, e.g. early intervention programs that develop language skills and coordinated movement. It is important they have physical therapy, speech therapy and occupational therapy to help develop skills and abilities, e.g. assistance with gross motor skills for walking, stair-stepping and being aware of body location in space and speech skills for enunciation and articulation.
11. The processes needed to study this gene involve using a restriction enzyme to cut the human genome into fragments and other restriction enzymes as markers. The segments are overlapped and then electrophoresis or Southern blotting is used to analyse the segments.
12. Research into the centromere region of chromosome 21 and its unusual alphoid DNA may provide information about the cause of trisomy 21 as the centromere is very important during cell division and the segregation of chromosomes. Since trisomy 21 is often caused by non-disjunction, this research could explain why the centromere region on chromosome 21 fails to separate and thus cause the extra copy to exist.
13. One of the main risk factors for Down syndrome is the age of the mother. The older the mother, especially for a first child, the greater the likelihood of a child with Down syndrome. The information shows that parents who both have normal karyotype and already have a Down syndrome child only have a 1% chance of a second child having Down syndrome.
14. B

31 Experiment – Investigating Meiosis

1. The students chose prepared slides of cross-section of testis as this is the male reproductive organ where sperm are produced.
2. First-hand data collection involved the students observing the prepared slide under the microscope and either drawing the cells they saw or taking photographs of the slides. Secondary data collection involved going to textbooks or to reliable websites to gather information that has been collected by other people.
3. Scientists make models so that a process or concept can be more easily understood. Models can give a 3-D or 2-D image to show what is happening so that people can 'see' the steps in the process. Models are tested, compared with real data and used to make predictions. The development of scientific understanding and scientific theories has been greatly aided by the construction of models.
4. A reliable secondary source needs to be trustworthy and dependable. The information gathered from the site should be similar to the information found at other trustworthy sites and by trustworthy authors. Trustworthy sites can be government sites, well recognised academic sites, peer-reviewed publications in scientific journals and textbooks.
5. An accurate secondary source provides data that has results or measurements that are close to true values. Accuracy of measurements is determined by the equipment being used and taking care with the measuring technique. Students would need to check any numerical data they gathered against accepted scientific values for this data to determine the accuracy of the data at the secondary site.
6. A valid secondary source has evidence supporting its contents. The students should check to find references to experimental data and references to other scientific reports, research papers or academic journals to show that the data was correctly derived from accepted premises that are supported by actual fact.
7. When assessing each other's work it was important to consider the 'completeness' of the presented information as meiosis is a step by step process involving two cell divisions which produces four haploid daughter cells. Each step needs to be clearly shown in the presentation to show how each daughter cell gets one complete set of chromosomes. It is also necessary to assess the 'correctness' of the presented work as misleading or wrong information leads to misunderstanding and the information would not be helpful to show the process of meiosis.
8. Making a model of meiosis is an effective way to show how a cell undergoes cell division in meiosis to produce four haploid daughter cells. In a model students can follow each individual chromosome to show what happens to it during the step by step process, including how genetic information can be swapped during crossing over. The hands-on and visual display of a model can have a bigger impact on some people who find it easier to learn if they can 'see' what is happening.
9. B
10. D

32 Comparing Meiosis and Mitosis

1. Meiosis is a type of cell division that produces gametes. A gamete is a sex cell (sperm or ovum) and has half the normal number of chromosomes. Mitosis is the process in which the cell nucleus divides into two.
2. Table to show differences between mitosis and meiosis.

Meiosis	Mitosis
Produces four daughter cells	Produces two daughter cells
Daughter cells are haploid	Daughter cells are diploid
Daughter cells are different to parent cells	Daughter cells are identical to parent cell
Occurs in body cells	Occurs in gonads
Produces cells for growth, repair and maintenance	Produces gametes for sexual reproduction
Involves one division	Involves two divisions
Chromosomes line up singly	Chromosomes line up in pairs for first division and then singly in the second division

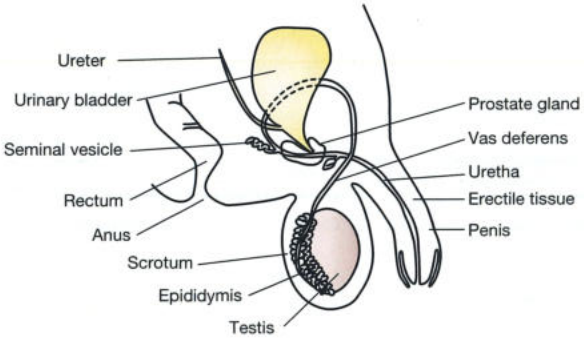
3. Meiosis is called the 'reduction division' as the daughter cells have half the number of chromosomes compared to the parent cell. Mitosis is called the 'equating division' as the daughter cells have identical chromosomes to the parent.
4. The diploid number is the 'normal' number of chromosomes found in body cells for that species and is given the symbol $2n$. The haploid number is half the 'normal' number of chromosomes and is given the symbol n.
5. An homologous pair are a corresponding pair of chromosomes, e.g. in humans the diploid number is 46 chromosomes which is 23 homologous pairs.
6. C
7. D
8. B
9. B
10. A

33 Fertilisation

1. Fertilisation is the union of two gametes.
2. A gamete is a sex cell and is haploid containing only one set of chromosomes.
3. A zygote is a fertilised egg.
4. When two haploid gametes fuse the result is a zygote with the diploid number – two sets of chromosomes. This means that fertilisation maintains the diploid number of a species.

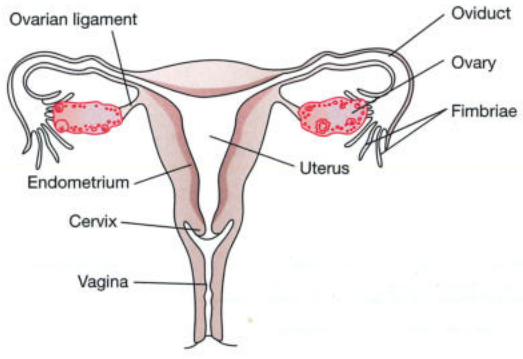
5. Fertilisation produces variation as it involves the joining of two haploid gametes. This means that each offspring receives half their chromosomes from their mother and half their chromosomes from their father. Each offspring is different to either parent and different to other offspring of that parental couple. Thus there is variation in the population.
6. Copulation is sexual union when sperm are deposited inside the female.
7. (a) The acrosome is a vesicle at the tip of a sperm cell that helps the sperm penetrate the egg.
   (b) The acrosomal reaction is a process that occurs when the sperm approach the zona pellucida of the egg. The acrosome releases hydrolytic enzymes that digest the jelly coat so that the acrosomal process can penetrate the coat.
8. Gonads are the sex organs in males and females.
9. An ovum is a large, inactive cell containing a store of food to provide energy for the zygote until it grows to a stage where it can feed itself.
10. Sperm are microscopic cells that swim using a relatively long tail. The nucleus in a sperm is in a small head piece with an acrosome at the tip which contains enzymes that help the sperm penetrate the egg. Behind the head a sperm has a middle piece with large numbers of mitochondria which provide the energy needed for the movements of the tail.
11. Advantages of sexual reproduction include: 1. In a changing environment variation in genotypes means that some individuals may have gene combinations that are favourable to the new environmental selective pressures leading to natural selection, evolution and survival. This is particularly beneficial if the population is small in size and the mutation rate is relatively high. 2. Sexual reproduction allows for multiple numbers of offspring which increases variation for natural selection in a changing environment. 3. The recombination of genes in sexual reproduction may lead to harmful genes being removed from a population.
12. D

34 The Male Reproductive System

1. The primary male gonad is the testes.
2. Sperm are not produced in undescended testes.
3. The male reproductive system – 1. Produces the male gamete (sperm) in the testes. 2. Delivers the sperm in a liquid medium (semen) to the vagina during copulation. 3. Produces the main male sex hormone (testosterone).
4. Each testis is subdivided into around 250 lobules each containing 1 to 4 tightly packed seminiferous tubules. Between the seminiferous tubules there are Leydig cells that secrete testosterone after puberty. The tubules contain spermatogenic cells and Sertoli cells.
5. The testes are an endocrine gland as they synthesise the hormone testosterone from cholesterol.
6. Testosterone begins the formation and the maintenance of secondary sexual characteristics, e.g. growth of gonads, enlargement of larynx, muscular development and body hair.
7. (a) Leydig cells secrete testosterone.
   (b) Leydig cells are interstitial cells found outside the seminiferous tubules.
   (c) Leydig cells are active in the foetus to direct development of a male child. They become inactive during childhood and become active again at puberty when they cause the development and then maintenance of secondary sexual characteristics.
8. 

9. (a) The scrotum is a pouch made of pigmented skin, fibrous tissue, connective tissue and smooth muscle. It is divided into two sections, which each hold one testis, one epididymis and the testicular end of the spermatic cord.
   (b) The scrotum allows the testes to be kept about 2°C below body temperature which is needed for sperm production.
10. Sperm move from the site of formation in the seminiferous tubule into the epididymis and stay there for two to three weeks to 10 becoming more motile. Those that are not ejaculated will be reabsorbed into the body usually within four weeks.
11. Semen consists of around 60% fluid from the seminal vesicles, 30% fluid from Cowper's gland and 10% sperm.
12. The spermatic cords suspend the testes in the scrotum with each containing vascular tissue (artery, vein and lymph vessels) and the vas deferens.
13. (a) In a vasectomy the sperm ducts are cut and each duct is tightly tied in two places and a short length removed so sperm cannot reach the urethra.
   (b) Many men choose to have a vasectomy as it allows the male to still have sexual intercourse but is a permanent method of birth control causing sterility.
14. The fluid from Cowper's gland contains a clotting enzyme which thickens the semen in the vagina and it is also slightly alkaline to protect the sperm in the acidic vagina.
15. (a) Cell X is most likely a Leydig cell.
   (b) Structure Y is a seminiferous tubule.
16. D
17. B
18. A
19. C
20. D

35 The Female Reproductive System

1. The primary female gonad is the ovaries.
2. The female reproductive system – 1. Produces the female gamete (ova) in the ovaries. 2. Delivers the ovum to the oviduct where fertilisation occurs. 3. Prepares the lining of the uterus for implantation and development of an embryo. 4. Produces main female sex hormones (oestrogens).
3. The vulva is the external female genitalia consisting of the labia majora, labia minora, clitoris, the vaginal orifice, the vestibule and hymen.
4. The ovaries are whitish oval bodies about 3 to 4 centimetres long that lie either side of the uterus.
5. The ovaries are held in place by ovarian ligaments.
6. The germinal epithelium is the outer coating of an ovary which holds the potential eggs in a female embryo.
7. (a) At birth a female has around 1 million primary oocytes per ovary.
   (b) At puberty a female has around 200 000 primary oocytes per ovary.
8. During ovulation the secondary oocyte (ovum) is released from the follicle and the remaining follicular material – the granulosa and theca cells form a solid mass called the corpus luteum.
9. 

10. Fertilisation usually occurs in the oviduct.
11. Fimbriae help collect the egg when it is released from the ovary.
12. The uterus is a thick muscular organ that is pear shaped and is about 80 millimetres long.

13. The endometrium is the inner layer of the uterus and consists of columnar epithelial cells and mucous secreting glandular cells in two layers – the functional layer and the basal layer. The surface section of the endometrium thickens and becomes rich in blood vessels in the first half of the menstrual cycle and is shed in menstruation if the egg is not fertilised. The basal layer will regenerate the thickening of the functional layer for the next menstrual cycle.
14. The cervix is the neck of the uterus and opens into the vagina.
15. The vagina – 1. Is a passageway for menstrual flow. 2. The site for insertion of the penis and sperm deposits. 3. Acts as the birth canal.
16. The mammary glands are exocrine sweat glands over the pectoral muscles. Each breast contains around 15 to 20 lobes each consisting of many lobules of glandular tissue with fibrous tissue and fatty tissue.
17. Breast development begins at puberty due to higher oestrogen and progesterone levels.
18. Lactation is milk production by the mammary glands.
19. A
20. B
21. C
22. A

36 Male and Female Hormones

1. Hormones are produced in specialised cells in endocrine glands.
2. Hormones travel in body fluids in the body to act on specific target cells.
3. The steroid hormones are lipid soluble and can easily pass through cell membranes. The receptors for steroid hormones can be in the cytoplasm or in the nucleus.
4. GnRH is gonadotropin releasing hormone that is produced by the hypothalamus and controls the release of luteinising hormone and follicle stimulating hormone from the anterior pituitary gland.
5. LH and FSH are hormones that affect the gonads (ovaries and testes) and are thus sometimes called gonadotrophic hormones.
6. Androgens are any steroid hormone that stimulates the development and maintenance of the male reproductive system and male secondary characteristics. Androgens can be natural or synthetic compounds.
7. Testosterone – 1. Needed for sperm production. 2. Causes the voice to deepen. 3. Causes body hair, e.g. facial and pubic hair to grow. 4. Causes the building of muscles. 5. Causes the strengthening of bones. 6. In males it is important in developing the prostate gland and male reproductive tissues, e.g. testes and penis.
8. Two main types of female hormones are the oestrogens and the progestins.
9. Oestrogens are responsible for the development of secondary female characteristics – 1. Stimulating breast growth. 2. Causing deposits of fatty tissue. 3. Thickening of the vagina and causing it to secrete mucus. 4. In the menstrual cycle cause the development of the endometrium (lining of the uterus). 5. Can affect skin appearance. 6. Can influence bone strength. 7. Can aid protection against heart disease.
10. Progesterone is important in maintaining pregnancy.
11. Hormone X is FSH (follicle stimulating hormone) and hormone Y is LH (luteinising hormone).

37 Spermatogenesis

1. Spermatogenesis is the production of sperm.
2. Spermatogonia are germinal epithelial cells that line the outside section of the seminiferous tubules.
3. In spermatogenesis a spermatogonium migrates to a Sertoli cell moving towards the lumen and centre of the seminiferous tubule. As it moves the spermatogonium gets larger to become a primary spermatocyte.
4. Spermatogonium → primary spermatocyte → secondary spermatocyte → spermatid → sperm.
5. The spermatogonium is the parent cell and the first meiotic division forms two primary spermatocytes. The second meiotic division forms the secondary spermatocytes.
6. Hormones include – 1. Luteinising hormone – secreted by anterior pituitary gland and stimulates Leydig cells to secrete testosterone. 2. Testosterone – Leydig cells secrete testosterone needed for growth and division of the spermatogonia. 3. Follicle stimulating hormone – secreted by anterior pituitary gland stimulates Sertoli cells to convert spermatids to sperm.
7. Sperm become fully mature and capable of swimming when they are in the epididymis.
8. D

38 Oogenesis

1. Oogenesis is the production of ova.
2. Oogonia are stem cells in the germinal epithelium of ovaries of a female foetus.
3. A primary oocyte is a diploid cell at prophase I in meiosis in an ovary and can be triggered to continue meiosis and development to become an ovum.
4. Oogonium → primary oocyte → secondary oocyte → ovum.
5. A primary follicle is a primary oocyte that is surrounded by a layer of follicle cells.
6. During female childhood the oocytes are resting at prophase I of meiosis.
7. First division of meiosis begins at puberty when several oocytes start to develop each month with usually only one maturing for ovulation and the rest degenerating.
8. The formation of polar bodies is a way to remove unneeded chromosomes without complete formation and then needing to dispose of a large amount of cytoplasm.
9. At ovulation the ovum is at metaphase II of meiosis.
10. The corpus luteum develops from a ruptured follicle that has released its ovum. It becomes an endocrine gland if pregnancy occurs or degenerates into the corpus albicans.
11. Hormones are – 1. Luteinising hormone – secreted by anterior pituitary gland and stimulates ovulation, the development of the corpus luteum and secretion of progesterone. 2. Follicle stimulating hormone – secreted by anterior pituitary gland stimulates maturation of ovarian follicles and the secretion of oestrogen for ovulation.
12. In spermatogenesis every spermatogonium will divide to produce four sperm cells while in oogenesis every oogonoim will produce one functional ovum and polar bodies which will disintegrate. During active sexual life spermatogenesis continuously produces sperm while oogenesis usually releases one ovum per month in ovulation.

39 Menstrual and Ovarian Cycles

1. The ovarian cycle refers to the monthly cycle of egg development in the ovary with follicular phase, ovulation and the luteal phase regulated by hormones. Whereas the menstrual cycle refers to the cyclic preparation of the endometrium of the uterus for pregnancy in females with discharge of blood and tissue through the cervix if implantation does not occur.
2. Ovarian cycle consists of – 1. Follicular phase – follicles take between 10 to 21 days to develop in the ovary. 2. Ovulation – a functional ovum is released from the ovary. 3. Luteal phase – the ruptured follicle develops into the corpus luteum which secretes hormones to prepare the endometrium of the uterus for pregnancy. If pregnancy does not occur the corpus luteum breaks down for the start of a new cycle.
3. Gonadotropic releasing hormone (GnRH) from the hypothalamus stimulates the release of follicle stimulating hormone (FSH) and luteinising hormone (LH) from the anterior pituitary gland and FSH and LH then stimulate the ovary to become active.
4. The menstrual cycle is usually between 25 and 35 days in length, with 28 days taken as the average.
5. Menarche is the beginning of menstruation at puberty and menopause is when the menstrual cycle stops and the ovaries do not release any more ova.
6. Menstruation is the shedding of the blood and tissues from the endometrium of the uterus when an ovum is not fertilised.

7. Three phases are – 1. Menstrual flow – is menstruation with the shedding of blood and tissue from the endometrium. 2. Proliferative phase – is the thickening of the endometrium caused by increasing levels of oestrogen. Secretory phase – begins after ovulation with oestrogens and progesterone secreted by the corpus luteum stimulating the continued development and maintenance of the endometrium. If fertilisation does not occur the corpus luteum breakdown causes a rapid drop in ovarian hormones and the endometrium disintegrates leading to the menstrual flow.
8. Oestrogen secreted by follicle cells in the follicular phase causes the endometrium to thicken. After ovulation the ruptured follicle becomes the corpus luteum which secretes oestrogens and progesterone which cause the endometrium to continue to thicken. If fertilisation does not occur these two hormones are no longer secreted and the rapid drop in hormone levels causes menstrual flow.
9. Endometriosis is the growth of endometrium tissue outside the uterus. Endometrial cells respond to sex hormones oestrogen and progesterone. After ovulation the increase in these hormones causes the misplaced endometrial cells to thicken and bleed but they cannot be shed in the menstrual flow. This causes pain and inflammation especially around the time of menstruation.

40 Conception and Pregnancy

1. Conception means an egg is fertilised.
2. A zygote is a fertilised egg.
3. Cleavage is the process of division of a zygote to form a blastocyst.
4. A blastocyst consists of a ball of cells around an inner cavity.
5. Implantation of a blastocyst occurs around day 21 of the average menstrual cycle if fertilisation occurs.
6. Step W shows fertilisation of an ovum with only one sperm penetrating the covering of follicle cells that form the corona radiata and also pass through the zona pellucida.
   Step X shows the fusion of the two nuclei.
   Step Y is the zygote – the fertilised egg.
   Step Z is the first division to form a two-celled embryo as part of the cleavage stage of development.
7. (a) Point A shows a ruptured follicle that has released an egg. It will develop into the corpus luteum.
   (b) Fertilisation occurs at point C.
   (c) Point G shows a blasocyst embryonic stage.
   (d) Point H shows implantation of the blastocyst in the uterine wall.
8. (a) Human chorionic gonadotropin (hCG) is an embryonic hormone produced by some of the cells in the blastocyst/embryo when it implants in the endometrium. hCG maintains the corpus luteum in the mother stimulating it to continue to produce oestrogen and progesterone through the first few months of pregnancy. As the embryo develops these cells become part of the placenta and it is the placenta which continues to produce hCG. If hCG is not present or levels are low the corpus luteum of the mother will break down, progesterone levels will drop and the mother will menstruate and the embryo will be lost in a miscarriage.
   (b) Since high levels of hCG are needed to maintain oestrogen and progesterone secretion, some hCG is excreted in the mother's urine and many early pregnancy tests involve checking for hCG.
9. Around the third month of pregnancy oestrogen and progesterone begin to be produced by the placenta so that only small amounts of hCG are needed after the third month of pregnancy.
10. The oestrogens produced during pregnancy suppress the secretion of FSH which means no follicles develop and there is no ovulation.
11. During pregnancy the mother's thyroid gland can enlarge up to 50% due to hCG and a specific thyroid stimulating hormone called human chorionic thyrotropin.
12. (a) Human gestation or pregnancy is on average 266 days (approximately 38 weeks).
    (b) Since timing is related to the menstrual cycle it is usually counted as 40 weeks since the start of the last menstrual cycle.
13. Monozygotic (MZ) twins come from a single fertilised egg while dizygotic (DZ) twins develop from two separate eggs which are each fertilised by different sperm.
14. The occurrence of multiple births is increasing mainly due to the increase in the use of fertility drugs.
15. Siamese twins are joined and share some organs. Depending on which organs are being shared some Siamese twins can be surgically parted after birth.
16. (a) An ectopic pregnancy occurs when the embryo develops outside the womb, e.g. in the fallopian tube.
    (b) In most cases the embryo dies as a suitable placenta cannot form and provide sufficient nutrients.
17. Infertility is the inability of a couple to achieve conception after at least one year of unprotected sex (and six months if a woman is over the age of 35).
18. Infertility can be due to – 1. Male related problems, e.g. blocked sperm ducts, too few numbers of sperm, defective sperm or incorrect chemical components. 2. Female related problems, e.g. blocked oviducts. 3. The reason is a combination of male/female problems or unexplained.
19. D
20. A
21. C

41 Play – Cell Productions

Scene 1 – The Gamete

1. Haploid means there is only one set of chromosomes present in the nucleus.
2. When a sperm makes contact with the jelly coat of an egg its acrosome releases enzymes that let it move through the outer jelly coat. The egg releases cortical granules in the cortical reaction to stop polyspermy.
3. Polyspermy means the egg is fertilised by more than one sperm which means there is three or more copies of each chromosome.
4. A diploid cell has two sets of chromosomes, one set inherited from each parent.
5. An allele is an alternative version of gene that produced distinguishable phenotypes.
6. Diploid number for humans is 46.

Scene 2 – The Blastocyst

1. The nuclear membrane controls what enters and leaves the nucleus.
2. The blastocyst is an embryonic stage that forms in humans about 5 days after fertilisation and cell division of the zygote.
3. The blastocyst consists of an inner mass of cells called the embryoblast, a cavity and an outer layer called the trophoblast.
4. An embryonic stem cell is part of the inner mass of cells in a blastocyst which is an unspecialised cell that can produce, during a single division, one identical daughter cell and one more specialised cell that can undergo further differentiation.
5. Gamma radiation is an extremely high frequency radiation with frequencies over 10^{19} hertz. It is high energy and has wavelengths less than 10 picometres which is less than the diameter of an atom.
6. Natural sources of gamma rays include naturally occurring radioisotopes and cosmic rays and artificial sources include nuclear reactors and high energy physics experiments.
7. In nuclear reactors the energy from nuclear fission is used to heat water to produce steam that drives turbines to make electricity.
8. The parts of the electromagnetic spectrum that cause mutation are gamma rays, X-rays and UV radiation.
9. Myosin is important in muscle contraction.
10. ChAT is the enzyme choline acetyltransferase This protein needs to be produced as it leads to the formation of acetylcholine which is a neurotransmitter released by neurons so that the nervous message can pass from one nerve cell to the next nerve cell.
11. The *INS* gene makes the preproinsulin precursor of insulin. Insulin is needed by the body as it is the hormone that controls the glucose levels of the blood.

Scene 3 – The Growing Embryo

1. The first step in gene expression is the changing of the chromatin of the gene that is going to be expressed.
2. Gene expression means that information coded in the DNA directs the production of proteins or in some cases the production of RNA.
3. No. The nucleus of an eukaryotic cell has the code for most of the proteins produced by the body but not all the proteins. The mitochondria have their own DNA which in humans has the codes for 37 genes. The mitochondria produce proteins, e.g. enzymes required for cellular respiration.
4. The study of mtDNA has enabled scientists to work out evolutionary relationships between different eukaryotic organisms.
5. DNA repair molecules can detect damage to the double helix structure and then repair the damage, e.g. by using the sister chromatid as a template to recover the original information in the code.
6. Acetylcholine is a neurotransmitter produced at the axon terminal and released into the synapse. Its function is to take the nervous message from one neuron to the next. It diffuses across the synapse and stimulates the next neuron.
7. During differentiation cells become specialised and only certain genes are expressed in these specialised cells.

42 Germ Layers and Embryonic Development

1. Developmental biology is the study of the process by which organisms grow and develop. It includes the study of the processes by which organisms grow and develop from a zygote to a fully grown adult, e.g. genetic control of cell growth, differentiation and morphogenesis.
2. A germ layer is a group of cells in an embryo that will give rise to tissues and organs of the body.
3. Diploblastic animals have two germ layers, e.g. sponges, jellyfish and corals while triploblastic animals have three germ layers, e.g. vertebrates.
4. A zygote is a fertilised egg.
5. The blastula is a hollow ball of cells formed from mitotic divisions of the zygote. Implantation is when the blastula embeds in the wall of the uterus. Gastrulation is the formation of the three germ layers – the ectoderm, endoderm and mesoderm.
6. Gastrulation is most important as it forms the three germ layers that are the basis of the tissues and organs of the human body. The ectoderm (outside layer) forms the skin and nervous system, the endoderm (inside layer) forms the gut and associated glands including the respiratory system, and the mesoderm (middle layer) forms all the other organ systems.
7.

Ectoderm	Mesoderm	Endoderm
Skin epidermis Sensory receptors in skin Epithelial lining of mouth and rectum Nervous system	Notochord Skeletal and muscular system Excretory system Circulatory and lymphatic systems Dermis of skin	Epithelial lining of digestive tract and respiratory system Liver and pancreas Thyroid and parathyroid glands Lining of urethra, urinary bladder and reproductive system

8. In humans the first two weeks is called the pre-embryonic period; the third to eighth week are called the embryonic period and the third to ninth month is called the foetal period.
9. Patterning is a process where genes cause embryonic cells to differentiate into body regions.
10. When genes become 'switched on' the cell will produce particular proteins. These proteins are often enzymes which control specific reactions which in turn alter the cell's metabolism and structure. Thus the 'switching on' of particular genes will lead to the formation of particular cell types, tissues and organs.
11. (a) Trophoblast cells of the blastocyst that contact the uterine lining form the chorion and part of the chorion forms the placenta.
    (b) The major function of the placenta is to provide food and oxygen to the embryo/foetus and remove excretory wastes from the embryo/foetus.
    (c) The diagram shows that oxygen and nutrients are delivered by the umbilical vein and excretory wastes are removed by the umbilical artery.
12. Layer A is the ectoderm, layer B is the mesoderm and layer C is the endoderm.
13. Positional information refers to the signals and molecular cues that regulate the genes controlling development and differentiation. The development of a cell is determined by its location in the cell with respect to the position of other cells in the developing embryo.
14. Apoptosis is one of the main types of programmed cell death by the deliberate suicide of an unwanted cell. It is very important in embryonic development as selective cell proliferation and selective apoptosis causes tissues to form their definitive shapes. For example, the separation of human fingers needs the cells in between the fingers to start apoptosis.
15. Genes have been identified that code for apoptosis-inducing proteins and other genes have been identified that protect the cells from programmed cell death.
16. The last five weeks of embryonic period are highly critical as this is when all the main organ systems are laid down, both externally and internally. If a problem occurs now, e.g. a gene mutation, it could lead to a major congenital malformation in one of the body systems.
17. The mesoderm forms the circulatory system so a congenital heart defect occurred during the development of the mesoderm layer.
18. A

43 Birth

1. Parturition means the birth of a baby.
2. Labour is a series of strong contractions of the uterus to push the baby out of the uterus, through the vagina for birth.
3. The three stages are – dilation, expulsion and placental stages.
4. The increased size of the foetus causes a stretching of the walls of the uterus and cervix which are mechanical factors that help initiate labour.
5. Labour is regulated by hormones, e.g. oestrogen and oxytocin and local regulators, e.g. prostaglandins.
6. Lanugo is downy hair produced by the foetus from about five months and is usually shed before birth.
7. (a) Vernix is a waxy/greasy substance which covers the skin of the foetus in the last few months of pregnancy.
   (b) Vernix acts to help moisturise and protect the skin and conserve heat.
8.

Stage of labour	Description
Dilation	The dilation of the cervix is usually the longest stage, e.g. usually from 2 to 16 hours. It usually begins with mild contractions of the uterus usually 15 to 30 minutes apart and at the end of the stage the contractions are more intense and can be 1 to 2 minutes apart. The cervix needs to open to around 10 centimetres to fit the head of the foetus. This stage is usually shorter for subsequent pregnancies.
Expulsion	When dilation is complete the amniotic membrane breaks and the amniotic fluid flows out. The head of the foetus is seen in 'crowning'. The baby is born with the most powerful contractions. This stage is usually about 30 minutes long in a first pregnancy and shorter, even just a few minutes in subsequent pregnancies.
Placenta expulsion	More uterine contractions expel the placenta as the 'afterbirth'. As the placenta separates the blood vessels in the endometrium close off to minimise blood loss.

9. Breech births occur when the foetus is presenting its bottom rather than the head.
10. After the sixteenth week of pregnancy the placenta releases increasing amounts of oestrogen and progesterone.
11. (a) Oxytocin is produced by the mother's posterior pituitary gland.
    (b) Oxytocin helps to regulate birth by stimulating rhythmic contractions of the muscles in the wall of the uterus and stimulating the placenta to produce prostaglandins.
12. It is positive feedback as oxytocin stimulates the uterus to contract as the response. This response stimulates more contractions and the head of the baby on the cervix causes the release of more oxytocin. Positive feedback involves the response reinforcing or amplifying the change.

13. (a) The foramen ovale is a small flap-like opening in the heart between the right atrium and the left atrium that is open during foetal development.
    (b) Before birth most of the blood bypasses the lungs with blood from the right atrium flowing directly into the left atrium via the foramen ovale.
    (c) If the foramen ovale does not close at birth, insufficient blood goes to the lungs for oxygenation and deoxygenated blood mixes with the oxygenated blood and insufficient oxygen is supplied to the tissues, giving the baby a bluish tinge.
14. When the placenta stops removing carbon dioxide the rise in carbon dioxide in the baby's blood stimulates the respiratory centre and triggers the first breath.
15. Trimester 1 is 0 to 14 weeks, Trimester 2 is 12 to 27 weeks and Trimester 3 is 28 to 40 weeks.
16. Ductus venosus is a foetal blood vessel that carries blood from the umbilical vein to the inferior vena cava. It is significant as most foetal blood bypasses the liver as the foetus obtains its oxygen and nutrients from the mother.
17. Ductus arteriosus is a foetal blood vessel that connects the left pulmonary artery with the aorta. It is significant as most foetal blood bypasses the lungs as they are not functional and oxygen is obtained from the mother. After birth the ductus arteriosus closes to become the ligametum arteriosum.
18. A
19. B

44 After the Birth

1. A neonate is a newborn child.
2. The APGAR score looks at appearance, pulse, grimace, activity and respiratory effort, e.g. a rating of 1 for appearance means the baby has a pink body but the limbs are blue and a rating of 2 for pulse means more than 100 beats per minute for heart rate.
3. Lactation is the production of milk from the mammary glands and begins 2 to 3 days after giving birth.
4. (a) Colostrum is the first milk released after birth. Unlike mature milk colostrum contains large amounts of white blood cells and antibodies, especially immunoglobulin A (IgA).
    (b) These components are important in preventing pathogens invading the baby's circulatory system through its immature intestinal walls. IgA also helps prevent food allergies. Colostrum also contains a mild laxative which helps the baby pass the first faeces called the meconium.
    (c) Mature milk has higher levels of carbohydrates, lipids and potassium compared to colostrum.
5. Newborn babies are born with reflexes such as sucking and swallowing which are necessary for feeding; crying which is needed so that needs can be met and blinking which protects the eye.
6. (a) Structure X is the cervix.
    (b) During the birth process the cervix needs to dilate sufficiently for the baby to pass from the uterus to the birth canal.
    (c) The body plates of a baby's skull are soft and flexible and have sutures which allow the bones to move during the birth process. While in the birth canal the pliable bones of the baby's skull can overlap causing the baby's head to change shape.
    (d) The most suitable presentation for birth is head first with the body facing towards the mother's back. This is the safest delivery position for passage of head and shoulders through the birth canal.
7. D

45 Contraception

1. Contraception is the deliberate use of methods to prevent pregnancy.
2. Condoms are used as a physical barrier to prevent gametes meeting and pregnancy and they can also be used to prevent the transmission of STDs such as HIV, syphilis. For other forms of contraception, e.g. steroid hormones, chemical spermicides and sterilisation there is no relationship between contraception and preventing the transmission of STDs.
3. (a) The rhythm method or natural family planning involves periodic abstinence with intercourse only during 'safe' times of the cycle when conception is unlikely.
    (b) The pregnancy rate for couples using natural family planning is around 10% to 20%. This means that 10 to 20 females out of 100 per year will become pregnant.
4.

Method	Description
Rhythm method	No intercourse 4 days before or 4 days after ovulation.
Temperature detection	There is a sharp drop in temperature at ovulation, then temperature rises again. Intercourse is safe 3 days after temperature rises.
Mucus detection	At ovulation cervical mucus is slippery, clearer so intercourse is safe when no mucus or 3 days after last clear mucus.
Symptothermal method	Combination of rhythm, temperature and mucus methods.

5. The main methods of contraception are – 1. Use of steroid hormones. 2. Use of physical barriers between gametes. 3. Use of chemical spermicides. 4. Use of sterilisation. 5. Function after coitus, e.g. emergency contraceptive pill and IUDs.
6. Female contraceptive pills use steroid hormones, e.g. synthetic forms of oestrogen and/or progesterone. A combined female contraceptive pill has both of these hormones while the mini pill only contains progesterone.
7. To be effective the contraceptive pill needs to be taken for 21 days of the cycle at the same time of day.
8. The contraceptive pill prevents ovulation and ensures a regular period. The hormones cause negative feedback in the ovarian cycle stopping the release of gonadotropin releasing hormone (GnRH) from the hypothalamus. This stops the release of LH and FSH from the pituitary gland so follicles do not develop and ovulation does not occur.
9. The pill can have side effects such as increased incidence of blood clots, heart attack, hypertension, weight gain, mood swings, headaches and regular smokers taking the pill have a greater risk of cardiovascular disease. There is also a possible increased risk of infertility.
10. Some women cannot take oestrogen and therefore they need to use the mini pill which only contains progesterone.
11. The male contraceptive pill contains hormone combinations including testosterone to cause the inhibition of the hypothalamus and pituitary hormones, which reduces testosterone production to levels below that necessary for sperm production. Research involves using hormone treatments that will supress the release of gonadotropin to stop spermatogenesis.
12.

Type of physical barrier	Description
Male condom	Is a thin sheath of rubber/plastic that slides over the erect penis before intercourse preventing sperm entering the vagina. The male condom also prevents the transmission of STDs, e.g. HIV. Disadvantages include a slightly decreased sensation for the male, foreplay is interrupted to put it on and a tear or slip during or after ejaculation stops protection.
Female condom	Is a lubricated pouch that lines the vagina with a closed end at the cervix. This prevents sperm entering the vagina and also prevents the transmission of STDs.
Cervical cap	Is a small thin rubber cap that fits directly over the cervix. Needs to be inserted ahead of time to prevent sperm reaching the uterus. It is effective, relatively cheap, must remain in place for 6 hours and used with a spermicide cream or gel.
Diaphragm	Is a large dome of soft rubber or silicone that covers the cervix and upper part of the vagina. Needs to be inserted ahead of time to prevent sperm reaching the uterus. It is effective, relatively cheap, must remain in place for 6 hours and used with a spermicide cream or gel.

13. Chemical spermicides include sponges, foams, jellies, aerosols and creams that immobilise sperm. Some react with moisture in the vagina to form bubbles of gas that are a physical barrier to sperm. They are relatively cheap, simple to use and most are effective in the vagina for up to 24 hours after insertion. They are unreliable on their own and are usually used with mechanical barriers.
14. Surgical cut A is a tubal ligation which involves cutting the oviduct so that the ovum cannot reach uterus. The ovary can still secrete hormones and there is no change in the menstrual cycle. Surgical cut B is a vasectomy which cuts the vas deferens on each side so that sperm cannot become part of the semen released in an ejaculation.
15. IUDs (intrauterine devices) are plastic or metal objects inserted into the lumen of the uterus by a doctor. Many are a small loop or coil and can be left in place until the woman wishes to have a child. They cause an inflammatory reaction in the endometrium so white blood cells engulf sperm or ovum and they prevent implantation of the embryo. Some contain chemicals, e.g. copper or hormones that also assist contraception.
16. The morning after pill is an emergency contraception that involves taking two pills up to 72 hours after intercourse and two more 12 hours later after having unprotected sex. Most morning after pills contain progestogen, a synthetic version of progesterone which will prevent ovulation or implantation.
17. Coitus interruptus is the withdrawal of the penis before ejaculation. It is not very reliable as sperm secreted in the fluid before ejaculation may lead to a pregnancy and it requires strong control to withraw just before orgasm.
18. A vasectomy cuts the vas deferens and stops the transport of sperm from the testes into the semen. Masculinity is not affected as the testes are still producing testosterone which travels in the blood and the other components of semen, e.g. from the seminal vesicles and prostate gland are produced and released in an ejaculation.
19. B

46 Lifestyle Choices and Foetal Development

1. Lifestyle choices, e.g. diet, illicit drugs, alcohol and nicotine may affect foetal development.
2. Mutagens are chemicals or substances that cause mutations.
3. Mutagens are chemicals or substances that cause mutations whereas teratogens are chemicals or agents that cause birth malformations and defect.
4. In the first two months of life embryonic development can be highly affected by exposure to teratogens, e.g. in the fourth and fifth weeks of development the limbs are rudimentary and exposure to several substances at this stage can lead to limb abnormalities, e.g. thalidomide babies.
5. Medications can be prescription, over the counter, recreational or illegal drugs. Examples include alcohol, tobacco, aminopterin which is used to treat cancerous tumours and isotretinoin (accutane) which is used to treat severe acne. Thalidomide was given to pregnant women to control nausea of early pregnancy and it prevented normal formation of foetal limbs with the arms which develop first more affected than legs.
6. If in a zika virus area a pregnant woman can take measures to reduce the likelihood of been bitten by an infected mosquito, e.g. wear covering clothing, sleep under mosquito netting, stay in air conditioned areas. She also needs to have protected sexual intercourse, e.g. using a condom to prevent transmission from her partner.
7. If a pregnant woman contracts rubella (German measles) during the fourth to twelfth weeks of pregnancy the virus can affect embryonic formation of the heart, the eye lens, the inner ear and brain depending upon when she contracted the disease. About 90% of infected babies have major congenital abnormalities.
8. Hot tubs and saunas will raise the body temperature of a woman. Since high persistent fever involves raised body temperature and is linked with multiple congenital abnormalities it is advisable for a pregnant woman to avoid hot tubs and saunas to help protect the developing foetus.
9. (a) TORCH syndrome refers to the infection of a developing foetus or newborn by any of a group of infectious agents. TORCH stands for T (toxoplasma), O (other, e.g. chickenpox), R (rubella), C (cytomegalovirus), H (herpes simplex).
   (b) TORCH symptoms include small birth weight, petechial rash, enlarged liver and spleen, jaundice, hearing impairment, vision impairment, mental retardation, autism and even death.
10. Smoking during pregnancy can lead to many birth defects, e.g. lower birth weight, increased risk of abortion, gastrointestinal problems, respiratory problems and the SIDS risk increases.
11. The four main problems caused by alcohol are mental retardation, impaired growth, facial malformations and organ defects, e.g. heart, kidneys, muscles and joints.
12. (a) According to the ABS there were 259 800 births in 2005, 247 300 births in 1985 and 299 700 births in 2014 with an increase in the average age of mothers from 27.3 years in 1985 to 30.7 in 2005 to 30.9 in 2014.
    (b) National figures show an change from 247 300 in 1985 to 299 700 in 2014 ($\frac{299\,700 - 247\,300}{247\,300} \times 100 = 21.2\%$ increase) while in Western Australia in 2004 there were 25 295 births and in 2014 there were 35 403 births ($\frac{35\,403 - 25\,295}{25\,295} \times 100 = 40\%$ increase). Thus WA shows a significantly higher increase in births that the national data.
12. (a) Infant mortality refers to the death of the child before their first birthday and the rate is calculated at deaths per 1000 live births.
    (b) The ABS recorded an infant mortality rate of 9.9 in 1985 and a rate of 5.0 in 2005.
    (c) The change in the infant mortality rate from 1985 to 2005 is related to improvements in medical care and technology and a reduction in the number of deaths from sudden infant death syndrome (SIDS).
14. The data shows that there is a slow decline in the percentage of women smoking during their pregnancy from 16% in 2008, to 12.7% in 2011 to 11.2% in 2013. Since smoking is linked to birth defects this trend should lead to a reduced percentage of birth defects.
15. B

47 Mutagens

1. A mutagen is a chemical or physical agent that causes a mutation changing DNA and the genetic information.
2. A mutation is the change in the DNA genetic make-up of a cell.
3. Mutagens can be environmental factors such as radiation (e.g. X-rays, cosmic rays, UV rays and alpha, beta and gamma radiation from radioisotopes) or chemical agents (e.g. benzene, formalin or Agent Orange) or biological agents, e.g. viruses and bacteria.
4. Chemical agents can disrupt DNA replication, e.g. deaminate nitrogenous bases, e.g. nitrous oxide or if the agent resembles a base pair it can lead to insertion or deletion of an extra base pair, e.g. ethidium bromide.
5. Spontaneous mutations occur naturally and randomly, e.g. due to errors in replication whereas induced mutations are caused by mutagens such as radiation or certain chemicals.
6. A somatic mutation occurs in a body cell and cannot be inherited by offspring whereas a germ cell mutation occurs in a gonad cell and the change will be inherited by the offspring and passed to the next generation.
7. Teratogens are any substance, e.g. chemical, physical or biological that can cause a birth defect affecting the developing embryo/foetus during the pregnancy.
8. Exposure to teratogens during the embryonic period in the first trimester of pregnancy when cell differentiation occurs and the major organs and systems are forming can lead to health problems in the child.

9. Thalidomide was a drug used to treat morning sickness in pregnant women. It was also used as a sedative to help them sleep. In the 1950s and 1960s after investigations into children with birth defects it was realised that there were environmental factors that were non-toxic to the mother that could affect the developing embryo which was not protected while in the womb. Thalidomide was identified as a teratogen that affects limb development in the growing embryo. Research showed that the sensitive period was from day 35 to day 49, e.g. if thalidomide was taken by the mother on days 35 to 37 the child was born without ears and was deaf, if taken from day 39 to day 41, the child was born with no arms.
10. *BRCA1* and *BRCA2* (genes named for BReast CAncer) are regulator genes of the cell cycle that are associated with breast cancer and less DNA repair.
11. B

48 Sexually Transmitted Diseases

1. Sexually transmitted infections (STIs) (also called venereal disease – VD) are diseases that are transmitted through unprotected sex or genital contact.
2. STIs include syphilis, gonorrhoea, chlamydia, AIDS, herpes, genital warts, pubic lice, hepatitis A and non-specific urethritis.
3. Early detection and treatment of the infection are important and reduce the likelihood of serious health consequences.
4. A chancre is a small, painless open sore that appears around 10 to 90 days after infection by the spirochete bacteria *Treponema pallidum* that causes syphilis. A chancre indicates the primary stage of infection and usually appears on the genital organs, rectum, cervix, fingers, lips or anywhere where there was direct contact.
5. In the secondary stage which occurs within 6 weeks to 6 months of initial contact and lasts up to 2 years there are flu-like symptoms with fever and sore throat then skin rashes, mouth ulcers, mild fevers, disorders of bones or eyes. While in the tertiary stage which occurs 5 to 40 years after initial contact there is damage to many body organs with syphilitic heart disease, brain damage and insanity, blindness, physical incapacity.
6. (a) Gonorrhoea is caused by the bacteria *Neisseria gonorrhoeae.*
   (b) It predominantly affects the penis in men, the vagina in women and the mouth and anus in both sexes and if left untreated becomes a blood infection causing heart problems, sterility and arthritis.
7. In 2011 there were 12 men/100 000 men diagnosed with syphilis compared to 1.7 women/100 000 women (total 13.7 people/100 000) diagnosed with syphilis compared with 40 people/100 000 diagnosed with gonorrhoea. This shows that gonorrhoea is more prevalent in society than syphilis. There are also differences in the age groups affected with middle aged men (40 to 44 years) most likely to be infected with syphilis while men aged 20 to 24 have the highest diagnosis rate of gonorrhoea.
8. (a) Herpes is caused by the virus herpes simplex type 2.
   (b) Herpes can be a threat to a foetus as it can pass across the placenta causing serious malformations of the foetus and can be transmitted during childbirth causing meningitis, brain damage in the newborn.
9. Chlamydia is caused by the bacteria *Chlamydia trachomatis* and many people do not show immediate symptoms. In males it causes a burning in the urethra, pain in urination and a yellow discharge from the penis. In some cases it can lead to epididymitis and occasionally infertility. In female there are often no symptoms though infection can cause bleeding between periods or after sex, pain when urinating and can lead to infertility (56% of females) if untreated, eye infections, arthritis, pelvic inflammatory disease and increase risk of premature or stillborn child.
10. Treatment involves antibiotics and prevention requires use of a condom as a preventive barrier.
11. The human papillomavirus is associated with some types of cancer especially cervical cancer and is thus part of the national immunisation program.
12. The statement is showing that becoming infected with HIV does not immediately cause all symptoms of AIDS. AIDS is a progressive disorder with the weakening of the immune system and increasing vulnerability to other diseases and infections over time. The late stages of infection are referred to as AIDS and are associated with serious diseases such as pneumonia, cancer, brain disorders.
13. Diagnosis involves HIV testing, e.g. ELISA (enzyme linked immunosorbent assay) test.
14. B

49 Reproductive Technologies

1. Assisted reproductive technologies (ART) are medical interventions with the use of any reproductive technology to help overcome infertility problems.
2. ART activities include in vitro fertilisation, artificial insemination, embryo transfer, hormone treatment, cryopreservation of sperm/eggs, embryo and fertility medications, mapping of a woman's ovarian reserve with the ability of the ovary to produce viable egg cells, follicular dynamics and associated biomarkers and semen analysis.
3. Infertility is defined as the failure to conceive a child after one year of regular sexual intercourse without birth control.
4. Sterility is the permanent, irreversible inability to have children while infertility is defined as the failure to conceive a child after one year of regular sexual intercourse without birth control.
5. A miscarriage is the natural or spontaneous end of a pregnancy where the embryo/foetus is unable to survive. It usually refers to events that happen in early pregnancy.
6. Different treatments are needed for different types of miscarriage as each type has differently affected the uterus and mother's body. In an incomplete miscarriage some pregnancy tissue is remaining in the uterus and medical help is needed to remove these tissues while in a spontaneous miscarriage there is complete loss and additional medical aid may not be needed.
7. Pregnancy testing involves using technologies that test blood or urine for human chorionic gonadotrophin (hCG) hormone.
8. In vitro fertilisation involves mixing oocytes with sperm in culture dishes and incubating for several days to ensure the beginning of gastrulation.
9. ZIFT (zygote intrafallopian transfer) involves fertilising the eggs in vitro and then immediately transferring the zygote to the oviducts whereas GIFT (gamete intrafallopian transfer) involves placing eggs and sperm in the oviducts hoping fertilisation will occur.
10. The main steps in the in vitro process are – 1. Hormone therapy to stimulate egg production. 2. Removal of multiple eggs from the ovary. 3. The sperm and eggs are mixed and fertilisation occurs in vitro. 4. The fertilised eggs are incubated. 5. The embryos are implanted in the uterus or frozen for future implantation.
11. (a) The ovarian reserve is the current supply of eggs in the ovary and associated with reproductive potential.
    (b) The age of the female is the most important factor in the size of the ovarian reserve.
12. Risks involved with IVF include – 1. The use of fertility drugs, e.g. hCG can cause ovarian hyperstimulation syndrome which causes the ovaries to become swollen and sore, nausea vomiting and diarrhoea. 2. If more than one embryo is implanted at a time there is a risk of multiple births which has a risk of early labour and low birth weight. 3. In a small percentage of cases IVF led to an ectopic pregnancy with the fertilised egg implanting in the fallopian tube.
13. (a) Cryopreservation is a process where biological material, e.g. cells, tissues, organs, embryos are preserved by cooling to very low temperatures, e.g. −80°C to maintain their viability.
    (b) Eggs, sperm and embryos can be cryopreserved and used in fertility programs.

14. In September 2016 fierce storms that caused power blackouts across the entire state of South Australia led to the destruction of embryos at Flinders Fertility in the Flinders Medical Centre. About 12 patients were affected with about 25 embryos lost per patient. This shows that the ART is limited by the power supply to the facility with the risk that embryos are lost when there is premature thawing due to power failure.
15. Donor sperm is quarantined for 6 months to confirm donor does not carry any viral diseases.
16. (a) Ovarian cysts are fluid filled sacs in the ovary that can interfere with normal ovulation or act as a mechanical obstacle for fertilisation.
    (b) Ovarian cysts can be detected by ultrasound and then using ultrasound as a guide the cyst can be collapsed (aspirated) through the vagina or by laparoscopy. Cysts need to be removed before starting IVF treatment.
17. FSH is a hormone from the anterior pituitary gland that stimulates gamete production in the gonads. An increase in FSH levels in a female occurs due to follicle depletion. As levels vary in different women, other indicators are also needed for accurate diagnosis.
18. C
19. A

50 Genetic Screening

1. Genetic screening is done for a particular condition in individuals, groups or populations without family history of the condition.
2.

Screening	Description
Maternal serum screening	(MSS) is a blood test for pregnant women for the detection of Down syndrome and neural tube defects, e.g. spina bifida or trisomy 18.
Newborn screening	Is a public health program for all newborn babies and tests for several metabolic disorders including phenylketonuria (PKU), hypothyroidism and cystic fibrosis. These conditions are treatable but not always clinically evident at birth.
Carrier screening	To find out if a person has an allele for a genetic disorder, e.g. for screening for cystic fibrosis (CF) – detects the presence of the abnormal CFTR allele. Fragile X syndrome and spinal muscular atrophy.
Thalassaemia screening	Usually for couples in antenatal care screened for haemoglobinopathies which have defects in the structure of the haemoglobin molecule in red blood cells.

3. The Western Australian state government provides many types of health services including population screening of newborns and second trimester foetuses for selected genetic conditions, diagnostic and predictive testing, and counselling services for people affected by, or at risk of birth defects, genetic disorders of childhood or later onset genetic disorders such as familial cancer.
4. Gene test results can be used to assist diagnosis and direct the medical practitioner towards the most suitable treatment or tests can identify people with high risk factor for particular diseases that may be preventable. Test results can be used to inform couples about the likelihood of offspring inheriting a disease. Genetic screening can identify if a person has a particular genetic disorder. Experimental treatments that remove defective genes, e.g. in patients with cystic fibrosis, bone marrow disorders and Huntington's disease show that there can be large impacts on the wellbeing of patients improving their quality of life and reducing their dependency on drugs and constant medical attention. This has an economic effect on the health care system and the time and effort of care staff, family and community. However, issues can arise over confidentiality of information about the results of genetic screening impacting on a person's right to privacy and how this data is used, e.g. by insurance companies and employers. Knowledge about the specific genes involved in genetic diseases has led to the development of specific drugs that can better target a disease. This aids patients but commercialisation of the drugs with associated costs and market distribution may mean only some people are able to access the new drugs. This can cause ethical issues of inequality.
5. The provision of genetic counsellors by GSWA is extremely important as many people need supportive and informative help to help make decisions about future actions after receiving the results of genetic testing. Many, both the individual involved, family members and close friends may also need help to adjust to the news.

51 Screening Embryos

1. Screening tests look for the increased risk of a birth defect and look at data from a blood test and an ultrasound, whereas diagnostic tests identify particular conditions and use data from chorionic villus sampling, amniocentesis and ultrasound.
2. Diagnostic tests include – 1. Testing for chromosome conditions, e.g. trisomy 13, trisomy 18. 2. Testing for conditions, e.g. neural tube defects, congenital heart conditions and congenital kidney conditions. 3. Check for sickle cell and thalassemia for at risk women.
3. Some of the blood tests carried out during a pregnancy include – 1. Full blood count of mother in first trimester. 2. Mother's glucose levels for maternal diabetes. 3. Mother's blood group and antibodies present. 4. Check for infectious diseases of mother, e.g. syphilis, rubella, hepatitis B, C, HIV antibodies, chlamydia screening.
4. (a) Amniocentesis is a procedure of prenatal diagnosis which removes about 10 to 20 mL of amniotic fluid from the womb of a pregnant woman by aspiration from a needle inserted into the uterus. The fluid is analysed for certain genetic and congenital defects in the foetus.
    (b) The test is usually carried out during weeks 16 to 20 of pregnancy.
    (c) The test is usually recommended for women over 35 years of age having their first child and those with a family history of a genetic disorder.
5. Risks associated with an amniocentesis include – 1. The procedure can involve foetal injury if carried out too early. 2. Potential miscarriage. 3. Possible Rh sensitisation if Rh positive foetal blood contacts the mother's Rh negative blood. 4. Potential infection transmission, e.g. HIV, hepatitis C from mother to foetus. 5. Potential risk of leakage of amniotic fluid. 6. There is a risk of the needle contacting the foetus causing needle injury.
6. (a) Chorionic villus sampling is a procedure of prenatal diagnosis that involves using a long narrow tube through the cervix to take foetal cells from the chorionic membrane.
    (b) A chorionic villus sample is usually taken during weeks 9 to 13 of pregnancy.
    (c) It is considered to be an invasive technique as it involves entry into the placenta by the insertion of an instrument.
7. Chorionic villus sampling has a higher risk of miscarriage and a higher risk of infection than an amniocentesis. There is the possibility of leakage of amniotic fluid which can affect foetal development and a risk of Rh sensitisation in both procedures.
8. Preimplantation genetic diagnosis (PGD) is a technique used to screen embryos before pregnancy.
9. A couple planning to have children who have family histories of a genetic disorder will use preimplantation diagnosis to check if the embryos that were formed in in vitro fertilisation are carrying the abnormal genes for the disorder. An embryo that is not carrying the mutant gene will be chosen for implantation and their child will not be born with the genetic disorder.
10. The ART guidelines by the Australian government and the National Health and Medical Research Council (NHMRC) restrict the use of PGD (preimplantation genetic diagnosis) for the use of sex selection except for cases where its use will restrict the transmission of a specific genetic disorder. This is important to maintain a gender balance in society. Some countries which have used technology in sex determination of children now have serious social issues with an imbalance of males to females.

11. Ethical concerns include – 1. Potential parents may face many ethical issues when embryo screening provides data about an embryo and the data may affect their decision about having children, terminating a pregnancy or to having a child with a genetic disorder. 2. In the process of applying assisted reproductive technologies a certain amount of data about an embryo can be compiled before implantation or birth. Ethical issues arise when potential parents may wish to use the data to choose the sex of their child or choose other features that are not related to avoiding their offspring inheriting a particular genetic disease. 3. Equity issues arise due to the cost of assisted reproductive technologies. Most procedures are expensive. Who should pay for the procedures? Should ART be available to all couples in all socioeconomic brackets? 4. Privacy issues arise with specific data being compiled about a person's genome. General access to this information could affect the ability of a person to get insurance for health care, get a job or get suitable life insurance. 5. There are several questions that need to be answered – What happens to unused frozen embryos, sperm, eggs? Who has legal rights to unused frozen embryos, sperm, eggs?
12. D

52 Pap Smears and Breast Screening

1. A Pap smear is a screening test that detects the early signs of cervical cancer.
2. The Pap smear was named after Dr Papanicolaou who discovered in 1928 that there is a change in the appearance of cervical cells before they become cancerous. He developed a stain to detect these cells.
3. Dr Papanicolaou published his book '*Diagnosis of Uterine Cancer by the Vaginal Smear*' in 1943 providing information about cervical cancer.
4. New testing procedures include the ThinPrep collection technique and liquid based cytology.
5. In a Pap smear cervical cells are swabbed from the cervix at the neck of the uterus. The cells are smeared onto a glass slide and sent to a pathology laboratory which analyses the cells looking for a change in the size and shape of the cells (dysplasia). The results are sent to the Cervical Screening Registry (CSR) of Western Australia with the results given as negative or normal, unsatisfactory or abnormal.
6. A negative result occurs for about 9 out of 10 Pap smear results and the woman is advised that another Pap smear is not needed for another 2 years.
7. (a) An unsatisfactory Pap smear result means there was a problem collecting cells which could occur due to a variety of reasons – 1. Mucus or blood obscured the cervical cells on the slide. 2. Not enough cells were obtained. 3. The smear was not properly prepared. 4.The slide was broken in transport.
   (b) The woman is advised that another Pap smear is required within 3 months.
8. A colposcope is an instrument used to examine the vagina and the cervix of the womb.
9. Other tests that are also carried out in conjunction with Pap smears include a test for HPV (human papillomarvirus) and optoelectronic screening.
10. Removal of abnormal cervical cells can be done by – 1. Using a wire loop to remove abnormal cells from the cervix. 2. Using a laser under a local anaesthetic or a short stay in hospital under a general anaesthetic. 3. By diathermy using heat to destroy the cells. 5. By a cone biopsy to remove a small piece of cervical tissue. 6. Having a hysterectomy which removes the uterus.
11. Mammography is a radiography screening method that uses low energy X-rays to check breasts for tumours.
12. BreastScreen Australia is a national program that promotes free two-yearly breast screening to women aged 50 to 74.
13. The screening process involves taking two X-rays of each breast one from the side and one from the top.
14. Women with high density breasts have a slightly greater risk of getting breast cancer and the white fibrous supportive tissue can obscure breast lesions in a mammogram. Thus additional checks, e.g. with the GP or ultrasound and/or MRI may be needed.
15. Further developments in technologies that have aided detection and diagnosis of breast cancer include digital mammograms, nuclear medicine techniques such as molecular breast imaging, ultrasonography and magnetic resonance imaging.
16. Risk factors associated with breast cancer include – 1. Genetic inherited predisposition and family history, e.g. they have mutations in the *BRCA1* or *BRCA2* genes (*BR*east *CA*ncer genes). 2. Exposure to ionising radiation. 3. Being childless or giving birth after 30 years of age. 4. Hormone replacement therapy during menopause. 5. Gaining a lot of weight in adulthood especially after menopause. 6. Drinking alcohol, e.g. more than 2 standard drinks a day. 7. Early age at first menstruation, e.g. before 12 years.
17. (a) *BRCA1* or *BRCA2* genes (*BR*east *CA*ncer genes). *BRCA2* are human genes that produce tumour suppressor proteins that repair damaged DNA.
    (b) Mutated forms of these genes are associated with several types of cancer, e.g. breast and ovarian cancers.
18. (a) A DNA chip is a DNA microarray DNA chip which is a small amount of a large number of single stranded DNA fragments of a specific gene fixed to a glass slide in a tightly spaced grid.
    (b) DNA chips are used in genetic testing to check for specific genes, e.g. mutant forms of *BRCA1* and *BRCA2*. They can analyse thousands of genes simultaneously.
19. C

53 Prostate Cancer

1. The prostate gland is situated in the pelvic cavity in front of the rectum.
2. The prostate gland consists of an outer fibrous covering, a layer of smooth muscle and glandular tissue composed of columnar epithelial cells.
3. Fluid from the prostate gland contains calcium, citrate ions, phosphate ions, a clotting enzyme, profibrinolysin and is slightly alkaline.
4. Being slightly alkaline means that the sperm are protected against the acid secretions from the vagina. The alkaline fluid will slightly neutralise the acids and the drop in pH to 6.5 to 6.0 helps sperm motility and fertility.
5. The clotting agents in the fluid from the prostate gland help thicken the semen in the vagina increasing the probability they will remain near the cervix.
6. Prostate cancer is the development of a tumour in the prostate gland of the male reproductive system.
7. (a) A DRE is a digital rectal examination.
   (b) In a DRE the doctor inserts a gloved finger into the rectum to check the size of the prostate gland and any other changes that may be present.
8. (a) PSA stands for prostate specific antigen which is a protein that helps nourish sperm.
   (b) The PSA test is a test that measures the amount of PSA in a man's blood in ng/mL.
9. Many of the symptoms of prostate cancer relate to urination, e.g. there is a need to urinate frequently, especially at night and difficulty beginning to urinate, incontinence, a slow, interrupted flow of urine, blood in urine or semen, lower back or pelvic pain and painful ejaculation.
10. After high PSA levels are detected regular blood tests are carried out to monitor PSA levels. This is to ensure that the raised PSA levels are not due to an infection of the prostate or the enlargement is a benign prostatic hyperplasia (BPH).
11. The cancerous area is usually removed by surgery.
12. If surgery is carried out then in many instances a pelvic lymph node dissection is carried out as well removing nearby lymph nodes to check if the cancer cells have entered the lymphatic system.

54 Genome, Gene and Allele

1. The genome is the complete genetic information of an organism.
2. Genomics is the study of whole sets of genes and their interactions, e.g. it includes how particular genes are expressed, how they are regulated, their interactions with other genes and how gene expression affects growth, development and evolution.
3. In many cases the genome of prokaryotes is a single DNA molecule not bound by a membrane while the genome of eukaryotes usually consists of a number of DNA molecules in the nucleus which is surrounded by a membrane.
4. A chromosome is a cellular structure that holds genetic information in the coding of the DNA molecule.
5. A gene is a heritable factor that determines a specific characteristic, e.g. gene for stem height in pea plants. It is a section of DNA coding for proteins that expresses itself as the phenotype for that trait.
6. An allele is an alternative for a particular trait, e.g. there are two alleles for height in pea plants – tall (T) or short (t).
7. If two alleles at a locus differ then one will be the dominant allele and will show in the phenotype and the other will be the recessive allele and will only show in the phenotype if there is no dominant allele present.
8. (a) PKU is phenylketonuria, an autosomal recessive genetic disorder that causes problems with brain development.
   (b) The gene involved with PKU is the PAH gene on chromosome 12. This gene codes for the enzyme phenylalanine hydroxylase which is needed for the reaction that breaks down the amino acid phenylalanine into the amino acid tyrosine.
   (c) PKU is detected by screening newborns 2 to 7 days after birth with tests for PKU and if a child is diagnosed with PKU it is prescribed a diet that is low in phenylalanine and high in tyrosine for the rest of their life, e.g. diet low in meat, chicken, nuts, cheese and potatoes with tyrosine supplements.
9. B
10. C

55 Gregor Mendel and His Experiments

1. Mendel studied the inheritance of seed shape in pea plants. He crossed purebreeding round seeds with purebreeding wrinkled seeds and found that the first generation were all round seeds. When he crossed two of the offspring he found that the ratio was 3 round : 1 wrinkled. Mendel concluded that there were two factors for a character and one (e.g. round seed shape) was dominant over the other (e.g. wrinkled seed shape).
2. Mendel repeated his experiments many times with a large number of crosses, which gives his results statistical verification. He was meticulous in making the crosses using purebreeding lines and transferring pollen from one plant to another using an artist's brush, preventing random pollination. In each experiment he investigated a limited number of characteristics, e.g. round or wrinkled seed shape.
3. Mendel worked out the law of segregation, which states that for each characteristic there are two factors which segregate, with one factor in each reproductive cell, and then recombine at fertilisation. The characteristics do not blend; one dominates over the other. He also worked out the law of independent assortment which states that pairs of factors segregate independently of other pairs of factors when forming reproductive cells.
4. Mendel's work was presented to a small group of scientists. Since he was an Augustinian monk and did not really move in scientific circles, the significance of his work was not recognised.
5. A monohybrid cross is when only one characteristic is considered. In such a cross an individual may be homozygous, i.e. be purebreeding with both alleles the same, e.g. TT or tt; or the individual may be heterozygous where the two alleles are not the same, e.g. Tt.
6. A gene refers to one characteristic, e.g. pea plants have a gene for height; and an allele is the two or more forms of that characteristic, e.g. the allele for tall height or the allele for dwarf height.
7. When both alleles are present in an individual, the dominant allele will show its characteristic, e.g. if T = tall allele and t = dwarf allele, in an individual Tt, the phenotype is tall.
8. Genotype refers to the actual genes present, e.g. TT, Tt, tt. Phenotype refers to the physical appearance, e.g. tall, dwarf.
9. A
10. D
11. C
12. D
13. A
14. B
15. B
16. D
17. C
18. B
19. C

56 Writing Genotypes

1. The genotype shows the genetic make-up or set of alleles of an organism that control a characteristic.
2. When writing the genotype for genotype it is customary to: 1. Represent the gene with a letter of the alphabet, e.g. for plant height the letter T can be chosen. It is a good idea to choose a letter of the alphabet that looks different in upper case and lower case. 2. The genotype will have two letters representing the two alleles – one from each parent. 3. The dominant allele is capitalised, e.g. T = tall pea plant height and the recessive allele is lower case, e.g. t = dwarf pea plant height. 4. The dominant allele is written first.
3. When both alleles are the same in the organism, e.g. TT or tt, the organism is homozygous for that trait. If the alleles are different, e.g. Tt, then the organism is heterozygous for that trait.
4. A monohybrid cross is a mating where a single pair of characteristics is studied.
5. (a) In a monohybrid test cross an organism of unknown genotype showing the dominant characteristic is crossed with an organism that is homozygous recessive for that trait.
   (b) A monohybrid test cross is carried out to find out the unknown genotype – if the organism showing the dominant trait is homozygous for that trait or heterozygous for that trait.
   (c) The unknown genotype is determined by looking at the phenotypes of the offspring. If any offspring show the recessive trait then the unknown genotype is heterozygous. If sufficient numbers of offspring are produced and none show the recessive trait then the genotype is likely to be homozygous.
6. Probability ratios are exactly that – theoretical ratios that will show in a large statistical sample. In real life situations, e.g. a couple have 4 children, the probability ratio is 2 girls to 2 boys. However, 4 offspring is a small sample size and the couple could have 4 girls and no sons. Thus probability models do not always happen in real life situations.
7. If you have a fruit fly with straight wings it could be homozygous (TT) or heterozygous (Tt) to show the dominant trait. To find its genotype you need to perform a test cross and mate it with a homozygous recessive (tt) curly wing fruit fly. With a significant number of offspring, e.g. over 20 offspring, if none are curly wing then your fly is homozygous (TT). If any offspring have curly wing then your fly must be heterozygous (Tt).
8. A

57 Probability Models

1. Gregor Mendel experimented with pea plants and his conclusions formed the basis of the study of heredity. For example, RC Punnett built on the work of Mendel, using statistics and the 'Punnett square' to predict the outcome of particular crosses. Mendel's work influenced the direction of following scientists as they tried to fully explain inheritance and the probability of a particular characteristic being present in the next generation.
2. To show if a tall pea plant was homozygous or heterozygous you would let it self-pollinate. If the offspring were all tall then the pea plant is most likely to be genotype TT; however, if there is a mixture of tall and short, then the genotype is Tt.

3. Let homozygous long-eared mouse = LL and homozygous short-eared mouse = ll.
Cross is LL (long-eared) × ll (short-eared)

	L	L
l	Ll	Ll
l	Ll	Ll

F_1 genotype = 100% Ll – all offspring are Ll
F_1 phenotype = 100% long-eared
Cross is Ll (long-eared) × Ll (long-eared)

	L	l
L	LL	Ll
l	Ll	ll

F_2 genotype = 1 LL : 2Ll : 1 ll
F_2 phenotype = 3 long-eared : 1 short-eared
4. Cross is Tt (heterozygous taster) × tt (non-taster)

	T	t
t	Tt	tt
t	Tt	tt

Offspring are 2Tt (taster) : 2 tt (non-taster)
There is a 50% chance that the offspring will be non-tasters.
5. If both parents have long eyelashes then they both must carry the dominant gene (L). For them to have a child with short eyelashes (ll), each parent must each carry one 'l'. Therefore the genotype of each parent must be 'Ll' – they are heterozygous for this trait.
6. (a) Normal parent rabbit = Aa, and the albino parent rabbit = aa.
Cross is Aa (normal) × aa (albino)

	A	a
a	Aa	aa
a	Aa	aa

This shows that the offspring will be a mixture of normal rabbits (Aa) and albino rabbits (aa) which is approximately 50 : 50.
(b) Both normal parent rabbits were heterozygous (Aa).
Cross is Aa (normal) × Aa (normal)

	A	a
A	AA	Aa
a	Aa	aa

Offspring are 3 normal : 1 albino.
7. (a) (i) Thin lips = tt;
(ii) Thick lips can be either TT or Tt.
(b) Two people with thin lips would both have the genotype tt, which means they can only form gametes with 't' and hence all offspring will be 'tt' – with thin lips. Two thick lip people could each have the genotype Tt, which means there is 3 thick lip : 1 thin lip probability of both phenotypes appearing in the offspring.
8. C
9. A
10. A
11. B

58 Variation in Genotype

1. Four processes that lead to variation in the genotypes of offspring include: 1. Mutation. 2. Crossing over in meiosis. 3. Random segregation in meiosis. 4. Random fertilisation.
2. Mutation can be caused by gamma rays, UV radiation, bromine, nitrous acid, X-rays and radioactive decay.
3. DNA can be changed by small scale alterations, e.g. point mutations with single nucleotide-pair substitution, insertions or deletions, or large scale chromosomal mutations with duplication and alteration of chromosome structure.
4. A mutation changes the permanent structure of the genetic information. If the change leads to a favourable characteristic that aids survival and the change is in a germ line cell, then those with the mutation will survive due to natural selection and will pass the changed characteristic on to their offspring. Over time the characteristic becomes more common in the population and the population changes. A change in a population is evolution and the change has led to evolution.
5. Random segregation is when the two alleles for each gene separate during meiosis and end up in different gametes.
6. If a cell is $2n = 6$ then it can make 2^3 different gametes = 8. (See the diagram in the chapter.)
7. The diagram shows crossing over of an homologous pair of chromosomes during meiosis. Crossing over involves the exchange of corresponding segments of DNA molecules. Crossing over leads to new gene combinations which is a source of variation in genotype.
8. Sexual reproduction involves the union of two gametes. It is random which gamete joins with which other gamete. This means one parent couple will produce a variety of different offspring.
9. C

59 Karyotyping

1. A karyotype is an arrangement of chromosome pairs according to size and shape. It is a property of a cell giving the number and type of chromosomes present in the nucleus and not a photograph or diagram of them.
2. A karyotype is prepared by collecting cells, culturing them and then halting cell division at metaphase. The cells are stained and then the chromosomes are sorted according to size and structure.
3. Karyotyping is carried out using cells that are dividing as this is when the chromosomes are clearly visible under the microscope.
4. A karyogram is a photograph or diagram of the chromosomes in a cell. The chromosomes are arranged in homologous pairs of decreasing length in a karyogram. The longest pair are chromosome 1.
5. Karyotyping is useful to look for chromosome anomalies, e.g. trisomies where there is an extra copy of a chromosome making three copies instead of two, or translocations where part of a chromosome has duplicated and been added to another chromosome. Karyotyping is also carried out if a female has recurrent miscarriages.
6. Klinefelter syndrome is a chromosomal disorder that occurs when males have two or more X chromosomes, e.g. a male has XXY. It is a congenital disorder detected by karyotyping. The additional X chromosome can come from the egg or the sperm or can be added during the early stages of embryo development. The additional X causes sterility and the male may have weaker muscles, poor coordination, less body hair, smaller genitals and breast growth.
7. Turner syndrome (TS) is a chromosomal disorder that occurs when females are missing part or a whole X chromosomes, e.g. a female has XO. The lack of the second X chromosome often causes low-set ears, low hairline, reduced breast development, heart defects, lack of menstrual periods and inability to produce children. There is no cure for Turner syndrome but treatments, e.g. hormone replacement therapy can aid growth development and breast development.
8. There is no cure for Turner syndrome but treatments, e.g. hormone replacement therapy can aid growth development and breast development and plastic surgery can remove the webbing on the neck.
9. Edwards syndrome is a chromosomal disorder when there is three copies of chromosome 18.
10. The three types of Edwards syndrome: 1. Full trisomy 18 with every cell in the body having the extra chromosome. 2. Partial trisomy 18 with only part of chromosome 18 additional and attached to another chromosome. 3. Mosaic trisomy 18 with some cells having the additional chromosome and some cells having the normal number of 46 chromosomes per cell.
11. (a) Trisomy 2 means that there is three copies of chromosome 2 present in cells.
(b) A mosaic means that some cells have an additional chromosome and some cells have the normal number of 46 chromosomes per cell.

12. Chromosome 2 is the second longest chromosome with many genes. The presence of the additional chromosome disrupts normal development, cell functioning and cell metabolism. Thus full trisomy 2 leads to a spontaneous abortion during pregnancy. Mosaics do not have the additional chromosome in every cell and the number and distribution of cells with the additional chromosome means they can survive and have a live birth.
13. C
14. B

60 Sex Chromosomes

1. The autosomes are chromosomes that are not directly involved in determining sex. The sex chromosomes are one pair of chromosomes that are directly responsible for determining the sex of an individual.
2. In humans the sex chromosomes are X and Y. A female is XX and a male is XY.
3. Karyotype P belongs to a male and karyotype Q belongs to a female. This can be determined by looking at the last pair of chromosomes. P has XY, which is male, while Q has XX, which is female.
4. Cross = XX × XY.

	X	X
X	XX	XX
Y	XY	XY

Offspring genotypes are 2XX : 2XY.
Phenotypes are 2 female : 2 male.
That is, there is a 50 : 50 chance of a child being male or female.

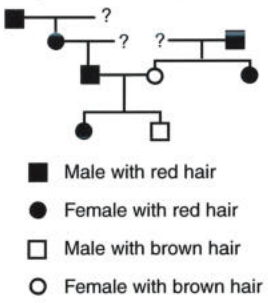
5. The father determines the sex of the child. The mother is XX, which means all eggs contain one X, while the sperm carry either an X or a Y. X-sperm will form daughters while Y-sperm will form sons.
6. Females are XX which means they would require two recessive alleles to show the trait, while males are XY and only need one recessive allele to show the trait. Thus males are more likely to show a recessive trait on the X chromosome, e.g. haemophilia, than females.
7. A hermaphrodite is an organism that has both male and female sexes. The term intersex is now used for humans.
8. The *SRY* gene is the sex-determining region on the Y chromosome. If this gene is present the gonads develop into testes. If this gene is not present the gonads develop into ovaries. The protein made by the *SRY* code starts male sex determination and regulates other genes which are also involved in developing fully functioning sex organs.
9. B
10. A
11. C

61 Polygenic Inheritance

1. Polygenic inheritance refers to a trait that involves several genes, e.g. human height is not the result of a single height gene.
2. Continuous variation occurs when there is a large variation in phenotypes between two extremes, e.g. when there is polygenic inheritance. Discontinuous variation is when there is only a few phenotypes, e.g. tall or dwarf pea plants as studied by Mendel.
3. The discontinuous graph, e.g. using Mendel's dominant and recessive trait will only have two columns, one for the recessive trait and one for the dominant trait. The continuous variation graph shows the typical bell-shaped curved of a normal distribution. Most individuals are in the middle range and fewer individuals have the traits found at either end.
4. A quantitative trait is a heritable feature that varies continuously over a range rather than an either-or manner. It depends upon the combination of each involved gene with each gene adding and contributing to the final appearance.
5. A polygene is a quantitative gene that is non-allelic and each gene affects the trait.
6. Pleiotropy is the ability of a single gene to have multiple effects.
7. QTL analysis is quantitative trait locus analysis and is a statistical method that links phenotypic data and genotypic data to explain the genetic basis for variation in complex traits.
8. The development of new technologies in the twenty-first century has rapidly increased our knowledge of quantitative traits. New technologies such as DNA sequencing methods and bioinformatics have increased both the amount of new data and the rate at which new data is gathered and analysed.
9. Three phenotypes you would expect to show the same phenotype would be AaBb, aaBB and AAbb.
10. Polygenic characteristics show a continuous variation from two extremes and are due to several genes contributing to the phenotype. Multiple alleles usually show discontinuous variation with a trait being displayed in more than two ways due to different combinations of alleles.
11. A trait in humans that is polygenic in origin is height, i.e. there is more than one pair of genes determining height and a trait that is determined by multiple alleles is blood groups, i.e. there are three possible alleles for blood group I^A, I^B and i.
12. C
13. A
14. B

62 Pedigrees

1. In a pedigree chart males are represented by squares and females are represented by circles. Shading indicates the particular phenotype being investigated. Each generation is set out in one line and the birth sequence in a family is read from left to right. The next generation will be on the line below.
2. (a)



(b) Man with red hair = rr; his wife with brown hair is Rr.

	R	r
r	Rr	rr
r	Rr	rr

Probability of future offspring having red hair is 50%.

3. To draw twin sons you would have two squares coming from the same point.

4. (a) There are 5 people in the family with the trait I2, II4, II5, III2 and III5.
   (b) The trait is recessive as parent II6 and II7 do not how the trait but they have a son III5 who does show the trait. Therefore the son has received a recessive allele from each of his parents to show the recessive trait.
5. Pedigree charts are drawn as they are graphical ways of picturing the ancestry of living things. They assist the study of the inheritance of genes by showing the genetic family history of specific traits.
6. Doctors and counsellors may draw pedigree charts for couples who are hoping to start a family if either or both of the pair have a family history of a particular hereditary disease. The pedigree chart will help explain to these couples about the chance of future children having a particular disease.

7. Queen Victoria was a carrier for the blood-clotting disease haemophilia and a study shows that it must be sex-linked as it only appears in males. Thus the family tree of the British royal family is often used to show how pedigrees can provide important information.
8. A
9. C
10. D
11. C
12. B
13. B
14. A

63 Codominance

1. Incomplete dominance occurs when the heterozygous form has a third phenotype – different to both homozygous forms.
2. (a) Codominance is when both alleles are expressed in the phenotype in the heterozygous form.
   (b) Codominance is a type of incomplete dominance as it has a third phenotype.
3. Codominance in humans occurs with Tay-Sachs disease, ABO blood group system and sickle cell anaemia.
4. The man is not the father of the child. The man is blood group AB and the woman is blood group A, which means she can be either homozygous AA, or heterozygous Ai. The child is blood group O and has the alleles ii. The child must have inherited one 'i' from its mother and the mother must therefore be Ai, and must have inherited the other 'i' from its father. Therefore the man with blood group AB cannot be the father of this child.
5. C
6. A
7. C

64 Multiple Alleles

1. An allele is a form of a gene for a particular characteristic having the code for a particular phenotype.
2. An example of multiple alleles in an organism is blood groups in humans with three alleles I^A, I^B and i.
3. Individuals can only have two alleles for a particular trait because each person inherits one allele from one parent and the other allele from the other parent, so there are two alleles for every trait in each individual.
4. In humans very few genes have multiple alleles – about 30% of the genes have two forms and about 70% exist in one form showing no variation.
5. Multiple alleles can be applied to society as they allow genetic fingerprinting. Many of the genes which have multiple alleles are concerned with tissue types and thus individual people will show the greatest variations from each other in these genes.
6.

Hair colour	Possible phenotypes
Dark brown	$C C$ $C c^m$ $C c^h$ $C c$
Mahogany brown	$c^m c^m$ $c^m c^h$ $c^m c$
Honey brown	$c^h c^h$ $c^h c$
Light brown	$c c$

7. (a) Monoallelic means there is only one form of the characteristics and there is no possible variation.
   (b) Around 70% of human genes are monoallelic.
8. D
9. A
10. B

65 Morgan and Sex-Linkage

1. Fruit flies are a suitable experimental organism as they have:
   1. Only four chromosomes which can be easily distinguished under a microscope. 2. Single mating can produce hundreds of offspring. 3. A new generation can be bred every two weeks.
2. TH Morgan investigated patterns of inheritance in fruit flies and found that the gene for eye colour was sex-linked and that it was located on the X-chromosome. This was the first gene to be linked to a specific chromosome. Morgan's results showed that the existing models for the inheritance of traits needed to be modified and a new model constructed to show sex-linkage. Mendel's ratio of 3 : 1 for dominant : recessive in a heterozygous cross does not hold if the sex of the offspring is considered. Morgan found there is still 3 red eye : 1 white eye in the F_2 generation but all the females have the dominant trait and half the males have the dominant trait and the other males have the recessive trait, i.e. only the males showed the recessive trait. Thus the work of TH Morgan was very important in contesting the existing model of inheritance and led to a new model that had greater explanatory power.
3. Let R = red (wild) allele and r = white allele.
   Parents' genotype = RR × rr.
   Parents' phenotype = purebreeding red, purebreeding white.

	R	R
r	Rr	Rr
r	Rr	Rr

   F_1 – all offspring have genotype Rr and red (wild) phenotype.
   Crossing two offspring to find F_2

	R	r
R	RR	Rr
r	Rr	rr

   F_2 offspring genotype = 1RR : 2Rr : 1rr.
   F_2 offspring phenotype = 3 red : 1 white.
4. (a) If all the F_1 were red-eyed, the allele for red eyes is dominant over the allele for white eyes.
   (b) With sex-linkage the alleles become X^R = red-eyed allele and X^r = white-eyed, and Y does not have this trait.
   Since all the F_1 had red eyes the parent would be X^RX^r and X^RY.

	X^R	X^r
X^R	X^RX^R – red-eyed female	X^RX^r – red-eyed female
Y	X^RY – red-eyed male	X^rY – white-eyed male

   (c) Yes, it is possible to have a white-eyed female, e.g. if a white-eyed male is crossed with a carrier female.

	X^R	X^r
X^r	X^RX^r	X^rX^r – white-eyed female
Y	X^RY	X^rY

5. (a) Number of different gametes = 2^4 = 16 different gametes.
   (b) Fertilisation is the union of two gametes and since it is random which sperm fertilises which ovum, each offspring is likely to have a unique combination of characteristics.
   (c) Mutation, e.g. by exposure to gamma rays, and certain chemicals will cause genetic changes and produce variants.
6. (a) Curly wing is the dominant trait as individuals 3 and 4 show the trait and they have offspring that do not, which means they are carrying the recessive gene (normal wing).
   (b) Using letter Q = curly allele and q = normal allele, individual 5 = Qq, individual 6 = qq and individual 20 could be either QQ or Qq.
   (c) Heterozygous cross = Qq × Qq

	Q	q
Q	QQ – curly wing	Qq – will develop wrong wings
q	Qq – will develop wrong wings	qq – normal wing

   Mendel's ratio should be 3 curly : 1 normal. In this instance the ratio will be 3 normal : 1 curly.

7. D
8. B
9. A
10. D
11. A
12. C

66 Human Blood Groups and Rhesus Factor

1.

Blood type	Possible allele combination for this blood type
A	I^AI^A or I^Ai
B	I^BI^B or I^Bi
AB	I^AI^B
O	ii

2. The antigens are on the surface of the red blood cells and the antibodies are in the blood plasma.
3. The phrase 'red blood cells agglutinate' means that the red cells clump together.
4. Blood type O is often called the universal donor as it is theoretically possible to use O blood in a transfusion to any other blood group. However, unless there is careful matching, antibodies in the donated plasma can damage the red blood cells of the recipient. AB is often called the universal recipient because blood type AB can receive blood from other all other groups.
5. To determine a person's blood group, you need a few drops of blood and to one sample add a serum containing anti-A antibodies and to another sample add a serum containing anti-B antibodies. Agglutination shows blood type, e.g. agglutination for anti-B antibody but not anti-A antibody means the person is blood type B and has antigen B present.
6. The rhesus factor was so named because it was first found in rhesus monkeys.
7. If you possess the main rhesus factor you are Rh+, and if you do not have this factor you are Rh−.
8. The Rh antigens seem to be needed for the transport of carbon dioxide and/or ammonia across the plasma membrane.
9. About 75% of the population is Rh+.
10. It is very important to know the Rh compatibility between the mother and foetus during a pregnancy. If the mother is Rh− and has had an Rh+ baby in a previous pregnancy it is highly likely that the mother has developed antibodies against the Rh antigen when there was leakage of the baby's blood into the mother's circulation during the childbirth of the previous baby. This means that during any *future* pregnancy for that mother, if the new foetus is Rh+, the mother's antibodies will cross the placenta and attack the foetus' red blood cells. This can cause anaemia and jaundice and even death of the foetus.
11. To solve haemolytic disease it is current practice to give the Rh− mother an injection of anti-Rh antibodies after the birth of an Rh+ baby. Once in the mother's bloodstream these acquired antibodies destroy any foetal cells that entered during childbirth and hence stop the mother's immune response becoming active. Thus the mother will not produce antibodies during her next pregnancy if the next foetus is Rh+ .
12.

Blood type of parents	Possible genotypes of parents	Possible blood types of children
A and A	AA and Ai Ai and Ai	A O
A and B	AA and BB Ai and Bi	AB AB, A, B, O
A and O	AA and ii Ai and ii	A A, O
A and AB	AA and AB Ai and AB	A, AB A, B

13. If the father is Rh+ he can have the alleles either Rh+ and Rh+ or Rh+ and Rh−. The mother only has Rh−. Therefore the children can be either Rh+ or Rh−.
14. Diagram A shows the 'before' adding anti-A antibody serum as you can see the individual red blood cells. Diagram B shows the 'after' adding anti-A antibody serum as the red blood cells have agglutinated – clumped together. The clumping is caused by the presence of the A antigen reacting to the addition of anti-A antibody.
15. To find out if claims of the environment influencing blood groups found in a population due to specific blood types being linked to particular diseases, you would need to undertake an epidemiologic study to compare the incidence of disease between the two blood groups (blood group A and O) and how this influenced the frequency of each blood group in the population.
16. The differences in the frequency of blood type O in European populations and North American Indian populations can be explained by the effects of isolation. It is believed that there was little variation in blood groups in the original migrants to North America and then many years of isolation with little interbreeding with other groups meant the gene pool did not change and blood group O increased. On the other hand there was extensive mingling between the peoples of Europe increasing the diversity of the gene pool and lowering the frequency of blood type O.
17. Marcus could be the father of Terumi's child. Terumi has blood group B which means she may have the alleles I^BI^B or I^Bi. Marcus has blood group A which means he may have the alleles I^AI^A or I^Ai. The following combination shows that it is possible for this couple to have a child with blood group O.

Parents	Terumi	×	Marcus
Genotypes	I^Bi	×	I^Ai
Gametes	I^B and i		I^A and i

	I^B	i
I^A	I^AI^B	I^Ai
i	I^Bi	ii

The child with ii will have blood group O and therefore Marcus could be the father.

18. To show how two parents with AB blood group can have three children with different blood groups, you need to look at the genotypes and possible gametes.

Parents genotype	AB	×	AB
Gametes	A and B		A and B

	A	**B**
A	AA	AB
B	AB	BB

Thus the children can have blood groups A, B or AB.

19. If Tom has blood group O his genotype is ii and thus each parent must have at least one 'i' allele. If Sam has blood group A one parent must have an 'I^A' and if Chris has blood group B one parent must have an 'I^B' allele. Mr and Mrs Chowdhury must have the genotypes I^Ai and I^Bi. There is insufficient data to assign a particular blood group to a particular parent.
20. A person could have shared blood with a non-identical twin before birth but due to some complication in the pregnancy the twin died early in gestation and was spontaneously aborted. Thus they were not born with a living twin but still have chimeric blood.
21. If both parents have blood group O, then they each only have the allele 'i'. This means that all their children must also be ii and thus have blood group O.
22. To transfuse blood, you need to match the blood groups or the red blood cells will agglutinate. Early attempts to transfuse blood did not take into account blood type and so they may or may not have been successful depending on the blood types involved in the transfusion.

23. When considering blood transfusions you must consider the ABO blood group and the Rh system. If the ABO group is incompatible clumping of red blood cells can occur and if Rh negative and Rh positive are mixed the Rh antibodies will attack the red blood cells.
24. In 1909 Karl Landsteiner developed the modern system of blood groups. This was a major advance in scientific understanding showing that the blood from each individual had particular properties. His work led to successful transfusions which were used in World War I to save many lives. His interest in pathology and immunology led to the discovery of Rh factor in 1940 and influenced the direction of future biological research into the action of antibodies and the immune response.
25. D

67 Autosomal Genetic Diseases

1. (a) Huntington's disease is an autosomal dominant disease.
   (b) Phenylketonuria is an autosomal recessive disease.
   (c) Tay-Sachs disease is autosomal codominant at a molecular level.
2. Tay-Sachs disease is caused by a mutation of the *HEXA* gene on human chromosome 15. Chromosome 15 is an autosome and thus it is an autosomal disease. Since a person with two Tay-Sachs alleles has the disease and people without the Tay-Sachs allele or only one Tay-Sachs allele do not have the disease, it can be considered to be an autosomal recessive disease on an organisational level. However, a biochemical analysis of the enzymes present in a person heterozygous for Tay-Sachs disease shows that they produce equal quantities of the normal enzyme and the dysfunctional enzyme which means that on a biochemical level the Tay-Sachs allele is codominant with the normal allele.
3. (a) Huntington's disease is a degenerative disease of the nervous system.
   (b) Huntington's disease is caused by the presence of a lethal dominant allele on the short arm of chromosome 4 on the Huntington gene (*HTT*). The defective allele has increased numbers of the trinucleotide repeat.
   (c) If a person has the Huntington allele, e.g. they are Aa with A = Huntington allele and a = normal allele, then a child will have a 50% chance of inheriting the A allele and having Huntington's disease.
4. Phenylketonuria (PKU) is a genetic disease that can lead to mental retardation and seizures if left untreated. Screening newborns for PKU, e.g. with the Guthrie test can identify babies with the disease. Once identified the child can go on the 'diet for life' restricting foods high in phenylalanine, e.g. meat, eggs, nuts and dairy products. The mutant allele that causes the disease prevents the production of a functional form of the enzyme phenylalanine hydroxylase (PAH). Without this enzyme phenylalanine cannot be converted into tyrosine and the person will have a build-up of phenylalanine and its by-product phenylpyruvate, as well as a deficiency in tyrosine. It is the build-up of phenylalanine and phenylpyruvate to toxic levels in the blood that leads to the intellectual disabilities, seizures and other medical problems that are characteristic of PKU. Thus the screening of newborn children for PKU is vitally important in detecting babies with the disease and putting them on a treatment program that will reduce the effects of being unable to produce a functional PAH enzyme. With a suitable diet or diet and medication a person detected at birth with PKU can lead a life with fewer of the problems caused by PKU.
5. Symptoms of PKU include stunted and delayed growth, hypersensitivity, poor bone strength, limb and hand tremors that lead to mental retardation and seizures. The damage is mostly irreversible.
6. (a) Cystic fibrosis is a genetic disorder caused by a mutation in the gene cystic fibrosis transmembrane conductance regulator (CFTR) which is on chromosome 7. The normal allele codes for a membrane protein that helps the transport of chloride ions across cell membranes and the presence of one normal allele prevents the disease.
   (b) Cystic fibrosis is a genetic disorder as it is caused by a mutant allele. It is an autosomal disorder as the CFTR gene is on chromosome 7. It is a recessive disorder as a person needs two mutant alleles to have the disease cystic fibrosis.
   (c) Genetic testing for cystic fibrosis (CF) before birth can diagnose infants with the disease. This allows treatment programs to be put in place once the child is born. As the disease does not produce a functional membrane protein that allows the transport of chloride ions across cell membranes, a person with CF has difficulty with chloride ions crossing the epithelium which covers many internal organs, e.g. lungs, pancreas and liver. These epithelia can become coated in mucus and many people with CF have difficulty breathing. Parents can pound on the chest of their child to clear mucus from blocked airways and daily doses of antibiotics can prevent lung infections. Early diagnosis can prolong the life of a person with CF and can help them have an improved quality of life.
7.

Type of disease	Name of disease	Symptoms of disease
Autosomal recessive	Cystic fibrosis	Thick mucus production causes respiratory problems which can lead to inflammation and infections of the respiratory system, e.g. lungs. Secretions from the pancreas affect the digestive system blocking the movement of digestive enzymes. This leads to persistent cough with difficulty breathing, tiredness and poor appetite.
Autosomal dominant	Huntington's disease	Is a degenerative brain disorder causing mental and physical disabilities, e.g. jerky random and uncontrollable movements, stiffness. The ability to solve abstract problems decreases as the disease progresses. Symptoms usually appear between ages 30 to 50 but can start outside this range.
Autosomal codominant	Tay-Sachs disease	In Tay-Sachs disease the body is unable to metabolise certain lipids. Without the functional enzyme the lipids accumulate in brain cells and the child will have blindness, seizures, deteriorating motor and mental functions and shortened life expectancy.

8. D
9. A

68 Sex-Linked Genetic Disease

1. A sex-linked genetic disease means that there are abnormal alleles on the sex chromosomes that cause a genetic disease.
2. Red-green colour blindness and haemophilia are X-linked genetic diseases.
3. The gene for red-green colour blindness is on the X chromosome while the gene for blue-yellow colour blindness is on an autosome. Thus red-green colour blindness is more common in males (XY) as they only have one allele and the recessive trait will always show in the phenotype while females are XX and have two alleles for red-green colour blindness which means they can be carriers and need to be homozygous for the trait to show red-green colour blindness phenotype. Since blue-yellow colour blindness is on an autosome it requires two defective alleles in both males and females and thus numbers of affected people are about the same for each gender.

4. When learning colours as a child and talking about colours, everyone will agree that something is 'green' for example. However, it is only when people carry out a colourblind test, e.g. Ishihara's test that they realise they cannot distinguish the path or the number among the coloured dots and are thus colourblind.
5. Congenital colour blindness means the disorder is a genetic mutation that has existed since birth while an acquired defect can occur at any time during a person's life, e.g. due to certain drugs (e.g. barbiturates, cocaine, caffeine and tobacco) and certain diseases of the retina or optic nerve that can harm normal colour vision.
6. Electrical wires are coded by colour to distinguish live, neutral and earth wires. Mistakes due to incorrect colour identification could be dangerous, even fatal.
7. D
8. A

69 DNA Profiling

1. DNA profiling is working out the DNA profile for a particular individual.
2. To make a DNA profile, you first need to take DNA from some cells or tissues, e.g. hair, skin or blood. Restriction enzymes are then used to cut the DNA at specific points and the segments are sorted by size by a sieving process called electrophoresis. The DNA segments are next transferred to a nylon sheet and radioactive dyes or coloured probes are added to the nylon sheet to produce a pattern called the DNA fingerprint. Several probes (e.g. 5 to 10 or more) are used to build the DNA fingerprint and when the pattern is exposed to X-ray film the result is a series of bars.
3. DNA polymorphisms are sites of variation in the population. Since over 99% of human DNA is identical among individuals, these areas are highly important in DNA fingerprinting as they are the areas studied to identify specific individuals.
4. Restriction enzymes cut up DNA in a process called restriction. Each restriction enzyme is very specific cutting up short distinctive nucleotide sequences into restriction fragments.
5. A 'reflip' is a restriction fragment length polymorphism used for DNA fingerprinting.
6. Male 3 is the father of her son. By analysing the DNA pattern it can be seen that the son has the allele frequencies that correspond to the mother and male 3.
7. Forensic science is any area of science that is used in the courts and justice system or in public. DNA profiling is an important area of forensic science as biological evidence, e.g. blood, semen, hair, can be used to obtain a DNA sample that will allow the forensic scientist to identify the individual and thus the evidence can be used in court to convict criminals.
8. Many people can have the same blood and tissue type while DNA profiling is unique to individuals, except identical twins. Thus in a court of law DNA fingerprinting provides stronger evidence than the same blood or tissue type.
9. STRs are short tandem repeats which are specific base sequences within the genome that are highly variable in length from person to person. They are found at many specific loci on the DNA molecule and are used as markers in DNA fingerprinting to identify individuals. Different ethnic groups may have particular frequencies of these markers within the group and knowledge about these frequencies can help aid investigations and provide a basis for statistical calculations.
10. DNA profiling is used by animal breeders to accurately identify parentage of offspring, e.g. DNA profiling of Merino sheep has meant more accurate pedigree records and it is predicted that there could be a 25% increase in the rate of genetic gain.
11. (a) The diagram shows gel electrophoresis.
    (b) Gel electrophoresis is used to separate DNA fragments.
    (c) Fragment X is larger than fragment Y as smaller fragments travel further from the well than larger DNA fragments.
12. (a) Process shows four samples W, X, Y and Z being tested.
    (b) The standard is used so that comparisons can be made and each fragment can be identified.
    (c) When the electric field is switched on the DNA fragments move from the negative end of the field towards the positive end of the field.
    (d) Sample Y has the shortest fragment as this is the fragment that has moved the furthest in the gel.
    (e) Sample Z has three long fragments which have not travelled far in the gel.
13. D
14. C
15. D.

70 Technology for DNA Profiling

1. DNA profiling identifies particular DNA sequences and the resulting pattern can be used to identify individuals.
2. (a) The Sanger method of DNA sequencing or chain termination method is a popular method developed by Frederick Sanger in 1977 and is used to determine the sequence of nitrogenous bases.
    (b) The process involves four basic steps – 1. Denaturation – the DNA is denatured into single strands using heat. 2. Primer attachment and extension of bases – the gene of interest is cloned using recombinant DNA techniques and polymerised chain reaction (PCR). The DNA fragments are mixed with four nucleotides, DNA polymerase and a radioactively labelled primer that attaches to a known sequence at the beginning of the fragment. The mixture is divided between four test tubes and a different nucleotide is added to each. 3. Termination – replication in the test tube stops when the a replication terminating nucleotide, e.g. dideoxynucleotide is added and is inserted into the growing polynucleotide chain. This results in a mixture of DNA copies of different lengths. 4. Gel electrophoresis and Southern blotting are then used to separate and show the resulting sequences. Analysis of the sequence allows you to determine the original template.
3. Polymerase chain reaction (PCR) is a method used to amplify a specific DNA sequence by repeated cycles of synthesis using the enzyme DNA polymerase and specific primers.
4. As high temperatures cause DNA to unwind into single strands the discovery of these bacteria was extremely important as the enzymes in these bacteria do not denature at high temperatures. The bacterial enzymes can then be used to be used in PCR at high temperatures.
5. DNA polymerase is an enzyme that catalyses the elongation of new DNA at the replication fork by adding DNA nucleotides to an existing chain, e.g. in DNA replication.
6. The basic steps in PCR are: 1. DNA is heated to separate the double helix into single strands. 2. The mixture is cooled to allow the primers (short pieces of synthetic single stranded DNA) to bind to the ends of the genes 3. DNA polymerase and free nucleotides are added and the DNA polymerase copies the DNA strands, starting with the primers 4. When the strands are copied, the mixture is heated again and the process is repeated leading to large amounts of particular strands of DNA being created.
7. Gel electrophoresis uses short fragments of DNA that have been cut using restriction enzymes. The fragments are placed at one end of a gel layer coated onto a glass plate. There are electrodes at each end of the gel and when the electrical current is switched on the fragments move through the gel with distance travelled dependent on fragment size – short fragments move farther and fastest. When the voltage is turned off the fragments stop moving. A standard mixture is provided in its own lane to allow comparison. A fluorescent dye or radioactive probes are added so the result can be studied under UV light or by X-ray respectively.
8. Gene probes are usually short sequences of DNA which are complementary to a mutated sequence that causes a particular disease. Oligonucleotides are short segments of RNA or DNA that are very useful as genetic probes as they readily bind to their complement.

9. A problem with using gene probes can occur if a young person, who has a strong family medical history of an adult onset disorder (e.g. Huntington's disease, Alzheimer's disease) or some cancers (e.g. breast cancer gene *BRCA1*) and the result is positive, it does not necessarily mean the young person will definitely develop the disorder in later life. The test gives a probability or likelihood for the development of the disorder. Receiving the genetic information at a young age may require psychological adjustment and counselling especially if there are no therapies or diagnostics for treatment of that particular disorder.
10. Different fragment analysis techniques used in profiling include restriction fragment length polymorphism analysis (RFLP), short tandem repeat (STR) analysis and amplified fragment length polymorphism (ampFLP) analysis.

71 Implications of Genetic Testing

1. Genetic screening is done for a particular condition in individuals, groups or populations without family history of the condition while genetic testing is done for a particular condition where an individual is suspected of being at increased risk due to their family history or the result of a genetic screening test.
2. If a family has a history of a particular disorder, e.g. cystic fibrosis (CF) prenatal screening tests may identify a baby at being at an increased risk of having this particular problem. However, the information given to the parents is limited, e.g. the presence of two CF mutations does not exactly predict the severity of CF in the baby. Thus the couple need to make a decision on how to proceed based on partial knowledge facing moral and ethical issues that will affect the lives of several people.
3. A young person can be tested for adult-onset disorders such as some cancers (e.g. breast cancer gene *BRCA1*) and Alzheimer's disease. If the result is positive, it does not necessarily mean the young person will definitely develop the disorder in later life. The test gives a probability or likelihood for the development of the disorder. There can be modifying factors, e.g. other genes, environmental factors that affect the expression of this gene. The positive result can cause problems associated with the psychological adjustment of the young person about their future life quality and length and there can be problems if the gene is identified and there are no therapies or diagnostics for treatment.
4. If genetic testing is used for sex selection then moral, social and economic problems arise in the community. Some cultures have a preference for sons which leads to discrimination against females and can lead to a distortion of males to females in the population. This can affect employment, crime rates, social attitudes and demographics. In Australia it is illegal for sex selection unless there is a medical reason, e.g. if there is a risk of the child having a genetic disorder and the NHMRC has ethical guidelines on the use of assisted reproductive technology.

72 The Human Genome Project

1. The Human Genome Project was a program which wanted to map all the genes in the human genome.
2. The human genome contains about three billion base pairs in a haploid nucleus.
3. Haploid means there is only one copy of each chromosome present while diploid means there are two copies of each chromosome present.
4. Technological advances have had a great impact on the Human Genome Project. It was only conceived after Frederick Sanger developed DNA sequencing techniques in the mid 1970s and the project was completed ahead of schedule due to the development of new techniques such as the use of the polymerase chain reaction, restriction fragment-length polymorphisms, and pulsed-field gel electrophoresis.
5. The complete human genome was finished in 2003.
6. The HGP could not be achieved by studying linkage maps as these maps only give relative locations of genes rather than absolute position. Another reason why linkage maps could not be used to determine the human genome is that linkage maps are created using the three-factor cross principle and you cannot use humans in such breeding experiments.
7. One limitation of the HGP relates to the way the collected information is used. There are many legal, ethical and social questions which have arisen due to the access to personal genetic information about individuals. For example, DNA information is now used in law courts in forensics to identify criminals, but can the DNA information of an individual be used for insurance purposes or by employers to discriminate against particular people?
8. One benefit of the HGP will occur when researchers who are using the data from the HGP enable physicians to be able to diagnose, treat and prevent disease. The data has already been used to create genetic tests that indicate a predisposition for particular diseases, e.g. breast cancer, cystic fibrosis and some haemostasis disorders.
9. Intellectual property rights of the data collected in the HGP could be claimed by the individual who provided the DNA, the researchers who collected the information, the employer(s) of the researchers, or the owners of the company/institution which carried out the research.
10. To protect individuals from the misuse of genetic information gathered in the Human Genome Project, governments can introduce guidelines and legislation to guard the rights and privacy of individuals.
11. Eugenics is the selective breeding of individuals that are considered 'superior'. Many people are concerned that the data collected in the HGP could be used to breed people with 'superior genes' which can lead to racial discrimination. For example, in the 20th century Hitler's desire to create a 'master race' led to the attempted extermination of Jews and gypsies.
12. Bioinformatics involves the construction, maintenance and use of databases to analyse the relationships in biological data, such as amino acid sequences or nucleotide sequences. The Human Genome Project (HGP) was a massive international scientific research project that aimed to map the complete DNA sequence of a human being and is an example of the rapid development of bioinformatics since the DNA sequencing process was automated in the 1980s. International cooperation, the sharing of new knowledge and technologies and the rapid advance of computing led to the HGP being completed ahead of schedule with the publication of the full sequence of the human genome in 2003. The work of the HGP is stored in databases on the internet with the human DNA sequences freely available on request and the data is extensively used by the international scientific community in a range of projects. Thus it was the cooperation of a vast international community involving specialists from a wide number of fields that developed the technologies and shared new information that led to the completion of the important bioinformatics project that was the HGP.

Topic Test

Section A – Multiple Choice (35 marks)

1. C	2. B	3. C	4. B	5. A
6. B	7. A	8. A	9. C	10. B
11. D	12. D	13. C	14. B	15. C
16. D	17. C	18. D	19. B	20. C
21. D	22. B	23. B	24. A	25. B
26. A	27. B	28. B	29. A	30. C
31. B	32. A	33. D	34. C	35. D

Section B – Written Response (65 marks)

36. (a) Marking Guidelines

Criteria	Marks
Correctly defines cell cycle AND Lists all the phases of the cell cycle	2
Correctly defines cell cycle OR Lists all the phases of the cell cycle	1

Sample answer:
The cell cycle is a series of events that occur in the life of eukaryotic cells. The cycle consists of mitosis, cytokinesis and the stages of interphase which are G_1 stage (growth stage 1), S stage (synthesis stage) and G_2 stage (growth stage 2).

(b) Marking Guidelines

Criteria	Marks
Outlines the significance of the G_1 restriction point AND Links it to the dividing or not dividing	2
Outlines the significance of the G_1 restriction point OR Links it to the dividing or not dividing	1

Sample answer:
The G_1 checkpoint is a significant restriction point in the cell cycle. If it receives a go-ahead then the rest of the cycle usually occurs and the cell divides. If the cell does not receive a go-ahead then the cell goes into a non-dividing phase (G_0 phase) from this G_1 restriction point.

37. (a) Marking Guidelines

Criteria	Marks
Correctly explains autosomal recessive	2
Correctly explains either autosomal or recessive	1

Sample answer:
A genetic disorder that is autosomal means that the gene involved is on an autosome – not a sex chromosome (X chromosome or Y chromosome). An autosomal recessive trait requires two recessive alleles for the recessive phenotype to show.

(b) Marking Guidelines

Criteria	Marks
Correctly links the genetic disorder to the breakdown of phenylalanine and relationship to restricted diet and symptoms of PKU	3
Two aspects of PKU, diet and symptoms	2
One aspect of PKU, diet and symptoms	1

Sample answer:
If a newborn infant is found to have phenylketonuria (PKU) the defective gene will prevent the breakdown of phenylalanine into the amino acid tyrosine. The build-up of phenylalanine will cause problems with brain development leading to mental retardation, stunted growth and limb and hand tremors. If the child is given a restricted diet that is low in phenylalanine and high in tyrosine, e.g. diet low in meat, chicken nuts and cheese they will not build up phenylalanine and have the symptoms of PKU.

38. (a) Marking Guidelines

Criteria	Marks
Gives correct definition of homologous chromosomes	1

Sample answer:
Homologous chromosomes are pairs of chromosomes of the same length and centromere position with one being inherited from the father and one from the mother. They look the same except for the chromosomes X and Y.

(b) Marking Guidelines

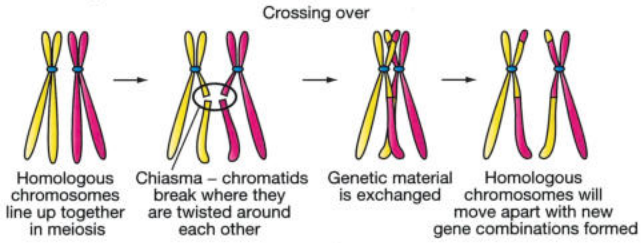
Criteria	Marks
Correctly identifies when homologous chromosomes pair in meiosis	1

Sample answer:
Homologous chromosomes pair during prophase I in meiosis.

(c) Marking Guidelines

Criteria	Marks
Draws a suitable labelled flow chart with at least two steps to show crossing over identifying chiasma and clearly showing swapped genetic material	3
Drawing not showing all elements in flow chart and/or reduced number of steps	1 to 2

Sample answer:



39. (a) Marking Guidelines

Criteria	Marks
Correctly explains codominance and multiple alleles	2
Correctly explains either codominance or multiple alleles	1

Sample answer:
Codominance is a type of incomplete dominance when both alleles are expressed. Multiple alleles is when there are more than two forms of a gene. An individual may only have two of those alleles.

(b) Marking Guidelines

Criteria	Marks
Draws a suitable Punnett grid and shows offspring with different blood groups	2
Draws some correct aspect showing parent and offspring inheritance of blood groups	1

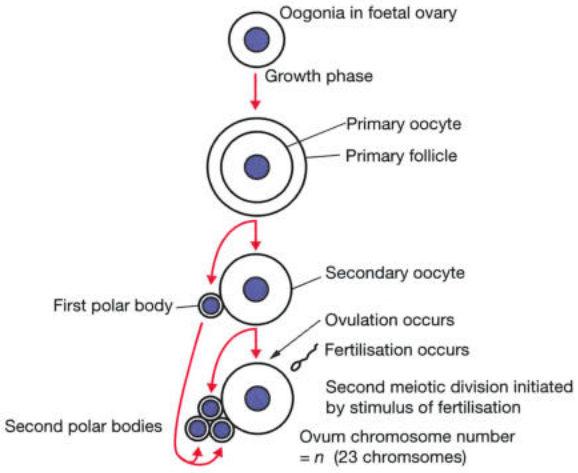
Sample answer:
Parents are I^Ai (blood group A) and I^Bi (blood group B)

	I^A	i
IB	I^AI^B Blood group AB	I^Bi Blood group B
i	I^Ai Blood group A	ii Blood group O

40. Marking Guidelines

Criteria	Marks
Correctly fills in all six missing labels	6
Correctly fills in some missing labels	1 to 5

Sample answer:



41. Marking Guidelines

Criteria	Marks
Correctly distinguishes between genetic testing and genetic screening	2
Defines either genetic testing or genetic screening	1

Sample answer:
Genetic screening is done for a particular condition in individuals, groups or populations without family history of the condition while genetic testing is done for a particular conditions where an individual is suspected of being at increased risk due to their family history or the result of a genetic screening test. Both genetic screening and genetic testing involve the same testing procedures.

42. Marking Guidelines

Criteria	Marks
Constructs a proper table with required headings and compares the causative agent, effects for males and females and treatment and control of two common STIs	8
Provides some comparison of two common STIs in a table with required headings	2 to 7
Draws a table with required headings and names at least one common STI	1

Sample answer:

STI	Causative agent	Effects	Treatment and control
Gonorrhoea	Bacteria – *Neisseria gonorrhoeae*	Males experience burning sensations in the urethra, pain in urination and yellow discharge from the penis usually within 2 to10 days of infection. The bacteria can spread to the prostate gland and male duct system causing inflammation. Females may show no symptoms with no pain, little or no discharge though some females have painful urination with cloudy urine, vomiting and diarrhoea. It can lead to pelvic inflammatory disease (PID) which can lead to infections of ovaries, oviducts and sterility. During birth passage through the birth canal can infect the child causing gonococcal conjunctivitis.	Treatment is with antibiotics, and control through safe sex practices, e.g. use condoms.
Chlamydia	Bacteria – *Chlamydia trachomatis*	In males it causes a burning in the urethra, pain in urination and a yellow discharge from the penis. In some cases it can lead to epididymitis and occasionally infertility. In female there are often no symptoms. Infection can cause bleeding between periods or after sex, pain when urinating and can lead to infertility (56% of females) if untreated, eye infections, arthritis, pelvic inflammatory disease and increase risk of premature or stillborn child.	Treatment involves antibiotics and prevention requires use of a condom as a preventive barrier.

43. Marking Guidelines

Criteria	Marks
Correctly defines metastasis and explains the formation of a breast cancer and how it can be detected earlier	4
Gives some details of development of breast cancer and detection	1 to 3

Sample answer:
Metastasis is the spread of cancer cells to a location away from their original site. The diagram shows a single abnormal cell that began dividing and formed a lump (tumour). A mammogram would have been able to detect this collection of abnormal cells in the lump and treatment could have begun before cells started to leave the original site. When the cells begin to migrate they move in the blood vessels and lymph vessels to other sites.

44. (a) Marking Guidelines

Criteria	Marks
Correctly identifies vessel A and vessel B	2
Correctly identifies vessel A or vessel B	1

Sample answer:
Vessel A is ductus arteriosus and vessel B is ductus venosus.

(b) Marking Guidelines

Criteria	Marks
Correctly identifies blood flow as through foramen ovale and explains why blood flow is needed across atria in foetus	4
Correctly explains some parts of foetal blood flow in the heart	1 to 3

Sample answer:
Blood flow X shows blood moving from the right atrium into the left atrium through the foramen ovale. While in the womb the foetus receives oxygen from the mother from across the placenta and the lungs are not functional until birth when the forman ovale closes up separating the atria.

45. (a) Marking Guidelines

Criteria	Marks
Correctly identifies family trees with suitable reasoning	2
Correctly identifies family trees OR Gives suitable reasoning	1

Sample answer:
Family 2 shows the inheritance of a recessive trait and must be the tree for PKU and the Lyle family. The other tree has insufficient information to determine inheritance but must be the Smith family and Huntington's disease which is a dominant trait.

(b) (i) Marking Guidelines

Criteria	Marks
Provides correct phenotypic ratios with working out that clearly supports the conclusion	3
Provides correct phenotypic ratios and some working out related to the cross	2
Provides some relevant information OR Provides incorrect phenotypic ratios with working out that supports this conclusion	1

Sample answer:
Family 2 shows the inheritance of a recessive autosomal trait PKU.
Let R = normal allele
R = PKU allele

	Alleles of partner with no history of PKU	
Parent 1 / Parent 2	R	R
Alleles of female with PKU: r	Rr	Rr
r	Rr	Rr

All offspring are genotype Rr which means there is zero chance that any of her children will have the recessive disease PKU. They will all be carriers of the disease.

(ii) Marking Guidelines

Criteria	Marks
Identifies late onset of symptoms of Huntington's disease showing dominant allele AND Links with reproductive age and production of next generation	3
Outlines some of the points relating to dominant allele, late onset of symptoms and link with reproductive age	1 to 2

Sample answer:
Huntington's disease is caused by a dominant allele, but since the symptoms do not show until the ages between 35 and 45 years, a person with one mutant allele will have the disease but is likely to have had children and passed the allele onto the next generation. This means that the frequency of a dominant mutant allele can stay high in a population.

46. Marking Guidelines

Criteria	Marks
Outlines the HGP and new technologies AND Shows how HGP is involved in disease AND Gives an evaluation	5 to 6
Outlines the HGP AND Some aspects of new technologies AND/OR Diseases AND/OR An evaluation	2 to 4
Makes a correct statement about HGP OR New technology	1

Sample answer:
The Human Genome Project (HGP) was a program which wanted to map all the genes in the human genome. Many new technologies have been used in the research work, e.g. many forms of recombinant DNA, e.g. DNA sequencing, restriction mapping and techniques such as gel electrophoresis and polymerase chain reaction (PCR) have been used to produce physical maps showing the position of genes and the sequences of bases in those genes. One of the many benefits of the HGP was the use of the data to diagnose, treat and prevent disease. The data combined with new technologies have enabled the creation of genetic tests that indicate a predisposition for particular diseases, e.g. breast cancer, cystic fibrosis and some haemostasis disorders. Newborn infants are tested for phenylketonuria (PKU), e.g. using the HPLC test or the Guthrie test as part of the national screening program for babies. Preventive medicine has also benefited as knowledge about an individual's genome can be used to provide a diet and lifestyle regime that will help maintain health and help find more effective drugs.
Thus the HGP and new technologies have been highly significant and highly beneficial in detecting harmful diseases.

47. Marking Guidelines

Criteria	Marks
Constructs a suitable table for the three reproductive technologies, explains how each works, when they are used during pregnancy and the information that is gained from the technology	8
Constructs a table for the three technologies with some description of why and when each technology is used	2 to 7
Constructs a table for the three technologies	1

Sample answer:

Technology	How it works	When it could be used during pregnancy	What information can be gained from it
Ultrasound	High frequency soundwaves are reflected off the foetal tissue to provide an image of the foetus.	Dating ultrasound at 8 to 10 weeks to confirm stage of pregnancy. 12 weeks and beyond for external probe.	Malformations of brain, head, limbs, other organs, spina bifida. Body dimensions can be measured.
Amniocentesis	Removing 10 to 20 mL of amniotic fluid which contains living cells from the foetus. Cells are checked, e.g. for chromosomal abnormalities and biochemical defects.	15 to 19 weeks	Tests for physical and intellectual problems that can affect the baby, e.g. Down syndrome, cystic fibrosis, neural tube defects, phenylketonuria, Duchenne muscular dystrophy, sickle cell disease.
Chorionic villus sampling	Foetal cells taken from chorionic membrane. Cells examined as for amniocentesis.	9 to 19 weeks	Diagnostic test for a range of physical and intellectual problems that can affect the baby. Genetic disorders, biochemical disorders, but not spina bifida.

Index

absolute risk 75
accuracy 5, 50, 84, 104, 109, 113
acetylcholine 63, 66
acquired immunodeficiency syndrome (AIDS) 77, 78
acrosome 52, 58, 63
actin 27, 52
adenine 2, 5, 7, 9, 11, 12, 18, 20
agglutination 104
albinism 91
alcohol 29, 74, 85
allele 29, 43, 46, 63, 81, 87, 90, 91, 96, 100, 101, 102, 104, 106, 108, 109, 113
alpha-helix 24
Alzheimer's disease 48, 81, 112, 113
amino acid 7, 9, 15, 18, 20, 22, 24, 26, 27, 87, 96, 106
amniocentesis 48, 82, 93
amphoteric 26
anaphase 32, 34, 36, 37, 43, 45, 51
androgens 57
aneuploidy 47
antibodies 28, 71, 82, 104
antiparallel 5
APGAR 71
apoptosis 65, 68
assisted reproductive technologies (ART) 79, 83, 107, 108, 113
Astbury, William 2
asymmetric division 37
Australian Bureau of Statistics (ABS) 75, 77, 78
autosome 47, 87, 93, 95, 102, 106, 108
Avery, Oswald 2, 5
bacteria 2, 3, 8, 10, 12, 14, 18, 28, 76, 77, 87, 106, 111
bacterial transformation 2
bacteriophage 3
basal cell carcinoma 42
basal ganglia 106
basal layer 55, 60
Beadle, George 2, 15
benign tumour 31, 33, 42, 86
bioinformatics 11, 96, 114
biomarker 79
biopsy 83, 84, 86
biotechnology 1
birth 31, 47, 53, 55, 59, 61, 69, 71, 74, 76, 77, 79, 81, 85, 87, 93, 98, 104, 106
biuret solution 24
bivalent 45
blastocyst 38, 40, 61, 63, 67, 70
blood chimera 105
blood group 29, 82, 100, 101, 103, 104
blood test 48, 77, 79, 81, 82, 86, 106
bonding 5, 12, 24
bone marrow 37, 38, 48, 81
BRCA1/*BRCA2* 76, 81, 85, 112, 113
breast 42, 49, 56, 57, 71, 76, 81, 84, 93, 98, 112, 113, 114
breast screening 84
BreastScreen Australia 84
breech birth 69
buffer 26
cancer 4, 11, 14, 17, 29, 31, 33, 40, 42, 74, 76, 78, 81, 84, 86, 105, 112, 114
cardiovascular disease 41, 72, 74
carrier proteins 27
carrier testing 81, 113
cartilage 27, 37, 39, 41
cell cycle 10, 32, 34, 37, 42, 65, 76
cell cycle checkpoints 32
cell cycle control system 32
cervical cap 72
Cervical Screening Registry 84
cervix 55, 60, 69, 72, 79, 82, 84, 86, 94
chancres 77
channel proteins 27
Chargaff, Erwin 2
Chase, Martha 3
chemotherapy 42
chiasmata 43, 45
chlamydia 77, 78, 82
chorion 38, 67
chorionic villus sampling (CVS) 31, 48, 82, 93
chromatid 11, 32, 34, 36, 37, 43, 45, 47, 51, 66
chromatin 9, 11, 17, 18, 29, 32, 34, 36, 84
chromatin-modifying enzymes 29
chromosome 1, 5, 7, 9, 11, 12, 17, 19, 30, 31, 34, 36, 37, 43, 45, 46, 47, 48, 51, 59, 76, 82, 87, 92, 93, 95, 102, 106, 107, 116
chromosome microarray 31, 113
circumcision 54
cleavage 35, 61, 64, 67
codominance 100, 104, 106
codon 8, 18, 20, 22
coexpressed genes 30
coitus interruptus 73
collagen 15, 25, 27
colostrum 71
colposcope 84
complement system 28
conception 61, 72, 79
condom 72, 77
conjugated protein 25
contraception 72
control elements 29
copulation 52, 53
copy number variations (CNV) 31
corpus albicans 59
corpus luteum 55, 57, 59, 60, 61
Cowper's gland 54, 86
Crick, Francis 3, 5, 7
cristae 9
crossing over 43, 45, 51, 92, 103
cryopreservation 79, 80
cyclins 32
cystic fibrosis 29, 31, 81, 107, 109, 113, 114
cytokinesis 32, 35, 36, 43, 45, 51
debate 2, 40
denaturation 25, 111
developmental biology 67
diabetes 41, 46, 74, 82, 113, 114
diagnostic test 31, 81, 82, 106, 113
diaphragm 72
diastema 90
differentiation 17, 32, 39, 40, 44, 68, 76
digital rectal examination (DRE) 86
dipeptide 24
diploid number 43, 45, 47, 51, 52, 63, 87, 92, 114
disulfide bridge 24, 28
DNA (deoxyribose nucleic acid) 1, 2, 4, 5, 7, 9, 11, 16, 18, 29, 32, 40, 48, 76, 85, 87, 92, 96, 106, 109, 111, 114
DNA damage removal 14
DNA damage reversal 14
DNA damage tolerance 14
DNA fingerprint 31, 109
DNA microassay 17
DNA polymerase 12, 14, 111
DNA repair 14, 65
DNA replication 1, 5, 7, 12, 31, 76, 111
double helix 3, 5, 7, 9, 11, 12, 17, 112
Down syndrome 47, 48, 81
Down, Langdon 48
Drosophila melanogaster 90, 102
ductus arteriosus 70
ductus venous 70
dysplasia 84
E site 20
ectoderm 38, 67

ectopic pregnancy 61
Edwards syndrome 94
electrophoresis 109, 111, 112, 114
electromagnetic spectrum 64
ELISA 78
elongation 20, 111
embryo 1, 27, 31, 35, 38, 40, 48, 52, 53, 55, 61, 66, 67, 71, 76, 79, 81, 82, 93
endocrine gland 53, 57, 59
endoderm 38, 67
endometriosis 60
endometrium 38, 55, 57, 59, 60, 62, 64, 67, 69, 72, 79
endosymbiont theory 9
enhancers 29
environment 1, 14, 25, 29, 42, 57, 74, 76, 77, 86, 92, 97, 105
enzyme 2, 8, 9, 12, 14, 15, 17, 18, 21, 22, 24, 27, 29, 52, 62, 67, 78, 81, 86, 87, 96, 100, 106, 109, 111
epididymis 44, 53, 58, 78, 86
epigenetic inheritance 29
epistasis 30
epithelial cell 30, 32, 35, 36, 45, 55, 58, 86
Escherichia coli 3, 12
ethical issues 4, 40, 81, 82, 106, 113, 114
eugenics 114
exocrine gland 56
exon 9, 16, 18, 29
experiment 2, 12, 15, 36, 40, 50, 81, 88, 91, 102, 104, 114
extranuclear genes 9
fertilisation 1, 38, 40, 43, 45, 47, 52, 59, 60, 61, 79, 81, 88, 92, 103
fibrous protein 24, 27
fimbriae 55
foetus 1, 31, 35, 38, 55, 59, 61, 67, 69, 74, 76, 77, 81, 82, 94, 104
follicle 55, 57, 59, 60, 61, 72, 80
follicle stimulating hormone (FSH) 57, 58, 59, 60, 80
fontanelles 70
foramen ovale 69
Franklin, Rosalind 3, 5
functional group 26
fungi 15
gamete 1, 43, 45, 47, 50, 51, 52, 55, 57, 63, 79, 91, 102
Garrod, Archibald 117
gastrula 67
gastrulation 67, 79
gene 1, 2, 5
gene expression 7, 9, 11, 16, 17, 18, 29, 65, 85, 87, 114
gene probe 112
gene test 81, 113
genetic predisposition 33
genetic profile 17, 31, 83, 107, 109, 111
genetic screening 81, 82, 113
genetics 45
genital warts 77, 78
genome 1, 11, 17, 19, 29, 31, 33, 42, 76, 83, 87, 96, 109, 111, 114
genome map 11
genome sequence 11, 114
genomics 87, 107
genotype 1, 29, 46, 88, 90, 91, 92, 97, 98, 100, 104, 107, 108
germ layer 1, 38, 67
germinal epithelium 55, 58, 59
gestation 1, 61
glans 54
globular protein 24, 27, 66
gluten 27
glycine 18, 20, 22, 24
gonad 43, 51, 52, 53, 55, 57, 76, 95
gonadotropin releasing hormone 57, 72
gonorrhoea 77
Griffith, Frederick 2
growth hormone 28, 89
Guthrie test 106
haematopoietic stem cell 39
haemoglobin 2, 15, 25, 27, 81, 100
haemolytic disease 104
haemophilia 31, 81, 95, 98, 102, 108
hair colour 96, 99, 101
haploid number 11, 15, 43, 45, 51, 63, 87, 92, 114
height 29, 87, 90, 94, 96, 97, 113
helicase enzyme 12
hermaphrodite 95
heroin 74
herpes 74, 77
Hershey, Alfred 3
Hershey-Chase experiment 3
heterozygous 46, 88, 90, 91, 98, 100, 103, 106
histone 9, 17
Homo sapiens 52
homologous chromosome 43, 45, 47
homozygous 88, 90, 91, 97, 100, 102, 108
hormones 18, 27, 37, 41, 53, 55, 57, 58, 59, 60, 61, 72, 79, 85
human chorionic gonadotropin (hCG) 61, 80
Human Genome Project 31, 114
human immunodeficiency virus (HIV) 78
human papillomavirus (HPV) 78, 84
Huntington's disease 31, 81, 106, 109, 112
hypothalamus 57, 60, 72
immunisation 75, 78
immunoglobulin 28, 71
implantation 38, 55, 60, 61, 67, 72, 79, 82
in vitro 40, 79, 81, 83
in vitro fertilisation (IVF) 40, 79, 81
induced pluripotent stem cell 38, 40
infant mortality 75, 113
infertility 62, 72, 78, 79, 93, 113
initiation 17, 20, 29
insulin 23, 28, 41, 63, 113
interferon 28
international 40, 96, 114
intersex 95
intrauterine device (IUD) 73
intron 9, 16, 18, 29
karyogram 47, 93, 95
karyotype 48, 82, 93, 95
keratin 15, 24, 27, 42, 84
keyhole surgery 86
kinases 32
Klinefelter syndome 47, 93
labour 69, 79
lactation 56, 71
lagging strand 12
Landsteiner, Karl 104
lanugo 69
law of independent assortment 43, 88
law of segregation 43, 88, 92
leading strand 12
Lejeune, Jerome 48
Levene, Phoebus 2
Leydig cell 53, 57
LGBT 95
light microscope 32, 36, 44, 47, 50, 54, 55
linked genes 43, 46, 102
luteinising hormone (LH) 28, 57, 58, 60, 61
lymphocyte 37
macular degeneration 41
magnetic resonance imaging (MRI) 85, 86
malignant tumour 33, 42, 46
mammary glands 56, 71
mammography 84
marijuana 75
maternal serum screening (MSS) 81
McCarty, Maclyn 2
McLeod, Colin 2
meiosis 1, 43, 45, 47, 50, 51, 59, 92, 94
melanocyte 30
membrane proteins 27
menarche 60
Mendel, Gregor 1, 88, 91, 96, 100, 102

menstruation 55, 57, 60, 61, 85
menstrual cycle 55, 57, 60, 61, 72
Meselson-Stahl experiment 12
mesenchymal stem cells 41
mesoderm 38, 67
metaphase 32, 34, 36, 43, 45, 51, 59, 93
metastasis 42
microfilament 27, 35
microtubules 24, 27, 35, 37, 43, 58
Miescher, Friedrich 2
milk duct 56, 85
mini pill 72
miscarriage 61, 75, 79, 82, 93
mitosis 1, 32, 34, 36, 37, 51, 64, 93
models 1, 3, 7, 12, 22, 36, 50, 90, 96, 100, 102, 106
monohybrid test cross 90
Morgan, Thomas Hunt 102
morning after pill 73
mosaics 48, 93
mRNA 7, 16, 17, 18, 20, 22, 29
mtDNA 9
multifactorial trait 29
multiple alleles 97, 101
multiple births 61
multiple sclerosis 41
multipotent 38
muscle 17, 27, 32, 35, 37, 38, 41, 45, 48, 53, 55, 57, 64, 67, 69, 71, 74, 85, 86, 93, 100, 106, 108
mutagen 1, 14, 33, 74, 76
mutation 1, 2, 5, 11, 14, 33, 42, 49, 74, 76, 84, 87, 92, 100, 106, 112, 113
myosin 27, 63, 66
natural family planning 72
neonate 71
neoplasm 33
nervous impulse 28, 66
neurosphere 39
Neurospora crassa 2, 15
newborn screening 81, 107
NHMRC 83, 113
nicotine 74
Nirenberg, Marshall 19
nitrogenous base 3, 5, 7, 11, 14, 18, 76, 111
Nobel Prize 3, 14, 15, 104
non-disjunction 47, 48, 51, 93
non-melanoma skin cancer 42
non-polar amino acid 26
norm of reaction 29
nuclein 2
nucleosome 9, 17
nucleotide 2, 5, 7, 11, 12, 16, 18, 20, 22, 29, 31, 92, 96, 106, 111, 114
oestrogen 55, 57, 59, 60, 61, 69, 72, 80
Okazaki fragment 12
oligonucleotide 112
oncogene 33
one gene-one polypeptide hypothesis 15
one gene-one protein hypothesis 15
oocyte 55, 59, 79, 82
oogenesis 59
organogenesis 38
origin of replication 12
ovarian cycle 60, 61, 72
ovarian reserve 79, 80
ovary 43, 51, 52, 55, 57, 59, 60, 72, 76, 77, 79, 95
oviduct 55, 62, 67, 72, 77, 79
ovulation 55, 59, 60, 72, 79
oxytocin 69
P site 20
Pap smear 78, 84
Papanicolaou, Georgios 84
parturition 1, 69
patterning 68
Pauling, Linus 2
pedigree 1, 98, 103, 106, 109
penis 53, 56, 72, 77, 86
peptide bond 21, 24
phenotype 8, 29, 46, 63, 75, 87, 90, 92, 96, 98, 100, 101, 102, 104, 106, 108, 114
phenylketonuria (PKU) 74, 81, 87, 106
phenylthiourea 91
phosphodiester linkage 5
pituitary gland 57, 59, 60, 69, 72
placenta 31, 38, 57, 61, 64, 67, 69, 74, 77, 82, 94, 104
pleated sheet 24
pleiotropy 29, 97
pluripotent 38, 40
polar amino acid 26
polar body 59
poly-A tail 18
polygenic trait 30
polymerase chain reaction (PCR) 111, 114
polypeptide 6, 7, 15, 16, 17, 18, 20, 22, 24, 29
polysome 21
polyspermy 52, 63
positional information 68
positive feedback 69
pregnancy 1, 31, 55, 57, 59, 60, 61, 69, 71, 72, 74, 76, 79, 82, 94, 104
pregnancy test 61, 73, 79
preimplantation genetic diagnosis (PGD) 82
pre-mRNA 16, 18
prenatal testing 31
primary structure 24
primer 12, 111
probability model 90, 91, 102, 106
progesterone 56, 57, 59, 60, 61, 69, 72, 80
prolamines 27
proliferative phase 60
promoter 17, 18, 22
prophase 32, 34, 36, 43, 45, 51, 55, 59
prostaglandins 69
prostate cancer 42, 86
prostate gland 53, 57, 77, 86
prostate specific antigen (PSA) 86
protease 29
protein 2, 5, 7, 9, 15, 16, 17, 20, 22, 24, 26, 27, 29, 32, 37, 49, 52, 68, 76, 85, 87, 95, 106
proten synthesis 5, 7, 9, 11, 16, 19, 20, 22, 29, 76
proto-oncogene 33
Punnett, Reginald 91
quantitative trait 96
quaternary structure 24
radiation therapy 42
random assortment 51
reading frame 7
receptor proteins 28
red-green colour blindness 95, 102, 108
relative risk 75
release factor 20
reliability 36, 50, 72, 109
replication fork 12
reproductive system, female 1, 55
reproductive system, male 1, 53, 57, 86
research 2, 6, 18, 22, 31, 40, 49, 57, 68, 72, 83, 96, 105, 113, 114
restriction enzyme 109, 112
RFLPs 109, 112
Rhesus factor 104
rhythm method 73
ribosome 9, 20, 22, 29, 35
RNA (ribose nucleic acid) 7, 12, 14, 16, 18, 20, 22, 29
RNA capping 18
RNA polymerase 17, 18
RNA primase 12
RNA processing 16, 18
RNA splicing 18
rubella 74, 82
Sanger method 111
Sanger, Frederick 111, 114
scrotum 53, 79, 86
secondary structure 24

secretory phase 60
semen 53, 58, 62, 72, 79, 86
semiconservative model 12
seminiferous tubule 44, 46, 47, 53, 58
Sertoli cell 44, 53, 58
sex chromosome 45, 47, 93, 95, 102, 108
sexually transmitted infection (STI) 72, 77, 79
short tandem repeat (STR) 31, 109, 112
sickle cell anaemia 2, 29, 31, 81, 82, 100, 109
skin colour 29, 97
somatic cell 1, 40, 51, 76
Southern blotting 111
sperm (spermatozoa) 1, 9, 38, 43, 45, 52, 53, 58, 61, 63, 72, 76, 79, 86, 93
spermatogenesis 44, 58, 72
spermatogonia 58
spermicide 72
spindle fibre 34, 36, 37, 43, 51
squamous cell carcinoma 42
SRY gene 95
stem cell 1, 37, 38, 40, 65
stem cell research 40
sterilisation 72
sterility 53, 73, 77, 79, 93, 95
steroid hormone 28, 57, 72
sucrase 27
syndactyly 116
synthetase 20
syphilis 77, 82
Tatum, Edward 2, 15
Tay-Sachs disease 31, 81, 100, 106
technology 1, 17, 18, 75, 79, 82, 86, 109, 111, 113, 114
telophase 32, 34, 36, 43, 45, 51
template strand 18
teratogen 74
termination 20, 111
tertiary structure 24
testis 44, 46, 47, 50, 53, 86
testosterone 28, 53, 57, 72, 79, 93
tetrad 43, 45
tetranucleotide hypothesis 2
thalassemia 81, 82
thalidomide 74, 76
theca cells 55
thermophilic bacteria 111
thymine 2, 5, 7, 9, 11, 12, 14, 16, 18, 20, 23, 33
thyroid gland 39, 61, 67
TORCH syndrome 74
totipotent 38, 83
transcription 9, 16, 18, 21, 22, 29
transcription factors 29
translation 7, 9, 16, 19, 20, 22, 29
translocation 42, 48, 76, 92, 93
transport proteins 27
tripeptide 24
triplet code 7, 9, 16, 18, 20
trisomy 21, 47, 48
tRNA 20, 22
trophoblast 38, 67, 79
tubal ligation 72
tubulin 27
tumour 11, 28, 33, 42, 76, 84, 86, 87
Turner syndrome 47, 93
twins 31, 61, 98, 105, 109
ultrasound 31, 48, 80, 82, 85, 86, 115
umbilical cord 38, 82
universal donor 105
universality of genetic code 9, 18
uracil 7, 14, 16, 18, 20
urethra 1, 39, 53, 55, 67, 77, 86
uterus 31, 35, 55, 57, 60, 61, 67, 69, 72, 79, 82, 84, 94
vaccination 75, 78, 114
vagina 53, 55, 69, 72, 77, 84, 86
validity 50, 105
variation 31, 33, 43, 45, 51, 52, 92, 96, 101, 113
vas deferens 53, 72, 86
vasectomy 53, 72
venereal disease 77
vernix 69
vulva 55
Watson, James 3, 5, 7
Watson-Crick model 1, 3, 7, 12
Weiner, Alexander S 104
Wilkins, Maurice 3, 5
X chromosome 45, 47, 93, 95, 102, 108
X-ray diffraction 2, 3, 5, 28
X-ray 2, 3, 5, 14, 15, 28, 74, 76, 84, 92, 109, 112
Y chromosome 45, 47, 93, 95, 102, 108
ZIFT 79
zika virus 74
zona pellucida 52, 55, 59, 67
zygote 38, 43, 52, 61, 63, 67, 70, 79, 92, 106